

Learning Scores for Markovian Dynamics in Generative Diffusion

Companion Compendium

The extended research record

MSc Research Thesis

Data Science and Business Analytics, Bocconi University

A.Y. 2025–2026

About this volume. This is the complete, extended version of the thesis of the same title; the submitted thesis is its concise form. It carries every derivation written out, the step-by-step toolbox derivations, the rotating-ring precursor study, and the full reproducibility and aggregation appendices, and it is the reference for any result the concise thesis states without proof. Its content is that of the extended build preserved at the repository tag `extended-thesis-pre-compression`; chapter, section and equation numbers therefore differ from those of the submitted thesis, and every quantitative result is produced by the same frozen-output pipeline. Where this volume mentions “the archived technical compendium” it refers to the earlier, superseded technical compendium (September 2026), kept in the `archive/` folder of the distribution.

Contents

Disclosure on the Use of Artificial Intelligence	9
1 Introduction	10
1.1 Topic and motivation	10
1.2 Positioning and gap	13
1.3 Research questions	14
1.4 Contributions	15
1.5 Structure of the thesis	16
2 Statistical Mechanics, Energy-Based Models, and Diffusion	18
2.1 What statistical mechanics is, and why it is here	18
2.2 Energy-based models: the same object, learned	20
2.3 Generative diffusion	22
2.3.1 The forward process is a relaxation to a Boltzmann equilibrium . . .	23
2.3.2 Why a path, and not a partition function	24
2.4 Three scores	31
2.5 Transitions along the generation trajectory	36
2.6 Locality, architecture, and message passing	39
2.7 Other generative families	41
3 The Model, and the Object of Study	43
3.1 Markov sequences	43
3.2 The Ornstein–Uhlenbeck corruption channel	46

3.3	How the object of study was arrived at	48
3.4	Why the joint score	49
3.5	Markov random fields and factor graphs	52
3.6	Belief Propagation	54
4	The Rotating Ring: A Two-Dimensional Dynamic Object	57
4.1	Model and notation	57
4.1.1	The clean process	57
4.1.2	The radial drift is a confining force	58
4.1.3	One trajectory is one data point	59
4.2	The gauge, and the exact score	60
4.2.1	The surrogate, and what it keeps	60
4.2.2	The known rotation is an orthogonal change of frame	61
4.2.3	Why the surrogate is worth solving	63
4.3	Does the score know the data are rotating?	67
4.3.1	One frame: no	67
4.3.2	The trajectory: yes, and here is how	68
4.4	Chapter conclusion	70
5	The Gaussian Chain by Linear Algebra	71
5.1	The joint law and the object of study	71
5.2	The clean law: from the recursion to the covariance	72
5.2.1	Unrolling the recursion	72
5.2.2	The stationary covariance is Toeplitz	75
5.3	The clean law: the precision is tridiagonal	76
5.3.1	Reading the precision off a short chain	76
5.3.2	The tridiagonal precision and what its zeros mean	77
5.4	The noisy law	81
5.5	The score, two ways	84
5.5.1	Linear algebra	84

5.5.2	Bayesian inference	85
5.6	The spectral form	89
5.6.1	The spectral theorem and its affine-family corollary	89
5.6.2	The forward map in the eigenbasis	90
5.6.3	What the spectral form means	90
5.6.4	Decoupling in the eigenbasis	90
5.6.5	Two computational corollaries	92
5.7	The lifecycle of the precision: losing tridiagonality	96
6	The Gaussian Chain by Belief Propagation	101
6.1	The posterior and its factor graph	102
6.1.1	The posterior is a hidden Markov model	102
6.1.2	The factor graph is a tree	102
6.2	The sum-product sweep and its cost	103
6.2.1	The two sweeps	103
6.3	The Gaussian closure	106
6.4	The price of locality	110
6.4.1	Two ways to go local	111
6.4.2	What the neighbours are worth	113
7	Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation	115
7.1	What changes, and what is preserved	115
7.2	Where the score stops being linear	116
7.3	Where closed forms stop	117
7.4	Grid belief propagation: the discretised recursion	120
7.4.1	The representation and the update	120
7.4.2	The three choices, and what fixes them	121
7.4.3	Validation	122
7.5	The Gaussian-message approximation, measured	123

7.5.1	Proof of Theorem 7.3	124
7.5.2	Three metrics, and which one is primary	126
7.5.3	The measurement	128
7.5.4	Varying the innovation law	130
8	Learning the Parameters: EM on the Chain	132
8.1	Model: an estimated mixture innovation	133
8.2	The EM algorithm, derived	134
8.2.1	Complete data and the surrogate	135
8.2.2	Fisher's identity: why BP is never differentiated	140
8.2.3	The E-step: which expectations BP supplies	141
8.2.4	The M-step: deriving the conditional maximisers	143
8.2.5	Generalised EM and monotonicity	147
8.3	The numerical representation: grid BP, again	148
8.4	Diagnostics	150
8.4.1	Implementation and estimator checks	151
8.4.2	What does EM recover?	151
9	Learning the Kernel, and What Convergence Costs	154
9.1	Why there are two rates	155
9.1.1	EM is a fixed-point map, and its Jacobian is missing information	155
9.1.2	From eigenvalues to iteration counts	156
9.1.3	The channel enters the identity twice	156
9.1.4	Which coordinate should be slower, and why	157
9.2	Measuring the two rates	158
9.3	Settling times and the stopping rule	160
9.4	What the channel costs in accuracy at equal data	161
10	What Structure Buys, and Where It Stops Paying	164
10.1	What is compared, and how it is scored	165

10.1.1	The estimand	165
10.1.2	The common protocol	167
10.2	The estimators being compared	168
10.2.1	From a denoiser to a score	168
10.2.2	The structured estimator	169
10.2.3	The unstructured denoiser	169
10.2.4	The three structure-aware architectures	170
10.3	Pointwise data efficiency	172
10.4	How much of the margin is the architecture?	173
10.4.1	The bulk posterior, in closed form	173
10.4.2	The screen	179
10.4.3	The confirmatory comparison	180
10.5	How much mixture capacity the estimator needs	181
10.6	Where the Markov assumption stops paying	183
10.6.1	Two violations, defined	183
10.6.2	Why one is partly absorbable and the other is not	184
10.6.3	A covariance scalar orders the tested conditions	185
10.7	Why the shape coordinate is the expensive one	189
10.7.1	What the cumulant law predicts	189
10.7.2	The measurement	190
10.8	What the chapter establishes	192
11	Discussion and Conclusions	199
11.1	Answers to the research questions	199
11.2	What is new	201
11.3	Limitations	202
11.4	Future work	204
	Bibliography	205

Acknowledgements	214
A Reproducibility	215
A.1 Figure-to-code map	215
A.2 The learning experiments	216
A.3 Environment	221
A.4 Build manifest	222
B How the Reported Ratios Were Aggregated	224
B.1 Does the headline depend on how the ratio was averaged?	225

Disclosure on the Use of Artificial Intelligence

During the preparation of this thesis the author used large language models — Anthropic’s Claude, OpenAI’s ChatGPT, and Google’s Gemini — as assistive tools, in the following capacities:

- **Software development.** Assistance in writing, refactoring and documenting the analysis, experiment-management and figure-generation code of the accompanying research repository, including the $\text{\LaTeX}$ build and consistency-checking tooling.
- **Manuscript review.** Reading drafts of the manuscript to flag potential mathematical slips, inconsistencies between text, tables and figures, and unclear passages, which the author then checked and resolved.
- **Language and typesetting.** Editing prose for clarity and concision, and assistance with $\text{\LaTeX}$ formatting.

No experimental result, number, table entry or figure in this document was produced by a language model: all reported quantities are computed by the version-controlled pipeline described in Appendix A from the frozen experiment outputs it documents, and the figures and generated tables are built by the scripts that appendix lists. The research questions, the mathematical derivations, the experimental design and the conclusions are the author’s, every AI-assisted change was reviewed by the author before inclusion, and the author takes full responsibility for the entire content of this thesis.

Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). The same construction underlies the image and video generators deployed at scale by the large technology companies, from latent diffusion (Rombach et al., 2022) onwards, and reaches beyond media: structure prediction in AlphaFold 3 places atomic coordinates with a diffusion module (Abramson et al., 2024). They were introduced not simply as an engineering construction, but through the language of *non-equilibrium* thermodynamics: a forward process progressively destroys the structure of the data distribution, while a learned reverse process attempts to reconstruct it, with the corresponding path measures connected through ideas from fluctuation theorems for driven systems (Sohl-Dickstein et al., 2015). Their central object is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x, t) = \nabla_x \log p_t(x), \tag{1.1}$$

which is the one object a model must learn in order to reverse a noising process: given

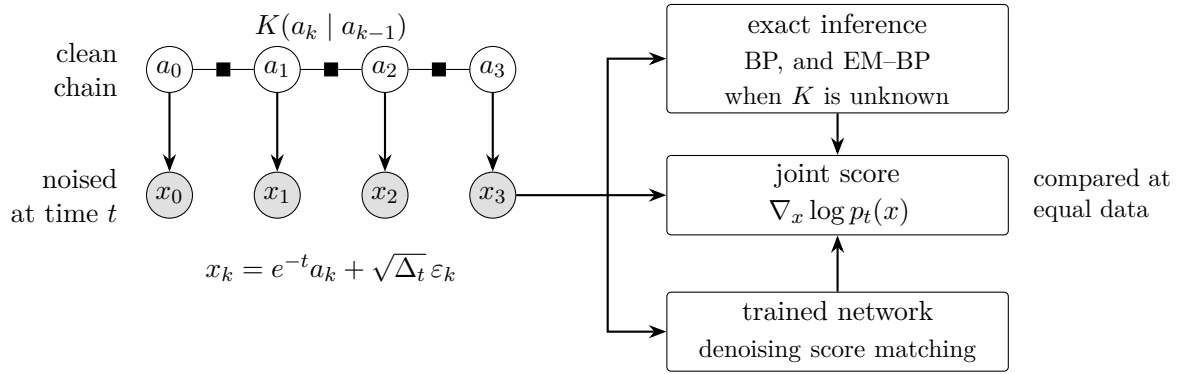


Figure 1.1: The object of the thesis and the two ways it is obtained. A Markov chain generates a clean trajectory; the Ornstein–Uhlenbeck channel corrupts every coordinate independently, so each observation attaches to the latent chain on its own and the posterior stays a chain. That is what makes the joint score exactly computable by message passing, with the transition kernel known (Chapters 5 and 7), or estimated from noised sequences alone when it is not (Chapter 8). The same score is what a denoising-score-matching network is trained to approximate without being given the chain structure. Chapter 10 measures what supplying that structure is worth, at an equal number of clean trajectories, and where it stops paying.

the forward process, its schedule and its terminal law, the score determines the reverse drift (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011), and what that training targets depends on which objective is meant. At the population level the unrestricted denoising objective is minimised by the score of the noised data law itself; conditional on a finite training set, the empirical objective is minimised by the score of the *empirical* distribution: a sum of point masses at the training samples, each blurred by the forward kernel. A network that reached that empirical minimiser and was integrated back to zero noise would reproduce its training set and nothing else; that diffusion models instead generate new data means the trained score departs from the minimiser of its own finite-sample objective, in a way that is still only partly charted (Chapter 2, Section 2.4). Deciding what a trained network has actually learned therefore requires holding it against the score it *should* have produced — and for essentially every distribution of interest that reference is unavailable, so a network’s error cannot be separated from the difficulty of the target it was chasing (Kadkhodaie et al., 2024).

A complementary programme, initiated in the statistical-physics community, replaces

the black box with *exactly solvable models*: data distributions simple enough that the score is available in closed form, yet rich enough to show the phenomena seen in real systems — speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024; Achilli et al., 2026).

This thesis works inside that programme, on a family chosen so that the reference score is both computable and nonlinear. The data are a trajectory $a_0, a_1, \dots, a_{L-1}$ of a first-order Markov process on $\mathbb{R}^L$ with continuous, non-Gaussian transitions, variance-matched so that competing innovation laws differ only beyond the second moment. Coordinatewise Ornstein–Uhlenbeck noising multiplies the prior by unary factors and adds no edges among the latent variables, so the posterior factor graph is still the prior’s chain — a tree, on which sum–product is exact. The joint score of the whole noisy sequence is then available as an exact functional recursion for a correlated, non-Gaussian model, and numerically to a validated tolerance once the continuous messages are discretised (Figure 1.1). Later chapters drop the assumption that the chain’s parameters are known and recover them from noised sequences alone.

Sequential data, such as videos, often exhibit strong local dependence: a frame is typically much more constrained by nearby frames than by distant ones. Modern architectures already exploit related forms of locality. Convolutional networks, and in particular the U-Net architectures commonly used in diffusion models, build in local receptive fields, weight sharing, and multiscale composition (Krizhevsky et al., 2012; LeCun et al., 2015). They therefore should not be viewed as structure-agnostic function approximators.

What remains less explicit is the probabilistic structure of the data. If the clean sequence is Markovian, its joint law admits a precise local factorisation, and Bayesian denoising can exploit this structure directly through message passing. This raises the central question studied here: *how much of the Markov structure of the clean dynamics remains visible after diffusion noising, how is it represented in the score, and to what extent can a learned score recover and exploit it from finite data?* The posterior representation of the score provides a convenient way to study this question. When the Markov prior

is known, the corresponding posterior quantities can be computed explicitly by belief propagation, providing a structured reference against which learned score estimators can be compared. The aim is therefore not to argue that neural score models ignore structure, but to understand how much structure they learn implicitly, how noising transforms it, and what is gained by making that structure explicit.

Two classical theories meet the modern one. For Gaussian Markov chains, the score is governed by structured matrix algebra; more generally, the score-to-posterior identity turns denoising on a chain into inference on a tree, where BP and Kalman filtering apply (Pearl, 1988; Mézard and Montanari, 2009; Kalman, 1960; Donoho et al., 2009). This thesis develops these two views, shows where they agree, and studies what is lost when the underlying structure is only approximated.

1.2 Positioning and gap

Chapter 2 reviews the literature this work builds on; the gap it leaves is stated here.

The statistical physics of generative diffusion has produced closed-form analyses of the backward dynamics and their transitions (Biroli and Mézard, 2023; Biroli et al., 2024; Achilli et al., 2026; Sclocchi et al., 2025b; Bonnaire et al., 2025). Exact posterior inference on Gauss–Markov chains is classical, through the Kalman filter and smoother and their formulation as sum–product message passing (Kalman, 1960; Rauch et al., 1965; Kschischang et al., 2001; Mézard and Montanari, 2009). And message passing with continuous variables closes its messages inside a parametric family, which is the step connecting belief propagation to AMP and to the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016).

The solvable-model analyses cited above treat data as unstructured collections of samples: the only time axis they carry is the diffusion time, and the *internal* time of a sequence, a frame correlated with its neighbours by a dynamical rule, does not appear in that line of work. The inference literature has the machinery for that internal time but does not ask how the exact posterior, and so the exact score, deforms as the noise level is swept

from zero to saturation. And the standard justification for closing messages in a Gaussian family is a central-limit argument that requires each node to receive many messages; on a chain each node receives exactly two, so the argument fails by construction.

The gap this thesis addresses is: the joint score of a noised Markov sequence, computed exactly where that is possible, approximated in a controlled and measured way where it is not, and learned from noised data when the dynamics are unknown.

Where no closed form exists, the reference used throughout is deterministic quadrature and not Monte Carlo. A particle smoother carries sampling error, which would add a second error term to every comparison and force the cost of an approximation and the cost of the reference to be disentangled before either could be read. Grid-based inference carries a discretisation error instead, which can be bounded separately and reported, and that is what makes it usable as a reference.

1.3 Research questions

The thesis addresses six questions. The first four concern scores under *known* dynamics and the last two concern learning dynamics that are unknown.

- RQ1.** For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?
- RQ2.** How can the same Gaussian-chain score be computed by belief propagation?
- RQ3.** For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?
- RQ4.** In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?

- RQ5.** Under what assumptions can an *unknown* local transition law be recovered from noised sequences alone, and which numerical, optimisation and statistical checks are required before such a recovery claim is reliable?
- RQ6.** On a family where the Markov assumption holds exactly, what is supplying that assumption worth, measured against denoising-score-matching networks trained on identical data and given none of it?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion scores, or about practical neural architectures. In particular RQ6 is a question about the value of a correct structural assumption on one family, not a comparison of learning algorithms in general: the two arms are asymmetric in supplied information by construction, which is what makes the measurement meaningful and also what bounds it.

Chapter 4 answers no numbered question. It comes first because it is where the work began, and because what it settles motivates the six: on a rotating-ring model, whose state space and estimand both differ from the chain's, it shows what joint observations can identify that no single-frame marginal can.

1.4 Contributions

Three contributions, in the order the body develops them.

1. **A Gaussian case worked out completely.** For the stationary AR(1) chain under coordinatewise OU corruption, we derive the score in closed form, recover the classical identification of belief propagation on it with Kalman filtering followed by RTS smoothing, and quantify what ignoring distant evidence costs; the closed-form influence-decay rate is derived where it is used, in Chapter 10 (Chapters 5 and 6).
2. **What belief propagation still gets right off the Gaussian case, and what it gives up.** The message-passing recursion stays exact for any innovation law; it

is the closure of its messages into a finite parametric family that Gaussianity buys, and Laplace innovations lose. We show that projecting the exact message onto the Gaussian family is not an approximation to shrug off but a plain object, the LMMSE estimator of the covariance-matched Gaussian model (Theorem 7.3), so the gap it leaves is attributable to the choice of family rather than to any one implementation (Chapter 7).

3. **Learning the chain itself.** Fisher’s identity lets us fit an unknown transition kernel from noised sequences by EM, each E-step and likelihood gradient supplied by one forward–backward pass at the current parameter value, with no derivative taken through the recursion (Chapter 8). Chapters 9 and 10 then measure what the recovered structure is worth against denoising score matching, and where that advantage runs out.

1.5 Structure of the thesis

Chapter 2 is the context: statistical mechanics and the variational principle behind the Boltzmann distribution, energy-based models as the same object with a learned energy, generative diffusion and the reverse-time SDE, and then the distinction the rest of the thesis turns on, between the exact, the empirical and the fitted score. It closes with the reading of U-Nets as belief propagation that makes exact inference a serious alternative to a trained network rather than a toy one. Chapter 3 fixes the model and the estimand: the Markov sequence, the Ornstein–Uhlenbeck channel, the argument that the joint score and not a per-frame score is the object worth studying, and the factor-graph language the computations are written in.

Chapter 4 begins the research where the research began, with a planar trajectory near a circle. It shows that the rotation rate carries zero Fisher information in every one-frame marginal and positive information in the trajectory. That is the argument for the joint score in a single exact statement, and it is what sends the following chapters to a model where the joint score can be computed in full.

Chapters 5 and 6 do that for the stationary Gaussian chain, first with matrices and then with messages: the score in closed form, the lifecycle of its couplings, the identification of belief propagation with the Kalman filter and the Rauch–Tung–Striebel smoother, and the measured price of locality; the closed-form geometric decay behind it is derived in Chapter 10, where it drives a design argument (RQ1, RQ2, RQ4). Chapter 7 leaves Gaussianity in the most controlled way available, through Laplace innovations, builds a validated numerical reference, and measures exactly what a Gaussian message family discards (RQ3).

Chapters 8 and 9 drop the assumption that the transition law is known and recover it from noised sequences alone, first two parameters and then a whole kernel, with the convergence schedule of the two coordinates (RQ5). Chapter 10 measures what supplying the Markov structure is worth against networks trained on identical data, and locates where it stops being worth anything (RQ6). Chapter 11 answers the six questions and states what is left open.

Chapter 2

Statistical Mechanics, Energy-Based Models, and Diffusion

2.1 What statistical mechanics is, and why it is here

Newton's second law is deterministic and complete. Given the positions and momenta of N particles and the forces between them, the trajectory of the system is fixed for all time. For N of order 10^{23} : the initial condition cannot be written down, the equations cannot be integrated, and the answer, if it were available, would be a list of 6×10^{23} numbers. Yet the gas in a room has a temperature, a pressure and a heat capacity, and these are stable, reproducible, and predictable from a handful of parameters.

Statistical mechanics is the resolution of that tension. It abandons the trajectory and keeps a probability distribution over microstates, and it fixes that distribution by a variational principle: among all distributions with a given mean energy, the one realised at equilibrium maximises the entropy

$$S[p] = - \sum_x p(x) \log p(x), \tag{2.1}$$

which is the unique measure of uncertainty consistent with a short list of consistency axioms (Jaynes, 1957). Carrying out the maximisation subject to $\sum_x p(x)E(x) = \bar{E}$

produces the Boltzmann distribution

$$p(x) = \frac{1}{Z(\beta)} e^{-\beta E(x)}, \quad Z(\beta) = \sum_x e^{-\beta E(x)}, \quad (2.2)$$

with the inverse temperature β appearing as the Lagrange multiplier of the energy constraint (Landau and Lifshitz, 1980; Goldstein et al., 2002).

The maximisation is Toolbox 2.1.

Toolbox 2.1 — From maximum entropy to the Boltzmann distribution

Maximise (2.1) subject to normalisation and the energy constraint using Lagrange multipliers λ and β :

$$\mathcal{L}[p] = - \sum_x p(x) \log p(x) - \lambda \left(\sum_x p(x) - 1 \right) - \beta \left(\sum_x p(x) E(x) - \bar{E} \right).$$

Differentiating pointwise in $p(x)$ for each x and setting the result to zero,

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -\log p(x) - 1 - \lambda - \beta E(x) = 0 \implies p(x) = e^{-1-\lambda} e^{-\beta E(x)}.$$

The prefactor does not depend on x ; call it $1/Z(\beta)$ and fix it by normalisation, which forces $Z(\beta) = \sum_x e^{-\beta E(x)}$ and recovers (2.2). The multiplier β is fixed in turn by demanding the constraint hold, $\bar{E} = -\partial_\beta \log Z(\beta)$: with the state space, the energy function and the reference measure fixed, and on the branch where that relation is invertible, one number — the mean energy — determines the whole distribution. $\square$

For the canonical family (2.2), the equilibrium thermodynamics follows from the normalising constant: the free energy is $F = -\beta^{-1} \log Z$, and its derivatives give the energy, the entropy and the response functions. Two features of that construction matter for what follows.

The first is that the whole of the macroscopic behaviour is encoded in a single scalar function of the parameters, $\log Z$, which is a sum over exponentially many configurations and, for broad classes of discrete interacting models, is #P-hard to compute (Barahona,

1982; Jerrum and Sinclair, 1993) — the tractable exceptions, trees and Gaussians, are exactly the structures the later chapters exploit. The second is that competition between energy, which orders, and entropy, which disorders, can produce *phase transitions*: the equilibrium state can change character abruptly as a parameter is varied, even though the microscopic law does not change at all. A system with a symmetry can settle into a state that lacks it, and which of the equivalent states it settles into is decided by fluctuations. The canonical example is the Ising model (Ising, 1925), whose two-dimensional case Onsager (1944) solved exactly, and which remains the reference picture for a transition driven by the competition between energy and entropy. Both features will reappear, unchanged in substance, in the generative models of the next sections.

2.2 Energy-based models: the same object, learned

Traffic between statistical mechanics and learning machines runs in both directions and is old. The Hopfield network stored memories as minima of an energy (Hopfield, 1982); its capacity was settled by spin-glass methods (Amit et al., 1985; Mézard et al., 1987; Gardner, 1988); and the Boltzmann machine and its restricted form made the Boltzmann distribution itself the model (Ackley et al., 1985; Smolensky, 1986). The 2024 Nobel Prize in Physics recognised that lineage (The Royal Swedish Academy of Sciences, 2024). What follows takes (2.2) and reads it in the opposite direction. Rather than deriving a distribution from a known energy, it posits a parametric energy E_θ and fits it so that the induced distribution matches data (Ackley et al., 1985; LeCun et al., 2006):

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \quad Z(\theta) = \int_{\mathbb{R}^d} e^{-E_\theta(x)} dx. \quad (2.3)$$

The microstate becomes a data point, the energy becomes a learnable scoring function, and the partition function keeps its role and its difficulty. Fitting by maximum likelihood gives a gradient with the structure of a competition,

$$\nabla_\theta \mathbb{E}_{x \sim p_{\text{data}}}[-\log p_\theta(x)] = \mathbb{E}_{x^+ \sim p_{\text{data}}}[\nabla_\theta E_\theta(x^+)] - \mathbb{E}_{x^- \sim p_\theta}[\nabla_\theta E_\theta(x^-)], \quad (2.4)$$

The derivation is Toolbox 2.2.

Toolbox 2.2 — The maximum-likelihood gradient of an energy-based model

From (2.3), $\log p_\theta(x) = -E_\theta(x) - \log Z(\theta)$, so

$$\nabla_\theta \log p_\theta(x) = -\nabla_\theta E_\theta(x) - \nabla_\theta \log Z(\theta).$$

The second term does not depend on x . Differentiating under the integral sign in $Z(\theta) = \int e^{-E_\theta(x)} dx$,

$$\begin{aligned} \nabla_\theta \log Z(\theta) &= \frac{1}{Z(\theta)} \int_{\mathbb{R}^d} \nabla_\theta e^{-E_\theta(x)} dx \\ &= - \int_{\mathbb{R}^d} \nabla_\theta E_\theta(x) \frac{e^{-E_\theta(x)}}{Z(\theta)} dx = -\mathbb{E}_{x^- \sim p_\theta} [\nabla_\theta E_\theta(x^-)]. \end{aligned}$$

Substituting back,

$$\nabla_\theta \log p_\theta(x) = -\nabla_\theta E_\theta(x) + \mathbb{E}_{x^- \sim p_\theta} [\nabla_\theta E_\theta(x^-)],$$

and averaging $-\nabla_\theta \log p_\theta(x)$ over $x = x^+ \sim p_{\text{data}}$ gives (2.4). □

in which the first term lowers the energy of observed data and the second raises the energy of the model's own samples. The second expectation is over p_θ , which cannot be sampled directly; the negative phase is what makes maximum-likelihood training of these models generically intractable, and the standard approach approximates it by Markov chain Monte Carlo, typically Langevin dynamics, whose stationary measure is $\propto e^{-E_\theta}$ (Hinton, 2002; Robert and Casella, 2004). This is expensive, and its cost is the reason the family was, for a long time, outperformed by models that give up the explicit density in exchange for cheap sampling (Goodfellow et al., 2014; Kingma and Welling, 2014; Papamakarios et al., 2021). Carbone (2025) surveys how the energy-based view relates to the rest of the generative family.

The learning dynamics of these models are themselves a statistical mechanics problem, and one with a structure worth carrying forward. The spectral dynamics of the weights

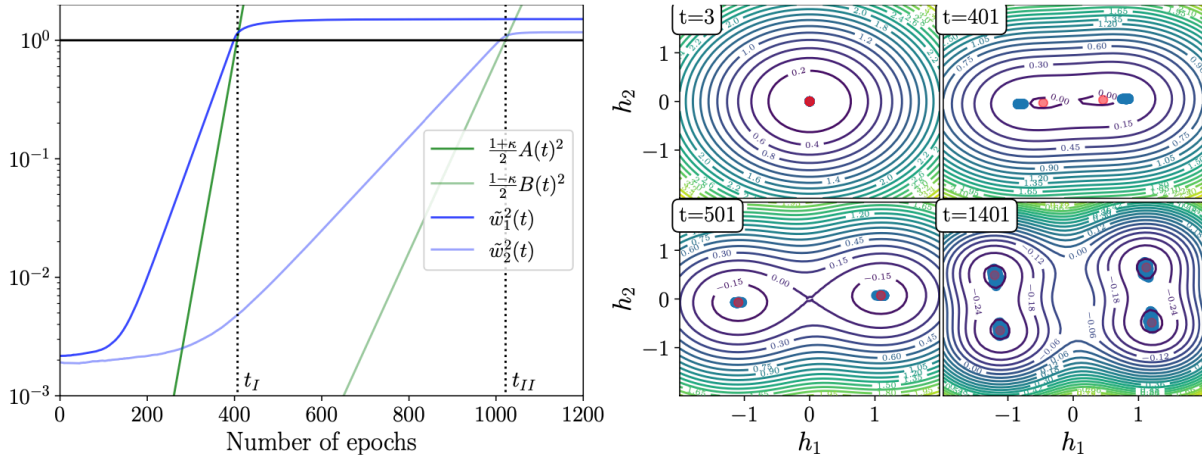


Figure 2.1: *Left:* the eigenvalues $\tilde{w}_\alpha^2(t)$ of the RBM’s weight matrix during training, each crossing $y = 1$ at a time (t_I , t_{II}) that the theory predicts from the initial conditions alone. *Right:* the free energy in the order-parameter plane (h_1, h_2) at four of those times. One minimum splits into two along the coarse feature η^1 , then each of those splits again, giving four — the diverging susceptibility at each split is the second-order transition. Reproduced from Bachtis et al. (2024), their Figure 6.

(Decelle et al., 2017; Mézard, 2017) and the emergence of compositional features (Tubiana and Monasson, 2017) are both statistical-mechanics accounts of training. Bachtis et al. (2024) show that training a restricted Boltzmann machine proceeds through a *cascade of second-order phase transitions* (Figure 2.1): the model first encodes a coarse, centre-of-mass feature, and then resolves progressively finer sub-modes, each emergence accompanied by a diverging susceptibility. The features are learned in order, from the largest scale down. Section 2.5 finds the same ordering in a different process, and Chapter 3 onwards works in a model where the ordering is analytic.

2.3 Generative diffusion

A diffusion model destroys structure in a controlled way and then learns to run the destruction backwards (Figure 2.2) (Sohl-Dickstein et al., 2015; Song and Ermon, 2019; Ho et al., 2020; Song et al., 2021). Its deterministic counterparts, neural ODEs and flow matching (Chen et al., 2018; Lipman et al., 2023), transport the same measures along a

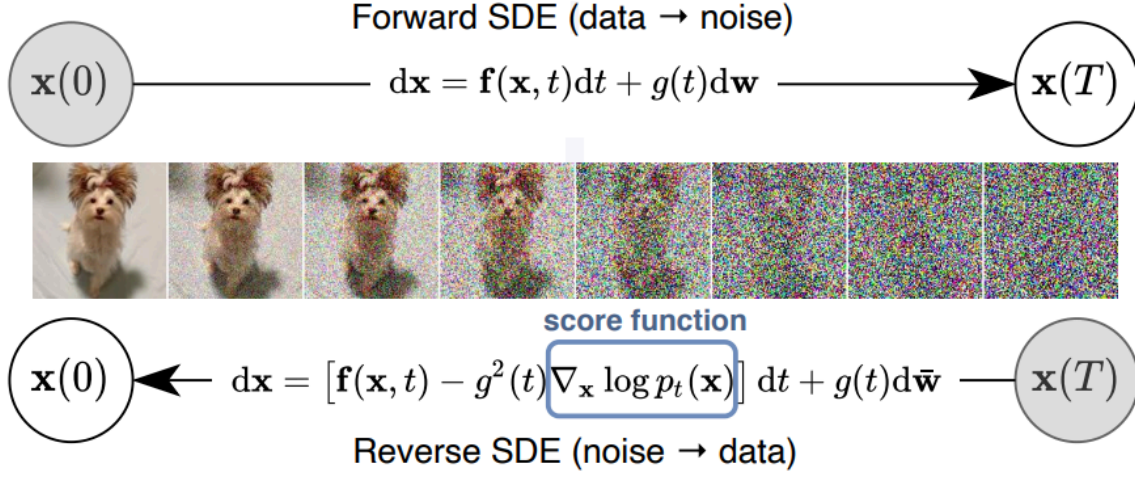


Figure 2.2: The picture the equations below make precise. The forward SDE corrupts a data point into noise; the reverse SDE run with the score $\nabla_x \log p_t$ undoes exactly that corruption, step for step. Reproduced from Song et al. (2021), their Figure 1.

probability-flow trajectory instead of a stochastic one. The forward process is a stochastic differential equation

$$dX_t = f(X_t, t) dt + g(t) dW_t, \quad (2.5)$$

started at $X_0 \sim p_0$, the data law, and producing marginals p_t that approach a simple terminal measure.

2.3.1 The forward process is a relaxation to a Boltzmann equilibrium

The choice of (2.5) used throughout this thesis is not an arbitrary corruption. Overdamped Langevin dynamics in a potential E ,

$$dX_t = -\nabla E(X_t) dt + \sqrt{2\beta^{-1}} dW_t, \quad (2.6)$$

has, for a confining potential — $e^{-\beta E}$ normalisable and the dynamics non-explosive and ergodic — a unique stationary density proportional to $e^{-\beta E(x)}$: the Boltzmann distribution (2.2) of Section 2.1, recovered as the equilibrium of a dynamics rather than as the solution

of a variational problem (Riskin, 1989; Gardiner, 2009). Setting $\beta = 1$ and $E(x) = \frac{1}{2}\|x\|^2$ gives $f(x, t) = -x$, $g = \sqrt{2}$: the Ornstein–Uhlenbeck channel, whose transition law is available in closed form,

$$x = e^{-t}a + \sqrt{\Delta_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad \Delta_t = 1 - e^{-2t}, \quad (2.7)$$

and whose stationary law is the standard Gaussian (Uhlenbeck and Ornstein, 1930; Gardiner, 2009).

So the terminal measure of the forward process is not merely “simple noise”: it is the equilibrium state of a quadratic Hamiltonian at unit temperature. The forward process takes a system prepared far from that equilibrium — in the data distribution, which is the thing with all the structure — and relaxes it to a reference equilibrium whose free energy is known exactly. That is the setting of non-equilibrium thermodynamics, and it is where the construction came from.

2.3.2 Why a path, and not a partition function

Section 2.1 exposed the central computational obstruction of equilibrium statistical mechanics: the Boltzmann law is simple once its partition function is known, but

$$Z = \sum_x e^{-\beta E(x)}$$

requires a sum over all microscopic configurations. Non-equilibrium statistical mechanics suggests a different way of comparing probability laws. Rather than characterising the endpoints through their partition functions, one drives the system between them and studies the probability of entire *trajectories*.

The physical prototype is the Jarzynski–Crooks theory. If a system is driven between equilibrium states, the random work W performed along its realised trajectory contains

the equilibrium free-energy difference:

$$\mathbb{E}\left[e^{-\beta(W-\Delta F)}\right] = 1. \quad (2.8)$$

This identity holds even when the protocol is finite-time and irreversible (Jarzynski, 1997; Crooks, 1999). The important conceptual point is that an equilibrium quantity, ΔF , can therefore be related to an average over *paths*, rather than obtained by evaluating the endpoint partition functions directly. The original diffusion construction of Sohl-Dickstein et al. (2015) uses precisely this non-equilibrium viewpoint: define a tractable forward process that destroys structure, and learn a reverse process that assigns probability to the same paths in the opposite direction.

Forward and reverse path measures. Discretise the forward diffusion into T steps. Since the corruption process is chosen by us, its transition probabilities are known. Starting from $x_0 \sim p_0$, its joint path law is

$$Q(x_{0:T}) = p_0(x_0) \underbrace{\prod_{k=1}^T q(x_k \mid x_{k-1})}_{q(x_{1:T} \mid x_0)}. \quad (2.9)$$

Generation instead starts from the tractable terminal law π introduced in Section 2.3.1 and moves backwards through learned conditionals,

$$P_\theta(x_{0:T}) = \pi(x_T) \prod_{k=1}^T p_\theta(x_{k-1} \mid x_k). \quad (2.10)$$

Thus the model is not specified by one global energy that must subsequently be normalised. It is built as a product of already normalised local transition probabilities.

The natural quantity comparing the two processes is their pathwise log-likelihood ratio (defined when the two path measures are mutually absolutely continuous, so that the ratio below exists; the Gaussian forward and reverse transitions used here have full support,

and every expectation taken of the ratio is assumed integrable),

$$\mathcal{W}_\theta(x_{0:T}) := \log \frac{Q(x_{0:T})}{P_\theta(x_{0:T})}. \quad (2.11)$$

When the two path measures are those of a physical forward protocol and its thermodynamic reversal, this log-ratio is the dimensionless dissipated work appearing in the Crooks and Jarzynski relations. For a learned P_θ , it is more generally a *path-space mismatch*; the same algebra nevertheless survives.

Indeed,

$$\begin{aligned} \mathbb{E}_Q[e^{-\mathcal{W}_\theta}] &= \int Q(x_{0:T}) \frac{P_\theta(x_{0:T})}{Q(x_{0:T})} dx_{0:T} \\ &= \int P_\theta(x_{0:T}) dx_{0:T} = 1. \end{aligned} \quad (2.12)$$

This has the Jarzynski form for dimensionless dissipated work. Taking the log-ratio itself in expectation gives

$$\mathbb{E}_Q[\mathcal{W}_\theta] = \int Q \log \frac{Q}{P_\theta} = \text{KL}(Q \| P_\theta) \geq 0. \quad (2.13)$$

The average mismatch therefore vanishes only if the learned reverse path law coincides with the forward path law.

From paths to the diffusion training objective. For learning, the relevant quantity is not the probability of the complete path but the likelihood assigned by the model to the observed starting point x_0 . Marginalising the intermediate states gives

$$p_\theta(x_0) = \int P_\theta(x_{0:T}) dx_{1:T}. \quad (2.14)$$

This integral is still difficult in general. The crucial advantage is that we possess a tractable distribution over the missing path: $q(x_{1:T} | x_0)$. Multiply and divide by it,

$$\begin{aligned} p_\theta(x_0) &= \int q(x_{1:T} | x_0) \frac{P_\theta(x_{0:T})}{q(x_{1:T} | x_0)} dx_{1:T} \\ &= \mathbb{E}_{q(x_{1:T} | x_0)} \left[\frac{P_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right]. \end{aligned} \quad (2.15)$$

Applying Jensen's inequality to the logarithm yields

$$\boxed{\log p_\theta(x_0) \geq \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{P_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right]}. \quad (2.16)$$

This is the evidence lower bound used by the diffusion construction. Expanding the two path measures makes its tractability explicit:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \pi(x_T) + \sum_{k=1}^T \log p_\theta(x_{k-1} | x_k) - \sum_{k=1}^T \log q(x_k | x_{k-1}) \right]. \quad (2.17)$$

The forward terms are known; learning acts only on the reverse conditionals.

The gap to the true likelihood is itself a KL divergence:

$$\log p_\theta(x_0) - \mathcal{L}_{\text{ELBO}} = \text{KL}(q(x_{1:T} | x_0) \| p_\theta(x_{1:T} | x_0)) \geq 0. \quad (2.18)$$

The thermodynamic analogy can therefore be stated precisely: irreversibility in the physical problem appears as a discrepancy between forward and reverse path measures; in the generative problem, the same path-space discrepancy appears as the slack of the likelihood bound. Training reduces that discrepancy by making the learned reverse transitions better approximate the reversal of the chosen forward process.

This is what replacing a partition function by a path buys. There is no global normalising constant analogous to $Z(\theta)$ to compute: the terminal law π is normalised, each forward and reverse transition is locally normalised, and the remaining marginalisation over trajectories is handled through the known forward path measure. The price is that

the reverse transition law must be learned. The next question is therefore: *what determines the reverse dynamics?*

Toolbox 2.3 — The Fokker–Planck equation, from Itô’s lemma

For a smooth test function φ , Itô’s lemma applied to (2.5) gives

$$d \mathbb{E}[\varphi(X_t)] = \mathbb{E} \left[f(X_t, t) \cdot \nabla \varphi(X_t) + \frac{1}{2} g(t)^2 \Delta \varphi(X_t) \right] dt.$$

Writing

$$\mathbb{E}[\varphi(X_t)] = \int \varphi(x) p_t(x) dx$$

and differentiating with respect to time,

$$\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \int \varphi(x) \partial_t p_t(x) dx.$$

On the right-hand side, integrate the drift term by parts once and the diffusion term twice:

$$\int f(x, t) \cdot \nabla \varphi(x) p_t(x) dx = - \int \varphi(x) \nabla_x \cdot [f(x, t) p_t(x)] dx,$$

and

$$\int \Delta \varphi(x) p_t(x) dx = \int \varphi(x) \Delta_x p_t(x) dx,$$

assuming the boundary terms vanish. Since the resulting identity holds for every test function φ ,

$$\partial_t p_t(x) = -\nabla_x \cdot [f(x, t) p_t(x)] + \frac{1}{2} g(t)^2 \Delta_x p_t(x). \quad (2.19)$$

This is the Fokker–Planck, or Kolmogorov forward, equation (Gardiner, 2009). The SDE describes random trajectories; the Fokker–Planck equation describes how their probability density evolves. $\square$

The forward equation contains no score. The score appears when the same evolution of probability measures is represented in the opposite direction.

Before reversing the stochastic process, it is useful to notice that (2.19) already admits

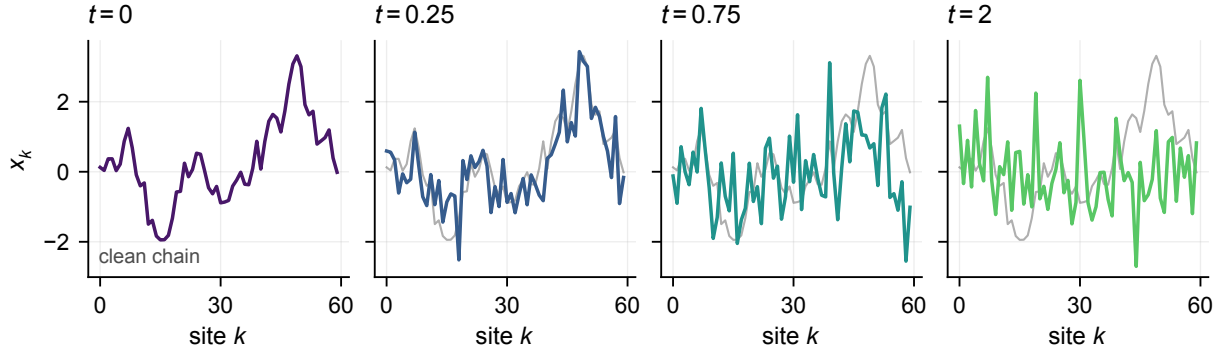


Figure 2.3: One clean stationary AR(1) trajectory with $\alpha = 0.85$, in grey, and the same trajectory after the Ornstein–Uhlenbeck channel (2.7) at four diffusion times. The signal contracts by e^{-t} while independent noise of variance Δ_t accumulates. Correlation between neighbouring sites survives longer than the original amplitude. Generation runs this evolution from right to left.

a deterministic representation. Using

$$\Delta_x p_t = \nabla_x \cdot (p_t \nabla_x \log p_t), \quad (2.20)$$

the Fokker–Planck equation becomes

$$\partial_t p_t = -\nabla_x \cdot \left[\left(f - \frac{1}{2} g^2 \nabla_x \log p_t \right) p_t \right]. \quad (2.21)$$

This is a continuity equation. Consequently the deterministic ODE

$$\dot{X}_t = f(X_t, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(X_t) \quad (2.22)$$

has the same one-time marginals p_t as the stochastic forward SDE. This is the *probability-flow ODE* of Song et al. (2021): individual trajectories differ from those of the diffusion, but at every time the distribution of states is the same.

For the Ornstein–Uhlenbeck process fixed in Section 2.3.1, Figure 2.3 shows this forward loss of structure explicitly. The clean trajectory is progressively contracted while independent Gaussian noise is accumulated; generation must undo this evolution.

The stochastic reversal is a classical result. For a state-independent diffusion coefficient

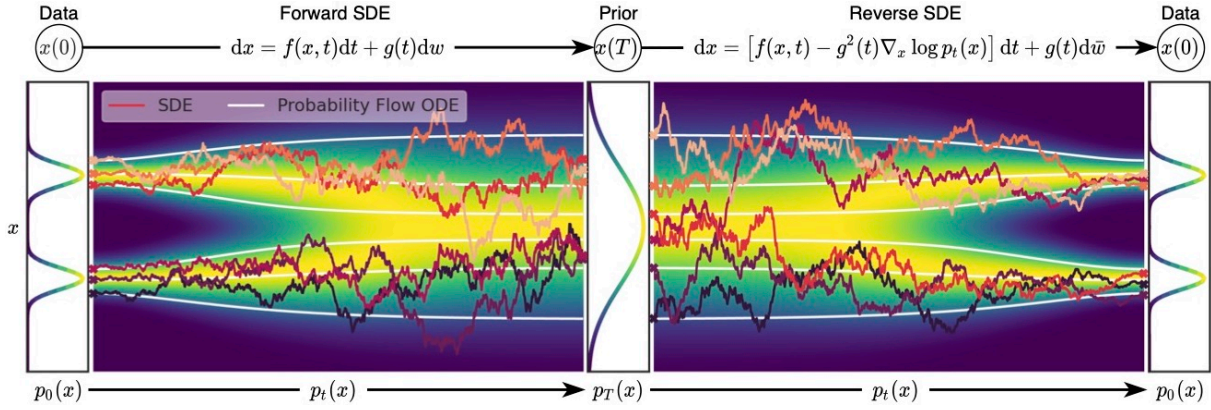


Figure 2.4: The forward SDE carries the data law p_0 to a simple terminal law p_T . Given the score $\nabla_x \log p_t$, the reverse SDE follows the same family of marginals in reverse. The white curves are trajectories of the deterministic probability-flow ODE, which has the same one-time marginals p_t despite having no Brownian term. Reproduced from Song et al. (2021).

$g(t)$, and under regularity conditions on the forward process — smooth, strictly positive marginals with an integrable score — Anderson (1982) shows that the reverse-time process is

$$dX_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t) \right] dt + g(t) d\bar{W}_t, \quad (2.23)$$

when integrated from T back to 0, starting from $X_T \sim p_T$. It exactly reproduces the forward marginals in reverse order; Figure 2.4 draws the forward SDE, the reverse SDE and the probability-flow ODE over one family of marginals.

The difference between (2.22) and (2.23) is worth noting. The deterministic probability-flow ODE contains $\frac{1}{2}g^2$ because its velocity field alone must reproduce the Fokker–Planck evolution. The reverse SDE contains g^2 because it must both change the drift and reverse the effect of the Brownian diffusion. Both depend on the same object,

$$s_t(x) := \nabla_x \log p_t(x),$$

the score of the noised distribution.

This also clarifies the parameterisation used in practical diffusion models. In the

discrete variance-preserving construction of Ho et al. (2020),

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (2.24)$$

The network is usually trained to predict the injected noise, $\varepsilon_\theta(x_t, t)$, rather than the score itself. These are only two parameterisations of the same conditional expectation. At the population optimum,

$$\boxed{s_t(x_t) = -\frac{\mathbb{E}[\varepsilon | x_t]}{\sqrt{1 - \bar{\alpha}_t}}, \quad s_\theta(x_t, t) = -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}.} \quad (2.25)$$

Thus DDPM noise prediction, continuous-time score matching, the reverse SDE and the probability-flow ODE all require the same underlying quantity. The next section separates three different senses in which that quantity can be meant: the exact score of the population law, the score of the empirical training distribution, and the fitted score produced by a finite model.

A book-length treatment of diffusion models and their variants is given by Lai et al. (2025).

2.4 Three scores

The word “score” is used for three different objects in this literature, and almost every difficulty in generative diffusion is a property of the gaps between them. The distinction is drawn sharply in Biroli and Mézard (2023) and organises what follows.

The exact score. The score of the true noised law, $\nabla_x \log p_t$ with $p_t = p_0 * \mathcal{N}(0, \Delta_t I)$ after contraction. It is the object (2.23) calls for. It has a form that will be used on every page of this thesis: writing p_t as the channel convolution and differentiating under the integral gives

$$\boxed{\nabla_x \log p_t(x) = \frac{e^{-t} \mathbb{E}[a | x] - x}{\Delta_t} \iff \mathbb{E}[a | x] = e^t (x + \Delta_t \nabla_x \log p_t(x)).} \quad (2.26)$$

The derivation is Toolbox 2.4.

Toolbox 2.4 — The score–posterior identity, from the channel convolution

Setup. The variance–preserving OU channel at time t is the linear–Gaussian map

$$x = e^{-t} a + \sqrt{\Delta_t} \xi, \quad a \sim P_0, \quad \xi \sim \mathcal{N}(0, I_d), \quad \xi \perp a,$$

so that $x \mid a \sim \mathcal{N}(e^{-t}a, \Delta_t I_d)$ with transition density $p(x \mid a) = G(x; e^{-t}a, \Delta_t)$, where $G(x; m, v) := (2\pi v)^{-d/2} \exp(-\|x - m\|^2/(2v))$. Marginalising over the prior,

$$P_t(x) = \int_{\mathbb{R}^d} P_0(a) G(x; e^{-t}a, \Delta_t) da. \quad (2.27)$$

We assume only $\int \|a\| P_0(a) da < \infty$. Since $G > 0$, we have $P_t(x) > 0$ for every x , and the score $S(x, t) := \nabla_x \log P_t(x)$ is well defined.

Step 1: log–gradient identity. For any differentiable $f > 0$ one has $\nabla \log f = \nabla f / f$. Applied to P_t ,

$$S(x, t) = \frac{\nabla_x P_t(x)}{P_t(x)}. \quad (2.28)$$

The whole computation is thus $\nabla_x P_t(x)$ divided by $P_t(x)$.

Step 2: differentiate the marginal. In (2.27) the prior $P_0(a)$ does not depend on x ; only G does. Both G and $\nabla_x G$ are bounded, locally uniformly in x , by a fixed Gaussian in a times a polynomial, which is P_0 –integrable. Hence differentiation passes under the integral (Leibniz’s Integral Rule / Dominated Convergence Theorem (DCT)):

$$\nabla_x P_t(x) = \int P_0(a) \nabla_x G(x; e^{-t}a, \Delta_t) da. \quad (2.29)$$

Step 3: gradient of the Gaussian kernel. From $\log G(x; m, \Delta_t) = -\|x - m\|^2/(2\Delta_t) + \text{const}$,

$$\nabla_x G(x; e^{-t}a, \Delta_t) = -\frac{x - e^{-t}a}{\Delta_t} G(x; e^{-t}a, \Delta_t). \quad (2.30)$$

The gradient of the channel is the channel itself times the displacement from its mean, $-(x - e^{-t}a)/\Delta_t$.

Step 4: substitute and split. Insert (2.30) into (2.29). The factor x is constant in a , so it pulls out and leaves $\int P_0(a)G \, da = P_t(x)$ by (2.27):

$$\begin{aligned}\nabla_x P_t(x) &= -\frac{1}{\Delta_t} \int (x - e^{-t}a) P_0(a) G(x; e^{-t}a, \Delta_t) \, da \\ &= -\frac{x}{\Delta_t} P_t(x) + \frac{e^{-t}}{\Delta_t} \int a P_0(a) G(x; e^{-t}a, \Delta_t) \, da.\end{aligned}\quad (2.31)$$

Both integrals are finite: $G \leq (2\pi\Delta_t)^{-d/2}$ and $\int \|a\| P_0 < \infty$.

Step 5: Bayes' theorem. The joint density of the clean-noisy pair is $P_0(a)G(x; e^{-t}a, \Delta_t)$; dividing by its marginal $P_t(x)$ gives the posterior

$$P_t(a \mid x) = \frac{P_0(a) G(x; e^{-t}a, \Delta_t)}{P_t(x)}, \quad (2.32)$$

which is a probability density in a *precisely because* $P_t(x)$ is its normalising constant. Therefore

$$\frac{1}{P_t(x)} \int a P_0(a) G(x; e^{-t}a, \Delta_t) \, da = \int a P_t(a \mid x) \, da =: \langle a \rangle_{x,t}. \quad (2.33)$$

Step 6: divide by $P_t(x)$. Combining (2.28), (2.31) and (2.33) (linearity of the integral; $P_t(a \mid x)$ a probability measure, so x integrates out):

$$\boxed{S(x, t) = -\frac{x}{\Delta_t} + \frac{e^{-t}}{\Delta_t} \langle a \rangle_{x,t}} \quad \langle a \rangle_{x,t} = \int a P_t(a \mid x) \, da. \quad (2.34)$$

Coordinate-wise this is $S_k(x, t) = (e^{-t}\langle a_k \rangle_{x,t} - x_k)/\Delta_t$, since ∇_x and the kernel both factorise over components.

Denoiser and noise forms. Solving (2.34) for the conditional mean,

$$\langle a \rangle_{x,t} = e^t \left(x + \Delta_t S(x, t) \right), \quad (2.35)$$

so the score is equivalent to the Bayes-optimal denoiser of the clean signal. Taking $\langle \cdot \rangle_{x,t}$ of the channel relation $\xi = (x - e^{-t}a)/\sqrt{\Delta_t}$ and using (2.35) (with x passing through the conditional expectation unchanged),

$$\langle \xi \rangle_{x,t} = -\sqrt{\Delta_t} S(x, t). \quad (2.36)$$

Tilted-measure form. Because $(x - e^{-t}a)^2 = e^{-2t}(a - e^t x)^2$, set $\tilde{x} := e^t x$ (the noiseless back-projection of x to $t = 0$). Then $G(x; e^{-t}a, \Delta_t) = (2\pi\Delta_t)^{-d/2} \exp\left(-\frac{e^{-2t}}{2\Delta_t} \|a - \tilde{x}\|^2\right)$, and (2.32)–(2.33) read

$$\langle a \rangle_{x,t} = \frac{\int a P_0(a) \exp\left(-\frac{e^{-2t}}{2\Delta_t} \|a - \tilde{x}\|^2\right) da}{\int P_0(a) \exp\left(-\frac{e^{-2t}}{2\Delta_t} \|a - \tilde{x}\|^2\right) da}. \quad (2.37)$$

This is a averaged under P_0 tilted by a quadratic pinning to $\tilde{x}$ with stiffness $\kappa_t := e^{-2t}/\Delta_t$. For large t , $\kappa_t \rightarrow 0$ and $\langle a \rangle_{x,t} \rightarrow \int a P_0(a) da$ (prior mean, no information); for $t \rightarrow 0^+$, $\Delta_t \simeq 2t$ gives $\kappa_t \simeq 1/(2t) \rightarrow \infty$ and, for a prior with a continuous positive density near x , $\langle a \rangle_{x,t} \rightarrow \tilde{x} \rightarrow x$ (data dominates). The regularity matters: under a discrete prior — the empirical measure below being the relevant case — the tilted mean concentrates on the nearest atoms instead, and need not approach an arbitrary fixed x .

Potential form. Let $P_0(a) = Z_0^{-1} e^{-V_0(a)}$ with $V_0 \in C^1$, and let $Q_{x,t}(a) \propto P_0(a) e^{-\kappa_t \|a - \tilde{x}\|^2/2}$ be the tilted density of (2.37). The Gaussian factor makes $Q_{x,t}$ and its gradient integrable with vanishing boundary terms, so $\int \nabla_a Q_{x,t}(a) da = 0$. Expanding the gradient and dividing by the normaliser,

$$\langle \nabla_a \log P_0(a) - \kappa_t(a - \tilde{x}) \rangle_{x,t} = 0 \implies \langle -\nabla_a V_0 \rangle_{x,t} = \kappa_t (\langle a \rangle_{x,t} - \tilde{x}).$$

Multiplying by e^t and using $e^t \kappa_t = e^{-t}/\Delta_t$, $e^t \tilde{x} = e^{2t}x$, the right side is exactly $-x/\Delta_t + (e^{-t}/\Delta_t)\langle a \rangle_{x,t} = S(x, t)$ by (2.34). Hence

$$\boxed{S(x, t) = e^t \left\langle -\nabla_a V_0 \right\rangle_{x,t}} \quad (2.38)$$

the optimal score is e^t times the expected original force $-\nabla_a V_0$ under the tilted measure centred at $\tilde{x} = e^t x$.

We call (2.26) the *score-posterior identity*; in the empirical-Bayes literature it is Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011; Vincent, 2011). It holds for any p_0 whatever, and it says something stronger than an analogy: computing a score *is* solving a Bayesian inference problem. The exact score is unavailable in practice for the plain reason that p_0 is unknown.

For *vector-valued* data $a \in \mathbb{R}^L$ noised coordinatewise, (2.26) holds componentwise but with the *full* posterior:

$$\left(\nabla_x \log p_t(x) \right)_k = \frac{e^{-t} \mathbb{E}[a_k \mid x_0, \dots, x_{L-1}] - x_k}{\Delta_t}. \quad (2.39)$$

The conditioning is on *every* coordinate. The channel is independent across frames, but the prior is not, so the posterior mean of frame k , and with it the k -th component of the score, depends on all L observations. Equation (2.39) is the exact sense in which the joint score of a trajectory differs from a stack of per-frame scores, and the reason this thesis studies the former.

The empirical score. Replace p_0 by the empirical measure of a training set $\{a^{(i)}\}_{i=1}^N$, which puts mass $1/N$ on each example. The convolution is then a finite Gaussian mixture and the score is available in closed form,

$$\nabla_x \log p_t^{\text{emp}}(x) = \frac{1}{\Delta_t} \left(e^{-t} \sum_{i=1}^N w_i(x) a^{(i)} - x \right), \quad w_i(x) = \frac{e^{-\|x - e^{-t} a^{(i)}\|^2 / 2\Delta_t}}{\sum_j e^{-\|x - e^{-t} a^{(j)}\|^2 / 2\Delta_t}}, \quad (2.40)$$

a softmax-weighted average of training points. This is exactly computable and exactly useless as a generative model: as $t \rightarrow 0$ the weights concentrate on the nearest training example and the reverse dynamics deposit the trajectory on top of it. Integrated exactly, (2.23) driven by (2.40) returns the empirical measure itself; run as any sampler actually is, from a Gaussian start and at finite step size, it returns points arbitrarily close to the training set. Either way nothing new is generated, which is the point.

The fitted score. A neural network $s_\phi(x, t)$ trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011) to minimise

$$\mathbb{E}_t \mathbb{E}_{a \sim p_{\text{data}}} \mathbb{E}_\varepsilon \left\| s_\phi \left(e^{-t} a + \sqrt{\Delta_t} \varepsilon, t \right) + \frac{\varepsilon}{\sqrt{\Delta_t}} \right\|^2. \quad (2.41)$$

Two optima must be kept apart. At the population level — the expectation over p_{data} taken literally — the minimiser of (2.41) over all measurable functions is the exact score of the noised population law (Vincent, 2011). Conditional on a finite training set, the empirical objective is minimised by the empirical score (2.40). A trained network reaches neither exactly, and one proposed mechanism of generalisation, in the controlled settings where it has been analysed, is precisely this gap: the network is constrained enough to be unable to represent (2.40), and that failure to fit the empirical optimum is what produces novel samples (Kadkhodaie et al., 2024; Kamb and Ganguli, 2024). On that account the mechanism of generalisation is an approximation-theoretic property of an architecture, which is a considerably less comfortable foundation than the variational principle of Section 2.1.

2.5 Transitions along the generation trajectory

Because generation is a dynamical process, its failures are dated. Two crossover times structure the reverse trajectory.

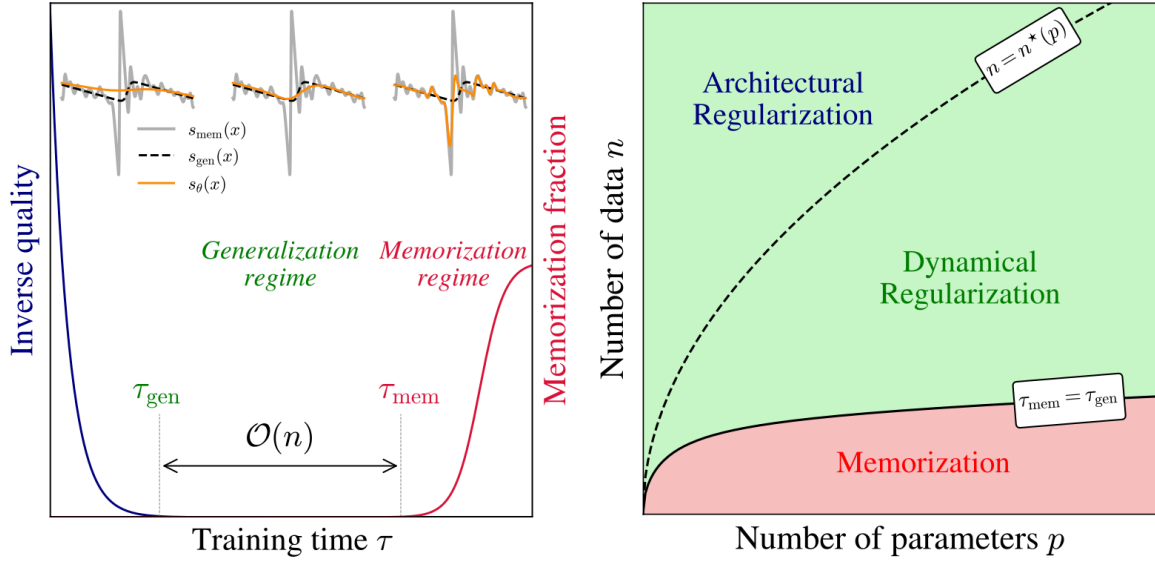


Figure 2.5: The qualitative picture behind Section 2.5. *Left:* sample quality (inverse, blue) and memorised fraction (red) against training epoch τ ; a generalisation window opens between τ_{gen} and τ_{mem} whose width grows with the training-set size n — along the training-epoch axis, where the mechanism is early stopping. *Right:* the corresponding regimes in the (data, parameters) plane. Reproduced from Bonnaire et al. (2025), their Figure 1.

Speciation. Run backwards from noise, the trajectory commits to a broad class of outcomes long before it commits to a specific sample. In the Gaussian-mixture models analysed by Biroli et al. (2024) and Achilli et al. (2026), for data whose covariance has top eigenvalue λ , the commitment happens at $t_S \sim \frac{1}{2} \log \lambda$ (in the variance-preserving convention used here), when the contracted signal along that direction becomes comparable with the accumulated noise. When the data carry coherent extensive structure — a spiked covariance whose top eigenvalue grows with the dimension — t_S diverges logarithmically with it. When the data are hierarchical there is not one such time but a *ladder* of them, one per level of the hierarchy, and features are resolved coarse-to-fine (Sclocchi et al., 2025b,a; Favero et al., 2025a,b): the same ordering that Bachtis et al. (2024) observe in the training of an energy-based model, appearing here along the generation trajectory instead.

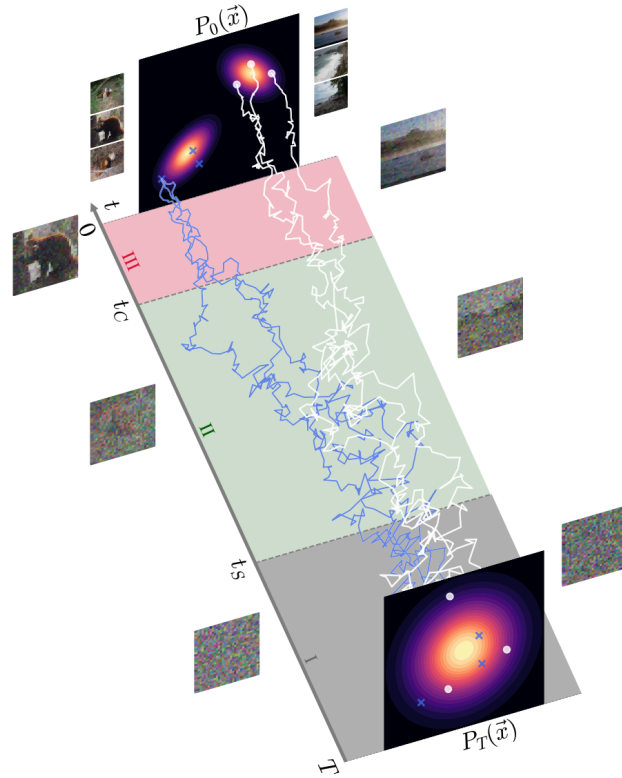


Figure 2.6: The two crossover times below, illustrated on a two-dimensional Gaussian-mixture example. Trajectories destined for different classes are indistinguishable above the speciation time t_S (regime I); between t_S and the collapse time t_C they commit to a class without yet matching any training point (regime II, where a model genuinely generalises); below t_C they are captured by a single training example (regime III). Reproduced from Biroli et al. (2024), their Figure 1.

Collapse and memorisation. Later in the reverse trajectory a second transition occurs, and it is the one that limits the whole construction (Figures 2.6 and 2.5). Biroli et al. (2024) and Bonnaire et al. (2025) show that dynamics driven by the empirical score are captured by a single training example past a time t_C , and that avoiding this requires a training set exponentially large in the data dimension. Memorisation need not be a collapse onto one example: Achilli et al. (2024) identify a geometric form in which dimensions are lost without any single training point capturing the trajectory, and Garnier-Brun et al. (2026) show how the resulting generalisation can be biased. Solvable models of the learning problem itself (Cui et al., 2025) complete the picture. The criterion is entropic:

collapse occurs when the excess entropy of the noised law relative to the terminal measure falls below $\log N$, so the dataset size needed to postpone it grows as $e^{(\text{excess entropy})}$.

The two transitions bracket the useful part of the trajectory. Above t_S nothing has been decided; below t_C the empirical score reproduces its training data.

2.6 Locality, architecture, and message passing

If the fitted score is doing the work, the natural question is what the architecture contributes, and here a second literature meets the first. The network in almost every image or video diffusion model is a U-Net (Figure 2.7) (Ronneberger et al., 2015): a convolutional encoder that repeatedly downsamples, a decoder that upsamples back, and long skip connections joining each encoder stage to the decoder stage of the same resolution. The design is almost universal and was, for a long time, justified anecdotally.

Mei (2024) gives it a precise reading, and it is the reading this thesis works inside. For data generated by a tree-structured hierarchical model, the exact denoiser is computed by belief propagation, which consists of an upward pass that accumulates evidence from the leaves and a downward pass that redistributes it, the second reusing the messages formed by the first. A U-Net has exactly this shape: the encoder is the upward pass, the decoder is the downward pass, and the long skip connections are the reuse of the intermediate messages. On this account, for the hierarchical generative models analysed there, a U-Net is not merely a good inductive bias for denoising but an *approximate implementation of the exact inference algorithm*, a reading that yields sample-complexity bounds for learning the denoiser in those models; nothing in it certifies an arbitrary U-Net on arbitrary data as message passing. Convolutional classifiers admit the corresponding reading for the upward pass alone.

Two consequences are useful for the setting considered in this thesis. First, sum-product is exact on trees (Theorem 3.4), so for a chain the posterior marginals can be computed without introducing an additional approximation at the inference stage. The chain structure makes the problem particularly tractable: together with the homogeneous

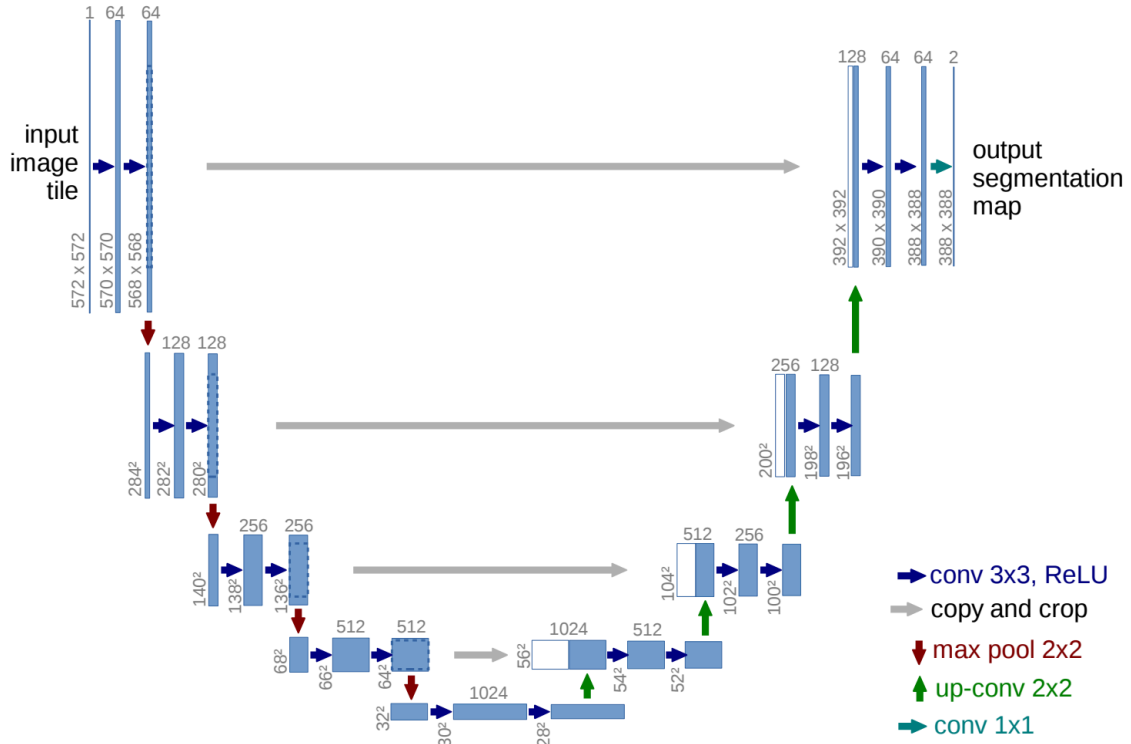


Figure 2.7: The architecture behind the reading below. Resolution halves at each down-sampling step on the encoder side (left) and doubles again on the decoder side (right); the grey arrows are the skip connections, copying each encoder stage’s feature map across to the decoder stage of matching resolution before its convolutions run. Reproduced from [Ronneberger et al. \(2015\)](#), their Figure 1.

transition law, it leads in the Gaussian case to the closed-form recursions derived in Chapter 6. This provides a simple reference point against which learned denoisers can be compared.

Second, the same inference formulation offers a way to study how much of the surrounding sequence is relevant for estimating the score at a given site. Rather than treating the receptive field only as an architectural hyperparameter, one can relate it to the decay of the influence of increasingly distant observations. Chapter 6 computes this decay rate explicitly for the chain studied here. A related result on sequences is [Garnier-Brun et al. \(2024\)](#), who find that a transformer trained on hierarchically generated sequences reconstructs correlations sequentially, shortest length scale first, and who use exact belief propagation as the reference against which the network is measured.

2.7 Other generative families

To *generate* an object is to draw a sample from the data law p_0 , not to evaluate p_0 at a point: there is no single best image, sequence, or molecule, only a distribution that puts more mass on the ones that look right, and generation is sampling from it. A training set $\{a^{(i)}\}_{i=1}^N$ is, throughout this thesis (Section 2.4), exactly this: finitely many independent draws from p_0 , standing in for a law nobody can write down. What every family shares, whatever the recipe, is the same computational strategy for turning that goal into an algorithm: start from a reference law that is trivial to sample, almost always an isotropic Gaussian, and construct a map, deterministic or stochastic, carrying samples of the reference law to samples that look drawn from p_0 . Diffusion builds that map by running (2.23) backwards from the terminal Gaussian; the families below build it differently.

Generative adversarial networks sample by a single push-forward, $x = G_\theta(z)$ with z simple noise, and train G_θ against a discriminator in a minimax game (Goodfellow et al., 2014): no likelihood is ever computed, samples can be sharp, and the adversarial objective is notoriously unstable to train. *Variational autoencoders* sample by the same kind of push-forward through a stochastic decoder, $z \sim p(z)$, $x \sim p_\theta(x \mid z)$, and train by maximising a tractable lower bound on the likelihood (Kingma and Welling, 2014); reconstructions are often blurred, an effect usually attributed to the Gaussian decoder’s conditional averaging rather than to the bound itself. *Normalizing flows* push a simple base measure through an invertible network, so the change-of-variables formula gives $\log p_\theta$ exactly (Rezende and Mohamed, 2015; Papamakarios et al., 2021), at the price of every layer needing a tractable Jacobian. *Autoregressive models* sample one coordinate at a time, $x_i \sim p_\theta(x_i \mid x_{<i})$, factoring the joint by the chain rule and training each factor by ordinary classification; transformer-based language models (Vaswani et al., 2017) and, for images, PixelCNN-style networks (van den Oord et al., 2016) both work this way, with an exact likelihood but a sampling cost sequential in the data dimension. *Diffusion models*, the subject of this chapter, instead sample by repeated denoising (Section 2.3); sampling is

sequential in the noise level rather than the dimension, and iterative rather than one-shot. [Ghio et al. \(2024\)](#) places flows, diffusion and autoregressive networks under one spin-glass analysis and finds the same first-order phase transition governing sample quality in all three, so the comparison between them is not merely architectural.

Chapter 3

The Model, and the Object of Study

This chapter fixes what the thesis computes and the language it computes in. A Markov chain generates a clean trajectory; an Ornstein–Uhlenbeck channel corrupts every frame independently; the object of study is the joint score of the corrupted sequence. Two justifications come before any analysis. The first is that the quantity worth studying is the joint score of a whole trajectory and not a per-frame score; the second is that the stationary AR(1) chain is a convenient model in which to study it.

3.1 Markov sequences

We briefly recall the notions of stationarity, Markov dependence, and Toeplitz covariance used throughout, together with the one point among them that is easy to get wrong and matters later (Brockwell and Davis, 1991).

A process $(X_k)_{k \in \mathbb{Z}}$ is *strictly stationary* if its finite-dimensional laws are invariant under shifts in time: for every $n \in \mathbb{N}$, every $k_1 < \dots < k_n$, and every $h \in \mathbb{Z}$,

$$(X_{k_1+h}, \dots, X_{k_n+h}) \stackrel{d}{=} (X_{k_1}, \dots, X_{k_n}). \quad (3.1)$$

A process with finite second moments is *weakly*, or *covariance-*, *stationary* if its mean is

constant,

$$\mathbb{E}[X_k] = \mu, \quad (3.2)$$

and its covariance depends only on the distance between two indices,

$$\text{Cov}(X_j, X_k) = \gamma(j - k). \quad (3.3)$$

Strict stationarity implies weak stationarity whenever second moments exist. For a Gaussian process the converse also holds, because a Gaussian law is determined completely by its mean and covariance.

Weak stationarity alone gives the covariance matrix of any segment $(X_0, \dots, X_{L-1})$ of length L a *Toeplitz* form, constant along every diagonal:

$$\Sigma_{ij} = \gamma(i - j), \quad i, j = 0, \dots, L - 1, \quad \Sigma = \begin{pmatrix} \gamma(0) & \gamma(1) & \gamma(2) & \cdots \\ \gamma(1) & \gamma(0) & \gamma(1) & \cdots \\ \gamma(2) & \gamma(1) & \gamma(0) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (3.4)$$

using $\gamma(-d) = \gamma(d)$ for a real-valued process.

Definition 3.1 (Markov chain). A sequence $(X_k)_{k \geq 0}$ on a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}(X_{k+1} \in A \mid X_k, X_{k-1}, \dots, X_0) = \mathbb{P}(X_{k+1} \in A \mid X_k). \quad (3.5)$$

It is *time-homogeneous* if a single *transition kernel* $K(x' \mid x)$, with $\int K(x' \mid x) dx' = 1$ for every x , governs every step; every chain in this thesis is of this kind.

The Markov property (3.5) is exactly a statement about *factorisation*: the joint law of the first L states is

$$p(x_0, x_1, \dots, x_{L-1}) = p_0(x_0) \prod_{k=0}^{L-2} K(x_{k+1} \mid x_k). \quad (3.6)$$

This is the most heavily used equation of the thesis. It says the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables: the structure that makes the chain a tree in the factor-graph sense of Chapter 3, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 5, and that makes belief propagation exact. Marginals evolve by integrating the kernel, $p_{k+1}(x') = \int K(x'|x) p_k(x) dx$, and a law π with $\pi = K\pi$ is *stationary*: a chain started in π remains in it, and its trajectory is then strictly stationary.

Example 3.2 (The Gaussian AR(1) chain). The running example throughout is the scalar Gaussian autoregression of order one,

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad k = 0, \dots, L-2, \quad (3.7)$$

with kernel $K(a' | a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$. Taking $\sigma_\eta^2 = 1 - \alpha^2$ and $a_0 \sim \mathcal{N}(0, 1)$ gives $a_k \sim \mathcal{N}(0, 1)$ for every k and

$$\text{Cov}(a_j, a_k) = \alpha^{|j-k|}, \quad \Sigma = \begin{pmatrix} 1 & \alpha & \alpha^2 & \dots & \alpha^{L-1} \\ \alpha & 1 & \alpha & \dots & \alpha^{L-2} \\ \alpha^2 & \alpha & 1 & \dots & \alpha^{L-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha^{L-1} & \alpha^{L-2} & \alpha^{L-3} & \dots & 1 \end{pmatrix}, \quad (3.8)$$

Toeplitz by (3.4). Because a_0 is chosen to be the fixed point of K itself — $\mathcal{N}(0, 1)$ solves $\pi = K\pi$ exactly when the innovation is $\mathcal{N}(0, \sigma_\eta^2)$ — the chain is started in its own stationary law, hence strictly stationary by the general fact above, and not merely weakly stationary by the covariance computation. In this Gaussian baseline the three notions coincide completely: the same model is a time-homogeneous Markov chain, a strictly stationary process, and a weakly stationary process with Toeplitz covariance, all at once. Every chapter from Chapter 5 onwards concerns (3.7), one of its non-Gaussian variants, or what an estimator can recover of it through noise.

The coincidence is special to the Gaussian case, and breaks the moment Chapter 7

keeps $a_0 \sim \mathcal{N}(0, 1)$ and α fixed but replaces η_k with a Laplace innovation of matched variance $1 - \alpha^2$. Only the weak notion is available for the non-Gaussian chains this thesis studies.

3.2 The Ornstein–Uhlenbeck corruption channel

The noising process used by diffusion models, and by every result of this thesis, is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 = a. \quad (3.9)$$

The drift and diffusion constants are the variance-preserving normalisation of the diffusion literature (Song et al., 2021); any other choice rescales time. The process is the canonical stationary Gaussian Markov diffusion: by Doob’s characterisation (Doob, 1942) it is, up to affine rescaling of state and time, the only one-dimensional stationary Gaussian Markov process with continuous sample paths and nondegenerate stationary variance. Stationary there describes the process started from its invariant law $\mathcal{N}(0, 1)$; the run from the fixed $X_0 = a$ of (3.9) merely relaxes to it. It is exactly solvable: the derivation is Toolbox 3.1.

Toolbox 3.1 — Solving the Ornstein–Uhlenbeck SDE by an integrating factor

Integrating factor. Multiply (3.9) by e^t and apply the product rule to $e^t X_t$ — ordinary, not Itô’s, because e^t carries no stochastic part and contributes no quadratic-variation term:

$$d(e^t X_t) = e^t X_t dt + e^t dX_t = e^t X_t dt + e^t (-X_t dt + \sqrt{2} dW_t) = \sqrt{2} e^t dW_t.$$

The drift term cancels exactly, leaving a pure stochastic differential.

Integration. Integrating from 0 to t and using $X_0 = a$,

$$e^t X_t - a = \sqrt{2} \int_0^t e^s dW_s \quad \implies \quad X_t = a e^{-t} + \sqrt{2} e^{-t} \int_0^t e^s dW_s.$$

The stochastic integral is Gaussian. Write $I_t := \sqrt{2} e^{-t} \int_0^t e^s dW_s$. This is an Itô integral of a *deterministic* integrand against Brownian motion: it is the L^2 limit of Riemann-sum increments $\sum_i \sqrt{2} e^{-t} e^{s_i} (W_{s_{i+1}} - W_{s_i})$, each term Gaussian and independent of the others, so the limit I_t is Gaussian too, with mean zero by linearity of the expectation. Its variance follows from the Itô isometry,

$$\begin{aligned} \text{Var}(I_t) &= \mathbb{E} \left[\left(\sqrt{2} e^{-t} \int_0^t e^s dW_s \right)^2 \right] = 2e^{-2t} \int_0^t e^{2s} ds \\ &= 2e^{-2t} \cdot \frac{e^{2t} - 1}{2} = 1 - e^{-2t} = \Delta_t. \end{aligned}$$

Conclusion. $X_t = a e^{-t} + I_t$ with $I_t \sim \mathcal{N}(0, \Delta_t)$ independent of a , so

$$X_t \mid X_0 = a \sim \mathcal{N}(a e^{-t}, \Delta_t), \quad \Delta_t = 1 - e^{-2t},$$

which is (3.10) below. □

Corrupting a data point a for a time t therefore produces

$$x = a e^{-t} + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1) : \quad (3.10)$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$ the channel is the identity; as $t \rightarrow \infty$, $x \sim \mathcal{N}(0, 1)$ regardless of a . The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0 and diverges as $1/2t$ for small t . The channel interpolates between the data distribution and a standard Gaussian, which is what a diffusion model needs.

3.3 How the object of study was arrived at

The estimand of this thesis was not chosen at the outset. It was reached by getting it wrong first; the correction is a consequential step in the project.

The propagator hypothesis. The work began from a structural guess about sequences. If a trajectory is generated by a fixed dynamical rule, the score at one frame ought to be obtainable from the score at the previous one by a fixed operator. Writing $S_{:,u}$ for the block of the score at internal time u , the conjecture was that some *propagator* P_t satisfies

$$S_{:,u}(x) \approx P_t(S_{:,u-1}(x)). \quad (3.11)$$

If such an operator existed and were cheap, generation over a long sequence would cost one solve plus $L - 1$ applications of P_t . The programme adopted to test (3.11) was deliberately austere: exactly solvable models only, closed forms wherever possible, so that the focus is on understanding

Six toy models, and a mistake they shared. A sequence of solvable objects was built to probe (3.11): a static Gaussian mixture under the channel; a two-frame joint law; a two-dimensional rotating process $a_{u+1} = R_{\Delta\theta} a_u + \xi_u$; a billiard in a box; a scalar AR(1) chain examined on the whole (u, t) plane; and a diffusion on the circle. The rotating model and the circle are the direct ancestors of Chapter 4, and the AR(1) study is the ancestor of Chapters 5 and 6.

All but one of them computed the wrong object. What was evaluated was the *per-frame marginal* score

$$s(x_u, u, t) = \nabla_{x_u} \log p_t(x_u | u), \quad (3.12)$$

which integrates out every other frame. Doing so is not a mild approximation. Driving generation with the stacked marginal scores is operationally equivalent to replacing the joint score by the score of the product law $\prod_u p_0(a_u | u)$ — the model one would have if the clean law factorised over frames, which it does not for any correlated process and, in

particular, for no model in the list. The quantity being studied had the frames' dependence removed from it before the study began, so no propagator could have been found in it: there was nothing left for P_t to propagate.

The correction. The object was corrected to the *joint* score of the whole trajectory, $S(x, t) = \nabla_{(x_0, \dots, x_{L-1})} \log P_t(x_0, \dots, x_{L-1})$, whose k -th block is the posterior expectation

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x_0, \dots, x_{L-1}] - x_k}{\Delta_t}, \quad (3.13)$$

taken over the full posterior of the clean trajectory given all L observations.

3.4 Why the joint score

A dynamic datum is a sequence $x = (x_0, \dots, x_{L-1})$, so its reverse diffusion is driven by the *joint* score

$$s_t(x) = \nabla_x \log p_t(x_0, \dots, x_{L-1}), \quad (3.14)$$

rather than by the collection of one-frame marginal scores

$$s_{t,k}^{\text{marg}}(x_k) = \partial_{x_k} \log p_{t,k}(x_k).$$

The distinction is structural:

$s_{t,k}(x_0, \dots, x_{L-1})$ may depend on the other frames, whereas $s_{t,k}^{\text{marg}}(x_k)$ cannot.

(3.15)

The smallest example already contains the whole argument. Consider two consecutive states of the stationary Gaussian AR(1) chain,

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \sim \mathcal{N}(0, \Sigma_0), \quad \Sigma_0 = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}, \quad (3.16)$$

and corrupt both coordinates independently through

$$x_i = e^{-t} a_i + \sqrt{\Delta_t} z_i, \quad \Delta_t = 1 - e^{-2t}, \quad z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (3.17)$$

Since the added noise is independent,

$$\text{Var}(x_i) = e^{-2t} \text{Var}(a_i) + \Delta_t = e^{-2t} + 1 - e^{-2t} = 1, \quad (3.18)$$

$$\text{Cov}(x_0, x_1) = e^{-2t} \text{Cov}(a_0, a_1) = \alpha e^{-2t}. \quad (3.19)$$

Writing

$$r_t := \alpha e^{-2t}, \quad (3.20)$$

the noisy pair is therefore

$$\begin{pmatrix} x_0 \\ x_1 \end{pmatrix} \sim \mathcal{N}(0, \Sigma_t), \quad \Sigma_t = \begin{pmatrix} 1 & r_t \\ r_t & 1 \end{pmatrix}. \quad (3.21)$$

The two marginals are thus

$$x_0, x_1 \sim \mathcal{N}(0, 1), \quad s_{t,0}^{\text{marg}}(x_0) = -x_0, \quad s_{t,1}^{\text{marg}}(x_1) = -x_1, \quad (3.22)$$

with no α anywhere.

The joint law is different. Since

$$\Sigma_t^{-1} = \frac{1}{1 - r_t^2} \begin{pmatrix} 1 & -r_t \\ -r_t & 1 \end{pmatrix}, \quad (3.23)$$

its density is

$$p_t(x_0, x_1) = \frac{1}{2\pi\sqrt{1 - r_t^2}} \exp\left[-\frac{x_0^2 - 2r_t x_0 x_1 + x_1^2}{2(1 - r_t^2)}\right]. \quad (3.24)$$

Differentiating its logarithm gives

$$s_{t,0}(x_0, x_1) = -\frac{x_0 - r_t x_1}{1 - r_t^2}, \quad (3.25)$$

$$s_{t,1}(x_0, x_1) = -\frac{x_1 - r_t x_0}{1 - r_t^2}, \quad (3.26)$$

or

$$s_t(x_0, x_1) = -\frac{1}{1 - r_t^2} \begin{pmatrix} x_0 - r_t x_1 \\ x_1 - r_t x_0 \end{pmatrix}. \quad (3.27)$$

This is the missing information: the marginal score of frame 0 is $-x_0$, whereas its joint score contains the observation x_1 through $r_t x_1$. The temporal coupling has disappeared from the marginal law but remains explicitly in the joint score.

For a length- L stationary Gaussian AR(1) chain the same calculation becomes

$$\boxed{\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I, \quad (\Sigma_0)_{ij} = \alpha^{|i-j|}, \quad s_t(x) = -\Sigma_t^{-1}x.} \quad (3.28)$$

Hence

$$(\Sigma_t)_{ij} = \begin{cases} 1, & i = j, \\ e^{-2t}\alpha^{|i-j|}, & i \neq j. \end{cases} \quad (3.29)$$

The diagonal, and therefore every one-frame Gaussian marginal, is independent of α ; all information about the temporal coupling sits in the off-diagonal structure.

This also makes the Fisher-information statement immediate. A one-frame density is simply

$$p_{t,k}(x_k; \alpha) = \frac{1}{\sqrt{2\pi}} e^{-x_k^2/2},$$

which does not contain α . Therefore

$$\partial_\alpha \log p_{t,k}(x_k; \alpha) = 0 \quad \implies \quad \boxed{I_k(\alpha) = \mathbb{E} \left[\left(\partial_\alpha \log p_{t,k}(X_k; \alpha) \right)^2 \right] = 0.} \quad (3.30)$$

Hence, its Fisher information is exactly zero.

Remark 3.3 (What changes outside the Gaussian case). The conclusion that the marginal is always $\mathcal{N}(0, 1)$ is special to Gaussian data. For a general scalar input A ,

$$X_t = e^{-t}A + \sqrt{\Delta_t} Z \implies \varphi_{X_t}(u) = \varphi_A(e^{-t}u) \exp\left(-\frac{\Delta_t u^2}{2}\right). \quad (3.31)$$

If $A \sim \mathcal{N}(0, 1)$, the two Gaussian factors combine to give $\varphi_{X_t}(u) = e^{-u^2/2}$ exactly. By contrast, for a unit-variance Laplace variable, $\varphi_A(u) = (1 + u^2/2)^{-1}$, and therefore

$$\varphi_{X_t}(u) = \frac{\exp[-\Delta_t u^2/2]}{1 + e^{-2t}u^2/2}, \quad (3.32)$$

which is not Gaussian for any finite t . Gaussian corruption therefore preserves the Gaussian family exactly, while for non-Gaussian data it only drives the marginal progressively towards Gaussianity as t increases.

The lesson is therefore not that marginal scores are always trivial, but that *one-frame marginals do not determine dependence*. Figure 3.1 shows precisely this contrast.

Finally, by the score-posterior identity (3.13),

$$s_{t,k}(x) = \frac{e^{-t}\mathbb{E}[a_k \mid x_0, \dots, x_{L-1}] - x_k}{\Delta_t}, \quad (3.33)$$

so computing the joint score is an inference problem over the whole trajectory. This is the point at which the graphical structure of the Markov chain becomes useful, and motivates the belief-propagation construction developed below.

3.5 Markov random fields and factor graphs

Graphical models assign structure to distributions the way a precision matrix assigns it to a Gaussian, and for arbitrary variables. A collection $x = (x_i)_{i \in V}$ on an undirected graph G is a *Markov random field* if each variable is conditionally independent of the rest given its neighbours, $p(x_i \mid x_{V \setminus \{i\}}) = p(x_i \mid x_{\partial i})$. For a strictly positive density the Hammersley–Clifford theorem (Besag, 1974) makes that equivalent to factorisation over

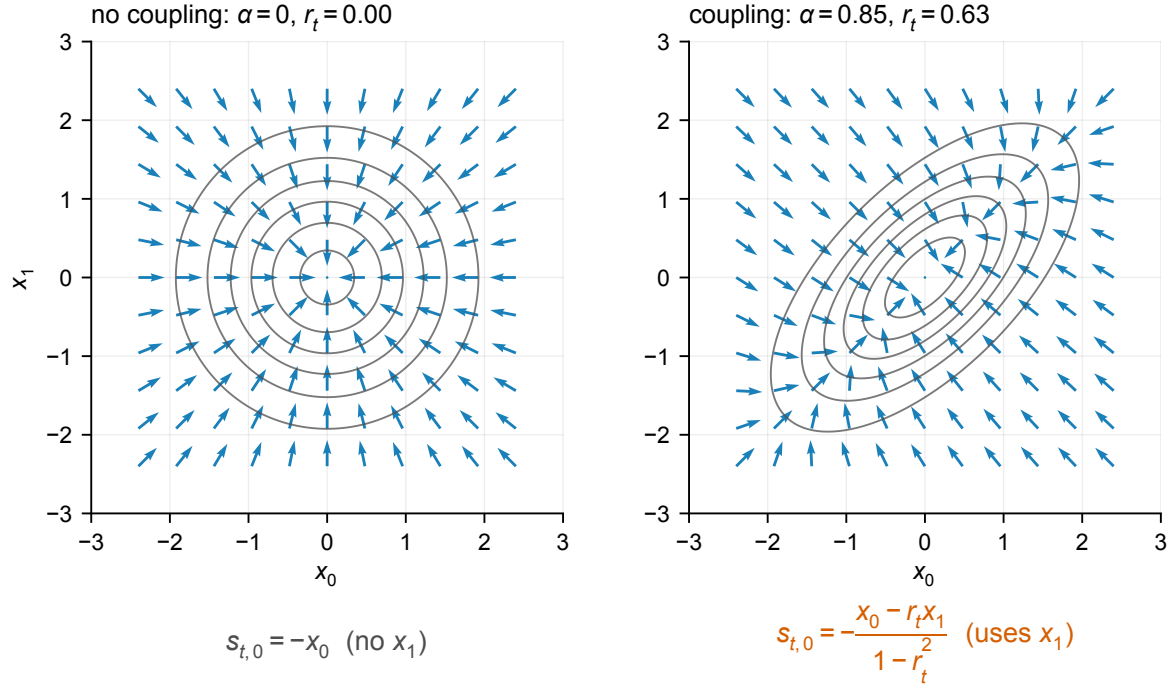


Figure 3.1: Same one-frame marginals, different joint score. In both cases $x_0, x_1 \sim \mathcal{N}(0, 1)$, so a frame-wise model sees the same marginal densities and the same scores $(-x_0, -x_1)$. When $\alpha = 0$ the joint density is circular and the joint score is indeed radial. When $\alpha > 0$, the off-diagonal covariance $r_t = \alpha e^{-2t}$ tilts the joint density and changes each score component through the other coordinate, as in (3.27). The dependence is therefore visible in the joint score and invisible in either marginal.

the cliques of G ,

$$p(x) = \frac{1}{Z} \prod_{C \in \text{cliques}(G)} \psi_C(x_C) = \frac{1}{Z} \exp \left[- \sum_C E_C(x_C) \right], \quad (3.34)$$

so conditional independence, graphical locality and additive local energies are one thing seen three ways. For Gaussian fields it becomes matrix-concrete: the energy is $\frac{1}{2}a^\top J a - h^\top a$, and the graph is exactly the sparsity pattern of J . A Gaussian Markov chain therefore has a tridiagonal precision, which Chapter 5 derives and uses throughout.

If each variable is observed through its own independent channel, the likelihood factors touch one variable each and no clique is enlarged: *the posterior of an MRF under independent per-node observations is an MRF on the same graph*. Independent unary observation changes the potentials, never the graph; correlated observation noise would

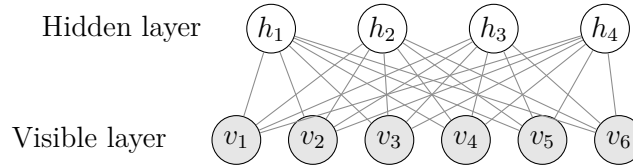


Figure 3.2: The restricted Boltzmann machine’s graph: every hidden unit connects to every visible unit, and no edge runs within a layer. Because the energy has only visible–hidden cliques, conditioning on one layer leaves the other in disjoint unary cliques, so $p(h \mid v) = \prod_a p(h_a \mid v)$ and $p(v \mid h) = \prod_i p(v_i \mid h)$: the same conditional-independence argument as the general MRF case above, specialised to a graph with no within-layer edges at all.

add edges.

An MRF records which variables interact but not how the joint factorises: a triangle of pairwise terms and one triple term have the same graph. The *factor graph* makes the factorisation itself graphical (Kschischang et al., 2001). Given $p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a})$, it is the bipartite graph with a variable node per x_i , a factor node per ψ_a , and an edge whenever $x_i \in x_{\partial a}$. A connected factor graph with one fewer edge than nodes is a *tree*, and trees are the tractable case: on them marginals and partition functions are exact under the recursion of the next section, whatever the alphabet. The factor graph of a Markov chain observed through per-node noise is a tree of a particularly simple shape, a spine of variables and transition factors with one evidence leaf per variable, as Chapter 5 verifies by counting. Figure 3.2 draws the bipartite structure used in RBMs as an example.

3.6 Belief Propagation

On a factor graph, sum–product belief propagation (BP) passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). Variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages:

$$\nu_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(x_i), \quad \hat{\nu}_{a \rightarrow i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(x_j) \, dx_{\partial a \setminus i}, \quad (3.35)$$

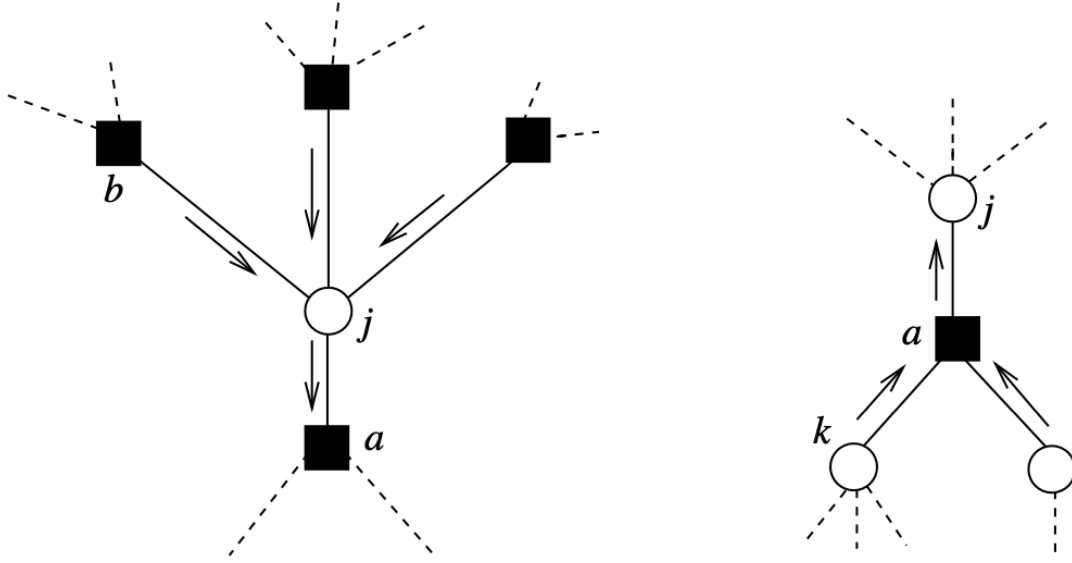


Figure 3.3: The two halves of (3.35), drawn on the local neighbourhood each one uses. *Left:* the message into j that will be sent on to a , a function of the incoming factor messages $\hat{\nu}_{b \rightarrow j}$ from every other factor neighbour $b \neq a$. *Right:* the message out of a towards j , a function of the incoming variable messages $\nu_{k \rightarrow a}$ from every other variable neighbour $k \neq j$. In the notation of this thesis, j here is i and k is j ; the original also carries iteration superscripts $(t)/(t+1)$; because this thesis's graphs are trees (Theorem 3.4 below), one outward sweep and one inward sweep of each rule is exact, and no iteration index is needed here. Reproduced from Mézard and Montanari (2009), their Figure 14.3.

with the belief at node i proportional to $\prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(x_i)$. The “exclude the recipient” rule is the heart of the algorithm: the message $i \rightarrow a$ summarises everything node i knows *except* what it learned from a , so that no piece of evidence is ever counted twice along a path.

Figure 3.3 draws the two rules of (3.35).

Theorem 3.4 (Tree exactness of sum–product). *On a tree-shaped factor graph, one inward and one outward pass of (3.35) compute the exact node and factor marginals of p (Mézard and Montanari, 2009, Theorem 14.1). The resulting beliefs are unique; the individual unnormalised messages are defined only up to positive multiplicative constants.*

Proof. (Not formal) Each edge of a tree splits the graph into two disjoint halves, so the message crossing it summarises exactly the half behind it and no evidence reaches a

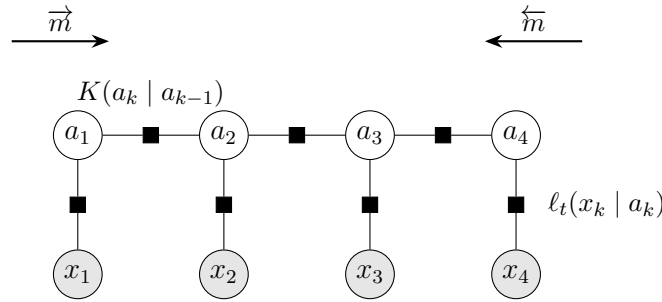


Figure 3.4: The posterior factor graph of a Markov chain observed through the coordinatewise channel. Open circles are latent variables, shaded circles observations, squares factors. Each transition factor $K(a_k | a_{k-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor attached to a single latent variable, because the channel corrupts every coordinate independently. Conditioning therefore reweights node potentials without creating any edge among the a_k , so the posterior graph is still a chain and Theorem 3.4 applies to it. Arrows show the two message sweeps of (3.35). The construction fails as soon as the forward noise is correlated across coordinates: the likelihood then touches several a_k at once and the graph acquires edges.

node twice. The recursion terminates with every marginal having absorbed each factor once. $\square$

On graphs with cycles the same updates, iterated, define *loopy* BP, which is extraordinarily useful but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none of that is needed in this thesis, because our graphs are trees.

The representation problem. For *discrete* variables each message is a finite probability vector and each update in (3.35) is finite arithmetic. For *continuous* variables each message is a probability *density*, an infinite-dimensional object, and the update integrals must be carried out in function space. Tree exactness (a property of the graph) says nothing about representability of the messages (a property of the model’s function class); keeping the two apart is essential, and the distinction is where all the difficulty of the non-Gaussian research chapter lives. There are three ways to make continuous BP computable: restrict to models whose messages close in a finite-dimensional family (the Gaussian route, next section); represent messages numerically on a grid (the route our non-Gaussian reference solver takes, Chapter 7); or project each message onto a tractable family and accept the error (the route whose cost this thesis measures).

Chapter 4

The Rotating Ring: A Two-Dimensional Dynamic Object

This is where the work started.

The object is a planar trajectory near a circle: a point that rotates by a fixed angle each step, held near the unit radius by a confining drift, and observed through the Ornstein–Uhlenbeck channel of Section 2.3. It is a dynamic object in the sense of Chapter 3, but a nonlinear one, and it was chosen before the scalar chain precisely because rotation is a simple dynamic whose signature is not visible in a single frame.

4.1 Model and notation

4.1.1 The clean process

The clean state is a radius and a phase, evolving in the internal time u as two independent diffusions,

$$dR_u = -\kappa(R_u - 1) du + \sqrt{2D_r} dB_u^{(r)}, \quad (4.1)$$

$$d\Theta_u = \omega du + \sqrt{2D_\theta} dB_u^{(\theta)}, \quad (4.2)$$

observed through the polar embedding

$$a_u = R_u \begin{pmatrix} \cos \Theta_u, \sin \Theta_u \end{pmatrix} \in \mathbb{R}^2. \quad (4.3)$$

We use the simplest constant-angular-drift reading of the model; richer readings — state-dependent drift, coupled radius and phase — change none of the structural conclusions below and are not pursued.

A point of $\mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis, and at each point the plane carries the orthonormal moving frame

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-\sin \theta, \cos \theta), \quad (4.4)$$

in which e_θ is e_r turned by a quarter revolution. Moving along e_r changes the radius at fixed angle, moving along e_θ changes the angle at fixed radius, and the area element is $dx dy = r dr d\theta$. The decomposition matters because the two directions are treated very differently by (4.1) and (4.2): the radius is pulled back towards one, the phase drifts and diffuses freely.

4.1.2 The radial drift is a confining force

Equation (4.1) is an overdamped Langevin equation, $dx_u = -\nabla V(x_u) du + \sqrt{2D} dB_u$, for the quadratic potential

$$V(r) = \frac{\kappa}{2} (r - 1)^2. \quad (4.5)$$

A potential stores energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa (r - 1). \quad (4.6)$$

The sign carries the content: along the *deterministic* drift flow $\dot{r} = -V'(r)$ the energy descends, $\frac{d}{du} V(r_u) = V'(r) \dot{r} = -V'(r)^2 \leq 0$. Under the full stochastic dynamics (4.1)

Itô's formula adds a fluctuation source,

$$dV(R_u) = \left[-V'(R_u)^2 + D_r V''(R_u) \right] du + \sqrt{2D_r} V'(R_u) dB_u^{(r)}, \quad (4.7)$$

so V is decreasing only on average and only away from the minimum; the descent statement is a property of the drift, not of sample paths. Reading the two cases of (4.6), if $r > 1$ then $F < 0$ and the force points inward, and if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards one from either side at a rate proportional to the displacement, with κ setting the stiffness. Overdamped Langevin dynamics equilibrates to $p_\infty \propto e^{-V/D}$, so the stationary radial law is $\mathcal{N}(1, D_r/\kappa)$.

One modelling caveat belongs here rather than in a footnote. A Gaussian law on $\mathbb{R}$ puts mass on $R_u < 0$, so (4.1) defines a *signed* radial coordinate, not a geometric radius: through the embedding (4.3) a negative R_u is the point at radius $|R_u|$ and phase $\Theta_u + \pi$. A strictly positive radius would require a reflected or Bessel-type process with a repulsive $1/R$ Itô term. We keep the linear SDE because it is exactly solvable and because, in the confining regime $D_r \ll \kappa$ used throughout, the stationary probability of a negative coordinate, $\Phi(-\sqrt{\kappa/D_r})$, is negligible — below 10^{-6} already at $\kappa/D_r = 23$ — so origin crossings play no role in anything measured here.

4.1.3 One trajectory is one data point

The trajectory is sampled at internal times $u_k = kh$ for $k = 0, \dots, L-1$, and the complete datum is the stacked vector

$$a = (a_0, a_1, \dots, a_{L-1}) \in \mathbb{R}^{2L}. \quad (4.8)$$

The L frames are correlated coordinates of one data point, not L samples. Relative to Chapter 5 the only change of setting is that each frame is now two-dimensional, so the score has L blocks of size two rather than L scalar components; the forward channel is

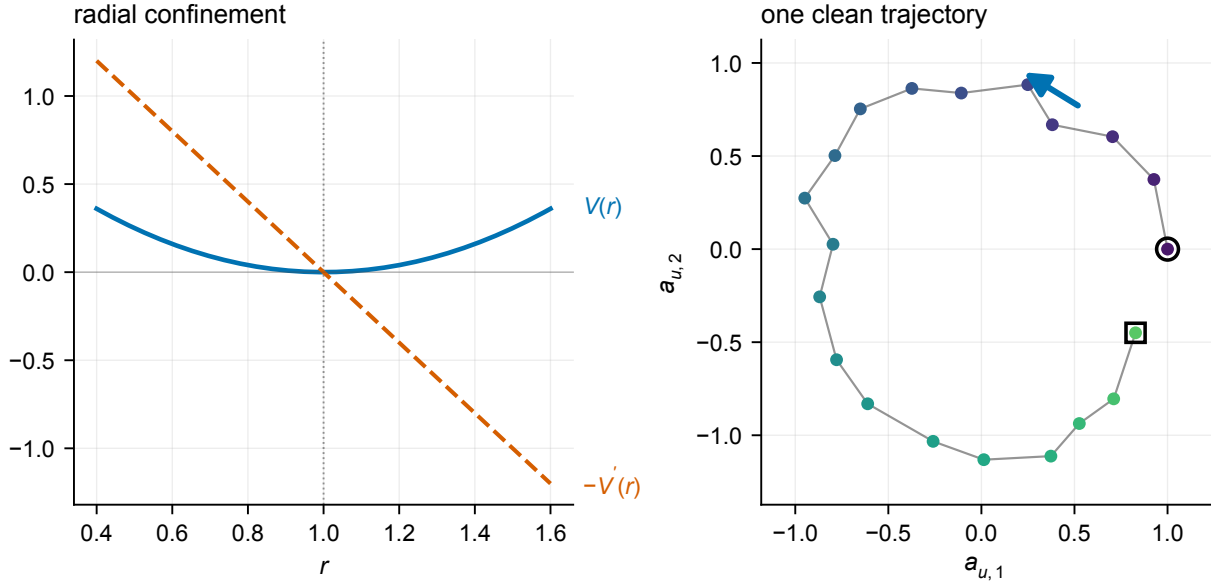


Figure 4.1: Radial confinement and angular motion. (a) The quadratic potential $V(r) = \frac{\kappa}{2}(r - 1)^2$ has its minimum at $r = 1$; the force $-V'(r) = -\kappa(r - 1)$ points towards that stable radius from both sides. (b) One trajectory generated by the polar model. Colour advances with the internal frame index; the open circle and square mark the first and final frames, respectively, and the curved arrow indicates the sense of the angular drift.

unchanged,

$$x = e^{-t}a + \sqrt{\Delta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_{2L}), \quad (4.9)$$

which is the channel of Section 3.2 acting on all $2L$ coordinates at once. Figures 4.1 and 4.2 show the model and the two clocks: internal time orders the frames along the path, diffusion time corrupts them simultaneously.

4.2 The gauge, and the exact score

4.2.1 The surrogate, and what it keeps

The surrogate retains a ring, a coherent rotation, and dependence along the trajectory, and drops the rest:

$$z_0 \sim p_\lambda(z) \propto \exp\left[-\frac{(\|z\| - 1)^2}{2\lambda}\right], \quad (4.10)$$

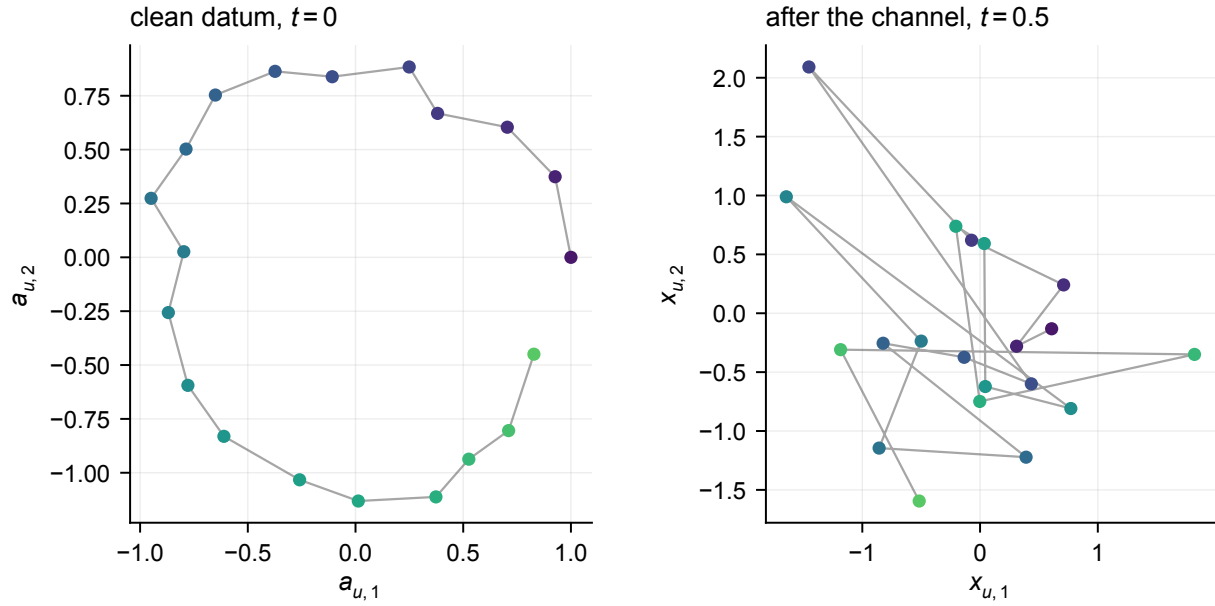


Figure 4.2: One trajectory before and after the Ornstein–Uhlenbeck channel. (a) A clean datum consisting of $L = 20$ ordered planar frames. (b) The same datum after independent coordinatewise corruption at diffusion time $t = 0.5$. Matching colours identify corresponding frame indices in the two panels. The channel obscures the circular path, but the complete $2L$ -dimensional trajectory remains the object whose joint score is sought.

$$z_{k+1} = R_\psi z_k + \sigma \eta_{k+1}, \quad \eta_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad (4.11)$$

with R_ψ the planar rotation through the one-frame angle $\psi = \omega h$.

Remark 4.1 (The surrogate is not a discretisation of the polar model). In (4.11) the ring acts only on z_0 . Later frames form a rotating Cartesian random walk and their radial variance grows without bound, whereas in (4.1) a confining force acts at every internal time. The surrogate is a solvable companion whose role is to exhibit the mechanism in closed form, not to approximate the model of Section 4.1.

4.2.2 The known rotation is an orthogonal change of frame

The rotation is the object to be removed, so we write it out:

$$R_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}. \quad (4.12)$$

Its columns are the standard basis vectors, each turned through ψ . They are unit vectors, since $\cos^2 \psi + \sin^2 \psi = 1$, and orthogonal, since $\cos \psi(-\sin \psi) + \sin \psi \cos \psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \quad \det R_\psi = 1, \quad R_\psi^\top = R_{-\psi}, \quad R_\psi R_\varphi = R_{\psi+\varphi}, \quad (4.13)$$

and orthogonality gives $\|R_\psi z\|^2 = z^\top R_\psi^\top R_\psi z = \|z\|^2$: lengths and inner products are preserved.

Lemma 4.2 (Rotations preserve the standard isotropic Gaussian). *If $\eta \sim \mathcal{N}(0, I_2)$ and R is orthogonal, then $R\eta \sim \mathcal{N}(0, I_2)$.*

Proof. A linear map of a Gaussian is Gaussian, so $R\eta$ is fixed by its first two moments: $\mathbb{E}[R\eta] = R\mathbb{E}[\eta] = 0$ and $\text{Cov}(R\eta) = R I_2 R^\top = R R^\top = I_2$. Equivalently, in terms of densities, $(2\pi)^{-1} e^{-\|z\|^2/2}$ depends on z only through $\|z\|$, and the change of variables $z \mapsto Rz$ changes neither $\|R^{-1}z\| = \|z\|$ nor $|\det R^{-1}| = 1$: the level sets are circles centred at the origin and a rotation maps each circle to itself. $\square$

The isotropy is essential. A rotation would not preserve $\mathcal{N}(0, \Sigma)$ for general Σ , and the reduction below is exact only because both the innovation in (4.11) and the channel (4.9) are isotropic.

Lemma 4.3 (Gauge reduction). *Define the de-rotated frame $y_k = R_{-k\psi} z_k$ and stack the per-frame rotations into $U_\psi = \text{diag}(I_2, R_{-\psi}, \dots, R_{-(L-1)\psi})$. Then*

$$y_{k+1} = y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} \stackrel{iid}{\sim} \mathcal{N}(0, I_2), \quad (4.14)$$

U_ψ is orthogonal, and the noised density and score satisfy

$$P_t(z | \psi) = P_t^{(0)}(U_\psi z), \quad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t), \quad (4.15)$$

where the superscript zero denotes the quantities of the non-rotating model.

Proof. Substituting (4.11) and using the composition rule of (4.13),

$$y_{k+1} = R_{-(k+1)\psi} \left(R_\psi z_k + \sigma \eta_{k+1} \right) = \underbrace{R_{-(k+1)\psi} R_\psi}_{=R_{-k\psi}} z_k + \sigma R_{-(k+1)\psi} \eta_{k+1} = y_k + \sigma \tilde{\eta}_{k+1},$$

and by Lemma 4.2 each $\tilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$; they stay independent because each is a fixed linear map of a different independent η . Next, U_ψ is block diagonal with orthogonal blocks, so $U_\psi^\top U_\psi = \text{diag}(I_2, \dots, I_2) = I_{2L}$ and $\det U_\psi = \prod_k \det R_{-k\psi} = 1$. Applying U_ψ to (4.9) gives $U_\psi x = e^{-t} U_\psi a + \sqrt{\Delta_t} U_\psi \xi$ with $U_\psi \xi \sim \mathcal{N}(0, I_{2L})$, so $U_\psi x$ has exactly the law of the noised non-rotating trajectory: the de-rotation commutes with the channel. For the densities, $z = U_\psi^\top y$ and $P_t(z | \psi) = P_t^{(0)}(U_\psi z) |\det U_\psi|$, the Jacobian factor being one. Differentiating that identity componentwise, with $\partial y_b / \partial z_c = (U_\psi)_{bc}$,

$$\frac{\partial}{\partial z_c} \log P_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{bc} \left(\nabla \log P_t^{(0)} \right)_b (U_\psi z) = \left[U_\psi^\top S^{(0)}(U_\psi z, t) \right]_c.$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$ while a point transforms with U_ψ . $\square$

Equation (4.15) says that a known rotation adds no statistical difficulty: de-rotate the data, solve the non-rotating problem, rotate the score back. Since $U_\psi^\top = U_\psi^{-1}$, the operation is exactly invertible and destroys no information. It would not hold for an *unknown* ψ , which is a genuine latent variable. Figure 4.3 shows the two frames.

4.2.3 Why the surrogate is worth solving

The surrogate's payoff is three closed-form objects:

$$S_k^{(0)}(\tilde{x}, t) = -\left(C_t^{-1} \tilde{x}\right)_k + e^{-t} q_k \mathbb{E}[A | \tilde{x}], \quad (4.16)$$

$$J_{kj}^{(0)}(\tilde{x}, t) = -\left(C_t^{-1}\right)_{kj} I_2 + e^{-2t} q_k q_j \text{Cov}(A | \tilde{x}), \quad (4.17)$$

$$S_\psi(x, t) = U_\psi^\top S^{(0)}(U_\psi x, t), \quad (4.18)$$

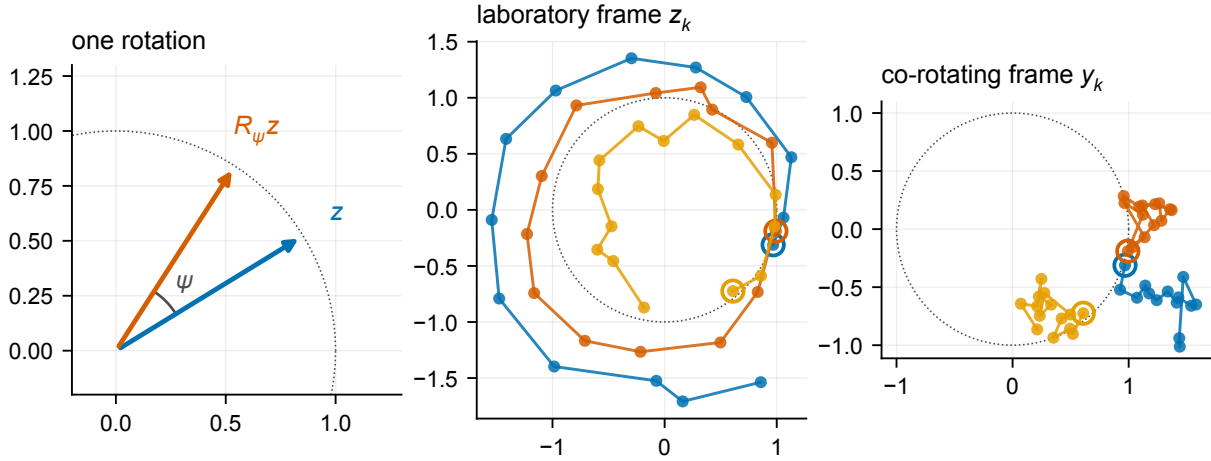


Figure 4.3: The rotation and the gauge. (a) R_ψ turns every vector through the same angle without changing its length, which is the statement $R_\psi^\top R_\psi = I_2$. (b) Three surrogate trajectories in the laboratory frame ($L = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; the marked point is $k = 0$), coherently circulating. (c) The same trajectories after U_ψ : a random walk anchored on the ring.

the exact score and response of the de-rotated surrogate, and the gauge map that carries them back to the laboratory frame. The first two are new; the third is Lemma 4.3 restated for the anchor-marginalised law, not a further derivation. The derivation of (4.16)–(4.17) is Toolbox 4.1.

Toolbox 4.1 — From the anchor posterior to the exact score and response

In the de-rotated frame the surrogate is the anchored random walk (4.14): $y_0 = A$ and $y_k = A + \sigma \sum_{r=1}^k \tilde{\eta}_r$. Conditionally on the anchor the walk increments are the only randomness, so $\text{Cov}(y_i, y_j \mid A) = \sigma^2 \min(i, j) I_2$ under the zero-based indexing used here — the first frame is pinned to the anchor exactly, $\min(0, j) = 0$. Passing through the channel (4.9) scales the conditional covariance by e^{-2t} and adds white noise, which defines the matrix used throughout this toolbox:

$$C_t := e^{-2t} \sigma^2 M + \Delta_t I_L, \quad M_{ij} = \min(i, j), \quad i, j = 0, \dots, L-1, \quad (4.19)$$

positive definite for every $t > 0$ because of the $\Delta_t I_L$ term even though M is singular. For $t > 0$, conditionally on the anchor $A = \alpha$, the de-rotated noisy trajectory is

therefore Gaussian,

$$\tilde{x} \mid A = \alpha \sim \mathcal{N}(e^{-t}\mathbf{1} \otimes \alpha, C_t \otimes I_2). \quad (4.20)$$

Write

$$q = C_t^{-1}\mathbf{1}, \quad h(\tilde{x}) = e^{-t}(q^\top \otimes I_2)\tilde{x} = e^{-t} \sum_{j=0}^{L-1} q_j \tilde{x}_j. \quad (4.21)$$

Here and below the action of C_t^{-1} on a block vector is understood as

$$\left[(C_t^{-1} \otimes I_2)\tilde{x}\right]_k = \sum_{j=0}^{L-1} (C_t^{-1})_{kj} \tilde{x}_j.$$

Score. For a latent-variable model

$$p_t(\tilde{x}) = \int p(\tilde{x} \mid \alpha) p(\alpha) d\alpha,$$

differentiation under the integral gives Fisher's identity,

$$\nabla_{\tilde{x}_k} \log p_t(\tilde{x}) = \mathbb{E} \left[\nabla_{\tilde{x}_k} \log p(\tilde{x} \mid A) \mid \tilde{x} \right]. \quad (4.22)$$

For the Gaussian law (4.20),

$$\begin{aligned} \nabla_{\tilde{x}_k} \log p(\tilde{x} \mid \alpha) &= - \left[(C_t^{-1} \otimes I_2)\tilde{x} \right]_k + e^{-t} (C_t^{-1}\mathbf{1})_k \alpha \\ &= - \sum_{j=0}^{L-1} (C_t^{-1})_{kj} \tilde{x}_j + e^{-t} q_k \alpha. \end{aligned} \quad (4.23)$$

Averaging over the anchor posterior therefore gives

$$\boxed{S_k^{(0)}(\tilde{x}, t) = - \sum_{j=0}^{L-1} (C_t^{-1})_{kj} \tilde{x}_j + e^{-t} q_k \mathbb{E}[A \mid \tilde{x}]} \quad (4.24)$$

Response. Expanding the Gaussian exponent in (4.20) shows that the dependence of the anchor posterior on the full trajectory is only through the two-dimensional statistic

$h(\tilde{x})$:

$$p(\alpha \mid \tilde{x}) \propto \exp\{g_t(\alpha) + h(\tilde{x})^\top \alpha\}, \quad (4.25)$$

where g_t contains the prior and all terms independent of $\tilde{x}$ once h is fixed.

Let

$$Z(h) = \int \exp\{g_t(\alpha) + h^\top \alpha\} d\alpha.$$

Then the posterior mean is

$$m(h) := \mathbb{E}[A \mid h] = \nabla_h \log Z(h),$$

and its Jacobian is the posterior covariance,

$$\nabla_h m(h) = \nabla_h^2 \log Z(h) = \text{Cov}(A \mid h). \quad (4.26)$$

Since

$$\frac{\partial h}{\partial \tilde{x}_j} = e^{-t} q_j I_2,$$

the chain rule gives

$$\frac{\partial}{\partial \tilde{x}_j} \mathbb{E}[A \mid \tilde{x}] = e^{-t} q_j \text{Cov}(A \mid \tilde{x}). \quad (4.27)$$

Differentiating (4.16) once more therefore yields

$$\boxed{J_{kj}^{(0)}(\tilde{x}, t) = -(C_t^{-1})_{kj} I_2 + e^{-2t} q_k q_j \text{Cov}(A \mid \tilde{x})}. \quad (4.28)$$

Equivalently, for the complete $2L$ -dimensional score and response,

$$S^{(0)}(\tilde{x}, t) = -(C_t^{-1} \otimes I_2) \tilde{x} + e^{-t} (q \otimes I_2) \mathbb{E}[A \mid \tilde{x}], \quad (4.29)$$

$$J^{(0)}(\tilde{x}, t) = -(C_t^{-1} \otimes I_2) + e^{-2t} (qq^\top \otimes \text{Cov}(A \mid \tilde{x})). \quad (4.30)$$

Thus marginalising a single two-dimensional anchor modifies the Gaussian response through a term that is rank one in the frame indices and has rank at most two in the

full $2L$ -dimensional space.

Three things follow from having these in closed form rather than as an intractable integral. It gives an exact score to check numerical inference against. It identifies the sufficient statistic: every frame feeds into h , but all the remaining nonlinearity concerns only the two-dimensional anchor A , not the full $2L$ -dimensional trajectory. And it separates two mechanisms inside the response $J_{kj}^{(0)}$: the C_t^{-1} term is the ordinary Gaussian-walk covariance, while the rank-one term $q_k q_j$ is what one shared latent variable, marginalised, does to every pair of frames at once. This structured loss of locality is what carries over to the model that cannot be solved in closed form.

4.3 Does the score know the data are rotating?

With a uniform initial phase, one frame's law is a rotationally symmetric annulus — it does not itself look like "a rotation." Rotation is a property of the transition law between ordered frames, not of any single frame's distribution. So the question is: given only the noised data, does the score — built without ever being told the generative process turns — carry a usable signal that it does, and if so, where does that signal live?

4.3.1 One frame: no

Theorem 4.4 (Marginal blindness). *With the initial phase uniform on the circle, every one-frame marginal is independent of the angular drift, at every diffusion time. This holds for the polar model in ω and for the surrogate in ψ . Consequently, every one-frame score carries zero Fisher information about the rotation.*

Proof. For the polar model, $\Theta_u = \Theta_0 + \omega u + \sqrt{2D_\theta} B_u^{(\theta)}$. Adding any independent angle to a uniform circular variable returns a uniform variable, so Θ_u is uniform for every u and every ω ; R_u carries no ω and is independent of Θ_u , so the law of $a_u = R_u(\cos \Theta_u, \sin \Theta_u)$ does not depend on ω either. The surrogate's initial law and Gaussian innovations are rotationally invariant for the same reason, and the channel is isotropic and parameter-free,

so noising preserves the independence. Hence $\partial_\omega \log p_t(x_u) = 0$ and $\partial_\psi \log p_t(x_u) = 0$ for every frame and every t . $\square$

Therefore, as expected, no amount of per-frame looking, however much data, recovers the one thing rotation is: a direction of turning.

4.3.2 The trajectory: yes, and here is how

The mechanism is visible in closed form for the clean surrogate. The derivation is Toolbox 4.2; for an interior frame it gives

$$S_k(z, 0) = \frac{1}{\sigma^2} \left(R_\psi z_{k-1} + R_\psi^\top z_{k+1} - 2z_k \right). \quad (4.31)$$

Toolbox 4.2 — The clean surrogate score at an interior frame

The joint law of the clean surrogate factorises along the chain,

$$p(z_0, \dots, z_{L-1}) = p_\lambda(z_0) \prod_{i=0}^{L-2} \mathcal{N}(z_{i+1}; R_\psi z_i, \sigma^2 I_2),$$

so, up to additive terms not involving z_k ,

$$\log p(z) = -\frac{1}{2\sigma^2} \|z_k - R_\psi z_{k-1}\|^2 - \frac{1}{2\sigma^2} \|z_{k+1} - R_\psi z_k\|^2 + \text{const.}$$

For an interior frame $1 \leq k \leq L-2$, z_k appears in exactly these two terms — once as the point being predicted from z_{k-1} , once as the point doing the predicting of z_{k+1} — and nowhere else, since $p_\lambda(z_0)$ does not involve z_k when $k \neq 0$.

The first term. Differentiating directly,

$$\nabla_{z_k} \left[-\frac{1}{2\sigma^2} \|z_k - R_\psi z_{k-1}\|^2 \right] = -\frac{1}{\sigma^2} (z_k - R_\psi z_{k-1}).$$

The second term. Writing $u = z_{k+1} - R_\psi z_k$, the Jacobian of u with respect to z_k is $-R_\psi$, so $\nabla_{z_k} \|u\|^2 = 2(-R_\psi)^\top u$, and

$$\nabla_{z_k} \left[-\frac{1}{2\sigma^2} \|z_{k+1} - R_\psi z_k\|^2 \right] = \frac{1}{\sigma^2} R_\psi^\top (z_{k+1} - R_\psi z_k) = \frac{1}{\sigma^2} (R_\psi^\top z_{k+1} - z_k),$$

using $R_\psi^\top R_\psi = I_2$ from (4.13) to collapse $R_\psi^\top R_\psi z_k$ to z_k .

Adding the two. $S_k(z, 0) = \nabla_{z_k} \log p(z)$ is the sum of both contributions:

$$S_k(z, 0) = \frac{1}{\sigma^2} \left[-z_k + R_\psi z_{k-1} \right] + \frac{1}{\sigma^2} \left[R_\psi^\top z_{k+1} - z_k \right] = \frac{1}{\sigma^2} (R_\psi z_{k-1} + R_\psi^\top z_{k+1} - 2z_k),$$

which is (4.31). □

Therefore, the score at frame k does not compare z_k to its neighbours directly, it compares z_k to the previous frame rotated forward by one step and the next frame rotated back by one step. The polar model cannot be reduced this cleanly, but the same mechanism operates through the posterior. The k -th joint-score block is the denoising identity

$$S_{\omega,k}(x, t) = \frac{e^{-t} \mathbb{E}_\omega[a_k | x] - x_k}{\Delta_t}, \quad (4.32)$$

where

$$p_\omega(a | x) \propto p(a_0) \prod_{u=0}^{L-2} K_\omega(a_{u+1} | a_u) \prod_{u=0}^{L-1} \mathcal{N}(x_u; e^{-t} a_u, \Delta_t I_2). \quad (4.33)$$

K_ω favours angular increments consistent with the drift, so it pulls the posterior mean of a_k away from "wherever x_k alone suggests" and towards "wherever the neighbouring frames say a turning trajectory should be."

Two scope notes complete the statement. The parameter enters only through the one-step rotation, so from discrete frames it is identifiable at most modulo 2π in the per-step angle (ψ and $\psi + 2\pi$ generate identical laws); within a fundamental interval $(-\pi, \pi]$ the joint law does distinguish ψ from $-\psi$, which is the sense in which the direction of turning is knowable. And what this section establishes is that the joint law *carries information* about the drift — the marginal Fisher information is zero while the joint

likelihood depends on ω — not a recovery theorem: no estimator of ω , and no rate, is constructed here.

4.4 Chapter conclusion

The rotating ring separates two kinds of information that a static object never distinguishes. The one-frame score reports where probability mass sits — here, an annulus. The joint score also reports how frames relate to one another — here, an oriented turn through internal time, invisible to any one frame and identifiable (modulo the angular aliasing above) from the joint law of the trajectory.

The surrogate is what makes this exact: its score is a Gaussian-walk term plus a rank-one correction generated by one shared unknown, and the gauge shows precisely how the sense of rotation enters through forward and backward comparisons between frames. Returning to the polar model shows the same split holding at the level of information itself: every one-frame marginal is exactly blind to ω ; the joint law carries information identifying it through the transition kernel, until the noise wins.

one-frame score $\longrightarrow$ geometry, joint trajectory score $\longrightarrow$ geometry and dynamics.	(4.34)
---	--------

Chapter 5

The Gaussian Chain by Linear Algebra

The stationary Gaussian AR(1) chain is the one model in this thesis whose joint score can be written down in closed form and checked by hand. This chapter does that with matrices: the clean law, its dense covariance and sparse precision, the score in two equivalent forms, and the lifecycle the coupling structure passes through as the diffusion time grows.

5.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \dots, a_{L-1})$ of the stationary AR(1) chain (3.7) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (3.10) for a time t ,

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (5.1)$$

and the object of study is the joint score of the noisy sequence,

$$S(x, t) = \nabla_x \log P_t(x) \in \mathbb{R}^L, \quad P_t(x) = \int_{\mathbb{R}^L} P_0(a) \prod_{k=0}^{L-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t) da. \quad (5.2)$$

The gradient is taken with respect to all L noisy frames simultaneously, under the law of the whole noisy trajectory (Section 2.4).

5.2 The clean law: from the recursion to the covariance

5.2.1 Unrolling the recursion

The AR(1) chain (3.7) is the first-order recursion

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad \sigma_\eta^2 = 1 - \alpha^2, \quad (5.3)$$

with

$$a_0 \sim \mathcal{N}(0, 1), \quad a_0 \perp (\eta_0, \eta_1, \dots), \quad |\alpha| < 1.$$

The parameter α controls how strongly one frame is inherited by the next: $\alpha = 0$ gives independent frames, while $|\alpha|$ close to one produces long persistence along the trajectory.

The recursion can be expanded explicitly. The first step gives

$$a_1 = \alpha a_0 + \eta_0.$$

Substituting this expression into the next step,

$$a_2 = \alpha a_1 + \eta_1 = \alpha(\alpha a_0 + \eta_0) + \eta_1 = \alpha^2 a_0 + \alpha \eta_0 + \eta_1.$$

Repeating once more,

$$a_3 = \alpha a_2 + \eta_2 = \alpha^3 a_0 + \alpha^2 \eta_0 + \alpha \eta_1 + \eta_2.$$

The pattern is therefore

$$\boxed{a_k = \alpha^k a_0 + \sum_{j=0}^{k-1} \alpha^{k-1-j} \eta_j} \quad (k \geq 1). \quad (5.4)$$

Each innovation is propagated forward by one factor of α for every subsequent step: the

most recent innovation η_{k-1} enters with coefficient one, whereas an innovation generated m steps earlier enters with coefficient α^m .

Equation (5.4) follows formally by induction. Suppose it holds at step k . Then

$$a_{k+1} = \alpha a_k + \eta_k \quad (5.5)$$

$$= \alpha \left(\alpha^k a_0 + \sum_{j=0}^{k-1} \alpha^{k-1-j} \eta_j \right) + \eta_k \quad (5.6)$$

$$= \alpha^{k+1} a_0 + \sum_{j=0}^{k-1} \alpha^{k-j} \eta_j + \eta_k \quad (5.7)$$

$$= \alpha^{k+1} a_0 + \sum_{j=0}^k \alpha^{k-j} \eta_j, \quad (5.8)$$

which is exactly (5.4) with $k+1$ in place of k .

This representation already exposes the probabilistic structure of the whole trajectory. Define

$$\varepsilon = (a_0, \eta_0, \dots, \eta_{L-2})^\top.$$

Its entries are independent Gaussian variables, so ε is jointly Gaussian. Equation (5.4) shows that there exists a deterministic lower-triangular matrix G_α such that

$$a = (a_0, \dots, a_{L-1})^\top = G_\alpha \varepsilon. \quad (5.9)$$

A deterministic linear transformation of a Gaussian vector is Gaussian; hence the complete trajectory a is jointly Gaussian.

The choice $\sigma_\eta^2 = 1 - \alpha^2$ makes this Gaussian chain stationary with unit marginal variance. Indeed, if $\text{Var}(a_k) = 1$, independence of a_k and η_k gives

$$\text{Var}(a_{k+1}) = \text{Var}(\alpha a_k + \eta_k) \quad (5.10)$$

$$= \alpha^2 \text{Var}(a_k) + \text{Var}(\eta_k) \quad (5.11)$$

$$= \alpha^2 + (1 - \alpha^2) = 1. \quad (5.12)$$

Since $\text{Var}(a_0) = 1$, induction gives

$$a_k \sim \mathcal{N}(0, 1) \quad \text{for every } k. \quad (5.13)$$

Finally, (5.4) makes the loss of memory explicit. Conditional on the initial state,

$$\mathbb{E}[a_k \mid a_0] = \alpha^k a_0, \quad (5.14)$$

because every innovation has zero mean. Since $|\alpha| < 1$,

$$|\alpha|^k \longrightarrow 0 \quad \text{as } k \longrightarrow \infty.$$

Thus the influence of the initial condition decays geometrically. More precisely, for $j \leq k$, iterating the recursion only from frame j gives

$$a_k = \alpha^{k-j} a_j + \sum_{r=j}^{k-1} \alpha^{k-1-r} \eta_r.$$

The second term is independent of a_j , and therefore

$$\text{Cov}(a_j, a_k) = \alpha^{k-j} \text{Var}(a_j) = \alpha^{k-j}. \quad (5.15)$$

By symmetry,

$$\boxed{\text{Cov}(a_j, a_k) = \alpha^{|k-j|}}. \quad (5.16)$$

The same geometric factor therefore controls both notions of memory: the effect of the initial condition and the correlation between frames separated by a given lag.

5.2.2 The stationary covariance is Toeplitz

With mean zero, the joint law is fixed by the covariances $\text{Cov}(a_i, a_j)$. Take $j = k$ first. The terms of (5.4) are independent, so their variances add:

$$\text{Var}(a_k) = \alpha^{2k} \underbrace{\text{Var}(a_0)}_1 + \sigma_\eta^2 \sum_{j=0}^{k-1} \alpha^{2(k-1-j)} = \alpha^{2k} + \sigma_\eta^2 \sum_{i=0}^{k-1} (\alpha^2)^i, \quad (5.17)$$

the last step substituting $i = k-1-j$. The remaining sum is geometric with ratio $\alpha^2 \neq 1$,

$$\sum_{i=0}^{k-1} (\alpha^2)^i = \frac{1 - \alpha^{2k}}{1 - \alpha^2}, \quad (5.18)$$

and with $\sigma_\eta^2 = 1 - \alpha^2$ the prefactor cancels the denominator:

$$\text{Var}(a_k) = \alpha^{2k} + (1 - \alpha^2) \frac{1 - \alpha^{2k}}{1 - \alpha^2} = \alpha^{2k} + (1 - \alpha^{2k}) = 1, \quad (5.19)$$

for every k : the pairing $a_0 \sim \mathcal{N}(0, 1)$, $\sigma_\eta^2 = 1 - \alpha^2$ is exactly what holds the marginal variance constant along the chain. For the cross-covariance, split (5.4) at index k ,

$$a_{k+d} = \alpha^d a_k + \sum_{j=k}^{k+d-1} \alpha^{k+d-1-j} \eta_j \quad (d \geq 0), \quad (5.20)$$

where the sum uses only $\eta_k, \dots, \eta_{k+d-1}$, all independent of a_k . Hence

$$\text{Cov}(a_k, a_{k+d}) = \alpha^d \text{Var}(a_k) = \alpha^d, \quad \text{i.e.} \quad (\Sigma_0)_{ij} = \alpha^{|i-j|}. \quad (5.21)$$

Collecting mean and covariance, the clean joint density is

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{L-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2) = \mathcal{N}(a; 0, \Sigma_0), \quad (5.22)$$

the product being the chain factorisation (3.6) and the right-hand side its covariance form,

with

$$\Sigma_0 = \begin{pmatrix} 1 & \alpha & \alpha^2 & \cdots & \alpha^{L-1} \\ \alpha & 1 & \alpha & \cdots & \alpha^{L-2} \\ \alpha^2 & \alpha & 1 & \cdots & \alpha^{L-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha^{L-1} & \alpha^{L-2} & \alpha^{L-3} & \cdots & 1 \end{pmatrix}. \quad (5.23)$$

Proposition 5.1 (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$: a symmetric Toeplitz matrix — constant along every diagonal — with geometrically decaying bands, autocovariance $\gamma(d) = \alpha^{|d|}$. It is dense for $\alpha \neq 0$; at $\alpha = 0$ the chain is white and $\Sigma_0 = I$.

The Toeplitz form is the matrix print of weak stationarity: by (3.3)–(3.4) the covariance of a stationary sequence depends only on the index difference, so the d -th diagonal of (5.23) carries the single number $\gamma(d) = \alpha^d$. Three readings of that line are used later, all for the coupled case $\alpha \neq 0$ (at $\alpha = 0$ the chain is white and $\Sigma_0 = I$). The matrix is *dense*: every pair of frames is correlated and no entry of Σ_0 is exactly zero. The correlation magnitude *decays geometrically*, $|\gamma(d)| = e^{-d/\tau}$ with correlation length $\tau = -1/\ln|\alpha|$ — for $\alpha < 0$ the sign alternates with the lag while the modulus follows the same law — so far-apart frames are weakly but not trivially coupled. And the coupling is *global though the mechanism is local*: (5.3) touches only neighbours, yet its covariance reaches across the whole chain. The locality is still there — in the inverse.

5.3 The clean law: the precision is tridiagonal

5.3.1 Reading the precision off a short chain

The score will depend not on Σ_0 but on its inverse, the *precision* $Q_0 := \Sigma_0^{-1}$. For short chains it is an exercise. At $L = 2$,

$$\Sigma_0 = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}, \quad Q_0 = \frac{1}{1 - \alpha^2} \begin{pmatrix} 1 & -\alpha \\ -\alpha & 1 \end{pmatrix} = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 & -\alpha \\ -\alpha & 1 \end{pmatrix}, \quad (5.24)$$

from the 2×2 rule $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = (ad - bc)^{-1} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ with determinant $1 - \alpha^2$. At $L = 3$ the cofactor inverse gives

$$\Sigma_0 = \begin{pmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{pmatrix}, \quad Q_0 = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 & -\alpha & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha \\ 0 & -\alpha & 1 \end{pmatrix} \quad (5.25)$$

(one line of $Q_0 \Sigma_0$ confirms it). Two features of (5.25) hold for every L : the corner entry $(Q_0)_{0,2}$ is *exactly* zero, and the interior diagonal carries an extra α^2 that the two endpoints do not.

5.3.2 The tridiagonal precision and what its zeros mean

Proposition 5.2 (Tridiagonal precision). *The precision matrix $Q_0 = \Sigma_0^{-1}$ is*

$$Q_0 = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 & -\alpha & & & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha & & \\ & \ddots & \ddots & \ddots & \\ & & -\alpha & 1 + \alpha^2 & -\alpha \\ 0 & & & -\alpha & 1 \end{pmatrix}. \quad (5.26)$$

Equivalently,

$$(Q_0)_{kk} = \begin{cases} \frac{1}{\sigma_\eta^2}, & k \in \{0, L-1\}, \\ \frac{1 + \alpha^2}{\sigma_\eta^2}, & 0 < k < L-1, \end{cases} \quad (Q_0)_{k,k+1} = -\frac{\alpha}{\sigma_\eta^2}, \quad (5.27)$$

with $(Q_0)_{ij} = 0$ whenever $|i - j| \geq 2$.

There is no need to invert Σ_0 entry by entry. For a centred Gaussian $a \sim \mathcal{N}(0, \Sigma_0)$,

$$P_0(a) = (2\pi)^{-L/2} |\Sigma_0|^{-1/2} \exp\left(-\frac{1}{2} a^\top Q_0 a\right), \quad Q_0 = \Sigma_0^{-1}, \quad (5.28)$$

and therefore

$$-2 \log P_0(a) = a^\top Q_0 a + L \log(2\pi) + \log |\Sigma_0| \quad (5.29)$$

$$= a^\top Q_0 a + \text{const.} \quad (5.30)$$

The precision matrix can thus be read directly from the quadratic part of the factorised negative log-density.

Toolbox 5.1 — Reading Q_0 off $-2 \log P_0$

The Markov factorisation (5.22) is

$$P_0(a) = p(a_0) \prod_{k=0}^{L-2} p(a_{k+1} \mid a_k),$$

with

$$a_0 \sim \mathcal{N}(0, 1), \quad a_{k+1} \mid a_k \sim \mathcal{N}(\alpha a_k, \sigma_\eta^2).$$

Discarding terms independent of a gives

$$-2 \log P_0(a) = a_0^2 + \frac{1}{\sigma_\eta^2} \sum_{k=0}^{L-2} (a_{k+1} - \alpha a_k)^2 + \text{const} \quad (5.31)$$

$$= a_0^2 + \frac{1}{\sigma_\eta^2} \sum_{k=0}^{L-2} (a_{k+1}^2 - 2\alpha a_k a_{k+1} + \alpha^2 a_k^2) + \text{const.} \quad (5.32)$$

We compare this with the quadratic form of a symmetric matrix,

$$a^\top Q_0 a = \sum_{m=0}^{L-1} (Q_0)_{mm} a_m^2 + 2 \sum_{0 \leq m < n \leq L-1} (Q_0)_{mn} a_m a_n. \quad (5.33)$$

Off-diagonal entries. The only products of distinct coordinates are

$$-\frac{2\alpha}{\sigma_\eta^2} \sum_{k=0}^{L-2} a_k a_{k+1}. \quad (5.34)$$

Matching coefficients with $2(Q_0)_{k,k+1}a_k a_{k+1}$ gives

$$(Q_0)_{k,k+1} = (Q_0)_{k+1,k} = -\frac{\alpha}{\sigma_\eta^2}, \quad k = 0, \dots, L-2, \quad (5.35)$$

$$(Q_0)_{ij} = 0, \quad |i-j| \geq 2. \quad (5.36)$$

The precision is therefore tridiagonal.

Interior diagonal. For $1 \leq m \leq L-2$, the term a_m^2 appears in two adjacent transition factors:

$$\frac{1}{\sigma_\eta^2} (a_m - \alpha a_{m-1})^2 = \frac{1}{\sigma_\eta^2} a_m^2 - \frac{2\alpha}{\sigma_\eta^2} a_m a_{m-1} + \frac{\alpha^2}{\sigma_\eta^2} a_{m-1}^2, \quad (5.37)$$

$$\frac{1}{\sigma_\eta^2} (a_{m+1} - \alpha a_m)^2 = \frac{\alpha^2}{\sigma_\eta^2} a_m^2 - \frac{2\alpha}{\sigma_\eta^2} a_{m+1} a_m + \frac{1}{\sigma_\eta^2} a_{m+1}^2. \quad (5.38)$$

Summing the coefficients of a_m^2 yields

$$(Q_0)_{mm} = \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad 1 \leq m \leq L-2. \quad (5.39)$$

Boundaries. The left endpoint also appears in its prior:

$$a_0^2 + \frac{\alpha^2}{\sigma_\eta^2} a_0^2 = \frac{\sigma_\eta^2 + \alpha^2}{\sigma_\eta^2} a_0^2 \quad (5.40)$$

$$= \frac{1}{\sigma_\eta^2} a_0^2, \quad (5.41)$$

where the last equality uses $\sigma_\eta^2 = 1 - \alpha^2$. Thus

$$(Q_0)_{00} = \frac{1}{\sigma_\eta^2}. \quad (5.42)$$

The final coordinate appears only in the last transition,

$$\frac{1}{\sigma_\eta^2} (a_{L-1} - \alpha a_{L-2})^2, \quad (5.43)$$

so similarly

$$(Q_0)_{L-1,L-1} = \frac{1}{\sigma_\eta^2}. \quad (5.44)$$

Putting the diagonal and off-diagonal coefficients together gives (5.26). The calculation also explains the pattern: an interior variable belongs to two transition factors, whereas an endpoint belongs to only one; and no factor ever contains two non-neighbouring variables. $\square$

The zeros of (5.26) have a direct probabilistic meaning. For a jointly Gaussian vector,

$$(Q_0)_{ij} = 0 \iff a_i \perp a_j \mid a_{\setminus\{i,j\}}. \quad (5.45)$$

Thus tridiagonality is the matrix expression of the first-order Markov structure: once the neighbouring frames are known, an interior frame is conditionally independent of all more distant frames.

The origin of the sparsity is more general than the Gaussian model itself. For any smooth first-order Markov density,

$$-\log P(a) = -\log p(a_0) - \sum_{k=0}^{L-2} \log p(a_{k+1} \mid a_k), \quad (5.46)$$

and each summand contains only neighbouring coordinates. Consequently,

$$\frac{\partial^2[-\log P(a)]}{\partial a_i \partial a_j} = 0, \quad |i - j| \geq 2. \quad (5.47)$$

For a general Markov chain this is a sparse, possibly state-dependent Hessian. In the Gaussian case the Hessian is constant and equals the precision matrix Q_0 .

The same matrix also tells us how one hidden frame is predicted from the others. For a centred Gaussian,

$$\mathbb{E}[a_k \mid a_{\setminus k}] = -(Q_0)_{kk}^{-1} \sum_{l \neq k} (Q_0)_{kl} a_l. \quad (5.48)$$

For an interior frame, only the two neighbouring entries of row k are nonzero, so

$$\mathbb{E}[a_k \mid a_{\setminus k}] = -\frac{\sigma_\eta^2}{1 + \alpha^2} \left(-\frac{\alpha}{\sigma_\eta^2} a_{k-1} - \frac{\alpha}{\sigma_\eta^2} a_{k+1} \right) \quad (5.49)$$

$$= \boxed{\frac{\alpha}{1 + \alpha^2} (a_{k-1} + a_{k+1})}. \quad (5.50)$$

All information supplied by the clean chain about a_k is therefore mediated by its two neighbours.

This exact locality will not survive the diffusion channel at the level of the observed variables. Although the clean trajectory is Markov, after adding independent Gaussian noise and marginalising the latent clean states, the noisy precision is generally dense. Distant noisy frames can therefore contribute to the optimal estimate, and hence to the joint score, of a given frame. This loss of sparsity is one of the central effects of diffusion that we study next.

5.4 The noisy law

Each frame is corrupted independently by the Ornstein–Uhlenbeck channel of Section 3.2. Solving the OU SDE (3.9) once (Toolbox 3.1) gives the Gaussian transition $x_k \mid a_k \sim \mathcal{N}(e^{-t}a_k, \Delta_t)$, $\Delta_t = 1 - e^{-2t}$, which is (3.10). Stacking the L frames,

$$x = e^{-t}a + \sqrt{\Delta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_L), \quad \xi \perp a, \quad (5.51)$$

with ξ white because the channels use independent noises.

Proposition 5.3 (Noised law). *Under the Ornstein–Uhlenbeck channel, the noisy trajectory remains Gaussian:*

$$P_t(x) = \mathcal{N}(x; 0, \Sigma_t), \quad \boxed{\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_L}. \quad (5.52)$$

Proof. Both a and ξ are Gaussian and independent, so their linear combination $x =$

$e^{-t}a + \sqrt{\Delta_t}\xi$ is again Gaussian. Its mean is

$$\mathbb{E}[x] = e^{-t}\mathbb{E}[a] + \sqrt{\Delta_t}\mathbb{E}[\xi] = 0. \quad (5.53)$$

For the covariance,

$$\text{Cov}(x) = \text{Cov}\left(e^{-t}a + \sqrt{\Delta_t}\xi\right) \quad (5.54)$$

$$= e^{-2t} \text{Cov}(a) + \Delta_t \text{Cov}(\xi) + e^{-t}\sqrt{\Delta_t} \left(\text{Cov}(a, \xi) + \text{Cov}(\xi, a) \right). \quad (5.55)$$

Independence gives $\text{Cov}(a, \xi) = 0$, while $\text{Cov}(a) = \Sigma_0$ and $\text{Cov}(\xi) = I_L$. Hence

$$\text{Cov}(x) = e^{-2t}\Sigma_0 + \Delta_t I_L, \quad (5.56)$$

which proves (5.52). $\square$

Because

$$(\Sigma_0)_{ij} = \alpha^{|i-j|},$$

the covariance can also be read entrywise:

$$(\Sigma_t)_{ij} = e^{-2t}\alpha^{|i-j|} + \Delta_t \delta_{ij}, \quad (5.57)$$

where δ_{ij} is the Kronecker delta. This separates the effect of the channel on variances and correlations.

On the diagonal, using $\Delta_t = 1 - e^{-2t}$,

$$(\Sigma_t)_{ii} = e^{-2t}(\Sigma_0)_{ii} + \Delta_t \quad (5.58)$$

$$= e^{-2t} + 1 - e^{-2t} = 1. \quad (5.59)$$

Thus every frame remains marginally $\mathcal{N}(0, 1)$ at every diffusion time: the channel is variance preserving.

For two distinct frames, $i \neq j$, the added white noise contributes nothing because $(I_L)_{ij} = 0$. Therefore

$$(\Sigma_t)_{ij} = e^{-2t} \alpha^{|i-j|}, \quad i \neq j. \quad (5.60)$$

Since the marginal variances remain one, these entries are also the correlations:

$$\text{Corr}(x_i, x_j) = e^{-2t} \alpha^{|i-j|}. \quad (5.61)$$

The channel therefore leaves each one-frame distribution unchanged while exponentially attenuating the dependence between different frames.

In matrix form,

$$\Sigma_t = \begin{pmatrix} 1 & e^{-2t}\alpha & e^{-2t}\alpha^2 & \cdots \\ e^{-2t}\alpha & 1 & e^{-2t}\alpha & \ddots \\ e^{-2t}\alpha^2 & e^{-2t}\alpha & 1 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}. \quad (5.62)$$

Thus Σ_t remains symmetric Toeplitz. For $\alpha \neq 0$ it is dense at every finite t : diffusion does not create new zeros in the covariance, but uniformly shrinks every off-diagonal entry by e^{-2t} .

The two limiting cases make the interpolation explicit:

$$\Sigma_t \Big|_{t=0} = \Sigma_0, \quad (5.63)$$

$$\lim_{t \rightarrow \infty} \Sigma_t = I_L. \quad (5.64)$$

At $t = 0$ we recover the clean correlated trajectory; as $t \rightarrow \infty$, all cross-frame correlations vanish and the frames approach independent standard Gaussians.

The important change, however, appears in the *precision*, not directly in the covariance. Define

$$Q_t := \Sigma_t^{-1}. \quad (5.65)$$

At $t = 0$,

$$Q_t = Q_0,$$

which is tridiagonal by Proposition 5.2. For positive diffusion time, however, adding $\Delta_t I_L$ to the covariance and then inverting does not preserve that tridiagonal structure: in the nontrivial correlated case $\alpha \neq 0$, Q_t is generally dense. Hence noisy frames that were conditionally independent at the clean level become conditionally coupled after the latent clean trajectory has been corrupted and marginalised.

This distinction is central:

clean precision Q_0 is local and tridiagonal $\Downarrow$ diffusion and marginalisation noisy precision Q_t is generally nonlocal and dense.	(5.66)
--	--------

Section 5.7 quantifies how this loss of locality develops with diffusion time.

5.5 The score, two ways

5.5.1 Linear algebra

Since $\log P_t(x) = -\frac{1}{2}x^\top Q_t x + \text{const}$ and $\nabla_x(x^\top Q_t x) = 2Q_t x$ for symmetric Q_t ,

$S(x, t) = \nabla_x \log P_t(x) = -Q_t x, \quad Q_t := \Sigma_t^{-1}.$	(5.67)
--	--------

The joint score is linear in x — the signature of a Gaussian P_t — and its whole content is Q_t : which frames the denoiser of frame k draws on, and how strongly, is row k of Q_t . Already at $L = 2$, with $r := \alpha e^{-2t}$, (5.67) gives $S_0(x, t) = -(x_0 - r x_1)/(1 - r^2)$: the denoiser of frame 0 uses frame 1 with weight α at $t = 0$, decaying as e^{-2t} , while the per-frame marginal score stays $-x_k$ at every t and carries nothing about the dynamics. The lifecycle of that coupling, for all L , is Sections 5.6–6.4.

5.5.2 Bayesian inference

The score can be obtained in two equivalent ways. Since the noisy law is Gaussian, the previous section gives it directly using linear algebra. There is, however, a second route that is more useful beyond the Gaussian case: infer the clean trajectory a from the noisy observation x , and express the score through its posterior mean.

For the Ornstein–Uhlenbeck channel, the score–posterior identity (2.39) gives, componentwise,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x] - x_k}{\Delta_t}. \quad (5.68)$$

Thus computing the score is equivalent to computing the posterior mean of the clean chain.

Proposition 5.4 (Posterior of the clean chain). *The posterior distribution of the clean trajectory is Gaussian,*

$$P(a | x) = \mathcal{N}(a; m, J^{-1}), \quad (5.69)$$

with posterior precision

$$J = Q_0 + \frac{e^{-2t}}{\Delta_t} I_L \quad (5.70)$$

and posterior field

$$h = \frac{e^{-t}}{\Delta_t} x, \quad Jm = h. \quad (5.71)$$

Equivalently,

$$m = \mathbb{E}[a | x] = J^{-1}h. \quad (5.72)$$

Toolbox 5.2 — Posterior precision and field by completing the square

Bayes' rule gives

$$P(a | x) \propto P_0(a) P(x | a). \quad (5.73)$$

The prior is

$$P_0(a) \propto \exp\left(-\frac{1}{2}a^\top Q_0 a\right), \quad (5.74)$$

while the channel likelihood is

$$P(x | a) \propto \exp \left[-\frac{1}{2\Delta_t} \|x - e^{-t}a\|^2 \right]. \quad (5.75)$$

Multiplying the two densities amounts to adding their negative logarithms:

$$-2 \log P(a | x) = a^\top Q_0 a + \frac{1}{\Delta_t} (x - e^{-t}a)^\top (x - e^{-t}a) + \text{const.} \quad (5.76)$$

Expanding the likelihood term,

$$(x - e^{-t}a)^\top (x - e^{-t}a) = x^\top x - 2e^{-t}x^\top a + e^{-2t}a^\top a. \quad (5.77)$$

The term $x^\top x$ is constant with respect to a and can therefore be absorbed into the normalising constant. Hence

$$-2 \log P(a | x) = a^\top Q_0 a + \frac{e^{-2t}}{\Delta_t} a^\top a - 2 \frac{e^{-t}}{\Delta_t} x^\top a + \text{const} \quad (5.78)$$

$$= a^\top \left(Q_0 + \frac{e^{-2t}}{\Delta_t} I_L \right) a - 2 \left(\frac{e^{-t}}{\Delta_t} x \right)^\top a + \text{const.} \quad (5.79)$$

A Gaussian written in information form has exponent

$$-\frac{1}{2} \left(a^\top J a - 2h^\top a \right). \quad (5.80)$$

Matching terms therefore identifies

$$J = Q_0 + \frac{e^{-2t}}{\Delta_t} I_L, \quad (5.81)$$

$$h = \frac{e^{-t}}{\Delta_t} x. \quad (5.82)$$

Completing the square makes the posterior mean explicit:

$$a^\top J a - 2h^\top a = (a - J^{-1}h)^\top J (a - J^{-1}h) - h^\top J^{-1}h. \quad (5.83)$$

The last term is independent of a , so

$$P(a \mid x) = \mathcal{N}(a; J^{-1}h, J^{-1}), \quad (5.84)$$

and therefore

$$\boxed{\mathbb{E}[a \mid x] = J^{-1}h.} \quad (5.85)$$

What the two terms in J mean. The posterior precision is the sum

$$\underbrace{J}_{\text{posterior precision}} = \underbrace{Q_0}_{\text{prior information}} + \underbrace{\frac{e^{-2t}}{\Delta_t} I_L}_{\text{information from the channel}}. \quad (5.86)$$

The prior term couples neighbouring clean frames through the tridiagonal matrix Q_0 . The likelihood contributes only to the diagonal because, conditional on a , each observation x_k depends only on the corresponding clean variable a_k . Hence J remains tridiagonal for every finite diffusion time.

This should be kept distinct from

$$Q_t = \Sigma_t^{-1},$$

which is the precision matrix of the *noisy marginal* $P_t(x)$ and is generally dense. Thus

J : posterior precision of $a \mid x$ — tridiagonal, Q_t : marginal precision of x — generally dense.
--

The first retains the local Markov graph because the clean variables are still present; the second results after those variables have been marginalised out. $\square$

We can now insert the posterior mean into the score identity. Since

$$m = J^{-1}h = \frac{e^{-t}}{\Delta_t} J^{-1}x, \quad (5.87)$$

equation (5.68) gives

$$S(x, t) = \frac{e^{-t}m - x}{\Delta_t} \quad (5.88)$$

$$= \left(\frac{e^{-2t}}{\Delta_t^2} J^{-1} - \frac{1}{\Delta_t} I_L \right) x. \quad (5.89)$$

This expression must equal the direct Gaussian score $-Q_t x$. The equality can be seen without performing another matrix inversion. From

$$J = \Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t} I_L \quad (5.90)$$

$$= \frac{1}{\Delta_t} \Sigma_0^{-1} (\Delta_t I_L + e^{-2t} \Sigma_0) \quad (5.91)$$

$$= \frac{1}{\Delta_t} \Sigma_0^{-1} \Sigma_t, \quad (5.92)$$

we obtain

$$J^{-1} = \Delta_t \Sigma_t^{-1} \Sigma_0 = \Delta_t Q_t \Sigma_0. \quad (5.93)$$

Therefore

$$\frac{e^{-2t}}{\Delta_t^2} J^{-1} - \frac{1}{\Delta_t} I_L = \frac{1}{\Delta_t} (e^{-2t} Q_t \Sigma_0 - I_L) \quad (5.94)$$

$$= \frac{1}{\Delta_t} [Q_t (e^{-2t} \Sigma_0) - I_L] \quad (5.95)$$

$$= \frac{1}{\Delta_t} [Q_t (\Sigma_t - \Delta_t I_L) - I_L] \quad (5.96)$$

$$= -Q_t. \quad (5.97)$$

Hence the two routes agree exactly:

$$\boxed{\frac{e^{-t}\mathbb{E}[a \mid x] - x}{\Delta_t} = -\Sigma_t^{-1}x.} \quad (5.98)$$

The direct route computes the score from the noisy marginal covariance; the Bayesian route computes it by first denoising the latent clean chain. Their equality is exact, but the second formulation is the one that will continue to make sense when the clean model is no longer jointly Gaussian.

5.6 The spectral form

Between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$ lies the whole lifecycle of the coupling structure. One tool resolves it: the spectral theorem, applied to a family of matrices that is affine in Σ_0 .

5.6.1 The spectral theorem and its affine-family corollary

Theorem 5.5 (Spectral theorem, real symmetric case). *Every real symmetric $M \in \mathbb{R}^{n \times n}$ has an orthonormal eigenbasis: there is an orthogonal U ($U^\top U = I$) and reals $\omega_1, \dots, \omega_n$ with*

$$M = U \operatorname{diag}(\omega_i) U^\top = \sum_i \omega_i u_i u_i^\top, \quad (5.99)$$

the u_i the columns of U .

Corollary 5.6 (Affine family). *For scalars c, d and $M = U \operatorname{diag}(\omega_i) U^\top$,*

$$cM + dI = U \operatorname{diag}(c\omega_i + d) U^\top. \quad (5.100)$$

The eigenvectors are untouched; each eigenvalue moves along the same affine map $\omega \mapsto c\omega + d$. If every $c\omega_i + d \neq 0$ then $(cM + dI)^{-1} = U \operatorname{diag}((c\omega_i + d)^{-1}) U^\top$.

Proof. $I = UU^\top = U \operatorname{diag}(1) U^\top$ because U is orthogonal, so I is diagonal in *every*

orthonormal basis. Hence $cM + dI = U \operatorname{diag}(c\omega_i)U^\top + U \operatorname{diag}(d)U^\top = U \operatorname{diag}(c\omega_i + d)U^\top$, and the inverse statement is the diagonal case applied to $\operatorname{diag}(c\omega_i + d)$. $\square$

5.6.2 The forward map in the eigenbasis

The forward map (5.52), $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$, is exactly of the form (5.100) with $c = e^{-2t}$, $d = \Delta_t$.

Theorem 5.7 (Spectral form). *Let $\Sigma_0 = U \operatorname{diag}(\omega_i)U^\top$. Then for every $t \geq 0$,*

$$\Sigma_t = U \operatorname{diag}(\lambda_i(t))U^\top, \quad Q_t = U \operatorname{diag}\left(\frac{1}{\lambda_i(t)}\right)U^\top, \quad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t. \quad (5.101)$$

Proof. The first equality is (5.100). Since $\Sigma_0 \succ 0$ gives $\omega_i > 0$, every $\lambda_i(t) = e^{-2t}\omega_i + \Delta_t \geq \Delta_t > 0$ for $t > 0$ (and $\lambda_i(0) = \omega_i > 0$), so the inverse clause of Corollary 5.6 applies. $\square$

5.6.3 What the spectral form means

Diffusion never rotates the problem. The eigenvectors of Σ_t and Q_t are those of Σ_0 , frozen for all t ; only the eigenvalues move, each along the scalar path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$. As $t \rightarrow \infty$ every $\lambda_i(t) \rightarrow 1$: modes with $\omega_i > 1$ descend to 1, modes with $\omega_i < 1$ climb to 1. In eigen-coordinates the noisy covariance is an ellipsoid whose axes are fixed in direction and whose lengths all relax to the unit sphere — ... anisotropic at $t = 0$, isotropic at $t = \infty$ (Figure 5.1).

5.6.4 Decoupling in the eigenbasis

Write $\tilde{x} = U^\top x$ for the noisy trajectory in the eigenbasis of Σ_0 . Because U is orthogonal this is a rigid rotation — no stretching, no loss of information — and it turns the joint problem into scalars.

Theorem 5.8 (Decoupling). *In the coordinates $\tilde{x} = U^\top x$, with $\lambda_i(t)$ as in (5.101):*

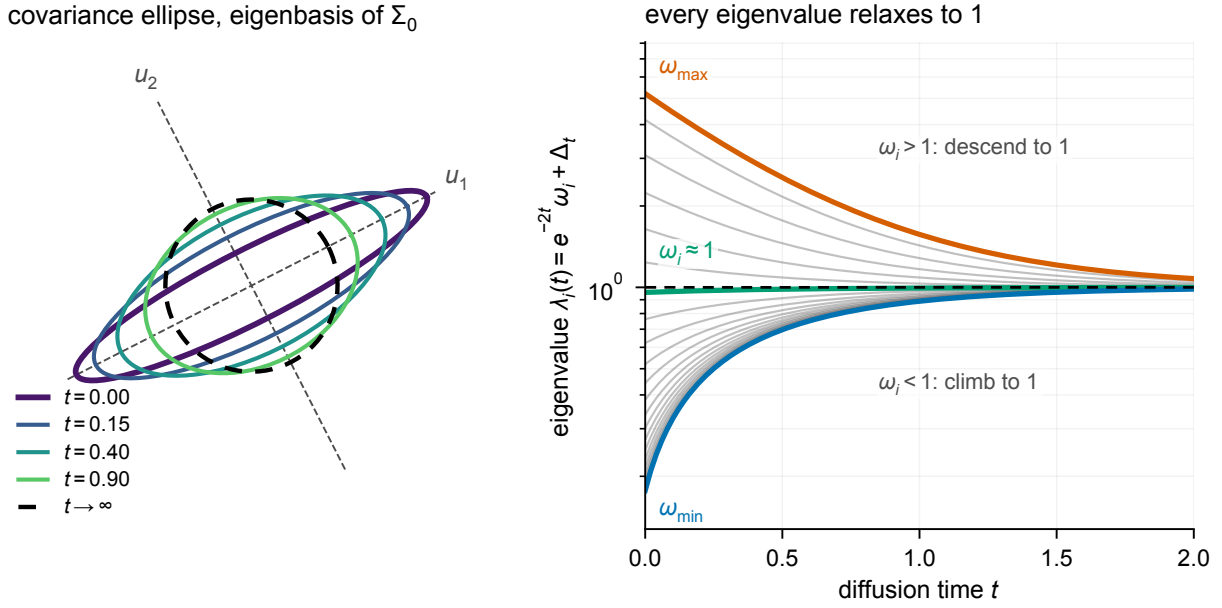


Figure 5.1: The spectral form (5.101) as geometry, for the AR(1) chain ($\alpha = 0.7$, $L = 24$). *Left:* the covariance ellipse $\{x : x^\top \Sigma_t^{-1} x = 1\}$ in the eigenbasis of Σ_0 , at four diffusion times and in the limit. Its axes point along the fixed eigenvectors u_1, u_2 at every t ; only the semi-axes $\sqrt{\lambda_i(t)}$ change, contracting the long axis and stretching the short one until, as $t \rightarrow \infty$, the ellipse is the unit circle. *Right:* the eigenvalue flow $\lambda_i(t) = e^{-2t}\omega_i + \Delta_t$ for all L modes (grey), with $\omega_{\max}$, $\omega_{\min}$ and a mode $\omega_i \approx 1$ highlighted. Every $\lambda_i(t) \rightarrow 1$ — modes above 1 descend, modes below 1 climb. Diffusion never rotates the problem; it only drives the spectrum to isotropy.

1. the noisy law factorises, $P_t(\tilde{x}) = \prod_i \mathcal{N}(\tilde{x}_i; 0, \lambda_i(t))$;
2. the score splits into independent scalar components, $\tilde{S}_i(\tilde{x}, t) = -\tilde{x}_i/\lambda_i(t)$, each depending on its own coordinate only;
3. the reverse-time denoising dynamics decouple into L independent scalar linear Gaussian diffusions, one per mode — OU-type but time-inhomogeneous, since the reverse drift coefficient depends on t through $\lambda_i(t)$.

Proof. (i) $\text{Cov}(\tilde{x}) = U^\top \Sigma_t U = \text{diag}(\lambda_i(t))$ by (5.101), and an uncorrelated Gaussian is independent. (ii) $\tilde{S} = U^\top S(U\tilde{x}, t) = -U^\top Q_t U \tilde{x} = -\text{diag}(1/\lambda_i(t)) \tilde{x}$. (iii) The reverse dynamics of Section 2.4 are driven by the score and by isotropic noise; under $U^\top$ the score is diagonal by (ii) and an orthogonal image of white noise is again white, so the vector equation separates coordinate by coordinate. $\square$

The eigenbasis is where the Gaussian chain is trivial: L scalars, each a one-dimensional denoising problem with its own signal-to-noise ratio $e^{-2t}\omega_i/\Delta_t$. Everything non-trivial in the frame basis — the dense Q_t , the band that fills and empties — is this diagonal object seen through the fixed rotation U .

5.6.5 Two computational corollaries

The covariance formula

$$\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_L$$

admits two useful rearrangements. They contain the same information, but emphasise different aspects of the diffusion: the first makes the signal-to-noise interpolation explicit, while the second explains how non-local entries appear in the precision matrix.

Signal-to-noise form. Factor Δ_t from the covariance:

$$\Sigma_t = \Delta_t \left(I_L + \frac{e^{-2t}}{\Delta_t} \Sigma_0 \right). \quad (5.102)$$

Defining

$$\gamma_t := \frac{e^{-2t}}{\Delta_t}, \quad (5.103)$$

and inverting gives

$$Q_t = \frac{1}{\Delta_t} \left(I_L + \gamma_t \Sigma_0 \right)^{-1}. \quad (5.104)$$

The scalar γ_t is the ratio between the variance carried by the attenuated signal and that injected by the channel. Since $\Delta_t = 1 - e^{-2t}$,

$$\gamma_t = \frac{e^{-2t}}{1 - e^{-2t}}, \quad (5.105)$$

so

$$\gamma_t \longrightarrow \infty \quad (t \downarrow 0), \quad \gamma_t \longrightarrow 0 \quad (t \rightarrow \infty).$$

These limits recover the two ends of the diffusion. For small t , $\gamma_t \Sigma_0$ dominates I_L ,

hence

$$Q_t \approx \frac{1}{\Delta_t} (\gamma_t \Sigma_0)^{-1} \quad (5.106)$$

$$= \frac{1}{\Delta_t \gamma_t} Q_0 \quad (5.107)$$

$$= e^{2t} Q_0. \quad (5.108)$$

Thus the noisy precision is initially close, up to an overall scale, to the clean tridiagonal precision.

At large t , instead, $\gamma_t \ll 1$, so

$$Q_t \approx \frac{1}{\Delta_t} I_L. \quad (5.109)$$

Since $\Delta_t \rightarrow 1$, this tends to I_L . The precision is therefore local at both extremes: tridiagonal near $t = 0$, and diagonal as $t \rightarrow \infty$. The non-local structure appears at intermediate diffusion times.

Resolvent form. A second rearrangement is more informative about how this non-locality is created. Factor e^{-2t} instead:

$$\Sigma_t = e^{-2t} (\Sigma_0 + e^{2t} \Delta_t I_L). \quad (5.110)$$

Because

$$e^{2t} \Delta_t = e^{2t} (1 - e^{-2t}) = e^{2t} - 1, \quad (5.111)$$

define

$$c_t := e^{2t} - 1. \quad (5.112)$$

Then

$$\Sigma_t = e^{-2t} (\Sigma_0 + c_t I_L), \quad (5.113)$$

and therefore

$$Q_t = e^{2t} (\Sigma_0 + c_t I_L)^{-1}. \quad (5.114)$$

Using $\Sigma_0 = Q_0^{-1}$,

$$\Sigma_0 + c_t I_L = Q_0^{-1} + c_t I_L \quad (5.115)$$

$$= Q_0^{-1} (I_L + c_t Q_0). \quad (5.116)$$

Hence

$$\boxed{Q_t = e^{2t} Q_0 (I_L + c_t Q_0)^{-1}}. \quad (5.117)$$

Since Q_0 commutes with any polynomial, and therefore with $(I_L + c_t Q_0)^{-1}$, the order of the two factors is immaterial.

The inverse in (5.117) can be expanded locally using a *Neumann series*. This is the matrix analogue of the scalar geometric series

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + \cdots, \quad |z| < 1. \quad (5.118)$$

For a matrix A , the corresponding identity is

$$(I_L + A)^{-1} = \sum_{m=0}^{\infty} (-A)^m = I_L - A + A^2 - A^3 + \cdots, \quad (5.119)$$

whenever the series converges. Applying it with $A = c_t Q_0$ gives

$$(I_L + c_t Q_0)^{-1} = I_L - c_t Q_0 + c_t^2 Q_0^2 - c_t^3 Q_0^3 + \cdots. \quad (5.120)$$

Substituting into (5.117),

$$Q_t = e^{2t} Q_0 (I_L - c_t Q_0 + c_t^2 Q_0^2 - \cdots) \quad (5.121)$$

$$= \boxed{e^{2t} \sum_{m=0}^{\infty} (-c_t)^m Q_0^{m+1}} \quad (5.122)$$

$$= e^{2t} (Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \dots). \quad (5.123)$$

For symmetric positive-definite Q_0 , this expansion converges whenever

$$c_t \lambda_{\max}(Q_0) < 1, \quad (5.124)$$

and in particular holds in a sufficiently small neighbourhood of $t = 0$, because $c_t = e^{2t} - 1 \sim 2t$.

The reason for introducing the Neumann series is not merely computational. It reveals how the dense precision is built from powers of the clean sparse precision. Since Q_0 is tridiagonal,

$$Q_0 \quad \text{couples frames at distance at most 1,}$$

while

$$Q_0^2 \quad \text{can couple frames at distance at most 2,}$$

and, more generally,

$$(Q_0^m)_{ij} = 0 \quad \text{whenever} \quad |i - j| > m. \quad (5.125)$$

Thus the expansion

$$Q_t = e^{2t} (Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \dots) \quad (5.126)$$

has a direct spatial interpretation. The first term contains only nearest-neighbour couplings; the second can create next-nearest-neighbour couplings; the third can reach one frame farther again; and so on.

For two frames at graph distance $d = |i - j|$, the first power capable of connecting them is Q_0^d . Since Q_0^d first appears with coefficient c_t^{d-1} , their precision coupling is generated,

near $t = 0$, only at order

$$(Q_t)_{ij} = O(c_t^{d-1}) = O(t^{d-1}), \quad d \geq 1. \quad (5.127)$$

The dense noisy precision therefore does not appear everywhere at once: diffusion fills the bands outward from the diagonal, one graph distance at a time. This is the band-fill law studied in the next section.

5.7 The lifecycle of the precision: losing tridiagonality

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each t . Figure 5.2 shows the whole arc at a glance.

When: immediately. For $\alpha \neq 0$, every off-diagonal entry of Q_t is non-zero at every $t > 0$: no exact zero survives any positive diffusion time. Analyticity alone does not give this — each entry is analytic in t by (5.101), but an analytic function that is not identically zero can still vanish at isolated times, so the band-fill law below only guarantees non-zero entries *near* $t = 0$. The all- t statement needs the structure of the resolvent. From (5.117),

$$Q_t = \frac{e^{2t}}{c_t} \left[(I + c_t Q_0) - I \right] (I + c_t Q_0)^{-1} = \frac{e^{2t}}{c_t} \left[I - (I + c_t Q_0)^{-1} \right], \quad (5.128)$$

so for $i \neq j$ the entry $(Q_t)_{ij}$ is $-e^{2t}/c_t$ times an off-diagonal entry of B^{-1} , $B := I + c_t Q_0$. Now B is symmetric positive definite and tridiagonal with off-diagonal entries $-c_t \alpha / \sigma_\eta^2 \neq 0$: an *irreducible* tridiagonal matrix. The explicit inverse formula for such matrices gives,

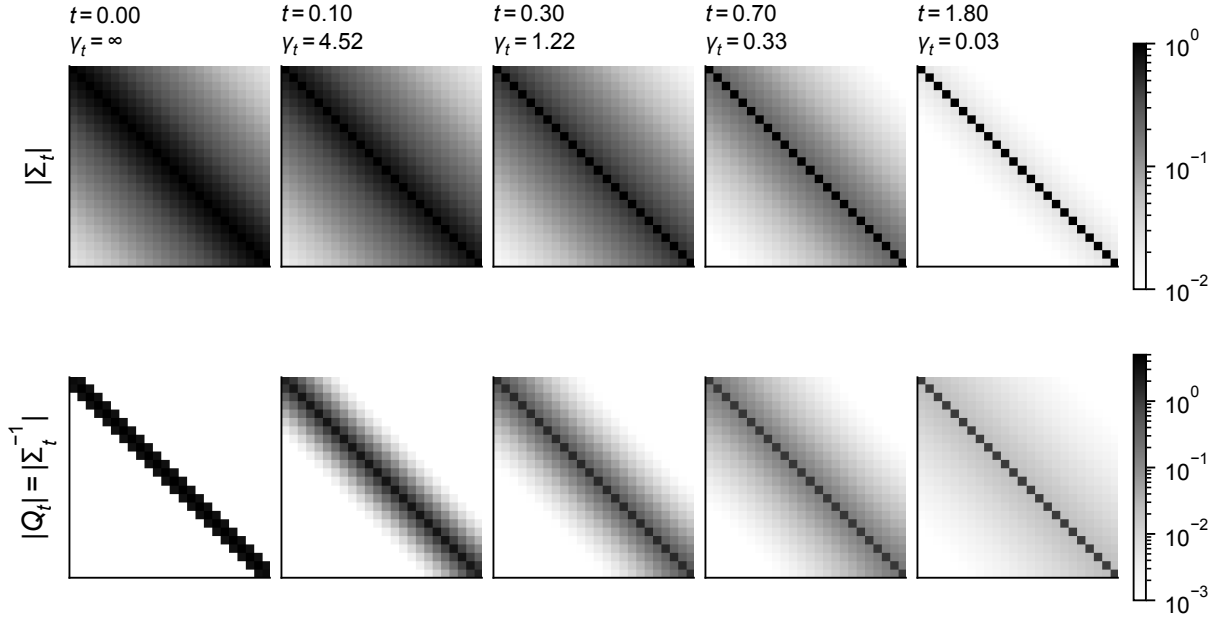


Figure 5.2: Magnitude heatmaps (log grey scale) of the noisy covariance $|\Sigma_t|$ (top) and precision $|Q_t|$ (bottom), $\alpha = 0.85$, $L = 24$, at five diffusion times with their signal-to-noise ratios $\gamma_t = e^{-2t}/\Delta_t$. At $t = 0$ the covariance is dense and the precision is tridiagonal. As t grows the covariance contracts onto its diagonal while the precision band *fills* outward, then fades: by $\gamma_t \ll 1$ both are within $O(e^{-2t})$ of the identity.

for $i \leq j$,

$$(B^{-1})_{ij} = (-1)^{i+j} \frac{\theta_{i-1} \phi_{j+1}}{\theta_{L-1}} \prod_{k=i}^{j-1} b_k, \quad (5.129)$$

where b_k are the off-diagonal entries and θ_m , ϕ_m are the leading and trailing principal minors of B — all strictly positive because $B \succ 0$. Every factor is non-zero, so every entry of B^{-1} , and hence every off-diagonal entry of Q_t , is non-zero for every $t > 0$. The formula also gives the sign pattern: for $\alpha > 0$ all off-diagonal entries of Q_t are negative (attractive couplings at every distance), while for $\alpha < 0$ they alternate in sign with the distance $|i - j|$. At $\alpha = 0$, by contrast, $\Sigma_t = (e^{-2t} + \Delta_t)I = I$, so $Q_t = I$ and every density claim is empty.

How: one diagonal per order in t . The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on — visible in Figure 5.2 as the precision band widening by roughly one diagonal between successive panels.

Theorem 5.9 (Band-fill leading order). *For small t and frame distance $d \geq 1$,*

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (5.130)$$

Toolbox 5.3 — Band fill from the Neumann series

In the resolvent form (5.123), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = (1 + O(t)) \left[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots \right].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m : the entry at distance d receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1} (2t)^{d-1} (1 + O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t : the band fills outward, one diagonal per order. $\square$

How much: the off-tridiagonal mass. Structure is measured here with the Frobenius norm: for $A \in \mathbb{R}^{L \times L}$,

$$\|A\|_F := \left(\sum_{i,j=0}^{L-1} A_{ij}^2 \right)^{1/2} = \sqrt{\text{tr}(A^\top A)}, \quad (5.131)$$

the Euclidean length of A read as a list of its L^2 entries. The one property used below is additivity over disjoint entry sets: if $A = B + C$ with B and C supported on disjoint index pairs, then $\|A\|_F^2 = \|B\|_F^2 + \|C\|_F^2$.

Split Q_t by that rule along the sparsity pattern of the clean precision. Let Π be the projection onto the *Markov band* — the tridiagonal set $\{(i, j) : |i - j| \leq 1\}$ that Q_0 occupies,

$$(\Pi A)_{ij} = A_{ij} \mathbf{1}\{|i - j| \leq 1\}, \quad (5.132)$$

so ΠQ_t is the nearest-neighbour approximation of the score operator and $Q_t - \Pi Q_t$ is everything it is forced to drop. The two are supported on disjoint entries, so (5.131) splits

Q_t into three orthogonal pieces,

$$\|Q_t\|_F^2 = \underbrace{\|\text{diag } Q_t\|_F^2}_{\text{diagonal}} + \underbrace{\|\Pi Q_t - \text{diag } Q_t\|_F^2}_{\text{nearest-neighbour band}} + \underbrace{\|Q_t - \Pi Q_t\|_F^2}_{\text{off-tridiagonal}}, \quad (5.133)$$

and the two coupling pieces are the curves of Figure 5.3, each divided by the clean nearest-neighbour mass $\|\Pi Q_0 - \text{diag } Q_0\|_F = \sqrt{2(L-1)} |\alpha|/\sigma_\eta^2$. The object of interest is the *off-tridiagonal mass*

$$M_\perp(t) := \|Q_t - \Pi Q_t\|_F = \left(\sum_{|i-j| \geq 2} (Q_t)_{ij}^2 \right)^{1/2}, \quad (5.134)$$

the norm of exactly the couplings a Markov model may not carry.

At small t the band-fill law (5.130) gives $(Q_t)_{i,i+2} = -2t(Q_0^2)_{i,i+2} + O(t^2)$ and $(Q_t)_{i,i+d} = O(t^{d-1})$ for $d \geq 3$, so $M_\perp(t)$ leaves zero linearly in t , dominated by the $d = 2$ diagonal. At large t , write $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ and expand the inverse — the Neumann series converges once $e^{-2t}(\omega_{\max} - 1) < 1$, i.e. for all large enough t —

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \quad (5.135)$$

so $\|Q_t - I\|_F \simeq e^{-2t} \|\Sigma_0 - I\|_F$: every deviation from the identity, off-band and on-band alike, decays at the single rate e^{-2t} , and the score approaches the isotropic $-x$. In between, $M_\perp(t)$ rises, peaks, and falls (Figure 5.3): the noisy sequence is most non-Markov at intermediate diffusion times, yet even at the peak $M_\perp$ stays below one-tenth of the clean nearest-neighbour mass. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d ; maximally coupled — in the Frobenius measure used here, at the displayed (α, L) — at intermediate t ; isotropic as $t \rightarrow \infty$ at rate e^{-2t} .

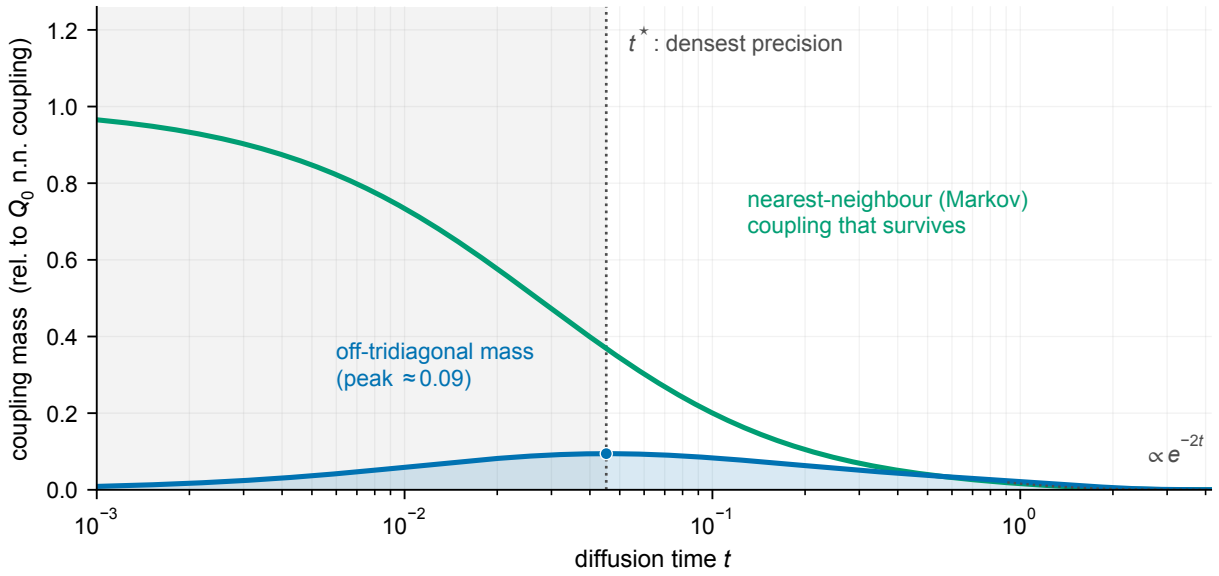


Figure 5.3: Frobenius lifecycle of the precision’s inter-frame coupling ($\alpha = 0.9$, $L = 48$); every mass is a Frobenius norm of a set of diagonals of Q_t , normalised by the nearest-neighbour coupling mass of Q_0 . The nearest-neighbour (Markov) mass (green) decays monotonically, $\propto e^{-2t}$ once noise dominates. The off-tridiagonal mass (blue) rises from zero, peaks at $t^* \approx 0.05$ — the densest precision, where the sequence is least Markov — then decays with the rest of the structure. Its peak is below 10% of the original coupling: in this norm, at this (α, L) , the Markov band carries almost all of the score operator’s inter-frame coupling mass at every diffusion time.

Chapter 6

The Gaussian Chain by Belief Propagation

Chapter 5 obtained the joint score through the marginal covariance: forming and inverting the dense $L \times L$ matrix naively costs $O(L^3)$. That cost is a property of the naive route, not of the problem — the posterior precision is tridiagonal, so a banded solver already delivers the same answer in $O(L)$ — and this chapter obtains the score by the message-passing form of exactly that structured solve. The construction is generic — it is the sum-product algorithm run on the factor graph of the posterior — and for a chain of K -state variables it costs $O(LK^2)$ time and $O(LK)$ memory. For the Gaussian AR(1) chain a message collapses to two numbers, and the answer is exact; run in mean-variance arithmetic the recursion is the Kalman filter and the Rauch–Tung–Striebel smoother, update for update (Kalman, 1960; Rauch et al., 1965). The chapter closes with the operational cost of locality: what truncating the evidence range does to the score, measured exactly. (The closed form of the bulk posterior itself, and the geometric influence-decay rate $q(\alpha, t)$ it implies, are derived in Chapter 10, Section 10.4.1, where they are used as a design calculation.)

6.1 The posterior and its factor graph

The matrix route is global: one $L \times L$ inversion. The score-posterior identity (2.39) says the same object is a posterior expectation, and that posterior has a structure — a Markov chain observed through independent per-frame noise — built for local computation.

6.1.1 The posterior is a hidden Markov model

Bayes' rule multiplies the chain prior (3.6) by the per-frame likelihoods (5.1). Nothing in the likelihood couples two frames — each x_k sees only its own a_k — so the product keeps the prior's two-variable structure:

$$p(a | x) \propto \underbrace{\psi_{\text{pr}}(a_0)}_{\mathcal{N}(a_0; 0, 1)} \prod_{k=0}^{L-2} \underbrace{\psi_{\text{tr}}^{(k)}(a_k, a_{k+1})}_{\mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2)} \prod_{k=0}^{L-1} \underbrace{\psi_{\text{ob}}^{(k)}(a_k)}_{\mathcal{N}(x_k; e^{-t} a_k, \Delta_t)}. \quad (6.1)$$

This is a hidden Markov model (Rabiner, 1989): a latent chain with nearest-neighbour transitions, each state reported once through an independent channel. The data x_k are fixed numbers absorbed into the observation factors, so $\psi_{\text{ob}}^{(k)}$ is a function of the single variable a_k . By (2.39), the k -th component of the joint score is fixed by one number from this posterior — the smoothing mean $\mathbb{E}[a_k | x]$.

6.1.2 The factor graph is a tree

The factor graph (Section 3.5) of (6.1) has

- L *variable* nodes, one per a_k ;
- $2L$ *factor* nodes: one prior ψ_{pr} , $L - 1$ transitions $\psi_{\text{tr}}^{(k)}$, L observations $\psi_{\text{ob}}^{(k)}$;
- $3L - 1$ *edges*: each transition factor joins two variables ($2(L - 1)$ edges), the prior and each observation factor join one ($L + 1$ edges).

A connected factor graph with one fewer edge than node ($3L - 1$ against $3L$) is a tree. Its shape is a spine of variables a_k linked by transition factors, with one observation leaf

per spine node (Figure 3.4). Conditioning on the data did not change the graph of the prior: per-node evidence only re-weights an existing factor and hangs a leaf, and never joins two spine nodes (Section 3.5).

6.2 The sum–product sweep and its cost

6.2.1 The two sweeps

Specialising the sum–product rules (3.35) to this tree, the schedule collapses to one forward and one backward pass (Theorem 3.4).

Proposition 6.1 (Chain sweeps). *Define messages*

$$\mu_{\rightarrow 0}(a_0) := \psi_{\text{pr}}(a_0), \quad \mu_{\rightarrow k+1}(a_{k+1}) := \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) \, da_k, \quad (6.2)$$

$$\mu_{\leftarrow L-1}(a_{L-1}) := 1, \quad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) \, da_k. \quad (6.3)$$

Then for every k ,

$$p(a_k \mid x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) : \quad (6.4)$$

the forward message into a_k carries the evidence $x_0, \dots, x_{k-1}$ only, the backward message carries $x_{k+1}, \dots, x_{L-1}$ only, and the local evidence x_k enters once, at combination time.

Toolbox 6.1 — Why the two sweeps give the exact smoothing marginal

Fix a site k . Starting from the posterior factorisation (6.1), separate all factors into those lying to the left of a_k , the local observation at a_k , and those lying to its right:

$$\begin{aligned} p(a \mid x) &\propto \underbrace{\psi_{\text{pr}}(a_0) \prod_{j=0}^{k-1} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=0}^{k-1} \psi_{\text{ob}}^{(j)}(a_j)}_{\text{left of } a_k} \\ &\quad \times \psi_{\text{ob}}^{(k)}(a_k) \end{aligned}$$

$$\times \underbrace{\prod_{j=k}^{L-2} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=k+1}^{L-1} \psi_{\text{ob}}^{(j)}(a_j)}_{\text{right of } a_k}. \quad (6.5)$$

To obtain $p(a_k | x)$, integrate over every variable except a_k . Because the factor graph is a chain, the variables on the two sides of a_k occur in disjoint factors once a_k is held fixed. The integral therefore separates:

$$\begin{aligned} p(a_k | x) &\propto \left[\int \psi_{\text{pr}}(a_0) \prod_{j=0}^{k-1} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=0}^{k-1} \psi_{\text{ob}}^{(j)}(a_j) da_0 \cdots da_{k-1} \right] \\ &\times \psi_{\text{ob}}^{(k)}(a_k) \\ &\times \left[\int \prod_{j=k}^{L-2} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=k+1}^{L-1} \psi_{\text{ob}}^{(j)}(a_j) da_{k+1} \cdots da_{L-1} \right]. \end{aligned} \quad (6.6)$$

The first bracket is precisely the information arriving at a_k from the left. Define it as

$$\mu_{\rightarrow k}(a_k) := \int \psi_{\text{pr}}(a_0) \prod_{j=0}^{k-1} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=0}^{k-1} \psi_{\text{ob}}^{(j)}(a_j) da_0 \cdots da_{k-1}. \quad (6.7)$$

For $k = 0$ there is nothing to integrate, so

$$\mu_{\rightarrow 0}(a_0) = \psi_{\text{pr}}(a_0). \quad (6.8)$$

To obtain the message one step farther to the right, separate the last integration variable a_k :

$$\mu_{\rightarrow k+1}(a_{k+1}) = \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) da_k, \quad (6.9)$$

which is exactly the forward recursion (6.2).

Similarly, define the information arriving from the right by

$$\mu_{\leftarrow k}(a_k) := \int \prod_{j=k}^{L-2} \psi_{\text{tr}}^{(j)}(a_j, a_{j+1}) \prod_{j=k+1}^{L-1} \psi_{\text{ob}}^{(j)}(a_j) da_{k+1} \cdots da_{L-1}. \quad (6.10)$$

At the right endpoint there are no factors beyond a_{L-1} , so

$$\mu_{\leftarrow L-1}(a_{L-1}) = 1. \quad (6.11)$$

Peeling off the first variable on the right gives

$$\mu_{\leftarrow k-1}(a_{k-1}) = \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k, \quad (6.12)$$

which is the backward recursion (6.3).

Substituting the two partial marginalisations into (6.6) yields

$$\boxed{p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k)}. \quad (6.13)$$

The separation of the evidence is now explicit:

$$\mu_{\rightarrow k} \text{ depends only on } x_0, \dots, x_{k-1}, \quad (6.14)$$

$$\psi_{\text{ob}}^{(k)} \text{ contains only } x_k, \quad (6.15)$$

$$\mu_{\leftarrow k} \text{ depends only on } x_{k+1}, \dots, x_{L-1}. \quad (6.16)$$

Thus every observation enters the marginal exactly once. Because the factor graph is a tree, no approximation has been made. $\square$

Computational cost. For a discrete chain in which each a_k takes K values, every message is a vector in $\mathbb{R}^K$. Consider, for example, the forward update

$$\mu_{\rightarrow k+1}(a_{k+1}) = \sum_{a_k} \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k).$$

For each of the K possible values of a_{k+1} , the sum runs over the K possible values of a_k , so one transition message costs $O(K^2)$. A forward-backward sweep sends $O(L)$ such

messages along the chain; therefore

$$\boxed{\text{time} = O(LK^2).} \quad (6.17)$$

Each stored message contains K numbers. Keeping the messages required to construct all L smoothing marginals therefore costs

$$\boxed{\text{space} = O(LK).} \quad (6.18)$$

Thus sum-product is linear in the sequence length L ; the K^2 factor is the local cost of marginalising across one pairwise transition factor.

6.3 The Gaussian closure

Two facts close the recursion: a product of Gaussians in the same variable is Gaussian, and an affine map of a Gaussian, marginalised, is Gaussian. The observation factor, read as a function of the clean variable, is itself Gaussian,

$$\psi_{\text{ob}}^{(k)}(a_k) \propto \exp\left(-\frac{e^{-2t}}{2\Delta_t}(a_k - e^t x_k)^2\right) = \mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t}),$$

a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t .

Theorem 6.2 (Gaussian closure of BP on the chain). *For the Gaussian AR(1) chain observed through the OU channel, every sum-product message and every one-node belief is Gaussian. A message can therefore be represented by two scalar natural parameters, and each message update requires $O(1)$ arithmetic. Since a forward-backward sweep contains $O(L)$ messages, all posterior marginals, and hence the complete joint score, are obtained exactly in*

$$\boxed{\text{time} = O(L), \quad \text{space} = O(L)} \quad (6.19)$$

when all smoothing marginals are retained.

Toolbox 6.2 — Why Gaussian BP closes exactly

It is convenient to write an unnormalised scalar Gaussian in natural form,

$$\mu(a) \propto \exp\left(-\frac{1}{2}\lambda a^2 + ra\right), \quad (6.20)$$

where λ is the precision and r the field. When $\lambda > 0$,

$$v = \frac{1}{\lambda}, \quad m = \frac{r}{\lambda}. \quad (6.21)$$

Thus a complete message is represented by only the pair (λ, r) .

The observation factor at site k is itself Gaussian as a function of a_k :

$$\psi_{\text{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) \quad (6.22)$$

$$\propto \exp\left[-\frac{(x_k - e^{-t}a_k)^2}{2\Delta_t}\right] \quad (6.23)$$

$$\propto \exp\left[-\frac{1}{2} \frac{e^{-2t}}{\Delta_t} a_k^2 + \frac{e^{-t}x_k}{\Delta_t} a_k\right]. \quad (6.24)$$

Hence its natural parameters are

$$\lambda_{\text{ob}} = \frac{e^{-2t}}{\Delta_t}, \quad r_{\text{ob},k} = \frac{e^{-t}x_k}{\Delta_t}. \quad (6.25)$$

1. Products preserve Gaussianity. Suppose an incoming message has parameters (λ, r) . Multiplying it by the local observation factor gives

$$\mu(a_k)\psi_{\text{ob}}^{(k)}(a_k) \propto \exp\left[-\frac{1}{2}(\lambda + \lambda_{\text{ob}})a_k^2 + (r + r_{\text{ob},k})a_k\right]. \quad (6.26)$$

Thus the product remains Gaussian, with

$$\bar{\lambda} = \lambda + \lambda_{\text{ob}}, \quad \bar{r} = r + r_{\text{ob},k}. \quad (6.27)$$

In natural parameters, multiplying Gaussian factors is therefore just addition.

2. Passing through the transition preserves Gaussianity. The AR(1) transition is

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2). \quad (6.28)$$

After incorporating the observation at k , the variable a_k has Gaussian parameters $(\bar{\lambda}, \bar{r})$, hence

$$a_k \sim \mathcal{N}\left(\frac{\bar{r}}{\bar{\lambda}}, \frac{1}{\bar{\lambda}}\right). \quad (6.29)$$

An affine transformation of a Gaussian plus independent Gaussian noise is again Gaussian, so

$$a_{k+1} \sim \mathcal{N}\left(\alpha \frac{\bar{r}}{\bar{\lambda}}, \sigma_\eta^2 + \frac{\alpha^2}{\bar{\lambda}}\right). \quad (6.30)$$

Therefore the outgoing forward message is Gaussian with natural parameters

$$\lambda_{\rightarrow k+1} = \frac{\bar{\lambda}}{\alpha^2 + \sigma_\eta^2 \bar{\lambda}}, \quad (6.31)$$

$$r_{\rightarrow k+1} = \frac{\alpha \bar{r}}{\alpha^2 + \sigma_\eta^2 \bar{\lambda}}. \quad (6.32)$$

The map

$$(\lambda_{\rightarrow k}, r_{\rightarrow k}) \mapsto (\lambda_{\rightarrow k+1}, r_{\rightarrow k+1})$$

therefore uses only a fixed number of scalar arithmetic operations.

The backward recursion is closed for exactly the same reason. If the observation at k and the incoming right message combine to $(\bar{\lambda}, \bar{r})$, then integrating

$$\int \mathcal{N}(a_k; \alpha a_{k-1}, \sigma_\eta^2) \exp\left(-\frac{1}{2} \bar{\lambda} a_k^2 + \bar{r} a_k\right) da_k \quad (6.33)$$

gives another Gaussian function of a_{k-1} , with

$$\lambda_{\leftarrow k-1} = \frac{\alpha^2 \bar{\lambda}}{1 + \sigma_\eta^2 \bar{\lambda}}, \quad (6.34)$$

$$r_{\leftarrow k-1} = \frac{\alpha \bar{r}}{1 + \sigma_\eta^2 \bar{\lambda}}. \quad (6.35)$$

3. Combining the two sweeps preserves Gaussianity. At node k the exact smoothing belief is

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k).$$

All three factors are Gaussian, so their natural parameters add:

$$\lambda_k^{\text{post}} = \lambda_{\rightarrow k} + \lambda_{\text{ob}} + \lambda_{\leftarrow k}, \quad (6.36)$$

$$r_k^{\text{post}} = r_{\rightarrow k} + r_{\text{ob},k} + r_{\leftarrow k}. \quad (6.37)$$

Consequently

$$\mathbb{E}[a_k | x] = \frac{r_k^{\text{post}}}{\lambda_k^{\text{post}}}. \quad (6.38)$$

Substitution into the score-posterior identity gives

$$S_k(x, t) = \frac{e^{-t r_k^{\text{post}} / \lambda_k^{\text{post}} - x_k}}{\Delta_t}. \quad (6.39)$$

The closure is exact: no Gaussian approximation has been introduced. Gaussian factors are closed algebraically under the two operations used by sum-product, multiplication and marginalisation, and the factor graph is a tree. $\square$

Remark 6.3 (Why not TAP or approximate message passing?). The Gaussian closure above should not be confused with the approximations used in TAP or approximate message passing (AMP). TAP equations originate as mean-field equations for densely interacting systems and introduce an Onsager reaction correction to account for correlations neglected by naive mean-field theory. AMP turns closely related ideas into an iterative message-passing algorithm for high-dimensional, typically dense random models: when each variable receives many individually weak contributions, the full collection of BP messages can be compressed into approximately Gaussian scalar updates together with an

Onsager correction.

None of those approximations is needed here. Our factor graph is a one-dimensional tree of finite degree, so ordinary sum-product is already exact. Moreover, in the Gaussian model the messages are Gaussian because Gaussian multiplication and marginalisation are exact algebraic operations, not because many weak contributions become Gaussian in a large-system limit. Using TAP or AMP would therefore replace an exact algorithm with an approximation designed for a different, typically dense and loopy, regime.

Numerical agreement of the two routes. The audit suite runs BP against the matrix route over 80 configurations ($L \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x): maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores — double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix A). One qualitative detail from a worked $L = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

6.4 The price of locality

Belief propagation on the chain is exact, but it is not local: the message reaching frame k has travelled the whole sequence. Chapter 5 showed that the exact score operator Q_t is tridiagonal only at $t = 0$ and dense thereafter, and measured how much coupling mass sits outside the Markov band (Figure 5.3). That is a statement about a matrix. This section asks the operational question behind it: *what does it cost to ignore that mass?* — how much of the score survives if evidence is only allowed to travel a few frames.

6.4.1 Two ways to go local

Truncate the answer. Take the exact operator Q_t , keep only the entries within a band, and use what is left:

$$\widehat{S}^{(b)}(x, t) = -Q_t^{(b)}x, \quad \left(Q_t^{(b)}\right)_{ij} = (Q_t)_{ij} \mathbf{1}\{|i - j| \leq b\}. \quad (6.40)$$

At $b = 1$ this is the tridiagonal — Markov — truncation of the score.

Truncate the algorithm. Let the messages of Proposition 6.1 travel at most r hops before frame k is read off: beyond the cut, the incoming message is replaced by what it was before any evidence arrived, the stationary prior marginal. Because a contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, this is not an approximation of the window's inference but exact inference on the window,

$$m_k^{(r)} = \mathbb{E}[a_k | x_{k-r}, \dots, x_{k+r}], \quad \widehat{S}_k^{(r)}(x, t) = \frac{e^{-t}m_k^{(r)} - x_k}{\Delta_t}, \quad (6.41)$$

Both approximations are linear in the noisy trajectory. Writing either of them as

$$\widehat{S}(x, t) = -Mx, \quad (6.42)$$

while the exact Gaussian score is

$$S(x, t) = -Q_t x, \quad (6.43)$$

their error on a particular trajectory is

$$\widehat{S}(x, t) - S(x, t) = -(M - Q_t)x. \quad (6.44)$$

Its size depends on the particular realization of x , so we average the squared error over the actual noisy data law $x \sim \mathcal{N}(0, \Sigma_t)$. To make errors comparable across diffusion times

and locality budgets, we normalize by the average squared magnitude of the exact score itself. This gives the relative RMS score error

$$\varepsilon(M) := \sqrt{\frac{\mathbb{E}\|(M - Q_t)x\|^2}{\mathbb{E}\|Q_t x\|^2}}. \quad (6.45)$$

Thus $\varepsilon(M) = 0$ means an exact score, while, for example, $\varepsilon(M) = 0.1$ means that the RMS approximation error is 10% of the RMS size of the exact score.

Because both estimators are linear, the expectations can be evaluated without Monte Carlo sampling. For any matrix A and any centred random vector x with covariance Σ_t ,

$$\mathbb{E}[x^\top A x] = \mathbb{E}[\text{tr}(A x x^\top)] \quad (6.46)$$

$$= \text{tr}(A \mathbb{E}[x x^\top]) \quad (6.47)$$

$$= \text{tr}(A \Sigma_t). \quad (6.48)$$

Applying this identity with $A = (M - Q_t)^\top (M - Q_t)$ gives

$$\mathbb{E}\|(M - Q_t)x\|^2 = \text{tr}[(M - Q_t)^\top (M - Q_t) \Sigma_t] \quad (6.49)$$

$$= \text{tr}[(M - Q_t) \Sigma_t (M - Q_t)^\top], \quad (6.50)$$

where the second equality uses cyclicity of the trace.

The denominator simplifies further because $Q_t = \Sigma_t^{-1}$ and both matrices are symmetric:

$$\mathbb{E}\|Q_t x\|^2 = \text{tr}(Q_t \Sigma_t Q_t) \quad (6.51)$$

$$= \text{tr}(Q_t). \quad (6.52)$$

Hence

$$\boxed{\varepsilon(M) = \sqrt{\frac{\text{tr}[(M - Q_t) \Sigma_t (M - Q_t)^\top]}{\text{tr} Q_t}}}. \quad (6.53)$$

Equation (6.53) is therefore a population error, not a sampled estimate: once M , Σ_t , and Q_t are known, the expected score error is deterministic.

6.4.2 What the neighbours are worth

Table 6.1 evaluates (6.53) for both families.

Table 6.1: Relative RMS score error (6.53) of the two truncations ($L = 40$, $\alpha = 0.8$). Both estimators are linear in x , so every entry is closed-form, not sampled. The last two rows are derived: the fraction of the no-neighbour error removed by admitting a single hop, and the penalty for truncating the answer rather than the algorithm at equal budget.

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
no neighbours, $r = 0$	0.815	0.567	0.194	0.0045
one hop, $r = 1$	0.123	0.235	0.130	0.0035
two hops, $r = 2$	0.020	0.107	0.089	0.0028
banded answer, $b = 1$	0.210	0.302	0.135	0.0035
banded answer, $b = 2$	0.032	0.134	0.092	0.0028
removed by the first hop	85%	59%	33%	21%
penalty, $b=1$ over $r=1$	$1.71\times$	$1.28\times$	$1.03\times$	$1.00\times$

The first row is the cost of treating the frames as independent. At $t = 0.05$ the isolated-frame score is wrong by 82% in relative RMS: almost all of the score at small diffusion times *is* inter-frame structure, which is the quantitative form of the point made with the marginal score in Section 5.5. This error does not vanish as $t \rightarrow 0$: the $r = 0$ estimator converges to $-x$ while the exact score converges to $-Q_0x$, so its low-noise limit is set by the clean coupling itself. The second row is what one hop of message passing recovers: the error falls to 0.123, so a single nearest-neighbour exchange — the Markov band, and nothing more — removes 85% of it. Two hops remove 98%.

That fraction falls with diffusion time (85%, 59%, 33%, 21%), which sounds like the Markov band mattering less; the first row shows why it does not. By $t = 3$ the entire inter-frame content of the score is worth 0.0045 in relative error, so a model that ignores

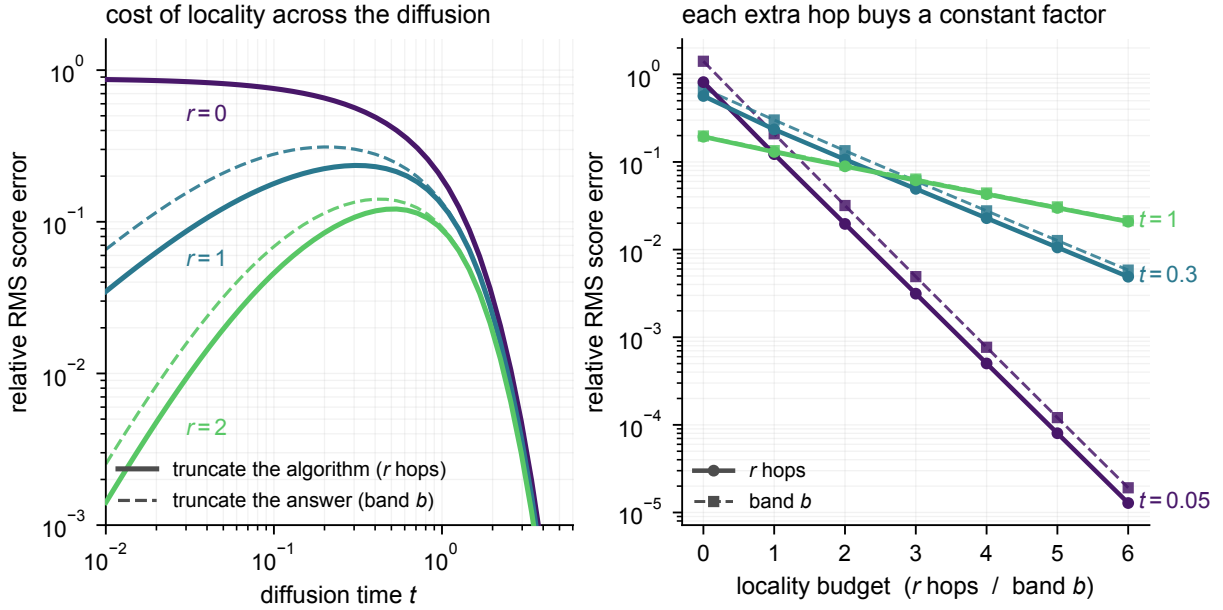


Figure 6.1: The two truncations ($L = 40$, $\alpha = 0.8$; every value closed-form by (6.53), none sampled). *Left:* relative RMS score error against diffusion time, for message range r (solid) and matrix band b (dashed). For $r \geq 1$ both are worst at intermediate t and vanish at both ends — at $t \rightarrow 0$ because the clean score is tridiagonal, so any window that retains the Markov blanket becomes exact; the no-neighbour estimator $r = 0$ is the exception, since the clean score of frame k genuinely uses its neighbours whenever $\alpha \neq 0$, and its error tends to a positive limit instead. The gap between the families closes as t grows. *Right:* the same error against the locality budget at three diffusion times: geometric decay, at a rate that worsens with t , with the banded answer (squares) tracking above the windowed inference (circles) at every budget.

every neighbour is already accurate to half a percent and there is nothing left for any hop to recover. The band carries the largest share of the coupling exactly where the coupling is large.

Chapter 7

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 5 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable.

7.1 What changes, and what is preserved

The Laplace AR(1) chain replaces the Gaussian innovations of (3.7) with Laplace ones, keeping the initial law and the second-order structure fixed:

$$a_0 \sim \mathcal{N}(0, 1), \quad a_k = \alpha a_{k-1} + \varepsilon_k, \quad \varepsilon_k \sim \text{Laplace}(0, b) \text{ i.i.d.}, \quad k = 1, \dots, L-1, \quad (7.1)$$

observed through the same per-frame OU channel. The innovation scale is *not* free: $b = \sqrt{(1 - \alpha^2)/2}$, so that $\text{Var}(\varepsilon_k) = 1 - \alpha^2$ and the chain has unit marginal variance and $\text{Cov}(a_j, a_k) = \alpha^{|j-k|}$, exactly as in the Gaussian case. The Gaussian and Laplace chains therefore agree on every second moment and differ *only* beyond them.

The score-posterior identity (2.39) holds for any p_0 . The posterior factorises over the same tree-shaped factor graph, so the sum-product recursion (6.2)–(6.3) still computes exact smoothing marginals in one forward and one backward sweep. And the OU likelihood factors are still Gaussian in a_k . What is lost is that the messages are no longer Gaussian and the score ceases to be linear in x .

7.2 Where the score stops being linear

The score-posterior identity (2.39) is indifferent to the prior. Componentwise, for the chain (7.1) under the OU channel, with $\mu := e^{-t}$,

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k | x] - x_k}{\Delta_t}. \quad (7.2)$$

For Gaussian innovations $\mathbb{E}[a_k | x]$ is a linear function of x and (7.2) is the linear score $-Q_t x$; for Laplace innovations the posterior mean is nonlinear, and the score inherits the nonlinearity exactly.

The two chains agree on every second moment by construction (Section 7.1); they differ first at fourth order. The *innovation* carries excess kurtosis 3: for a centred Laplace variable of variance v , $\kappa_4 = 3v^2$, so the standardised fourth cumulant is 3, against 0 for a Gaussian innovation.

The chain *state* does not inherit that number unchanged, because a_k is a weighted sum of one Gaussian seed and k independent Laplace innovations, and cumulants of independent sums add with the fourth power of the coefficients. From (7.1), $a_k = \alpha^k a_0 +$

$\sum_{j=1}^k \alpha^{k-j} \varepsilon_j$ with $\kappa_4(a_0) = 0$ and $\kappa_4(\varepsilon_j) = 3(1 - \alpha^2)^2$, so

$$\kappa_4(a_k) = 3(1 - \alpha^2)^2 \sum_{m=0}^{k-1} \alpha^{4m} = 3 \frac{1 - \alpha^2}{1 + \alpha^2} (1 - \alpha^{4k}), \quad (7.3)$$

which is also the excess kurtosis of a_k since $\text{Var}(a_k) = 1$. Two consequences matter later. The state's non-Gaussianity is much milder than the innovation's — at $\alpha = 0.85$ the bulk value is $3 \times 0.2775/1.7225 \approx 0.48$, not 3 — and it depends on the position k : the chain is covariance-stationary by construction but *not* strictly stationary, because the Gaussian initial law is not the stationary law of the Laplace recursion, and the shape settles towards its bulk value (7.3) only as α^{4k} dies away.

Under the variance-preserving forward process, fourth cumulants of the independent sum $x_{t,k} = e^{-t}a_k + \sqrt{\Delta_t}z_k$ scale as the fourth power of the coefficients, and $\kappa_4(z) = 0$, so

$$\boxed{\kappa_4(x_{t,k}) = e^{-4t} \kappa_4(a_k)} \quad (7.4)$$

with $\kappa_4(a_k)$ as in (7.3). Hence Laplace-driven non-Gaussianity survives every finite diffusion time but decays as e^{-4t} , while the Gaussian case remains Gaussian throughout.

Figure 7.1 shows how this appears in the score.

7.3 Where closed forms stop

The tree structure is untouched by any of this, so the message decomposition survives in full.

Theorem 7.1 (Joint score from chain messages). *For any chain prior, in particular (7.1), and the OU likelihood, the joint score is (7.2) with*

$$\mathbb{E}[a_k \mid x] = \frac{\int a_k \vec{m}_k(a_k) \psi_{\text{ob}}^{(k)}(a_k) \overleftarrow{m}_k(a_k) da_k}{\int \vec{m}_k(a_k) \psi_{\text{ob}}^{(k)}(a_k) \overleftarrow{m}_k(a_k) da_k}, \quad (7.5)$$

the messages being those of Proposition 6.1.

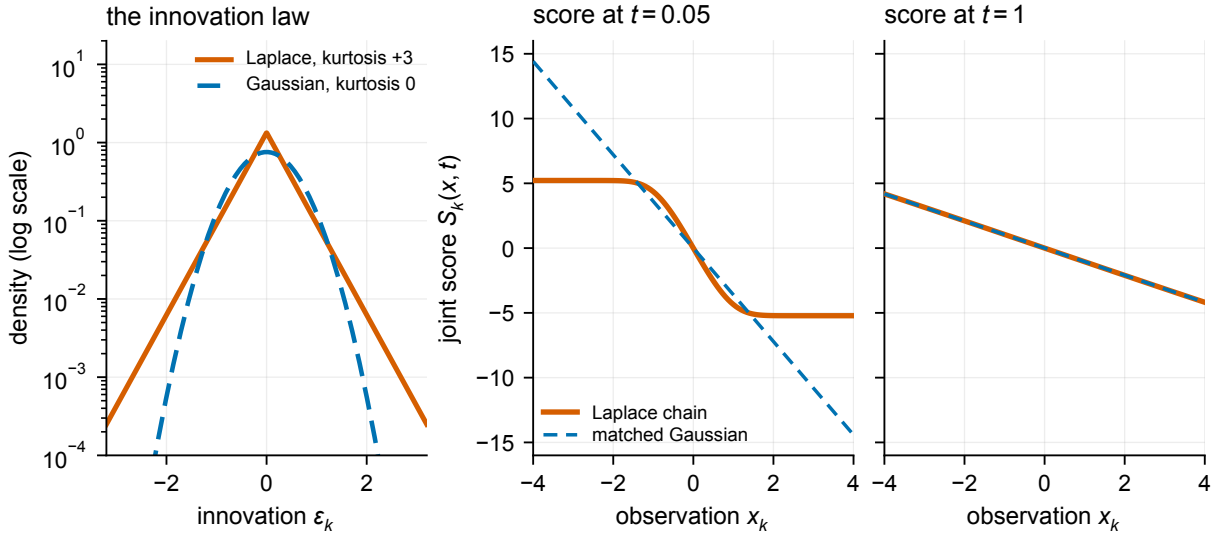


Figure 7.1: What changes when the innovations go Laplace ($\alpha = 0.85$, so $b = \sqrt{(1 - \alpha^2)/2}$). Vermilion and solid is Laplace throughout; blue and dashed is Gaussian. *Left:* the two innovation laws on a log scale. They carry the same variance $1 - \alpha^2$ by construction and differ only in shape — the kink at the origin and the exponential rather than Gaussian tail — which the fourth cumulant summarises as excess kurtosis +3 against 0. *Centre and right:* the joint score (7.2) at the middle frame of a chain of $L = 9$, as the observation x_k at that frame is swept with the others held at zero, computed by grid BP (Section 7.4) against the covariance-matched Gaussian chain — which is exactly the estimator the Gaussian closure returns (Theorem 7.3). At $t = 0.05$ the Laplace score saturates near ± 5.1 while the Gaussian one grows without bound; at $t = 1$, on the same vertical scale, the two coincide.

Computational locality survives. What does not survive is the collapse of message updates into two parameters.

Remark 7.2 (Where analytic closure breaks). The first message can be evaluated explicitly.

With

$$M(a' | a) = \frac{1}{2b} \exp\left(-\frac{|a' - \alpha a|}{b}\right), \quad p(x' | a') = \mathcal{N}(x'; \mu a', \Delta_t),$$

set $r = a' - \alpha a$ and $y = x' - \mu \alpha a$. Then

$$I(y) := \int_{\mathbb{R}} M(a' | a) \mathcal{N}(x'; \mu a', \Delta_t) da' \quad (7.6)$$

$$= \frac{1}{2b\sqrt{2\pi\Delta_t}} \int_{\mathbb{R}} \exp\left[-\frac{|r|}{b} - \frac{(y - \mu r)^2}{2\Delta_t}\right] dr. \quad (7.7)$$

Splitting at the single kink $r = 0$ and completing the square on each half-line gives

$$I(y) = \frac{e^{\Delta_t/(2\mu^2 b^2)}}{2\mu b} \left[e^{-y/(\mu b)} \Phi\left(\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\mu b}\right) + e^{y/(\mu b)} \Phi\left(-\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\mu b}\right) \right], \quad (7.8)$$

for $\mu > 0$, where Φ is the standard Gaussian cdf. Thus the first Laplace–Gaussian convolution closes in elementary exponentials and error functions.

Closure is already lost at the next update. The incoming message is now a sum of terms of the form

$$e^{ca} \Phi(da + e),$$

so multiplication by the Gaussian likelihood and the next Laplace transition requires integrals such as

$$\int_{-\infty}^{\alpha a'} e^{-qa^2 + \ell a} \Phi(da + e) da \quad \text{and} \quad \int_{\alpha a'}^{\infty} e^{-qa^2 + \ell a} \Phi(da + e) da.$$

These involve Gaussian cdfs inside truncated Gaussian integrals—equivalently, higher-dimensional Gaussian probabilities—and no longer belong to the one-dimensional normal–Laplace family. Each propagation step therefore increases the functional complexity, unlike the Gaussian chain, where Gaussian-times-Gaussian integration preserves the same two-parameter family indefinitely.

Changing coordinates does not remove the obstruction. Define the innovations

$$r_0 = a_0, \quad r_k = a_k - \alpha a_{k-1},$$

so that

$$a_k = \sum_{j=0}^k \alpha^{k-j} r_j, \quad a = Br, \quad B_{kj} = \alpha^{k-j} \mathbf{1}_{\{k \geq j\}}.$$

The prior is then factorised, but the Gaussian likelihood contains

$$r^\top B^\top B r,$$

with

$$(B^\top B)_{ij} = \alpha^{|i-j|} \frac{1 - \alpha^{2(L-\max\{i,j\})}}{1 - \alpha^2},$$

which is dense whenever $\alpha \neq 0$. The dependence has therefore only moved from the prior to the likelihood. Exact BP remains valid, but its messages must be represented numerically; the next section describes that representation.

7.4 Grid belief propagation: the discretised recursion

7.4.1 The representation and the update

A message is a density on $\mathbb{R}$. We represent it by its values on N_g equispaced points of a truncated interval,

$$g_i = -A + i h, \quad h = \frac{2A}{N_g - 1}, \quad i = 0, \dots, N_g - 1, \quad (7.9)$$

so that each message is a vector in $\mathbb{R}^{N_g}$ and every integral in (6.2)–(6.3) becomes a composite trapezoidal quadrature with weights

$$w_i = \begin{cases} h/2, & i \in \{0, N_g - 1\}, \\ h, & \text{otherwise,} \end{cases} \quad (7.10)$$

the halved endpoints being what distinguishes the trapezoidal from the rectangle rule (numerically immaterial here, since the truncation level of $A = 8$ keeps the endpoint values below 10^{-9} , but the implementation carries them and the equations should say what the implementation does). Two arrays carry the model. The transition enters as a matrix, the likelihood as a diagonal:

$$K_{ij} = M(g_i | g_j) = \frac{1}{2b} e^{-|g_i - \alpha g_j|/b}, \quad (\ell_k)_i = \mathcal{N}(x_k; \mu g_i, \Delta_t). \quad (7.11)$$

The forward and backward sweeps are then the trapezoidal quadratures of Proposition 6.1, with $\odot$ the elementwise product and $w = (w_i)$ the weight vector of (7.10):

$$f_0 \propto p_0 \odot \ell_0, \quad f_k \propto \ell_k \odot K(w \odot f_{k-1}), \quad k = 1, \dots, L-1, \quad (7.12)$$

$$\beta_{L-1} = \mathbf{1}, \quad \beta_k \propto K^\top(w \odot \ell_{k+1} \odot \beta_{k+1}), \quad k = L-2, \dots, 0. \quad (7.13)$$

The belief, posterior mean and score at frame k follow from (7.2), with the same weights in the moments:

$$\text{bel}_k = f_k \odot \beta_k, \quad \widehat{m}_k = \frac{\sum_i w_i g_i (\text{bel}_k)_i}{\sum_i w_i (\text{bel}_k)_i}, \quad \widehat{S}_k = \frac{\mu \widehat{m}_k - x_k}{\Delta_t}. \quad (7.14)$$

Each sweep step is one matrix–vector product, so the solver costs $O(LN_g^2)$ time and $O(LN_g)$ space: the cost of sum–product on a chain over a finite alphabet, with the alphabet size replaced by the grid resolution N_g . Nothing about the algorithm has changed; only the alphabet is now a discretisation rather than a given.

7.4.2 The three choices, and what fixes them

Renormalisation. The proportionalities in (7.12)–(7.13) are enforced after every step, the forward messages to unit discrete mass and the backward messages to unit maximum. This is free — sum–product messages are defined only up to a positive constant (Theorem 3.4) and the ratio (7.14) is invariant — and it is necessary: without it the message norms drift by a factor per frame and a chain of $L = 32$ underflows.

Truncation A . The messages are beliefs about a chain of unit marginal variance, so mass beyond $|a| = A$ is bounded by the tail the innovations can produce. We take $A = 8$, at which the exponential tail contributes below 10^{-9} of the mass for every α used here; the diagnostic is the mass in the outermost grid cells, logged per run.

Resolution N_g . The binding constraint is not the smoothness of the messages but the width of the likelihood. In a -coordinates ℓ_k has standard deviation $\sqrt{\Delta_t}/\mu$, which for

small t is $\sqrt{2t} + O(t^{3/2})$; if that falls below the grid spacing the likelihood is a spike between two nodes and the quadrature ceases to see it. We therefore require

$$\sqrt{2t} \gtrsim 3h = \frac{6A}{N_g - 1}, \quad (7.15)$$

implemented in code as `src/noising.py`'s `likelihood_resolution_ok` and recorded per cell by the grid validation of `exp_01`, which writes it as a column rather than refusing to run: the working configuration clears it with room to spare ($3h = 0.12$ against $\sqrt{2t} = 0.32$ at the smallest reported t), so it documents the margin rather than gating on it. It is (7.15), not the innovation law, that sets the smallest diffusion time this chapter reports.

7.4.3 Validation

Grid BP is a numerical approximation of the functional recursion and can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain, where the closed form of Chapter 5 gives the truth. For smooth, rapidly decaying integrands — the Gaussian validation case — the trapezoidal rule converges superalgebraically, so failure is abrupt rather than gradual, and the measurement shows exactly that: at $N_g = 101$, $A = 8$ the relative score error sits at the double-precision floor down to $t \approx 0.03$, has grown to 10^{-10} at $t = 0.02$ and 10^{-6} at $t = 0.012$, and reaches the percent scale by $t = 0.005$. The gate (7.15) is deliberately conservative relative to that onset — at $N_g = 101$ it already demands $t \gtrsim 0.115$ — so every configuration this thesis reports passes it with a wide margin rather than sitting at the failure edge. (For the Laplace kernel the integrand has a kink along $a' = \alpha a$ and no spectral statement is invoked; the non-Gaussian reference is certified by self-convergence instead.) The working configuration for every non-Gaussian experiment is $N_g = 401$, $A = 8$ — gate satisfied for all $t \geq 0.05$ with room to spare — with a reference solution at $N_g = 801$; the self-convergence error between the two remains 10^4 to 10^7 times smaller than every effect measured below (Figure 7.3, left).

Self-convergence answers whether the mesh is fine enough and cannot answer whether the domain is wide enough, since both resolutions share the same truncation. That ques-

tion is settled separately, by the half-width sweep of `exp_18` (144 cells over $A \in \{4, 6, 8, 10\}$ and three resolutions), which records the posterior mass reaching the outermost cells of the domain: 4.4×10^{-3} at $A = 4$, 4.2×10^{-4} at $A = 6$, 3.9×10^{-6} at the working $A = 8$, and 3.2×10^{-8} at $A = 10$. At the setting used here the truncated tails carry parts in 10^6 of the posterior, two orders below the smallest closure error reported in this chapter, so the domain is not a binding approximation. Within this regime — resolution gated, truncation measured — we treat grid BP as the reference score; outside it no claim is made.

Finally, one identity governs how all of this is reported. Every score error is an amplified posterior-mean error: for any two solvers with means $\widehat{m}$ and m_{ref} , (7.2) gives, exactly,

$$\widehat{s} - s_{\text{ref}} = \frac{e^{-t}}{\Delta_t} (\widehat{m} - m_{\text{ref}}), \quad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0). \quad (7.16)$$

Its numerical residual is logged in every experiment and stays between 10^{-15} and 10^{-12} . Equation (7.16) says in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

7.5 The Gaussian-message approximation, measured

We now test *Gaussian-projected BP*: the same forward–backward recursion as grid BP, but with every transition message projected onto the Gaussian family by moment matching. Write $\Pi_G f$ for the Gaussian density with the same mean and variance as $f/\int f$.

Theorem 7.3 (Gaussian closure is the LMMSE estimator). *Consider moment-projected sum-product on the chain (7.1), with $\alpha \neq 0$, Gaussian unary likelihoods, $\mathbb{E}[a_0] = 0$, innovation mean zero and variance σ_η^2 , and a Gaussian initial message, or one projected by Π_G . Then the posterior means coincide exactly with those of the covariance-matched Gaussian model:*

$$\widehat{a}_{\text{LMMSE}} = \text{Cov}(a, x) \text{Cov}(x)^{-1} x = e^{-t} \Sigma_0 (e^{-2t} \Sigma_0 + \Delta_t I)^{-1} x. \quad (7.17)$$

Hence the closure returns the same estimator for every innovation law sharing the moments $(0, \sigma_\eta^2)$.

7.5.1 Proof of Theorem 7.3

Proof. We show that every projected message is identical to the corresponding exact message of the covariance-matched Gaussian chain.

Let $\vec{m}_i$ denote the projected forward message at site i and ℓ_i its Gaussian unary likelihood.

Base case. The initial message is Gaussian, either directly or after projection, with the same first two moments as the Gaussian reference chain. The two initial messages therefore coincide.

Forward recursion. Assume the incoming message is Gaussian,

$$\vec{m}_{i-1}(a_{i-1}) = \mathcal{N}(a_{i-1}; m, v).$$

Multiplication by the Gaussian likelihood ℓ_{i-1} remains Gaussian, with the same precision-weighted update in both models. Propagation through

$$a_i = \alpha a_{i-1} + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma_\eta^2,$$

maps its moments as

$$(m, v) \longmapsto (\alpha m, \alpha^2 v + \sigma_\eta^2). \quad (7.18)$$

Crucially, this transformation depends on the innovation distribution only through $(0, \sigma_\eta^2)$. After applying Π_G , the resulting Gaussian is therefore exactly the transition message of the covariance-matched Gaussian chain.

Backward recursion. The same argument applies in reverse. If the incoming Gaussian factor has moments (m, v) , the backward transition integral, viewed as a density in the

preceding state, has moments

$$(m, v) \mapsto \left(\frac{m}{\alpha}, \frac{v + \sigma_\eta^2}{\alpha^2} \right), \quad (7.19)$$

again depending only on the innovation variance. The condition $\alpha \neq 0$ is used here. Since the terminal message $\overleftarrow{m}_{L-1} \equiv 1$ is not normalisable, the induction begins at the terminal Gaussian likelihood ℓ_{L-1} , which is the first object actually propagated and projected.

Thus all projected forward and backward messages, and hence all posterior beliefs, coincide with those of the covariance-matched Gaussian chain.

For jointly Gaussian zero-mean (a, x) ,

$$\mathbb{E}[a \mid x] = \text{Cov}(a, x) \text{Cov}(x)^{-1} x.$$

Under

$$x = e^{-t} a + \sqrt{\Delta_t} z, \quad z \perp a,$$

we have

$$\text{Cov}(a, x) = e^{-t} \Sigma_0, \quad \text{Cov}(x) = e^{-2t} \Sigma_0 + \Delta_t I. \quad (7.20)$$

Substitution gives (7.17). Since neither (7.18) nor (7.19) uses any property of the innovation law beyond $(0, \sigma_\eta^2)$, the same estimator is obtained for every innovation distribution with those moments. $\square$

Theorem 7.3 gives a precise interpretation of the approximation: Gaussian-projected BP is not merely a heuristic Gaussianisation of the messages; it computes the *best second-order linear estimator*. One asymmetry in the statement deserves emphasis: the *estimator* is the same for every innovation law with matched first two moments, but its *error* against the true posterior mean is not — each law has its own nonlinear posterior, and the sweep of Section 7.5.4 measures exactly how differently the same estimator misses different truths. Comparing it with grid BP therefore measures the loss from discarding non-Gaussian structure beyond the first two moments, law by law.

7.5.2 Three metrics, and which one is primary

The obvious quantity to report is the relative score error, and it is what the diffusion model ultimately consumes. But by (7.16) it is the posterior-mean error multiplied by e^{-t}/Δ_t and divided by the norm of the reference score, and both of those move by orders of magnitude across the schedule. A score error that falls with t therefore cannot, on its own, distinguish an approximation that is becoming more accurate from one whose errors are simply being scaled away. The two effects have to be separated, and the posterior mean is where they separate.

The primary metric. The exact posterior mean $m(x) = \mathbb{E}[a \mid x]$ is the MMSE estimator, so the distance to it has a risk interpretation that no other choice has.

Proposition 7.4 (Excess-risk decomposition). *Let $m(x) = \mathbb{E}[a \mid x]$ and let $\hat{a}(x)$ be any estimator with finite second moment. Then*

$$\underbrace{\mathbb{E}\|a - \hat{a}(x)\|^2}_{R(\hat{a})} = \underbrace{\mathbb{E}\|a - m(x)\|^2}_{R_{\text{MMSE}}} + \mathbb{E}\|m(x) - \hat{a}(x)\|^2, \quad (7.21)$$

so the squared distance between the two posterior means is exactly the excess denoising risk incurred by using $\hat{a}$ in place of the exact posterior mean.

Proof. Write $a - \hat{a} = (a - m) + (m - \hat{a})$ and expand. The cross term is $2\mathbb{E}[(a - m)^\top (m - \hat{a})]$; conditioning on x and using $\mathbb{E}[a - m(x) \mid x] = 0$, together with the fact that both $m(x)$ and $\hat{a}(x)$ are functions of x alone, kills it. $\square$

Applied with $\hat{a} = \hat{a}_{\text{LMMSE}}$, which Theorem 7.3 identifies as what the closure returns, (7.21) says that the gap between the two posterior means is not an arbitrary distance but the price in mean squared error of denoising with second moments only. We report it normalised by the signal energy, which for this chain is $\mathbb{E}\|a\|^2 = L$:

$$\varepsilon_{\text{PM}} = \sqrt{\frac{1}{L} \mathbb{E}\|\hat{a}_{\text{LMMSE}} - \mathbb{E}[a \mid x]\|^2}, \quad (7.22)$$

a dimensionless number, free of the prefactor, and stable at both ends of the schedule. Normalising by $\mathbb{E}\|m(x)\|^2$ instead would be equally natural at small t but degenerates as $t \rightarrow \infty$, where the posterior mean itself vanishes.

The secondary metrics. Two more are reported alongside. The *excess risk fraction* $\mathbb{E}\|m - \hat{a}\|^2 / R_{\text{MMSE}}$ turns (7.21) into a statement a practitioner can act on — the percentage by which closure raises the denoising error above the floor set by the posterior itself — and, because the floor grows with t while the excess shrinks, it is the sharpest of the three. And the relative score error is retained, because it is what (7.16) converts the inference error into, and therefore what the generative process actually sees.

7.5.3 The measurement

Table 7.1: Gaussian closure against validated grid BP on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t , reference $N_g = 801$); all columns from one sweep, sampled densely for $t \lesssim 0.3$. The left group measures the posterior approximation directly, before the prefactor is applied: ε_{PM} (7.22) with its 10–90% band, the same normalised to $t = 0.02$, and the excess denoising MSE of Proposition 7.4 as a percentage of the MMSE floor (also listed, since it grows while the excess shrinks). The last column is the resulting discrepancy in the diffusion score.

t	<i>posterior inference</i>					<i>score</i>
	ε_{PM}	10–90%	$\varepsilon_{\text{PM}}/\varepsilon_{\text{PM}}^{0.02}$	excess MSE [% floor]	MMSE floor	rel. score error
0.02	0.036	(0.029, 0.052)	1.00	3.9	0.032	0.38
0.032	0.040	(0.034, 0.057)	1.14	3.4	0.047	0.30
0.05	0.050	(0.036, 0.079)	1.42	3.9	0.066	0.27
0.072	0.053	(0.038, 0.082)	1.50	3.1	0.089	0.20
0.105	0.050	(0.037, 0.100)	1.41	2.1	0.114	0.14
0.153	0.043	(0.034, 0.085)	1.20	1.2	0.151	0.10
0.222	0.038	(0.033, 0.058)	1.06	0.74	0.193	0.065
0.322	0.033	(0.027, 0.059)	0.93	0.43	0.253	0.041
0.98	0.016	(0.012, 0.028)	0.45	0.045	0.584	7.3×10^{-3}
2.07	0.0018	(0.0013, 0.0038)	0.05	< 0.01	0.916	2.4×10^{-4}

Two readings that disagree. Read the last column and the closure improves steadily as noise is added: the relative score error falls by a factor of 3.6 between $t = 0.02$ and $t = 0.15$. Read ε_{PM} and it does not. The posterior-mean error *rises* by half over $t \in [0.02, 0.07]$, is still larger at $t = 0.15$ than at $t = 0.02$, and begins to decay only past $t \simeq 0.3$ (Figure 7.2, left and centre). By (7.16) the gap between the two readings is the prefactor $e^{-t}/\Delta_t \sim 1/2t$ and the shrinking norm of the reference score, neither of which reports on the quality of the inference. Relative score error measures the *consequence* of the approximation for the diffusion; ε_{PM} measures the inference itself, and only the latter is flat at small t .

As a denoiser. By Proposition 7.4 the squared posterior-mean discrepancy is exactly the excess MSE over the MMSE estimator. As a fraction of the MMSE floor it falls from 3.9% at $t = 0.05$ to 0.43% at $t = 0.32$ and below 0.05% once $t \gtrsim 1$, while the floor itself grows from 0.07 to 0.92 over the same range: the approximation error shrinks as the irreducible denoising uncertainty grows, and the ratio is the sharpest of the metrics for that reason.

Direction survives magnitude. The probability of a trial exceeding 10% relative score error is 1.00 through $t \simeq 0.07$, passes 0.55 at $t = 0.15$, and reaches zero by $t \simeq 0.7$. Direction is far more robust: the median cosine similarity to the reference score is 0.964 at $t = 0.05$, exceeds 0.99 by $t \simeq 0.11$, and is numerically one near $t = 1$ (Figure 7.3, right). Gaussian closure loses *how strongly* to move before it loses *where* to move.

Why the peak is where it is. The maximum in ε_{PM} separates two regimes. As $t \rightarrow 0$ each latent state is tightly pinned by its own near-noiseless observation, so inference is easy even for a non-Gaussian prior. At large t Gaussian noise suppresses the higher-order structure of the innovation law and the tilted posteriors approach Gaussianity. The largest error therefore sits between the limits, where the evidence is weak enough for the prior's shape to matter and strong enough that it has not yet been washed out. In the worst individual trials the reference posterior is strongly non-Gaussian — heavy-tailed and sharply kinked, though never bimodal: the negative log-posterior of the Laplace chain under Gaussian observations is strictly convex, so the posterior and its marginals are log-concave and unimodal at every $t > 0$ — whereas a single Gaussian retains only one mean and variance. (Genuinely bimodal posteriors do arise for the bimodal mixture innovations of Section 7.5.4, and they are that sweep's hardest case.) This is a qualitative mechanism for the aggregate curve, not a direct measurement of posterior non-Gaussianity.

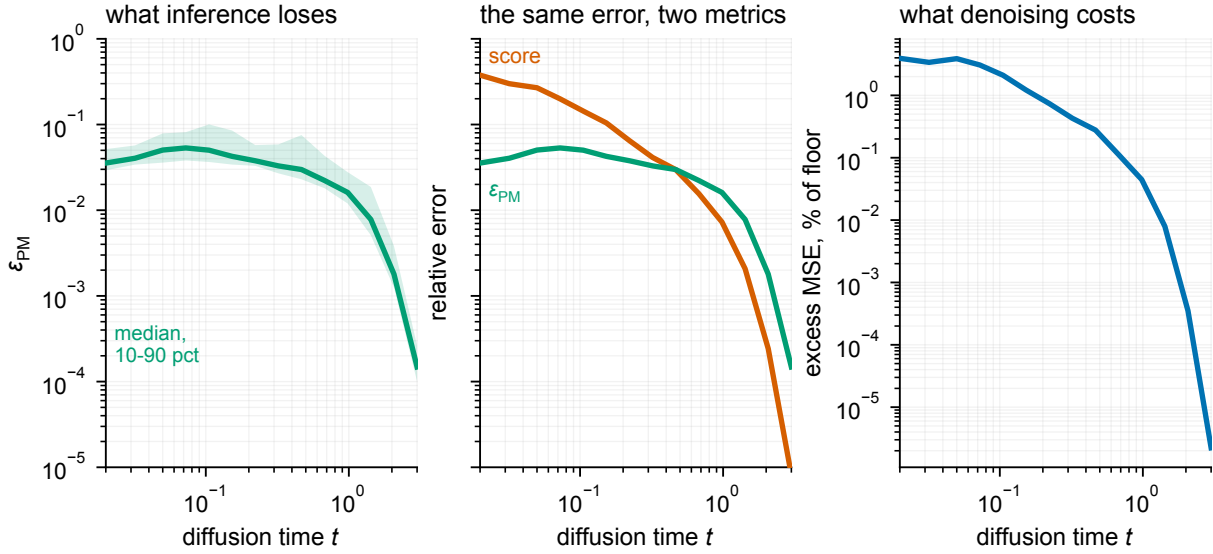


Figure 7.2: The same experiment under three metrics ($\alpha = 0.85$, $L = 32$, 60 trials per t , reference $N_g = 801$). *Left:* the posterior-mean error (7.22), median and 10–90% band. It is not monotone: it peaks at $t \simeq 0.07$ and is no smaller at $t = 0.02$ than at $t = 0.15$. *Centre:* the same runs seen through the relative score error, which falls by $3.6\times$ over exactly the interval on which ϵ_{PM} is flat. The gap between the curves is the prefactor of (7.16) together with the changing norm of the reference score; the two cross near $t \simeq 0.5$. *Right:* the excess denoising MSE of Proposition 7.4, as a percentage of the MMSE floor.

7.5.4 Varying the innovation law

Holding $\text{Cov}(a_i, a_j) = \alpha^{|i-j|}$ fixed, we vary the innovation distribution: Laplace, Student- t with $\nu \in \{5, 8\}$, and symmetric Gaussian mixtures of increasing separation. Across every family substantial closure error is confined to $t \lesssim 0.4$ and falls to about 10^{-3} or below by $t = 2.4$. At $t = 0.05$ and $\alpha = 0.5$:

innovation law	median rel. score error
strongly bimodal Gaussian mixture	0.94
Laplace	0.52
Student- t , $\nu = 5$	0.32
mildly bimodal Gaussian mixture	0.07

Strong bimodality is the hardest case: a single Gaussian cannot place mass on two separated posterior modes and sits between them. Stronger chain correlation helps — for Laplace innovations at $t = 0.05$ the median error falls from roughly one half at $\alpha = 0.5$

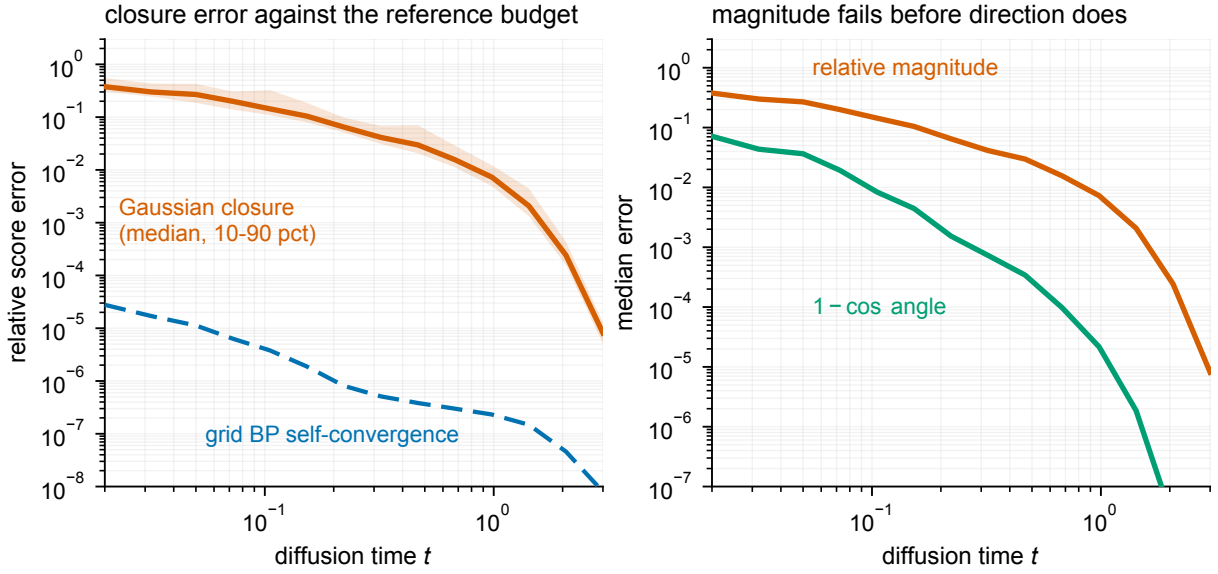


Figure 7.3: The score-side view of the same sweep. *Left:* median relative score error with its 10–90% band, against the grid-BP self-convergence error ($N_g = 401$ versus 801) on identical data. The closure error exceeds the numerical reference budget by at least three orders of magnitude everywhere and by four to five over most of the range, so the measurement is limited by the closure, not the quadrature. *Right:* relative score error against $1 - \cos$ of the angle between the projected and reference score vectors. Directional error is one to two orders of magnitude smaller than magnitude error at every diffusion time.

to 0.27 at $\alpha = 0.85$ and 0.13 at $\alpha = 0.95$ — so informative neighbours reduce the weight the local non-Gaussian shape carries, again an interpretation rather than a direct measurement of posterior Gaussianity.

Chapter 8

Learning the Parameters: EM on the Chain

Everything so far assumed the transition kernel was given. This chapter drops that assumption. The data are still a Markov chain observed through the Ornstein–Uhlenbeck channel, but the kernel that generated them is unknown and must be recovered from the noised sequences alone: no clean data, no supervision, and no differentiation through the inference.

The tool is expectation–maximisation, and the reason it is cheap here is Fisher’s identity. At each current parameter value, one complete forward–backward pass of belief propagation already contains the exact gradient of the marginal log-likelihood with respect to the parameters, so each E-step is the BP recursion of Chapter 6 with the kernel held fixed and the M-step moves the kernel; the full fit runs one such pass per iteration, and nothing is ever back-propagated through the recursion. Section 8.2 derives that claim in full, in the form actually implemented.

8.1 Model: an estimated mixture innovation

The clean chain is the one of (7.1), in the index convention of this chapter,

$$a_1 \sim \mathcal{N}(0, 1), \quad a_i = \alpha a_{i-1} + \varepsilon_i, \quad i = 2, \dots, L, \quad (8.1)$$

with innovations ε_i i.i.d., mean zero and variance $\sigma_\eta^2 = 1 - \alpha^2$: the same Gaussian initial state and the same variance-matching Chapter 7 used, but now drawn from a member of a *parametric family* rather than a fixed law. Forward noising is the coordinatewise Ornstein–Uhlenbeck channel of Section 3.2, $x = e^{-t}a + \sqrt{\Delta_t}\xi$, unchanged from every earlier chapter.

The estimated family is a C -component Gaussian mixture on the innovation. It is the transition kernel written $K(a' | a)$ in Chapters 3 and 7, carrying its parameters explicitly because they are now what is being estimated:

$$K_\theta(a' | a) = \sum_{c=1}^C \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad \theta = \{\alpha, (\pi_c, \nu_c, s_c)_{c=1}^C\}, \quad (8.2)$$

so both the autoregressive coefficient α and the shape of the innovation density are free, estimated jointly by maximum marginal likelihood from noised data alone. The parameter space is

$$\Theta = \left\{ \pi_c \geq 0, \sum_c \pi_c = 1, s_c^2 \geq s_{\min}^2, \nu_c \in \mathbb{R}, \alpha \in (-1, 1) \right\}, \quad (8.3)$$

the variance floor $s_{\min}^2$ being the guard of Section 8.3 against a component collapsing onto a single grid point, and $|\alpha| < 1$ being what gives the recursion a stationary law at all. It buys existence and no more: with (π_c, ν_c, s_c) free the fitted chain’s stationary variance is $\sum_c \pi_c (\nu_c^2 + s_c^2) / (1 - \alpha^2)$, which equals one only under a variance-matching constraint this family deliberately does not impose, and its stationary mean is zero only if $\sum_c \pi_c \nu_c = 0$, which it likewise does not impose — the asymmetry recorded below. The conditional maximiser for α derived below is a closed form that does not enforce its own constraint, so the boundary is a real possibility rather than an excluded one; in every fit reported in this thesis the estimate stays far inside the interval, near the generating value, and the

constraint never binds. Component labels are exchangeable throughout, so Θ is identified only up to permutation and every label-dependent diagnostic is reported under a fixed ordering convention (Section 8.4). At $C = 1$ this collapses to a single Gaussian, recovering the model of Chapter 5; larger C can represent skew, heavy tails and multimodality to the resolution of the grid.

One deliberate asymmetry between the generating and the fitted model should be on the record here rather than discovered later. The *generating* innovations are centred with variance $1 - \alpha^2$; the *fitted* family (8.2) enforces neither constraint — no $\sum_c \pi_c \nu_c = 0$ and no matched total variance. That is a choice, not an oversight: projecting the means back to zero after each update does not maximise the surrogate and measurably broke monotone ascent when tried (by $\sim 10^{-2}$ nats per edge), whereas the unconstrained ECM is exactly monotone, and on centred data the fitted total innovation mean lands at the 10^{-3} level on its own. The moment conditions are therefore treated as *diagnostics* of the fit, reported rather than imposed, and the free-parameter count of the family is $(C-1)+C+C+1 = 3C$.

8.2 The EM algorithm, derived

The parameters of the transition kernel are not observed directly. What we observe is the noised sequence

$$x = (x_1, \dots, x_L),$$

while the clean trajectory

$$a = (a_1, \dots, a_L)$$

that generated it is latent. In the Gaussian-mixture model there is a second source of missing information: for every transition $a_{i-1} \rightarrow a_i$, we do not observe which mixture component generated the innovation. Denote by $c_i \in \{1, \dots, C\}$ the label of the transition into a_i and write

$$c = (c_2, \dots, c_L),$$

one label per transition, so $L - 1$ of them rather than L : the first state is drawn from the initial law and no innovation generates it.

The statistical goal is still ordinary maximum likelihood:

$$\hat{\theta} \in \operatorname{argmax}_{\theta} \mathcal{L}(\theta), \quad \mathcal{L}(\theta) := \log p_{\theta}(x). \quad (8.4)$$

The difficulty is that $p_{\theta}(x)$ is obtained only after removing all the unobserved variables,

$$p_{\theta}(x) = \sum_c \int_{\mathbb{R}^L} p_{\theta}(a, c, x) \, da, \quad (8.5)$$

and hence

$$\mathcal{L}(\theta) = \log \left[\sum_c \int_{\mathbb{R}^L} p_{\theta}(a, c, x) \, da \right]. \quad (8.6)$$

The logarithm therefore sits *outside* an integral over the whole clean trajectory and a sum over all mixture-label configurations. Directly optimising this expression is substantially harder than fitting the model if a and c were known.

Expectation–maximisation (EM) exploits exactly this distinction. At iteration k , it alternates between two questions:

E-step. Given the current parameters $\theta^{(k)}$ and the observation x , which hidden trajectories and mixture labels are plausible?

M-step. If those possibilities are weighted according to their current posterior probabilities, which new parameters fit them best?

Thus EM does not replace the hidden variables by a single reconstruction. It averages over their entire posterior uncertainty.

8.2.1 Complete data and the surrogate

To formalise this idea, collect all latent variables into

$$z = (a, c) \quad (8.7)$$

and define the posterior under the current parameter value,

$$q^{(k)}(a, c) := p_{\theta^{(k)}}(a, c \mid x). \quad (8.8)$$

The distribution $q^{(k)}$ is computed in the E-step and is then held fixed while the parameters are changed in the M-step.

The central EM quantity is

$$Q(\theta \mid \theta^{(k)}) := \mathbb{E}_{q^{(k)}}[\log p_{\theta}(a, c, x)]. \quad (8.9)$$

This is called the EM *surrogate*. Its meaning is: imagine that (a, c) were observed. Then $\log p_{\theta}(a, c, x)$ would be the ordinary complete-data log-likelihood. Since (a, c) are not observed, EM averages that quantity over all their possible values, weighted by how plausible those values are under the current model.

The connection between this surrogate and the true likelihood can be derived exactly. Bayes' rule gives

$$p_{\theta}(a, c \mid x) = \frac{p_{\theta}(a, c, x)}{p_{\theta}(x)}, \quad (8.10)$$

and therefore

$$\log p_{\theta}(x) = \log p_{\theta}(a, c, x) - \log p_{\theta}(a, c \mid x). \quad (8.11)$$

Since the left-hand side does not depend on (a, c) , taking expectation with respect to the fixed distribution $q^{(k)}$ leaves it unchanged:

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_{q^{(k)}}[\log p_{\theta}(a, c, x)] - \mathbb{E}_{q^{(k)}}[\log p_{\theta}(a, c \mid x)] \\ &= Q(\theta \mid \theta^{(k)}) - \mathbb{E}_{q^{(k)}}[\log p_{\theta}(a, c \mid x)]. \end{aligned} \quad (8.12)$$

Now add and subtract $\mathbb{E}_{q^{(k)}}[\log q^{(k)}(a, c)]$. Using

$$\mathcal{H}(q) := -\mathbb{E}_q[\log q] \quad (8.13)$$

and

$$\text{KL}(q \parallel p) := \mathbb{E}_q \left[\log \frac{q}{p} \right], \quad (8.14)$$

we obtain

$$\boxed{\mathcal{L}(\theta) = Q(\theta \mid \theta^{(k)}) + \mathcal{H}(q^{(k)}) + \text{KL}(q^{(k)} \parallel p_{\theta}(a, c \mid x))}. \quad (8.15)$$

Equivalently,

$$\underbrace{Q(\theta \mid \theta^{(k)}) + \mathcal{H}(q^{(k)})}_{\mathcal{F}(\theta; q^{(k)})} = \mathcal{L}(\theta) - \text{KL}(q^{(k)} \parallel p_{\theta}(a, c \mid x)) \leq \mathcal{L}(\theta). \quad (8.16)$$

Hence $\mathcal{F}(\theta; q^{(k)})$ is a lower bound on the true observed-data log-likelihood. During the M-step $q^{(k)}$ is fixed, so its entropy is a constant with respect to θ . Maximising the lower bound $\mathcal{F}$ is therefore exactly equivalent to maximising Q .

This explains the word *surrogate*: rather than attacking $\mathcal{L}(\theta)$ directly, EM optimises the easier function $Q(\theta \mid \theta^{(k)})$ while keeping the current posterior fixed.

There is one further crucial property. At the current parameter $\theta = \theta^{(k)}$,

$$q^{(k)}(a, c) = p_{\theta^{(k)}}(a, c \mid x), \quad (8.17)$$

so

$$\text{KL}(q^{(k)} \parallel p_{\theta^{(k)}}(a, c \mid x)) = 0. \quad (8.18)$$

Consequently the lower bound touches the true likelihood at the current iterate,

$$\mathcal{F}(\theta^{(k)}; q^{(k)}) = \mathcal{L}(\theta^{(k)}). \quad (8.19)$$

The EM iteration can therefore be pictured as

$$\theta^{(k)} \longrightarrow q^{(k)}(a, c) \longrightarrow Q(\theta \mid \theta^{(k)}) \longrightarrow \theta^{(k+1)}.$$

The first arrow infers what the hidden clean process probably was; the second turns that

posterior into a weighted complete-data likelihood; the final arrow fits a new transition kernel to those posterior-weighted hidden trajectories.

In a generic latent-variable model, computing $q^{(k)}$ may itself require an approximation. Here the chain structure makes this step particularly simple.

Proposition 8.1 (The E-step is exact). *Since the posterior factor graph is a tree, one forward and one backward sum-product sweep return the exact single-site and pairwise posterior marginals, at the level of functional messages. Consequently the E-step requires neither a variational approximation nor Monte Carlo sampling: all conditional expectations needed by EM are exact under the model, before the numerical grid discretisation is introduced.*

We next write the complete-data likelihood explicitly. Introduce the mixture label c_i on the transition $a_{i-1} \rightarrow a_i$. Conditional on $c_i = c$, the mixture transition becomes a single Gaussian,

$$\widetilde{K}_\theta(a', c | a) = \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad K_\theta(a' | a) = \sum_{c=1}^C \widetilde{K}_\theta(a', c | a). \quad (8.20)$$

The purpose of introducing c is now visible. Without the label, one transition contributes

$$\log K_\theta(a_i | a_{i-1}) = \log \left[\sum_{c=1}^C \pi_c \mathcal{N}(a_i - \alpha a_{i-1}; \nu_c, s_c^2) \right], \quad (8.21)$$

so the logarithm couples all mixture components through a log-sum. Once c_i is exposed, the same transition is assigned to a single Gaussian component, giving a weighted Gaussian likelihood with closed-form conditional maximisers.

For one sequence, the complete joint distribution factorises as

$$p_\theta(a, c, x) = p(a_1) \prod_{i=2}^L \widetilde{K}_\theta(a_i, c_i | a_{i-1}) \prod_{i=1}^L \phi_i(a_i), \quad (8.22)$$

where

$$\phi_i(a_i) = p(x_i | a_i)$$

is the fixed OU observation factor. Taking logarithms,

$$\log p_\theta(a, c, x) = \underbrace{\log p(a_1)}_{\text{independent of } \theta} + \sum_{i=2}^L \log \widetilde{K}_\theta(a_i, c_i \mid a_{i-1}) + \underbrace{\sum_{i=1}^L \log \phi_i(a_i)}_{\text{independent of } \theta}. \quad (8.23)$$

Thus, under the parameterisation used here, all terms relevant to the M-step are local transition terms.

For one labelled transition,

$$\begin{aligned} \log \widetilde{K}_\theta(a_i, c_i \mid a_{i-1}) &= \log \pi_{c_i} + \log \mathcal{N}(a_i - \alpha a_{i-1}; \nu_{c_i}, s_{c_i}^2) \\ &= \log \pi_{c_i} - \frac{1}{2} \log(2\pi) - \frac{1}{2} \log s_{c_i}^2 - \frac{(a_i - \alpha a_{i-1} - \nu_{c_i})^2}{2s_{c_i}^2}. \end{aligned} \quad (8.24)$$

Using indicators,

$$\log \widetilde{K}_\theta(a_i, c_i \mid a_{i-1}) = \sum_{c=1}^C \mathbf{1}\{c_i = c\} \left[\log \pi_c - \frac{1}{2} \log(2\pi) - \frac{1}{2} \log s_c^2 - \frac{(a_i - \alpha a_{i-1} - \nu_c)^2}{2s_c^2} \right]. \quad (8.25)$$

Substituting this expression into (8.9) gives

$$\begin{aligned} Q(\theta \mid \theta^{(k)}) &= \mathbb{E}_{a,c|x,\theta^{(k)}} [\log p_\theta(a, c, x)] \\ &\stackrel{c}{=} \sum_{i=2}^L \sum_{c=1}^C \mathbb{E}_{\theta^{(k)}} \left[\mathbf{1}\{c_i = c\} \left(\log \pi_c - \frac{1}{2} \log s_c^2 - \frac{(a_i - \alpha a_{i-1} - \nu_c)^2}{2s_c^2} \right) \right], \end{aligned} \quad (8.26)$$

where $\stackrel{c}{=}$ means equality up to an additive constant independent of θ . In particular, the initial-state term, the observation-channel term and $-\frac{1}{2} \log(2\pi)$ do not affect the M-step.

Equation (8.26) is the central simplification: the difficult logarithm of the marginalisation in (8.6) has been replaced by the expectation of a sum of local Gaussian transition terms. The remaining question is how those expectations are computed.

8.2.2 Fisher's identity: why BP is never differentiated

The same construction also explains why parameter learning does not require differentiating through the BP recursion. Starting from

$$\mathcal{L}(\theta) = \log p_\theta(x), \quad (8.27)$$

differentiate directly. The interchange of ∇_θ with the sum and the integral below is the one regularity condition the identity needs: it holds because $p_\theta(a, c, x)$ is differentiable in θ with $\|\nabla_\theta p_\theta(a, c, x)\|$ dominated, locally uniformly in θ , by an integrable function of (a, c) — which the Gaussian likelihood factors and the mixture kernel of (8.2) satisfy on a compact subset of Θ containing every fit reported here — every component non-degenerate and the fitted α staying well inside $(-1, 1)$. Then

$$\begin{aligned} \nabla_\theta \mathcal{L}(\theta) &= \frac{1}{p_\theta(x)} \nabla_\theta p_\theta(x) \\ &= \frac{1}{p_\theta(x)} \sum_c \int \nabla_\theta p_\theta(a, c, x) \, da. \end{aligned} \quad (8.28)$$

Using

$$\nabla_\theta p_\theta(a, c, x) = p_\theta(a, c, x) \nabla_\theta \log p_\theta(a, c, x), \quad (8.29)$$

we obtain

$$\begin{aligned} \nabla_\theta \mathcal{L}(\theta) &= \sum_c \int \frac{p_\theta(a, c, x)}{p_\theta(x)} \nabla_\theta \log p_\theta(a, c, x) \, da \\ &= \sum_c \int p_\theta(a, c \mid x) \nabla_\theta \log p_\theta(a, c, x) \, da \\ &= \mathbb{E}_{a,c \mid x, \theta} [\nabla_\theta \log p_\theta(a, c, x)]. \end{aligned} \quad (8.30)$$

This is Fisher's identity.

Since only the transition factors in (8.23) depend on θ ,

$$\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}_{a,c \mid x, \theta} \left[\sum_{i=2}^L \nabla_\theta \log \widetilde{K}_\theta(a_i, c_i \mid a_{i-1}) \right]. \quad (8.31)$$

Evaluating at the current iterate gives

$$\nabla_{\theta} \mathcal{L}(\theta) \Big|_{\theta=\theta^{(k)}} = \nabla_{\theta} Q(\theta \mid \theta^{(k)}) \Big|_{\theta=\theta^{(k)}}. \quad (8.32)$$

The important point is what is *not* differentiated. BP is used only to compute

$$q^{(k)}(a, c) = p_{\theta^{(k)}}(a, c \mid x).$$

Once that posterior has been computed, it is treated as fixed. The derivative acts on the candidate kernel parameters inside $\log \widetilde{K}_{\theta}$, not on the BP messages that produced the posterior. Thus BP supplies the posterior expectations and Fisher's identity converts them into the likelihood gradient without back-propagating through the inference algorithm.

8.2.3 The E-step: which expectations BP supplies

The complete-data objective in (8.26) is a sum over transitions. Consequently we do not need the full joint posterior over the entire clean trajectory explicitly. It is sufficient to know, for each edge $i-1 \rightarrow i$, the pairwise posterior

$$b_i(a_{i-1}, a_i) := p_{\theta^{(k)}}(a_{i-1}, a_i \mid x). \quad (8.33)$$

On a chain this marginal is obtained exactly from the forward and backward messages:

$$b_i(a_{i-1}, a_i) \propto \vec{m}_{i-1}(a_{i-1}) \phi_{i-1}(a_{i-1}) K_{\theta^{(k)}}(a_i \mid a_{i-1}) \phi_i(a_i) \overleftarrow{m}_i(a_i), \quad (8.34)$$

with normalisation

$$\iint b_i(a_{i-1}, a_i) da_{i-1} da_i = 1. \quad (8.35)$$

The mixture label requires no additional BP recursion. Once a pair (a_{i-1}, a_i) is given, Bayes' rule on the finite mixture gives the responsibility

$$r_{i,c}^{(k)}(a_{i-1}, a_i) := \mathbb{P}_{\theta^{(k)}}(c_i = c \mid a_{i-1}, a_i)$$

$$\begin{aligned}
&= \frac{\pi_c^{(k)} \mathcal{N}(a_i - \alpha^{(k)} a_{i-1}; \nu_c^{(k)}, (s_c^2)^{(k)})}{\sum_{d=1}^C \pi_d^{(k)} \mathcal{N}(a_i - \alpha^{(k)} a_{i-1}; \nu_d^{(k)}, (s_d^2)^{(k)})} \\
&= \frac{\pi_c^{(k)} \mathcal{N}(a_i - \alpha^{(k)} a_{i-1}; \nu_c^{(k)}, (s_c^2)^{(k)})}{K_{\theta^{(k)}}(a_i | a_{i-1})}.
\end{aligned} \tag{8.36}$$

For every pair,

$$\sum_{c=1}^C r_{i,c}^{(k)}(a_{i-1}, a_i) = 1. \tag{8.37}$$

Define expectation with respect to the pairwise posterior by

$$\langle f \rangle_i := \iint f(a_{i-1}, a_i) b_i(a_{i-1}, a_i) da_{i-1} da_i. \tag{8.38}$$

Because

$$\mathbb{E}[\mathbf{1}\{c_i = c\} | a_{i-1}, a_i, x, \theta^{(k)}] = r_{i,c}^{(k)}(a_{i-1}, a_i), \tag{8.39}$$

every expectation in (8.26) can be reduced to an integral against b_i weighted by $r_{i,c}^{(k)}$.

For each component define the six sufficient statistics

$$\begin{aligned}
R_c &= \sum_i \langle r_{i,c}^{(k)} \rangle_i, & U_c &= \sum_i \langle r_{i,c}^{(k)} a_i \rangle_i, & V_c &= \sum_i \langle r_{i,c}^{(k)} a_{i-1} \rangle_i, \\
P_c &= \sum_i \langle r_{i,c}^{(k)} a_{i-1} a_i \rangle_i, & W_c &= \sum_i \langle r_{i,c}^{(k)} a_{i-1}^2 \rangle_i, & G_c &= \sum_i \langle r_{i,c}^{(k)} a_i^2 \rangle_i.
\end{aligned} \tag{8.40}$$

For several training sequences the same quantities are simply summed over all edges of all sequences.

R_c is the posterior effective number of transitions assigned to component c ; U_c and V_c are responsibility-weighted first moments of the child and parent states; G_c and W_c are the corresponding second moments; and P_c is the parent-child cross moment.

A single forward-backward pass therefore reduces the entire E-step to these posterior-weighted sufficient statistics.

8.2.4 The M-step: deriving the conditional maximisers

We now substitute (8.40) into the surrogate and derive the parameter updates explicitly.

First expand the squared residual:

$$(a_i - \alpha a_{i-1} - \nu_c)^2 = a_i^2 - 2\alpha a_{i-1} a_i + \alpha^2 a_{i-1}^2 - 2\nu_c a_i + 2\alpha \nu_c a_{i-1} + \nu_c^2. \quad (8.41)$$

Taking the posterior-weighted sums gives

$$S_c(\alpha, \nu_c) := G_c - 2\alpha P_c + \alpha^2 W_c - 2\nu_c U_c + 2\alpha \nu_c V_c + \nu_c^2 R_c. \quad (8.42)$$

Hence, up to constants independent of θ ,

$$Q(\theta \mid \theta^{(k)}) \stackrel{c}{=} \sum_{c=1}^C \left[R_c \log \pi_c - \frac{R_c}{2} \log s_c^2 - \frac{S_c(\alpha, \nu_c)}{2s_c^2} \right]. \quad (8.43)$$

All quantities $R_c, U_c, V_c, P_c, W_c, G_c$ are fixed during the M-step: they were computed using $\theta^{(k)}$ in the E-step. Only the candidate parameters $\pi_c, \nu_c, s_c^2, \alpha$ are varied.

Mixture weights. The part of Q depending on the weights is

$$Q_\pi = \sum_{c=1}^C R_c \log \pi_c, \quad (8.44)$$

subject to

$$\sum_{c=1}^C \pi_c = 1. \quad (8.45)$$

Introduce a Lagrange multiplier λ :

$$\mathcal{J} = \sum_c R_c \log \pi_c + \lambda \left(\sum_c \pi_c - 1 \right). \quad (8.46)$$

Differentiating,

$$\frac{\partial \mathcal{J}}{\partial \pi_c} = \frac{R_c}{\pi_c} + \lambda = 0, \quad (8.47)$$

so

$$\pi_c = -\frac{R_c}{\lambda}. \quad (8.48)$$

Summing over c and imposing $\sum_c \pi_c = 1$ gives

$$\lambda = -\sum_{d=1}^C R_d. \quad (8.49)$$

Therefore

$$\pi_c^+ = \frac{R_c}{\sum_{d=1}^C R_d}. \quad (8.50)$$

Thus the new mixture weight is simply the fraction of the total posterior transition mass assigned to component c .

Component means. Holding α and s_c^2 fixed, only $S_c(\alpha, \nu_c)$ depends on ν_c . From (8.42),

$$\frac{\partial S_c}{\partial \nu_c} = -2U_c + 2\alpha V_c + 2\nu_c R_c. \quad (8.51)$$

Since

$$\frac{\partial Q}{\partial \nu_c} = -\frac{1}{2s_c^2} \frac{\partial S_c}{\partial \nu_c}, \quad (8.52)$$

the stationarity condition is

$$-2U_c + 2\alpha V_c + 2\nu_c R_c = 0. \quad (8.53)$$

Hence

$$\nu_c^+ = \frac{U_c - \alpha V_c}{R_c}. \quad (8.54)$$

Component variances. Let

$$\tau_c = s_c^2.$$

For fixed α and ν_c , the relevant part of the surrogate is

$$Q_c(\tau_c) = -\frac{R_c}{2} \log \tau_c - \frac{S_c(\alpha, \nu_c)}{2\tau_c}. \quad (8.55)$$

Differentiating,

$$\frac{\partial Q_c}{\partial \tau_c} = -\frac{R_c}{2\tau_c} + \frac{S_c(\alpha, \nu_c)}{2\tau_c^2}. \quad (8.56)$$

Setting this equal to zero gives

$$R_c \tau_c = S_c(\alpha, \nu_c), \quad (8.57)$$

and therefore

$$(s_c^2)^+ = \frac{S_c(\alpha, \nu_c^+)}{R_c}. \quad (8.58)$$

Substituting $\nu_c^+ = (U_c - \alpha V_c)/R_c$ simplifies the weighted residual to

$$S_c(\alpha, \nu_c^+) = G_c - 2\alpha P_c + \alpha^2 W_c - R_c(\nu_c^+)^2, \quad (8.59)$$

so

$$(s_c^2)^+ = \frac{G_c - 2\alpha P_c + \alpha^2 W_c - R_c(\nu_c^+)^2}{R_c}. \quad (8.60)$$

Combining the three mixture updates,

$$\pi_c^+ = \frac{R_c}{\sum_d R_d}, \quad \nu_c^+ = \frac{U_c - \alpha V_c}{R_c}, \quad (s_c^2)^+ = \frac{G_c - 2\alpha P_c + \alpha^2 W_c - R_c(\nu_c^+)^2}{R_c}. \quad (8.61)$$

Remark 8.2 (Floors, empty components, and label order). The updates (8.61) are undefined at $R_c = 0$ and the mixture likelihood is unbounded as any $s_c^2 \rightarrow 0$ with a component pinned on one residual, so the implementation carries two explicit guards: the responsibility mass is floored at 10^{-300} before any division (an empty component keeps its parameters rather than producing indeterminate ones), and the variance update is floored at $s_c^2 \geq 10^{-6}$, a component standard deviation of 10^{-3} — far below the grid spacing, so the floor marks degeneracy rather than shaping any resolved fit. A separate reporting gate (Appendix A) excludes fits whose narrowest component falls under two grid cells,

where the quadrature likelihood can reward lattice artefacts. Component labels are exchangeable, so all label-dependent diagnostics order components deterministically before comparison; the effective component count $\exp[\text{entropy}(\pi)]$ is logged per fit so that collapse onto fewer live components is visible rather than silent.

Autoregressive coefficient. Finally hold the mixture parameters fixed and optimise with respect to α . From (8.42),

$$\frac{\partial S_c}{\partial \alpha} = -2P_c + 2\alpha W_c + 2\nu_c V_c. \quad (8.62)$$

Therefore

$$\begin{aligned} \frac{\partial Q}{\partial \alpha} &= -\sum_{c=1}^C \frac{1}{2s_c^2} \frac{\partial S_c}{\partial \alpha} \\ &= \sum_{c=1}^C \frac{P_c - \alpha W_c - \nu_c V_c}{s_c^2}. \end{aligned} \quad (8.63)$$

Setting the derivative to zero gives

$$\sum_c s_c^{-2} (P_c - \nu_c V_c) = \alpha \sum_c s_c^{-2} W_c, \quad (8.64)$$

and hence

$$\alpha^+ = \frac{\sum_c s_c^{-2} (P_c - \nu_c V_c)}{\sum_c s_c^{-2} W_c}. \quad (8.65)$$

The factors s_c^{-2} are precisions: components with smaller innovation variance provide more information about the regression coefficient and therefore receive greater weight.

At $C = 1$ and $\nu_1 = 0$ this reduces to

$$\alpha^+ = \frac{P_1}{W_1} = \frac{\sum_i \mathbb{E}[a_{i-1} a_i \mid x]}{\sum_i \mathbb{E}[a_{i-1}^2 \mid x]}, \quad (8.66)$$

which is the posterior analogue of the ordinary least-squares slope of a_i on a_{i-1} .

The complete EM iteration can therefore be summarised as

$$\begin{array}{l} \text{E-step: } \theta^{(k)} \xrightarrow{\text{BP}} \{R_c, U_c, V_c, P_c, W_c, G_c\}_{c=1}^C, \\ \text{M-step: } \{R_c, U_c, V_c, P_c, W_c, G_c\}_{c=1}^C \xrightarrow{\text{closed-form updates}} \theta^{(k+1)}. \end{array} \quad (8.67)$$

8.2.5 Generalised EM and monotonicity

The updates above are conditional maximisers. The mixture block (π, ν, s^2) is maximised while α is fixed, and then α is maximised while the mixture block is fixed. Alternating these operations increases the same surrogate Q , but need not find its joint maximiser in one step. The implementation therefore belongs to the generalised EM family, more specifically to expectation–conditional maximisation (ECM) (Meng and Rubin, 1993).

The monotonicity guarantee does not require an exact joint maximisation of Q . To see this, use (8.15) with $q^{(k)} = p_{\theta^{(k)}}(a, c \mid x)$. At the current parameter the KL term vanishes, so

$$\begin{aligned} \mathcal{L}(\theta) - \mathcal{L}(\theta^{(k)}) &= Q(\theta \mid \theta^{(k)}) - Q(\theta^{(k)} \mid \theta^{(k)}) \\ &\quad + \text{KL}(p_{\theta^{(k)}}(a, c \mid x) \parallel p_{\theta}(a, c \mid x)). \end{aligned} \quad (8.68)$$

The second term is non-negative. Consequently,

$$Q(\theta^{(k+1)} \mid \theta^{(k)}) \geq Q(\theta^{(k)} \mid \theta^{(k)}) \quad (8.69)$$

is sufficient to imply

$$\boxed{\mathcal{L}(\theta^{(k+1)}) \geq \mathcal{L}(\theta^{(k)})}. \quad (8.70)$$

Thus each conditional maximisation need only increase the surrogate; joint maximisation is unnecessary for likelihood monotonicity.

In the implementation, the two parameter blocks are alternated until either the gain in Q per edge falls below the prescribed tolerance or a maximum of sixteen inner sweeps is reached. The returned kernel records which stopping condition occurred. One distinc-

tion keeps the guarantee honest. The derivation above concerns the continuum model; the implementation evaluates densities on the grid of Section 8.3 without renormalising transition columns, so the objective it computes — and the one whose monotone increase Table 8.2 certifies — is the quadrature-defined surrogate of that discretised factor model. It approximates the continuum likelihood to the quadrature error measured in Table 8.1, but the exact monotonicity statement applies to the grid objective, not automatically to the continuum one.

Remark 8.3 (The inner budget of sixteen). The cap of sixteen inner sweeps is chosen from the numerical behaviour rather than from a theoretical convention. At $C = 8$, $N = 256$ and a fixed outer budget of 40 iterations, increasing the inner budget from four to sixteen sweeps moves the fitted innovation excess kurtosis from 1.68 to 2.14, a 28% change relative to the former value and still below the true Laplace value 3.0, whereas $\hat{\alpha}$ changes only in the fourth decimal place. The innovation shape therefore converges substantially more slowly than the correlation parameter. By roughly 60 outer iterations the outer and inner loops largely compensate for one another and the inner budget has little remaining effect. The choice of sixteen is consequently a practical safeguard against interpreting incomplete inner optimisation as a failure to recover the innovation shape.

8.3 The numerical representation: grid BP, again

As in Section 7.4, messages are represented on N_g equispaced points of $[-A, A]$ and updated by trapezoidal quadrature. The pairwise belief (8.34) is therefore an $N_g \times N_g$ array, and the expectations in (8.40) become weighted sums. One EM iteration costs $O(LN_g^2)$ per sequence, with an additional factor C only in the mixture responsibilities.

There are two distinct numerical errors. *Truncation* comes from replacing $\mathbb{R}$ by $[-A, A]$ and is monitored through forward-message boundary mass; because this depends on the sampled trajectory, it is reported over 256 chains. *Quadrature* comes from the spacing $h = 2A/(N_g - 1)$: in the interior the trapezoidal error scales as $O(h^2)$, with the measured residual decreasing by a factor 4.00 per halving. Near the boundary it does not converge,

because the remaining error is truncation rather than resolution. Table 8.1 and Figure 8.1 separate the two effects.

Table 8.1: Numerical error diagnostics. *Upper:* boundary mass versus domain half-width at $N_g = 401$, reported over 256 chains. *Lower:* column-mass residual versus grid spacing at $A = 8$. Interior error follows the expected $O(h^2)$ rate, whereas the edge residual is truncation-dominated. The working choice is $N_g = 401$, $A = 8$.

<i>truncation: boundary mass against half-width ($N_g = 401$)</i>				
half-width A	4	6	8	10
max over chains	2.3×10^{-3}	1.6×10^{-4}	1.2×10^{-6}	3.2×10^{-8}
90th percentile	2.7×10^{-4}	2.5×10^{-6}	1.5×10^{-8}	7.8×10^{-11}
median	3.8×10^{-5}	3.7×10^{-7}	3.0×10^{-9}	1.8×10^{-11}
max / median	61×	434×	392×	1796×
<i>quadrature: column-mass residual against spacing ($A = 8$)</i>				
N_g / h	201 / 0.080	401 / 0.040	801 / 0.020	rate
interior column	3.84×10^{-3}	9.61×10^{-4}	2.40×10^{-4}	4.00× per halving
edge column	1.62×10^{-2}	1.90×10^{-2}	1.97×10^{-2}	none

Which finite-grid model is fitted. The transition array is the continuum density $K_\theta(a' | a)$ evaluated on the grid and integrated with trapezoidal weights, without normalising each parent-conditioned column. BP is therefore exact, up to floating point, for the resulting discretised factor model, but this is not exactly a normalised finite-state Markov kernel. The discrepancy is measured by the column-mass residual in Table 8.1.

Column-wise renormalisation would remove this discrepancy, but its normaliser Z_θ would depend on the parameters. Terms $\nabla_\theta \log Z_\theta$ would then enter Fisher’s identity and the M-step, destroying the closed-form updates (8.61)–(8.65). The present convention therefore fits the continuum kernel through the quadrature rule, with truncation controlled separately.

Remark 8.4 (The boundary diagnostic is empirical, not a bound). Boundary mass depends on the realised trajectory, so a single chain cannot certify truncation error. At the working

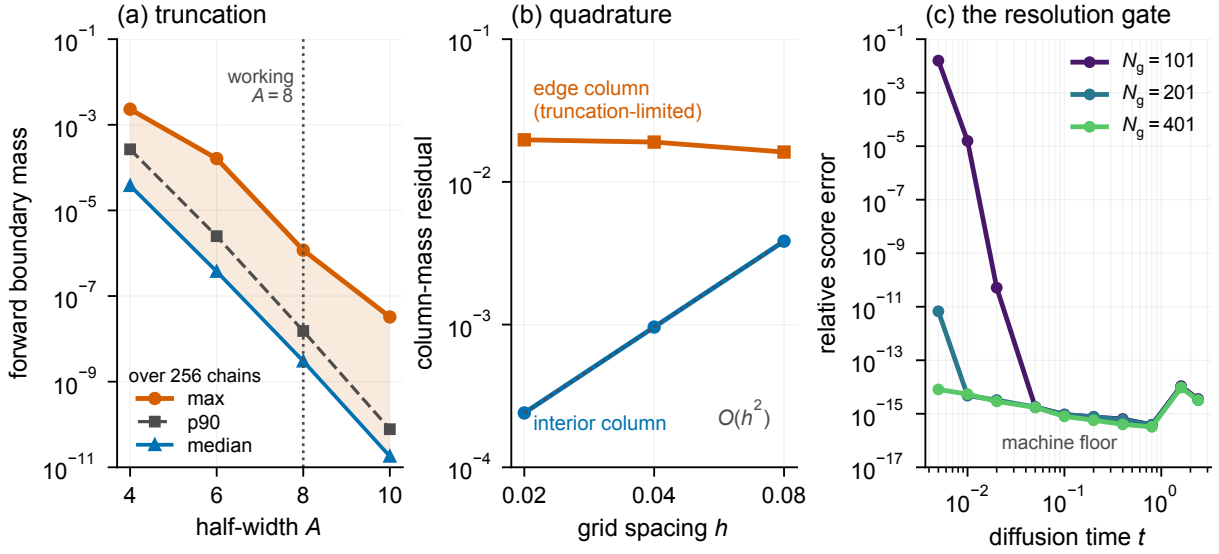


Figure 8.1: Numerical behaviour of the grid representation. (a) Boundary mass decreases rapidly with the domain half-width A , but varies substantially across sampled chains. (b) Interior column-mass error follows the $O(h^2)$ trapezoidal rate, whereas the edge error remains truncation-dominated. (c) Relative score error versus diffusion time for three resolutions; failure occurs when the resolution condition (7.15) is violated.

configuration, the batch maximum over 256 chains exceeds the single-chain reading by a median factor 86 and by as much as 3,281. This motivates reporting the distribution across chains rather than one value. The quadrature comparison, by contrast, depends only on (A, N_g) and reproduces to 0.000 relative difference under reseeding.

Finally, as in Section 7.4, posterior-mean errors are amplified in the score by

$$\frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t},$$

so numerical or estimation deficiencies are most visible at small diffusion times.

8.4 Diagnostics

The remaining checks answer two questions: whether the implementation is correct, and which parameters can actually be recovered from noised data.

8.4.1 Implementation and estimator checks

The BP recursion agrees with independent references to machine precision, and the surrogate never decreases in the logged runs.

Table 8.2: Correctness and monotonicity checks at $N_g = 401$, $A = 8$.

check	reference	discrepancy
posterior mean	closed-form Gaussian chain (Ch. 5)	9.2×10^{-15}
posterior mean	brute-force enumeration, no shared code	1.6×10^{-14}
log-evidence	brute-force enumeration, no shared code	1.8×10^{-15}
minimum increase of Q over twelve steps, Gaussian kernel		+2.61
the same, Laplace kernel		+4.34
the same, $C = 8$ mixture kernel		+4.18
decreases recorded in any logged run		0

Figure 8.2 adds two estimator-level checks. First, $\hat{\alpha}$ converges to the same value from initialisations spanning $[-0.4, 0.6]$, showing that it is genuinely estimated rather than supplied. Second, its RMSE and that of the innovation variance decay approximately at the $N^{-1/2}$ sampling rate.

8.4.2 What does EM recover?

The parameters are not recovered equally well. Table 8.3 shows a clear hierarchy: correlation is easiest, innovation variance is next, and higher-order shape is substantially harder.

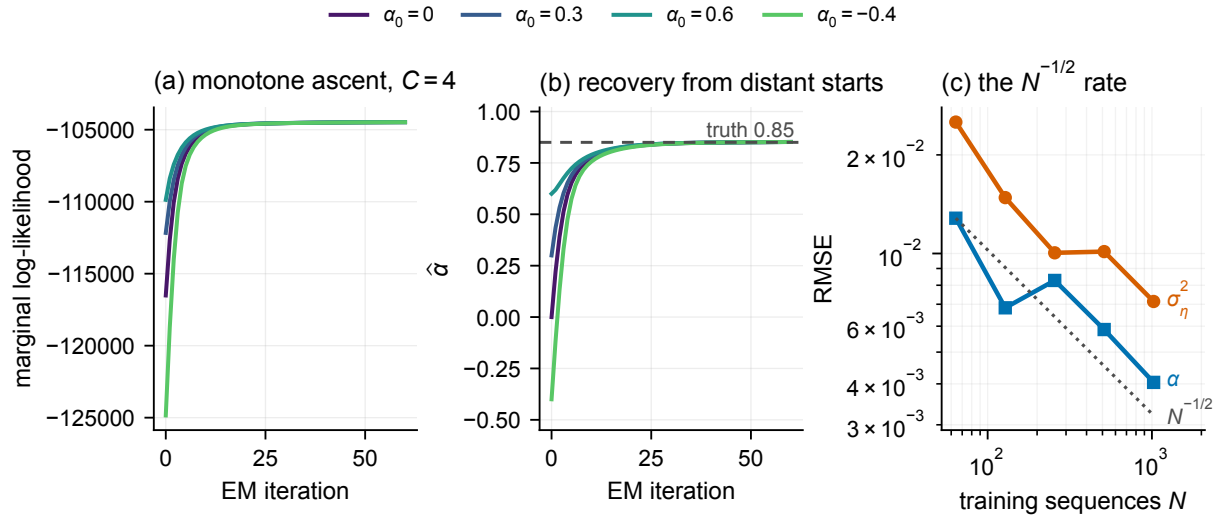


Figure 8.2: Optimisation diagnostics at $C = 4$. (a) Marginal log-likelihood increases monotonically from four initialisations of α . (b) $\hat{\alpha}$ converges to $[0.8517, 0.8520]$ from starts spanning $[-0.4, 0.6]$, against the true value 0.85. (c) RMSE of $\hat{\alpha}$ and the innovation variance versus N , over 16 seeds, with an $N^{-1/2}$ reference.

Table 8.3: Estimates after 61 EM iterations on noised Laplace chains ($\alpha = 0.85$, $\sigma_\eta^2 = 0.2775$, excess kurtosis 3.0, $L = 32$). No run recorded a monotonicity violation.

C	N	$\hat{\alpha}$	$\hat{\sigma}_\eta^2$	$\hat{\kappa}_4$	$ \Delta\alpha /\alpha$	$ \Delta\sigma_\eta^2 /\sigma_\eta^2$	$ \Delta\kappa_4 /\kappa_4$
4	128	0.8649	0.2686	1.61	1.8%	3.2%	46%
4	512	0.8518	0.2844	2.27	0.2%	2.5%	24%
4	2048	0.8465	0.2818	1.82	0.4%	1.5%	39%
8	128	0.8294	0.2824	3.18	2.4%	1.8%	6%
8	512	0.8503	0.2694	2.26	0.04%	2.9%	25%
8	2048	0.8507	0.2793	2.72	0.08%	0.7%	9%

At $C = 8$ and $N \geq 512$, α is recovered to below 0.1% relative error, while the innovation variance remains at the percent level and the fourth cumulant at the tens-of-percent level. Each row is a single fit, and the shape column is visibly not monotone in N at these budgets — the seed-to-seed spread of the fourth cumulant is of the same order as its error, so these rows demonstrate the recovery hierarchy, not a convergence law for the shape. The ordering is consistent with the OU channel: the fourth cumulant is attenuated as e^{-4t} , whereas the second moment is preserved by the variance-matched construction.

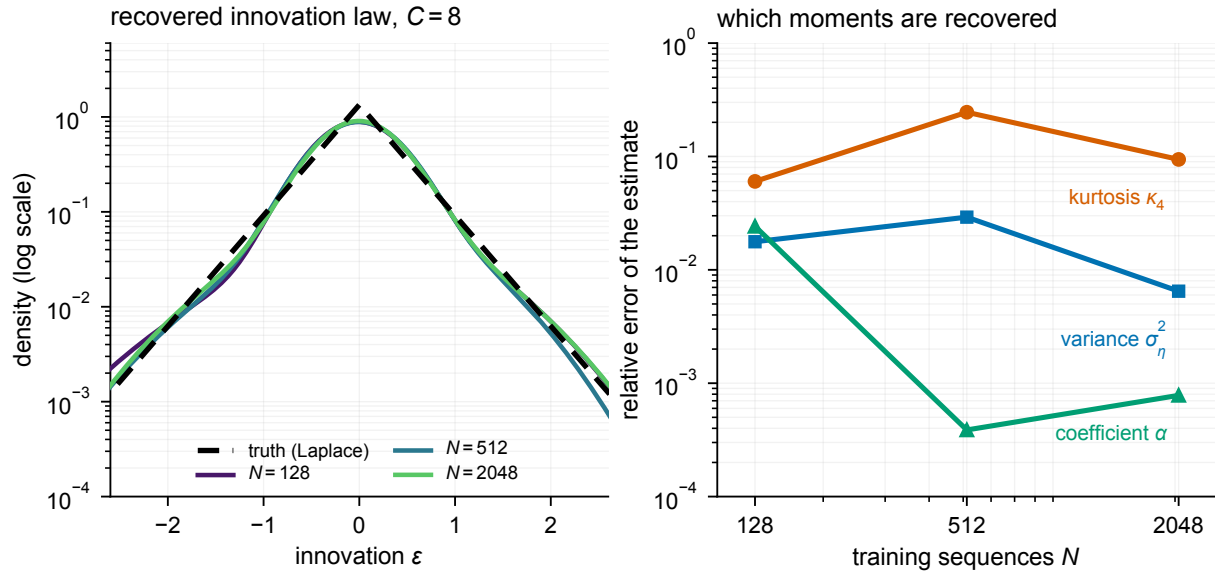


Figure 8.3: Recovery of the innovation law. *Left:* fitted $C = 8$ Gaussian-mixture innovation densities for three data budgets against the Laplace truth. The main discrepancies occur at the cusp and in the tails. *Right:* relative errors in α , innovation variance and fourth cumulant versus N , showing the hierarchy from correlation to scale to higher-order shape.

Higher-order shape is therefore much less visible in the observations. Throughout, “recovery” means what these experiments show: numerical agreement over the studied regime, with no identifiability or consistency theorem claimed for the mixture family.

The same conclusion appears directly in the fitted density.

A finite Gaussian mixture necessarily rounds the Laplace cusp, so the remaining shape error reflects both information lost through the OU channel and approximation by the mixture family. For the developments that follow, the estimated kernel is therefore most reliable in its correlation structure, reasonably accurate in scale, and substantially less informative about higher-order tail shape.

Chapter 9

Learning the Kernel, and What Convergence Costs

Chapter 8 established that the estimator is correct: the E-step is exact, the surrogate ascends, and the recovery error of the second-order coordinates falls approximately as $N^{-1/2}$ over the tested range. None of that says how long it takes, and this chapter is about the difference. Correctness is a statement about the fixed point; what follows is a statement about the *map* that reaches it, and the two are independent. An estimator can be exactly right in the limit and still be routinely reported before it arrives.

The finding is that the map does not converge at one speed. The autoregressive coefficient and the innovation *shape* contract at rates that differ by a factor of eight in the exponent, so a stopping rule calibrated on the first certifies a kernel that has not converged in the second. Every non-Gaussian claim in this thesis depends on the coordinate that settles last, which makes this a prerequisite for Chapter 10 and not a diagnostic beside it. Section 9.1 says analytically why two rates should be expected and which coordinate should be slower; the rest of the chapter measures both and reports where the prediction holds and where it does not.

9.1 Why there are two rates

9.1.1 EM is a fixed-point map, and its Jacobian is missing information

The M-step of Section 8.2 defines a map $\theta^{(k+1)} = \mathcal{M}(\theta^{(k)})$, and a stationary point θ^* of the marginal likelihood is a fixed point of it. Expanding about that point,

$$\theta^{(k+1)} - \theta^* = J(\theta^*) (\theta^{(k)} - \theta^*) + O(\|\theta^{(k)} - \theta^*\|^2), \quad J = \left. \frac{\partial \mathcal{M}}{\partial \theta} \right|_{\theta^*}, \quad (9.1)$$

so EM converges *linearly*, not quadratically, at a rate set by the spectrum of J . The identification of that Jacobian is the central result of [Dempster et al. \(1977\)](#). All three informations below are evaluated at θ^* and conditioned on the same realised observation x : $I_{\text{com}} = -\mathbb{E}[\nabla^2 \log p_\theta(a, c, x) \mid x]$ is the expected complete-data information given x , $I_{\text{obs}} = -\nabla^2 \mathcal{L}$ is the observed information, and I_{mis} is their difference — the information the latent variables retain. With that convention, and J written for a map acting on column vectors,

$$I_{\text{com}} = I_{\text{obs}} + I_{\text{mis}}, \quad J(\theta^*) = I_{\text{com}}^{-1} I_{\text{mis}} = \mathbb{I} - I_{\text{com}}^{-1} I_{\text{obs}}. \quad (9.2)$$

The rate of EM is the fraction of the information that the latent variables carry away. If nothing is missing, $J = 0$ and EM converges in one step; as $I_{\text{mis}} \rightarrow I_{\text{com}}$ the largest eigenvalue of J approaches one and progress per update approaches nothing.

One scope condition on (9.2) governs everything measured in this chapter. The identity is the Jacobian of the *exact* EM map, whose M-step maximises Q jointly. The mixture M-step of Chapter 8 does not: it cycles conditional maximisations and stops after at most the sixteen inner sweeps of Remark 8.3, which is ECM in the sense of [Meng and Rubin \(1993\)](#) and generalised EM when the cycle is cut short. In the quadratic idealisation with r inner sweeps the implemented map has Jacobian $G^r + (\mathbb{I} - G^r) J$, with G the block Gauss–Seidel matrix of the conditional maximisations, which is (9.2) only in the limit

$r \rightarrow \infty$. So (9.2) is the benchmark this chapter measures *against*, not a formula for the map that ran, and a contraction measured from a trace mixes missing-information slowness with conditional-maximisation slowness. It is treated throughout as an ordering heuristic for which coordinate should be slow, and never as a calibrated prediction of how slow.

9.1.2 From eigenvalues to iteration counts

Equation (9.1) says a coordinate aligned with an eigenvector of J with eigenvalue λ has its error contract as λ^k . Stopping that coordinate when its relative distance to the limit first falls below a tolerance τ therefore takes

$$k(\tau, \lambda) \approx \frac{\log \tau}{\log \lambda}, \quad \text{so} \quad \frac{k_2}{k_1} \approx \underbrace{\frac{\log \tau_2}{\log \tau_1}}_{\text{tolerance}} \cdot \underbrace{\frac{\log \lambda_1}{\log \lambda_2}}_{\text{rate}}. \quad (9.3)$$

Two coordinates of the same fit can thus require very different iteration counts, and (9.3) separates the two reasons cleanly: one is a choice of tolerance, the other is a property of the model. Two limits on the formula are stated now and verified below. It is an *asymptotic tail* statement — during the transient a trace can fall much faster than its terminal slope, so (9.3) upper-bounds rather than predicts a measured settling time. And it speaks about eigenmodes: a parameter *coordinate* generally mixes several modes of J , so a slope fitted to one coordinate's trace is an empirical contraction factor for that coordinate, identified with an eigenvalue only if a single mode dominates its window. Section 9.2 measures the coordinate slopes directly, under exactly those labels.

9.1.3 The channel enters the identity twice

Both effects of the observation channel are visible in (9.2), and they act on different terms.

On the rate. Fitting on clean transition pairs leaves only the mixture label c latent. Passing the same chains through the channel makes the entire state a latent as well, so I_{mis} grows by everything the posterior over a does not resolve, J moves towards the identity,

and the map slows. This is a statement about iterations and would persist at infinite sample size. Two things it is not. It is a comparison of what is latent in the two cases, not a claim that the clean case is fast in absolute terms: the components of a mixture fitted to a Laplace innovation overlap heavily, label information can then be a large share of I_{com} on its own, and no measurement here separates label slowness from state slowness. And Section 9.4 compares the two arms at their endpoints and by accuracy rather than by contraction rate, so it corroborates the direction of this argument without isolating it as the mechanism.

On the accuracy. The channel also shrinks I_{obs} , and the asymptotic covariance of the estimator is I_{obs}^{-1}/N . This is a statement about data, not iterations, and it does not vanish by iterating longer. Section 9.4 measures it directly.

9.1.4 Which coordinate should be slower, and why

The model supplies the answer before any fit is run. The forward channel is variance-preserving and the chain is normalised to unit marginal variance ($\sigma_\eta^2 = 1 - \alpha^2$, so $\kappa_2(A) = 1$): the second cumulant of the noised variable is then preserved *exactly* at every t , while the fourth is damped,

$$\kappa_2(X_t) = \kappa_2(A) \quad \text{for all } t, \quad \kappa_4(X_t) = e^{-4t} \kappa_4(A). \quad (9.4)$$

The autoregressive coefficient is a second-order quantity: it is identified by the lag-one covariance, which the channel attenuates in a known, invertible way and does not destroy. The innovation *shape* is a fourth-order quantity, and (9.4) says the channel removes an e^{-4t} fraction of exactly the structure that identifies it. The information the channel takes away is therefore concentrated on the shape coordinate, and by (9.2) that is the coordinate whose eigenvalue of J is pushed closest to one.

Prediction. $\lambda_{\kappa_4} > \lambda_\alpha$, with the gap governed by fourth- against second-order cumulant damping — and consequently, by (9.3), a settling time for the shape that is a large multiple of the settling time for the correlation.

9.2 Measuring the two rates

A dedicated sweep fits the mixture kernel through the channel at $N = 512$, $C = 8$ and the uniform noise design, over 16 seeds run to 800 updates, recording the full parameter trace. Figure 9.1(a) plots each coordinate’s relative distance to its own end-of-run value. Both are straight on a log axis over a wide window, which is (9.1) being obeyed, and the slopes differ sharply.

Table 9.1: Empirical *coordinate-wise* contraction factors λ read from the traces of Figure 9.1(a) (median over 16 seeds; least-squares slope of $\log |\theta^{(k)} - \theta^{(800)}|$ over the fitting window, relative to the end-of-run value, which is itself an estimate of the fixed point). A coordinate mixes eigenmodes of the ECM map and the excess kurtosis is a nonlinear functional of the mixture parameters, so these slopes are not identified with individual Jacobian eigenvalues; no eigenvector analysis is performed here. The asymptotic column applies (9.3) with the relative tolerance actually used; the last column is the measured median. The prediction is right about the ordering and the order of magnitude and over-states the multiple, for the reason given in the text.

coordinate	fitting window	λ	rel. tolerance τ	asymptotic $k(\tau, \lambda)$	measured median
α	20–150	0.9686	1.2×10^{-3}	211	80
σ_η^2	20–150	0.9671	—	—	68
κ_4	60–400	0.9960	6.7×10^{-3}	1256	229
ratio κ_4/α		log-ratio 8.0		5.9	2.8

The prediction of Section 9.1.4 is confirmed in direction and in magnitude of effect. The shape coordinate contracts by a factor 0.9960 per update against 0.9686 for the correlation: in the exponent that is a factor of 8.0. Read through (9.2), that ordering is *consistent with* a slow, shape-dominated eigenmode whose missing information sits far closer to one than the correlation’s, exactly the structure (9.4) predicts — consistent with, not a measurement of: turning the slopes into eigenvalues and missing-information fractions would require the Jacobian of the implemented map and its eigenvectors, which are not computed here. The innovation *variance* settles with the correlation, at $\lambda = 0.9671$, which is the same statement seen from the other side: it is a second-order quantity and the channel preserves it.

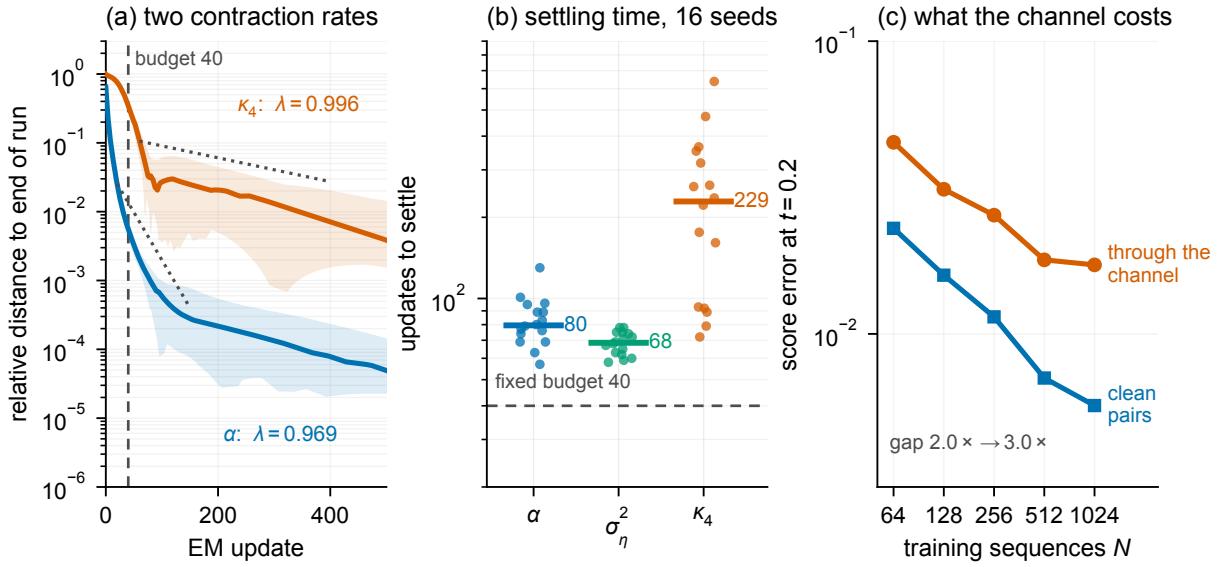


Figure 9.1: The two effects of Section 9.1.3, measured. (a) Relative distance of each coordinate to its own end-of-run value, median over 16 seeds with the 10–90% band; dotted lines are the fitted contraction rates of Table 9.1. The straightness on a log axis is (9.1); the difference in slope is (9.2) responding to the cumulant damping (9.4). (b) Settling times per seed, with medians; the correlation and the innovation variance settle together near 80 updates and the shape at 229, with a tail to 638. The dashed line is the fixed budget of 40 updates several experiments used as a default, which lies below *every* observed settling time of *every* coordinate. (c) The channel’s accuracy cost, which is not a rate: relative score error at $t = 0.2$ of a kernel fitted on clean transition pairs against one fitted on the same chains through the channel, both run to convergence. The clean arm improves fourfold over the range while the noised arm improves 2.6-fold, so the gap *widens* with data rather than closing.

Where the linear theory is not quantitative is the constant. Equation (9.3) predicts a settling ratio of 5.9 and the paired measurement gives 2.8; the asymptotic counts over-predict both individual settling times by a factor of two to three. The reason is visible in Figure 9.1(a): both curves fall considerably faster in their first hundred updates than their fitted slopes, so both tolerances are crossed during the transient rather than in the asymptotic regime that (9.1) describes. The eigenvalues bound the tail behaviour, not the crossing of a threshold, and this is the sense in which Table 9.1’s middle columns are an upper estimate.

9.3 Settling times and the stopping rule

Figure 9.1(b) shows the settling times the rates of Table 9.1 produce. Across 16 seeds the correlation settles at a median of 80 updates and the shape at 229, the largest observed value being 638. Forming the ratio *within* each fit and then taking the median gives 2.8; the quotient of the two medians is 2.9. The paired figure is the one reported, because both coordinates are measured on the same fit.

Two qualifications on what that statistic means. “Settled” is trace stability — the first update after which a coordinate stays within 10^{-3} of its end-of-run value for α , or 2×10^{-2} for the excess kurtosis — and not distance to an optimiser; a coordinate can sit still while the density it belongs to moves along a shallow direction. And the run has to be long enough for the statistic to mean anything: the sweep was run at two lengths, and these medians come from the 800-update shard, the only one in which no configuration settles near its own cap.

Against them, the fixed budget of 40 iterations that several experiments in Chapter 10 used as a default falls short of the shape coordinate’s median by about a factor of six and leaves 105 of the sweep’s 112 configurations — 94% — unsettled in it. That count pools both shards, which is conservative: censoring can only shorten a measured settling time, so it cannot inflate the count.

Table 9.2 makes the same point on a single configuration and adds the endpoints.

Table 9.2: What the observation channel costs in rate rather than in where the fit lands. Above: the through-channel fit at $N = 512$, $C = 8$, as a median over 16 seeds at four updates — the correlation is essentially in place by 30 while the innovation shape is still moving at 120. Below: where the two arms finish, fit on clean transition pairs and on the same chains once through the channel. Generating values are $\alpha = 0.85$ and excess kurtosis 3.0; a finite mixture does not contain the generating law exactly, so what is approached is the best fit within the family.

update	α	innovation excess kurtosis
1	0.4402	0.078
30	0.8412	1.275
120	0.8499	3.019
600	0.8504	2.980
<i>where each arm finishes</i>		
clean transition pairs	0.8509	2.939
through the channel	0.8526	2.729

The trace shows the coordinates separating exactly as Table 9.1 says they must: the correlation is essentially in place by thirty updates while the innovation shape is still climbing at a hundred and twenty. The mixture label is a latent variable of its own, and the channel compounds the missing information — the mechanism of [Dempster et al. \(1977\)](#) appearing in the convergence rate, where (9.2) places it.

9.4 What the channel costs in accuracy at equal data

The endpoints in Table 9.2 are close: the two arms finish within 0.002 of each other on the correlation and within a fifth of a unit on excess kurtosis, both near the generating values. Whether the channel is even *allowed* to leave the population optimum in place depends on specification. Under a correctly specified, identifiable model the clean and the noised likelihoods share their population maximiser, and shared endpoints are what shrinking I_{obs} predicts. The fitted family here is a finite Gaussian mixture that does not contain the Laplace innovation exactly, and under misspecification the data-processing

inequality does not preserve the *ranking* of candidate parameters, so the clean and noised pseudo-true optima need not coincide. The close endpoints above are therefore a *measured approximate invariance* for this family in this regime, not a theorem — and either way they are emphatically *not* evidence that the channel is free. The second effect lives in the variance, and it has to be measured on the variance.

Figure 9.1(c) does so, on the quantity the thesis actually uses: the relative score error of the fitted kernel at $t = 0.2$, for a kernel fitted on clean transition pairs against one fitted on the same chains through the channel, both run to convergence.

Table 9.3: Relative score error at $t = 0.2$ of the fitted kernel, clean transition pairs against the same chains through the channel, $C = 8$, mean over three replicates — an exploratory design: three replicates carry no usable uncertainty estimate, so the row-to-row shape of either curve should not be over-read. The penalty is real at every size; what the last two columns do and do not identify is discussed below.

N	64	128	256	512	1024
clean transition pairs	0.0229	0.0159	0.0114	0.0071	0.0057
through the channel	0.0451	0.0312	0.0254	0.0179	0.0172
ratio	2.0×	2.0×	2.2×	2.5×	3.0×

The penalty is real, and over the range measured it grows: the clean arm improves fourfold, close to the $N^{-1/2}$ behaviour of Figure 8.2(c), while the noised arm improves only 2.6-fold and is nearly flat between $N = 512$ and 1024, so the ratio moves from 2.0×

to 3.0×. What that flattening does and does not identify has to be stated carefully, because the asymptotic theory is unambiguous: under a regular, correctly specified, identifiable model *both* arms obey $\sqrt{N}(\hat{\theta} - \theta^*) \Rightarrow \mathcal{N}(0, I_{\text{obs}}^{-1})$, so a smaller observed information enlarges the *constant* of an $N^{-1/2}$ law and cannot, by itself, change the exponent or produce a plateau. A genuine flattening therefore requires something else: the finite-mixture approximation floor (the family cannot represent the Laplace cusp, so the score error cannot fall to zero), quadrature and stopping tolerances, a possible shift of the noised pseudo-true target under misspecification, or simply a pre-asymptotic range — and three replicates over five sizes

cannot separate these. What the table establishes is a larger finite-data cost through the channel, widening over the tested range; it does not identify a changed asymptotic rate, and the flattening is not attributed to Fisher information alone. Reduced I_{obs} remains the mechanism for the *level* of the penalty; the shape of its tail is unresolved here. Equal endpoints and unequal precision are consistent, and Table 9.2 and Table 9.3 are the two halves of that statement.

Chapter 10

What Structure Buys, and Where It Stops Paying

Chapter 8 built an estimator that is exact in its inference step and approximate only in its numerical representation and its optimisation budget. Chapter 9 measured what it costs to run. This chapter asks what it is *worth*, and the question is comparative: worth relative to a network trained on the same clean trajectories and given none of the structure. What each arm is allowed to observe of those trajectories differs, and Section 10.1 states how before any number is quoted.

How much does supplying the chain reduce denoising error at a fixed number of clean trajectories; how much of that reduction survives against a baseline that is given locality but not the model; how much mixture capacity does the estimator need; and how far can the Markov assumption itself be wrong before the structure stops paying?

The four are answered, at different strengths, and Section 10.8 states each at the strength its evidence supports.

10.1 What is compared, and how it is scored

10.1.1 The estimand

Every headline number in this chapter is a ratio of two relative L^2 score errors.

Fix a diffusion time t and a test set $\{x^{(j)}\}_{j=1}^J$ of noised sequences drawn from the model of Section 8.1. Let $S^*(x, t) = \nabla_x \log p_t(x)$ be the exact joint score, supplied by the validated grid reference of Section 7.4 under the true kernel. For an estimator $\hat{S}$ write

$$\mathcal{E}[\hat{S}; t] = \left(\frac{\sum_{j=1}^J \|\hat{S}(x^{(j)}, t) - S^*(x^{(j)}, t)\|^2}{\sum_{j=1}^J \|S^*(x^{(j)}, t)\|^2} \right)^{1/2}, \quad (10.1)$$

the ratio of two Euclidean norms over the whole test set, so that 0.1 means the estimate is wrong by a tenth of the signal's own size. It is scale free: multiplying the data by a constant leaves it unchanged. Pooling over the noise schedule $\mathcal{T}$ means pooling the two sums, not averaging the ratios,

$$\mathcal{E}^2[\hat{S}] = \sum_{t \in \mathcal{T}} w_t \mathcal{E}^2[\hat{S}; t], \quad w_t = \frac{\mathbb{E} \|S^*(X_t, t)\|^2}{\sum_{s \in \mathcal{T}} \mathbb{E} \|S^*(X_s, s)\|^2}. \quad (10.2)$$

The weights are not a free choice and they are not uniform. In the implementation they are the measured reference score energies of the frozen test bundle, level by level; the variance-matched Gaussian chain explains their shape in closed form. There, $S^* = -\Sigma_t^{-1}x$ with $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$, so

$$\mathbb{E} \|S^*(X_t, t)\|^2 = \text{tr } \Sigma_t^{-1}, \quad \Sigma_t^{-1} \rightarrow I \quad (t \rightarrow \infty), \quad (10.3)$$

and at the low-noise end the limit is *finite*: since $\Sigma_0 \succ 0$, $\Sigma_t \rightarrow \Sigma_0$ and

$$\text{tr } \Sigma_t^{-1} = \text{tr } Q_0 + 2t \left[\text{tr } Q_0 - \text{tr } Q_0^2 \right] + O(t^2) \quad (t \downarrow 0), \quad (10.4)$$

with no $1/t$ singularity — the quantity that does diverge like $1/(2t)$ is the likelihood precision e^{-2t}/Δ_t in latent coordinates, which is a different object. The trace nevertheless

falls steeply across the schedule (from $\text{tr } Q_0 = 193.4$ at $t \rightarrow 0$ to $L = 32$ at $t \rightarrow \infty$ for this α and L ; the linear coefficient in (10.4) is -3140), so the finite schedule still weights its lower-noise levels most heavily. Table 10.5 makes that concrete: under the closed-form weights the four smallest of the twelve levels carry 53.6% of the pooled weight, and under the measured Laplace energies actually used for pooling the same four carry 54.5%. This matters twice below — once when the window head’s optimal radius is predicted, and once when its residual error is discussed.

The two arms are the EM–BP estimator of Chapter 8, which is given the Markov factorisation, and a denoising-score-matching network trained by (2.41) with no structural information at all. Both are fitted from the same N clean trajectories, and that — not “the same data” — is the equality the protocol enforces. What each arm is allowed to see of those trajectories is not the same, and the difference runs *against* the result reported here, so it is stated plainly rather than left to the appendix:

- EM–BP sees one noised realisation x of each trajectory and never sees the clean trajectory a at all; the noised bundle is frozen before fitting begins.
- The network is trained on *paired* (a, x) data, as (2.41) requires, and redraws the diffusion time and the corrupting noise at every gradient step, so over a run it consumes many distinct noisy views of each clean trajectory rather than one.

Supplying the Markov factorisation is thus traded against supplying the clean signal and unlimited corruptions of it, and both asymmetries are deliberate: the baseline is the standard training recipe for this model class, and weakening it to match EM–BP’s view would measure a handicap rather than a structural advantage. The consequence for reading (10.5) is that the ratio is a statement about *value per clean trajectory* under the two standard recipes, not about generic data efficiency at a matched information interface. Table 10.2 records what each arm in the wider ladder is given. Writing $\mathcal{E}_{\text{base}}$ and $\mathcal{E}_{\text{EM}}$ for their errors on a common test set, the reported quantity is

$$R = \frac{\mathcal{E}_{\text{base}}}{\mathcal{E}_{\text{EM}}}, \quad R > 1 \text{ means the structured estimator is more accurate,} \quad (10.5)$$

evaluated cell by cell, a cell being one (training-set size, seed) pair, with both arms re-fitted independently in every cell.

One aggregation choice remains, and it matters because (10.5) is a ratio of the quantities (10.1) and not a linear functional of them. This chapter reports exactly the pooled definition above, paired at the seed level: for each seed and size, squared error and reference energy are summed over the twelve levels before the square root, giving one pooled error per arm; the ratio (10.5) is formed within the seed from those two pooled errors; and the reported number is the mean of the sixteen paired per-seed ratios, with its standard error. The seed is the inferential unit throughout — the twelve levels share a training set and a fitted model. Appendix B recomputes the same cells under four alternative summaries; the direction and the order of magnitude are the same in all of them, but the values are not interchangeable: the mean of per-level ratios, which earlier drafts reported, is systematically the largest (7.0–17.6× on these cells, against the pooled 6.6–10.9×), because it lets levels with little reference energy count as much as levels with a great deal.

10.1.2 The common protocol

Every comparison in Sections 10.3 to 10.5 runs on the same data-generating model and the same schedule. Table 10.1 fixes it once so that the sections need not repeat it; Section 10.6, which deliberately changes the prior, and Section 10.7, which is a likelihood calculation rather than a fit, state their own departures.

Table 10.1: The protocol shared by the comparisons of Sections 10.3–10.5. The generating prior is a Laplace AR(1) chain, so the reference score is genuinely nonlinear (Chapter 7) and no arm is being compared against a linear target. Validation, test and training bundles are disjoint, and no arm sees the test bundle before its budget is chosen.

generating prior	Laplace AR(1), $\alpha = 0.85$, unit marginal variance
innovation law	$\eta \sim \text{Laplace}(0, b)$, $b = \sqrt{\sigma_\eta^2/2}$, excess kurtosis +3
chain length	$L = 32$ sites
channel	variance preserving, $x = e^{-t}a + \sqrt{\Delta_t}\varepsilon$
noise schedule	12 levels, geometric, $t \in [0.05, 3.0]$, ratio 1.45
reference score	grid BP under the true kernel, $N_g = 401$, $A = 8$
fitted kernel	$C = 8$ mixture components, 24 free parameters
training sizes	$N \in \{32, \dots, 4096\}$ (headline) or $\{32, 128, 512, 2048\}$
replicates	16 seeds, both arms re-fitted per cell
validation / test	1,024 / 2,048 sequences (structured runs); 256 / 256 (headline)

10.2 The estimators being compared

10.2.1 From a denoiser to a score

Both network arms are trained by denoising score matching and both are read as scores, so the choice of output head is a parameterisation of the same object. If the network predicts the noise, $\hat{\varepsilon}_\phi$, or the clean signal, $\hat{a}_\phi$, then

$$\hat{S}_\varepsilon(x, t) = -\frac{\hat{\varepsilon}_\phi(x, t)}{\sqrt{\Delta_t}}, \quad \hat{S}_a(x, t) = \frac{e^{-t}\hat{a}_\phi(x, t) - x}{\Delta_t}, \quad (10.6)$$

and the two agree identically under the change of variable $\hat{a} = e^t(x - \sqrt{\Delta_t}\hat{\varepsilon})$. As *function classes* they are therefore the same; as *optimisation problems* they are not, because the substitution rescales the residual by $e^t\sqrt{\Delta_t}$ and so reweights the noise levels against one another. Both are offered to every network arm, and the choice is made on validation. The screen of Section 10.4.2 settles the question empirically and without ambiguity: within each architecture the best noise-predicting configuration beats every signal-predicting one under the pooled selection risk (0.148 against 0.224 for the window head, 0.226 against

0.427 for the dilated stack, 0.187 against 0.262 for the bidirectional recurrence). Where a parameterisation is quoted below, it is the noise head.

Throughout, $\tau(t) = (t, e^{-t}, \sqrt{\Delta_t})$ is the three-dimensional time embedding, broadcast to every site so that a local kernel can see it inside its own receptive field; a time feature supplied only at the sequence ends would be invisible to a convolution.

10.2.2 The structured estimator

EM-BP fits the mixture kernel of (8.2) by the procedure of Section 8.2 and evaluates the score through the score-posterior identity. Its free parameters are the $C - 1$ free mixture weights, C component means, C component variances and the autoregressive coefficient, so at $C = 8$

$$\dim \theta = (C - 1) + C + C + 1 = 3C = 24, \quad (10.7)$$

independent of L . Three structural facts are supplied and not learned: the Markov factorisation, the linear-plus-innovation form of the kernel, and homogeneity across sites. Everything else — the innovation law itself, and α — is estimated from the same noised data the networks see.

10.2.3 The unstructured denoiser

The baseline with no structural prior is a fully connected tanh network mapping the whole sequence and the time embedding to the whole output,

$$f_\phi(x, t) = W_3 \tanh(W_2 \tanh(W_1[x; \tau(t)] + b_1) + b_2) + b_3, \quad (10.8)$$

with $W_1 \in \mathbb{R}^{128 \times (L+3)}$, $W_2 \in \mathbb{R}^{128 \times 128}$, $W_3 \in \mathbb{R}^{L \times 128}$, giving 25,248 free parameters, a thousand times (10.7). Nothing in (10.8) knows that the sites are ordered: permuting the coordinates of x and the rows of the output is absorbed by permuting the columns of W_1 and the rows of W_3 . The network must rediscover the chain from data, and the comparison in Section 10.3 measures what that costs.

10.2.4 The three structure-aware architectures

Beating (10.8) says little, because a large part of the margin could be permutation-blindness rather than the Markov model. Three architectures that do encode sequence structure are therefore screened, spanning the receptive field from bounded to unbounded.

Weight-shared window head, radius r . One predictor applied at every site to a window of its neighbours,

$$\hat{y}_k = g_\phi(x_{k-r}, \dots, x_{k+r}; \tau(t)), \quad k = 1, \dots, L, \quad (10.9)$$

with g_ϕ a two-layer tanh network of width W , the same ϕ at every k , and zero padding outside the sequence. This is exactly a one-dimensional convolution of kernel width $2r + 1$ followed by pointwise layers, so it supplies locality *and* translation invariance and its parameter count, $(2r + 4)W + W^2 + 3W + 1$ — input layer $(2r + 4)W + W$, hidden layer $W^2 + W$, scalar readout $W + 1$, matching the implementation's own count — does not grow with L . Its receptive radius is r by construction: no amount of data lets it propagate information further.

Dilated convolutional stack. A residual stack of three-tap convolutions at geometrically increasing dilation,

$$h_k^{(\ell)} = h_k^{(\ell-1)} + P_\ell \tanh\left(\sum_{s \in \{-d_\ell, 0, d_\ell\}} W_{\ell,s} h_{k+s}^{(\ell-1)} + b_\ell\right), \quad d_\ell \in \{1, 2, 4, 8\}, \quad (10.10)$$

initialised from $h^{(0)} = \tanh(W_{\text{in}}[x; \tau(t)])$ and read out by a pointwise map. Its receptive radius is $\sum_\ell d_\ell = 15$, which spans an $L = 32$ chain end to end.

Bidirectional message passing. Forward and backward recurrences with shared local updates and a joint readout,

$$\vec{h}_k = \tanh(Az_k + F\vec{h}_{k-1} + a), \quad \overleftarrow{h}_k = \tanh(Bz_k + G\overleftarrow{h}_{k+1} + b), \quad \hat{y}_k = C[\vec{h}_k; \overleftarrow{h}_k; z_k] + c, \quad (10.11)$$

where $z_k = [x_k; \tau(t)]$. This is the same topology as the two-sweep sum-product schedule of Section 3.6: one local rule applied at every site, run once in each direction, then combined — with affine-plus-tanh updates in place of the exact message updates. Of the three it has the best claim to being able to *learn* what belief propagation computes, and its recurrent blocks are initialised orthogonally so that a perturbation at one end of the chain is not attenuated to nothing before it reaches the other.

Table 10.2: The five arms as a ladder of supplied structure, and what each is allowed to observe. “Order” means the architecture knows the sites are sequentially ordered; “sharing” means one rule is reused across sites; “radius” is the furthest a site’s prediction can depend on evidence. Only EM-BP is told the generating family, and only EM-BP is denied the clean trajectory: every network arm is trained on paired (a, x) data as (2.41) requires, redrawing t and the corrupting noise each gradient step, while EM-BP sees one frozen noised realisation per trajectory. The parameter ranges are those swept in the screen of Section 10.4.2; the window head at its selected setting ($r = 2$, $W = 64$) carries 4,801.

estimator	order	sharing	kernel form	radius	observes	free parameters
EM-BP, (8.2)	yes	yes	yes	L	x	24
window head, (10.9)	yes	yes	no	r	(a, x)	4.8k–21.4k
dilated stack, (10.10)	yes	yes	no	15	(a, x)	66.4k–264k
message passing, (10.11)	yes	yes	no	L	(a, x)	9.0k–34.3k
global MLP, (10.8)	no	no	no	L	(a, x)	25,248

Table 10.2 sets the five arms side by side as a ladder, which is the object the rest of the chapter climbs. All network arms are trained with Adam at batch size 64 and gradient clipping at 1, over a shared learning-rate grid and a shared checkpoint ladder from 500 to 64,000 steps, with both output heads of (10.6) offered to each. Each site contributes one training example to a weight-shared arm, so a minibatch of B sequences supplies BL examples: sharing is what makes a small receptive field cheap as well as structured.

10.3 Pointwise data efficiency

Against the unstructured denoiser (10.8), and under the structural assumptions of Table 10.1, the structured estimator attains lower relative denoising error by a factor of 6.6–10.9 across the 8 training-set sizes (Table 10.3, Figure 10.1a). It is more accurate in all but 2 of the 1,520 scored (seed, size, level) cells. The factor is not monotone in the budget; Appendix B reports the same cells under four alternative summaries, of which the retired mean-of-per-level-ratios is systematically the largest.

Table 10.3: Schedule-pooled relative score error against the deterministic grid-BP reference under the true kernel: squared error and reference score energy are summed over the twelve noise levels before the square root, per seed; 16 seeds $\pm$ one standard error. Both arms select their optimisation budget on a disjoint validation bundle. EM–BP is more accurate in 1,518 of 1,520 (seed, size, level) cells — a descriptive count, not 1,520 replications, since the twelve levels of a cell share a training set and a fitted model and the inferential unit stays the 16 paired seeds. The ratio column is the paired per-seed ratio of the two pooled errors, averaged over seeds, with a percentile bootstrap over those ratios beside it (seeds resampled, not cells) — close to, but not identical with, the quotient of the two columns beside it; Appendix B reports the same cells under five alternative aggregations. Protocol, cell exclusions and the budget calibration are in Appendix A.

sequences N	network	EM–BP (24 free)	ratio	95% CI
32	0.6024 ± 0.0046	0.0896 ± 0.0043	6.9 ± 0.3	[6.4, 7.5]
64	0.5025 ± 0.0042	0.0841 ± 0.0066	6.6 ± 0.5	[5.6, 7.6]
128	0.4238 ± 0.0024	0.0572 ± 0.0037	7.9 ± 0.5	[7.0, 8.8]
256	0.3494 ± 0.0017	0.0449 ± 0.0027	8.2 ± 0.4	[7.3, 9.0]
512	0.2891 ± 0.0010	0.0375 ± 0.0022	8.2 ± 0.5	[7.2, 9.2]
1024	0.2417 ± 0.0010	0.0294 ± 0.0025	9.0 ± 0.7	[7.7, 10.4]
2048	0.2033 ± 0.0007	0.0258 ± 0.0019	8.6 ± 0.7	[7.4, 9.9]
4096	0.1790 ± 0.0007	0.0180 ± 0.0015	10.9 ± 0.8	[9.4, 12.5]

The growth of the ratio has a one-line reading that the panel supports directly. If each arm’s error follows a power law $\mathcal{E} \approx c N^{-\varrho}$ over the tested range, then

$$R(N) = \frac{c_{\text{base}}}{c_{\text{EM}}} N^{\varrho_{\text{EM}} - \varrho_{\text{base}}}, \quad (10.12)$$

so the ratio grows precisely when the structured arm converges faster in data. The

fitted finite-range slopes of the pooled error curves are $\varrho_{\text{EM}} = 0.332 \pm 0.013$ and $\varrho_{\text{base}} = 0.256 \pm 0.001$ (log-log least squares over the 8 sizes — descriptive fits, not asymptotic rates), and (10.12) then predicts that a 128-fold increase in data multiplies the ratio by $128^{0.332-0.256} = 1.45$; the measured ratio moves from 6.9 to 10.9, a factor of 1.58, non-monotonically. Two remarks travel with that. Neither slope is the parametric $\varrho = 1/2$, so neither arm is behaving like a correctly specified finite-dimensional estimator over this range — for EM-BP the mixture is misspecified against a Laplace innovation and the grid is finite; for the network the function class is unrestricted. And (10.12) describes eight points; it identifies no mechanism.

Every tunable quantity either arm is allowed to choose is chosen on a validation bundle disjoint from both training and test, and the two arms are given the same privilege: the network selects its output head *and* its training length, EM-BP its iteration count. That choice coincides with the test-set optimum in 81.1% of cells for the network and 77.6% for EM-BP.

10.4 How much of the margin is the architecture?

10.4.1 The bulk posterior, in closed form

Everything in this subsection is exact for the variance-matched Gaussian chain and is used here as a *design calculation* for the Laplace chain, whose second-order structure it shares. How wide a window does a score head need? We tried to answer with arithmetic before any network is trained.

Step 1: the channel adds only to the diagonal. The prior is $a_k = \alpha a_{k-1} + \eta_k$ with $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$ and $\sigma_\eta^2 = 1 - \alpha^2$, so the marginal variance is one and $\Sigma_0 = (\alpha^{|i-j|})$. Its precision $Q_0 = \Sigma_0^{-1}$ is tridiagonal — the Markov property written in matrix form —

$$[Q_0]_{kk} = \frac{1 + \alpha^2}{\sigma_\eta^2} \text{ (interior),} \quad [Q_0]_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}, \quad [Q_0]_{kj} = 0 \quad (|k - j| \geq 2). \quad (10.13)$$

The channel contributes $-\frac{1}{2\Delta_t} \|x - e^{-t}a\|^2$ to the log-posterior. Collecting the quadratic and linear terms in a ,

$$-\log p(a | x) = \frac{1}{2} a^\top J a - h^\top a + \text{const}, \quad J = Q_0 + \frac{e^{-2t}}{\Delta_t} I, \quad h = \frac{e^{-t}}{\Delta_t} x, \quad (10.14)$$

so the posterior is Gaussian with mean $m = \mathbb{E}[a | x] = J^{-1}h$. The structural fact that drives the whole subsection is visible in (10.14): *the channel enters the diagonal alone*. It stiffens every site against its own observation and leaves the nearest-neighbour couplings exactly as the prior set them. In the bulk — away from the two ends, where (10.13) loses one neighbour — J is therefore tridiagonal Toeplitz,

$$J = \text{tridiag}(\beta, J_d, \beta), \quad J_d = \underbrace{\frac{e^{-2t}}{\Delta_t}}_{\text{evidence}} + \underbrace{\frac{1 + \alpha^2}{\sigma_\eta^2}}_{\text{prior}}, \quad \beta = -\frac{\alpha}{\sigma_\eta^2}. \quad (10.15)$$

Only J_d depends on t , and it is monotone: it diverges like $1/(2t)$ as $t \downarrow 0$ and falls to the prior value as $t \rightarrow \infty$.

Step 2: inverting a tridiagonal Toeplitz matrix. A matrix with three bands has an inverse that is dense, but on the bulk of a long chain that dense inverse is a single geometric sequence.

Proposition 10.1 (Bulk covariance of the chain posterior). *In the interior of a long chain the inverse of (10.15) is*

$$(J^{-1})_{k, k+d} = V q^{|d|}, \quad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2|\beta|} \in (0, 1), \quad V = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (10.16)$$

and the square root is real for every $|\alpha| < 1$ and every $t > 0$.

Toolbox 10.1 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Write $G = J^{-1}$ and look for a solution of the defining relation $\sum_j J_{ij} G_{j,k} = \delta_{ik}$ in geometric form, $G_{i, i+d} = V q^{|d|}$, symmetric in $\pm d$. With i interior, the sum has only

three terms, and setting $k = i + d$ with $d \geq 1$ gives

$$\beta V q^{d+1} + J_d V q^d + \beta V q^{d-1} = 0 \iff \beta q^2 + J_d q + \beta = 0,$$

a quadratic that is the same for every d — which is why one geometric sequence suffices. Its two roots multiply to $\beta/\beta = 1$, so they are q and $1/q$: *exactly one* lies inside the unit disc, and it is the one above. (For $\alpha > 0$ all entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus.) The remaining relation, at $d = 0$, fixes the prefactor: $J_d V + 2\beta V q = 1$, and since $2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$ this reads $V\sqrt{J_d^2 - 4\beta^2} = 1$.

Reality of the root needs $J_d > 2|\beta|$, and the margin is strictly positive at every operating point,

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} > 0,$$

both terms being non-negative and the first strictly positive for $t > 0$. So q is real, and $q < 1$ strictly. $\square$

Two checks. The formula is verified against a direct numerical inverse at $L = 32$, $\alpha = 0.85$: the relative disagreement over $d = 0, \dots, 5$ is 6×10^{-15} at $t = 0.05$ and 2.7×10^{-4} at $t = 3$, the growth being the two boundaries leaking into the “bulk” once the correlation length is an appreciable fraction of the chain — which is the one regime in which (10.16) should be read as approximate here. And the limits are the ones physical sense demands,

$$\begin{aligned} t \rightarrow 0 : \quad J_d &\rightarrow \infty, & q &\approx \frac{|\beta|}{J_d} \rightarrow 0, & V &\rightarrow 0 \quad (\text{sites pinned by their observations}); \\ t \rightarrow \infty : \quad J_d &\rightarrow \frac{1 + \alpha^2}{\sigma_\eta^2}, & q &\rightarrow |\alpha|, & V &\rightarrow 1 \quad (\text{posterior} \rightarrow \text{prior, unit variance}), \end{aligned} \tag{10.17}$$

the second because $J_d^2 - 4\beta^2 = (1 - \alpha^2)^2/\sigma_\eta^4 = 1$ once the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from zero to the prior’s own.

Step 3: the influence of distant evidence is exactly geometric. Because $m = \frac{e^{-t}}{\Delta_t} J^{-1} x$ is linear in the observation, Proposition 10.1 gives the sensitivity of one site's posterior mean to an observation d sites away in closed form,

$$\frac{\partial m_k}{\partial x_{k \pm d}} = \frac{e^{-t}}{\Delta_t} (J^{-1})_{k, k \pm d} = \frac{e^{-t}}{\Delta_t} V q^d = \frac{e^{-t} V}{\Delta_t} e^{-d/\xi}, \quad \xi := \frac{1}{\log(1/q)}. \quad (10.18)$$

This is the object a score head has to represent, and it is a pure exponential in the separation with a single length $\xi(\alpha, t)$. Nothing about it is fitted, and within the infinite-chain bulk nothing about it is asymptotic in t either; on a finite chain it is the bulk approximation, exact in the interior and corrected near the two ends, as Chapter 6 derives it.

Step 4: what a radius- r window must discard. A head of radius r cannot use $x_{k \pm d}$ for $d > r$, whatever it learns. In squared weight, using (10.18) and two geometric sums,

$$\frac{\sum_{|d| > r} q^{2|d|}}{\sum_{d \in \mathbb{Z}} q^{2|d|}} = \frac{2q^{2(r+1)}/(1-q^2)}{(1+q^2)/(1-q^2)} = \frac{2q^{2(r+1)}}{1+q^2}, \quad (10.19)$$

exactly. Pooling over the schedule with the closed-form weights of Table 10.5 gives one number per radius, the *coefficient-tail fraction*

$$\mathcal{T}(r) = \sum_{t \in \mathcal{T}} w_t \frac{2q(\alpha, t)^{2(r+1)}}{1 + q(\alpha, t)^2}. \quad (10.20)$$

What (10.20) is, and is not, has to be exact. It is a fraction of squared coefficient mass, and it sets the geometric *exponent* of everything local. It is *not* a prediction risk, for three reasons: the omitted observations are correlated with the retained ones, so their squared coefficients do not add to an error energy; a genuine windowed estimator reoptimises its retained coefficients — it is the posterior mean of the window model, not the exact coefficients with the tail set to zero; and a squared-mass fraction is not commensurable with a relative RMS error even in scale. It is used below only as a decay diagnostic.

The valid risk anchor exists in closed form and costs nothing. The Bayes-optimal

radius- r estimator of the design model is the windowed posterior mean $\mathbb{E}[a_k \mid x_{k-r}, \dots, x_{k+r}]$ — exactly the windowed inference whose price Chapter 6 computes — and its population relative score error, pooled over the schedule with the same weights, is

$$\mathcal{W}(r) = \left(\frac{\sum_t \text{tr} [(M_r(t) - Q_t) \Sigma_t (M_r(t) - Q_t)^\top]}{\sum_t \text{tr} Q_t} \right)^{1/2}, \quad (10.21)$$

— $M_r(t)x$ being the windowed score estimator. This is the exact local-oracle error of the Gaussian design calculation: no learning, no sampling, the best any radius- r estimator can do *on the covariance-matched Gaussian model*. Numerically, at this (α, L) and schedule,

r	0	1	2	4	8
$\mathcal{W}(r)$	0.657	0.202	0.089	0.029	0.007

It is a design anchor, not the experiment's own floor: the experiment is Laplace, whose local oracle is a different quantity. How different is a question of fact rather than of ordering theory — no result here ranks the Bayes risks of two priors that share a covariance — so it is measured. The radius- r Laplace oracle is $\mathbb{E}[a_k \mid x_{k-r:k+r}]$ under the chain of (7.1), which is exactly the L^2 -optimal radius- r estimator of S_k because x_k lies inside the window; window-restricted grid BP over the chain's true state marginals evaluates it, and Table 10.4 pools it exactly as (10.21) pools the Gaussian one.

Table 10.4: The exact Gaussian windowed-oracle error (10.21) and the Laplace chain's own, both pooled over the schedule at $\alpha = 0.85$, $L = 32$. The Laplace column is Monte Carlo over 1024 chains with a bootstrap standard error over draws; the Gaussian column is closed form. The Laplace oracle is the larger of the two wherever the difference is resolved — at $r = 2$ by 14.4 bootstrap standard errors — and the two are indistinguishable by $r = 8$.

r	0	1	2	4	8
Gaussian $\mathcal{W}(r)$	0.657	0.202	0.089	0.029	0.007
Laplace $\mathcal{W}_L(r)$	0.664	0.215	0.094	0.030	0.007

So $\mathcal{W}(r)$ tells us what the window is worth in the second-order geometry the two models share, and the measurement says the experiment's own floor sits slightly above it at every radius that matters here.

Table 10.5: The influence-decay rate of (10.16), the correlation length ξ it implies through (10.18), and the weight (10.3) each level carries in the pooled risk, at $\alpha = 0.85$ (eight of the twelve levels shown). The two right-hand columns pull in opposite directions, and that opposition is the whole design argument: evidence travels furthest exactly where the pooled score error is least sensitive. The four smallest levels carry 53.6% of the weight and have $\xi \leq 1.07$; the four largest carry 19.2% and have $\xi \geq 3.55$.

t	$q(\alpha, t)$	$\xi = 1/\log(1/q)$	weight w_t (%)
0.0500	0.203	0.63	16.7
0.0725	0.259	0.74	14.4
0.1053	0.323	0.88	12.2
0.1527	0.392	1.07	10.2
0.3215	0.541	1.63	7.1
0.6769	0.689	2.69	5.4
1.4250	0.806	4.65	4.8
3.0000	0.848	6.06	4.7

The prediction now follows without reference to any fit. Because the pooled risk is concentrated where $\xi \lesssim 1$, a small radius is enough where it matters: at $t = 0.1527$ a radius of 4 discards $q^5 = 0.9\%$ of the influence, the oracle error (10.21) has already fallen to 0.09 at $r = 2$ and 0.03 at $r = 4$, and the coefficient tail (10.20) confirms that what remains beyond such radii is contributed almost entirely by the high-noise levels, where the weight is a few percent. Enlarging the window much past the largest correlation length that carries appreciable weight therefore removes a quantity that is already small in the metric being optimised, while adding parameters that must be estimated from the same data.

Prediction. Under this protocol the best window radius is small — of order the largest correlation length that carries appreciable weight, a few sites — and the risk should *rise* again well beyond it, because past that point the receptive field is buying a quantity smaller than the estimation error it costs. And the residual error of the selected head should sit well above the Gaussian design oracle $\mathcal{W}(r)$ at its radius, because at small radii the window itself is no longer what limits a learned head.

10.4.2 The screen

Three candidates were screened on 4 development seeds, disjoint from the 16 used for the comparison: the window head (10.9), the dilated stack (10.10) and the bidirectional recurrence (10.11). Each ran over its own hyperparameter grid, a shared learning-rate grid, both output heads and a shared checkpoint ladder, 32 configurations in all, scored on the bulk region. **The window head won**, at radius 2 and width 64, with the bidirectional message passing network next (0.148 against 0.187).

Table 10.6: The architecture screen that selected Table 10.7’s baseline. Each candidate architecture was run over its own hyperparameter grid on 4 development seeds (disjoint from the confirmatory seeds), at $N \in \{32, 2048\}$, with a shared learning-rate grid, both output parameterisations, and a shared checkpoint ladder; the reported risk is the schedule-pooled relative score error, pooled over the development seeds and sizes on the interior (bulk) slice used for selection — the same estimand and accumulation as the selector inside the confirmatory job. ★ marks the winner carried into the confirmatory run. The dilated stack carries an order of magnitude more parameters than the winner and finishes last, so the outcome is not an artefact of one arm being given more capacity than another.

architecture	configurations	parameters	selection risk
weight-shared window head ★	16	4.8k–21.4k	0.148
dilated convolutional stack	8	66.4k–264k	0.226
bidirectional message passing	8	9.0k–34.3k	0.187

Table 10.6 reports the screen in full and Figure 10.2 shows every configuration in it, which is the more useful object: the three architectures separate by parameter count far more cleanly than by risk, and the two arms built to propagate further occupy the expensive half of the axis without buying anything there.

Both halves of the prediction hold. The measured radius sweep at width 64 is **0.148**, 0.149, 0.185, 0.214 at $r = 2, 4, 8, 16$ under the pooled selection risk: minimised at 2, in a near-tie with 4, and rising steadily beyond — the behaviour of a term that has stopped buying anything and started costing estimation error. And the selected head sits well clear of the local oracle at its own radius: 0.148 against the experiment’s own $\mathcal{W}_L(2) = 0.094$ (Table 10.4), and against the Gaussian design anchor’s $\mathcal{W}(2) = 0.089$. The first of those two differences is the one with a reading: it is what remains to be won *inside* the radius-2

information set — learning, capacity and optimisation together, not separated — and it says the receptive field is not the binding constraint at the selected radius. Because the measured Laplace oracle exceeds the Gaussian one here, the larger difference from $\mathcal{W}(2)$ bounds the first from above rather than replacing it; that ordering is a measurement at this (α, L) and schedule, not a theorem. No finer causal split of either gap follows from this design.

Two scoping conditions travel with the outcome. It is a model-selection fact: under this search space, protocol and development seeds, the simplest candidate scored best. It is not evidence that long-range propagation is useless here.

10.4.3 The confirmatory comparison

Against the screened baseline — the radius-2, width-64 window head, 4,801 parameters — at the headline protocol throughout: the same twelve noise levels and 16 seeds as Table 10.3, every site scored rather than the centre alone, the baseline’s architecture and radius fixed by the screen on separate seeds, and each arm’s checkpoint chosen on a validation bundle disjoint from test — the surviving margin is $2.3\text{--}6.1\times$.

Table 10.7: EM–BP against the screened structure-aware baseline, at the headline protocol throughout. The baseline is a weight-shared window head whose architecture and hyperparameters were fixed by the screen of Table 10.6 on development seeds; only the checkpoint is chosen here, on a validation bundle disjoint from test, and EM–BP’s iteration count is chosen the same way. Risks are all-site and averaged over the 16 seeds; the ratio is $\text{risk}(\text{window})/\text{risk}(\text{EM–BP})$ formed within each seed and then averaged, $\pm$ one s.e. *All* scores every site, and is the headline; *centre* is a single-site diagnostic. A third region, the predeclared interior slice, is scored in the stored output and used for architecture selection but not reported here.

N	risk, all sites		ratio	
	EM–BP	window	all sites	centre site
32	0.1005	0.2006	2.34 ± 0.26	2.04 ± 0.24
128	0.0652	0.1594	2.82 ± 0.30	2.35 ± 0.26
512	0.0394	0.1353	3.58 ± 0.18	3.25 ± 0.18
2048	0.0224	0.1267	6.14 ± 0.45	5.43 ± 0.39

Table 10.7 reports it, and Figure 10.1b and c place it beside the headline; the dotted curve in panel (b) is the two-parameter floor fit

$$\mathcal{E}(N)^2 = \mathcal{E}_\infty^2 + \frac{c}{N}, \quad (10.22)$$

a floor plus a $1/N$ estimation term, fitted to the window head’s four squared risks — descriptive, as a four-point fit must be. EM–BP is more accurate in every one of the 192 scored cells. That count is 16 seeds $\times$ 4 sizes $\times$ 3 scoring regions (all, bulk, centre), of which the table displays two; it is a descriptive diagnostic and not a replication count, since regions and sizes share fitted models and training data, so the inferential unit behind each displayed row is the 16 paired seeds. Appendix B reports the same ratio under 4 aggregations; none of them, at any size, falls below $1.98\times$.

A large fraction of the headline margin therefore does not survive supplying locality and weight sharing. That comparison does not decompose causally: architecture, parameter sharing, capacity and optimisation all change when the baseline is swapped, and no design here separates them. What it does establish is that the generic-MLP ratio overstates what can be attributed specifically to the supplied Markov model, and by roughly a factor of three.

10.5 How much mixture capacity the estimator needs

The estimator’s one nonparametric dial is the number of mixture components. The capacity sweep refits $C \in \{1, 2, 4, 8, 16\}$ on identical data, paired within seed against the single-Gaussian innovation $C = 1$, at $N \in \{128, 512\}$, and scores held-out log-evidence.

Table 10.8: Held-out log-evidence per edge at each mixture capacity, paired within seed against the single-Gaussian innovation $C = 1$ on identical training and evaluation data, 16 seeds, each fit run to an iteration cap placed at its measured per-edge-gain plateau (the tolerance stop never fires there, so the cap is the effective stopping rule). Positive would mean the mixture buys something. No capacity does, at either size, and the sign count is the more honest summary of the two columns: the mean is dominated by a few large-magnitude seeds while the direction is consistent across most of them. The last column counts cells, over both sizes, whose narrowest fitted component falls below the grid’s resolution floor of 2 spacings — a share that climbs with C , and a reason not to read the table as a pure statement about capacity. Means and standard errors are in units of 10^{-2} .

C	$N = 128$			$N = 512$			
	mean	s.e.	seeds better	mean	s.e.	seeds better	under-resolved
2	−1.58	0.68	4/16	−1.29	0.67	3/16	4/32
4	−2.35	1.25	3/16	−1.28	0.66	2/16	8/32
8	−2.30	1.18	3/16	−1.04	0.49	2/16	10/32
16	−2.35	1.18	3/16	−1.30	0.65	2/16	11/32

Table 10.8 and Figure 10.3 give the outcome and its caveat in one view. Under this protocol no capacity buys anything: every paired mean is negative, the $N = 128$ deficit at the largest capacity is *resolved* against zero (-4.4×10^{-4} , CI $[-9.2 \times 10^{-4}, -1.1 \times 10^{-4}]$), and only two to four seeds in sixteen ever prefer a mixture. But the sweep is not a clean statement about capacity in the abstract: above $C = 4$ every cell stops at the 1200-iteration cap rather than at tolerance, and the share of fits whose narrowest component falls below the grid’s resolution floor climbs with C , so a statistical disadvantage of large mixtures is entangled with an optimisation and a discretisation effect. The defensible conclusion is protocol-scoped: under this grid, M-step, iteration cap and validation rule, capacity beyond a single Gaussian innovation bought nothing — and the convergence economics of Chapter 9 say why more capacity is also *slower*: every added component deepens the missing-information problem the channel already creates.

That finding has a consequence for how the headline of Section 10.3 may be read, and it is worth stating against interest. The headline estimator runs at $C = 8$ because the protocol fixed it there in advance, not because any selection rule chose it; in this sweep the single-Gaussian innovation is at least as good on *both* criteria, the held-out

evidence and the score risk the thesis actually reports. A single Gaussian innovation with a fitted α is, by Theorem 7.3, exactly the covariance-matched linear estimator, so what that comparison says is that within the range tested here the margin over the network arms is earned by the Markov factorisation and its covariance, and not by the mixture's capacity to represent a non-Gaussian innovation shape. The two experiments run different bundles and scoring regions, so the risks are not comparable across them and no headline number is restated here; the transferable point is the attribution, not a substituted figure.

10.6 Where the Markov assumption stops paying

Every result so far assumes the clean chain is exactly first-order Markov. Two controlled violations are measured against an exact reference, so that the estimator being wrong is not confused with the reference being wrong.

10.6.1 Two violations, defined

A shared global latent, strength β . One standard normal variable is drawn per sequence and added to every site,

$$a = \frac{y + \beta g}{\sqrt{1 + \beta^2}}, \quad y \sim \text{chain prior}, \quad g \sim \mathcal{N}(0, 1), \quad (10.23)$$

variance preserving for every β and Markov only at $\beta = 0$. Its covariance is $\Sigma_\beta = (\Sigma_0 + \beta^2 \mathbf{1}\mathbf{1}^\top)/(1 + \beta^2)$ with Σ_0 the AR(1) covariance, so a Laplace and a Gaussian version differ *only beyond second moments* — the controlled comparison this thesis uses throughout. At $\beta = 1$ half the marginal variance of every site is one shared constant.

Long-range precision coupling, strength γ . Weak couplings are added directly to the precision at every separation of two or more,

$$Q_\gamma = Q_{\text{AR}} + \gamma Q_{\text{long}}, \quad [Q_{\text{long}}]_{ij} = -e^{-|i-j|/\ell} \quad (|i-j| \geq 2), \quad [Q_{\text{long}}]_{ii} = 1.05 \sum_{j \neq i} e^{-|i-j|/\ell}, \quad (10.24)$$

with $\ell = 8$ and the diagonal compensation making Q_γ strictly diagonally dominant, hence positive definite; the covariance is rescaled to unit diagonal so the prior stays variance preserving. Not Markov for $\gamma \neq 0$.

10.6.2 Why one is partly absorbable and the other is not

The two mechanisms are not two amounts of the same thing.

Proposition 10.2 (The global latent perturbs the chain in one direction). *Let $Q_{\text{AR}} = \Sigma_0^{-1}$ be the AR(1) precision and $Q_\beta = \Sigma_\beta^{-1}$. Then*

$$Q_\beta = (1 + \beta^2) Q_{\text{AR}} - c_\beta u u^\top, \quad u = Q_{\text{AR}} \mathbf{1}, \quad c_\beta = \frac{(1 + \beta^2) \beta^2}{1 + \beta^2 \mathbf{1}^\top Q_{\text{AR}} \mathbf{1}}, \quad (10.25)$$

so the departure from a rescaled chain precision has rank exactly one, in the single direction u . Moreover the lag-one correlation of (10.23), and hence the correlation of its best first-order chain approximation, is available in closed form:

$$\alpha_{\text{CL}}(\beta) = \frac{\alpha + \beta^2}{1 + \beta^2}. \quad (10.26)$$

Proof. Sherman–Morrison applied to $\Sigma_\beta = (\Sigma_0 + \beta^2 \mathbf{1} \mathbf{1}^\top) / (1 + \beta^2)$ gives (10.25) directly, the rank-one term being the correction that inverting a rank-one update produces. For (10.26), the covariance of (10.23) at separation $d \geq 1$ is $(\alpha^d + \beta^2) / (1 + \beta^2)$ and the marginal variance is 1; at $d = 1$ this is the stated value. The KL projection of a Gaussian onto a fixed tree matches that tree’s node and edge marginals (Chow and Liu, 1968), so on a path graph the fitted chain reproduces the true adjacent correlations. $\square$ $\square$

Numerically, (10.25) is exact to machine precision: the correction has one nonzero eigen-

value at every β tested, and its eigenvector agrees with $u/\|u\|$ to $|\langle \cdot, \cdot \rangle| = 1.000000$. The direction u is, up to boundary terms, the constant vector — the chain’s mean level. By contrast the correction in (10.24) is γQ_{long} , dense by construction, with numerical rank $L = 32$ and 30 to 31 eigenvalues above 1% of its largest at the strengths tested. *One direction against thirty* is the structural difference, and a chain that may reparameterise a single scalar can trade against the first in a way it cannot against the second — partly: matching the lag-one correlation absorbs the covariance change along that one direction, but a homogeneous first-order chain cannot represent the dense rank-one precision correction exactly, so “absorbable” means *partly absorbed through the refitted local parameter*, not reproduced.

The fitted coefficient does exactly that. Under (10.23) the fitted $\hat{\alpha}$ climbs from 0.848 on the clean chain to 0.939 at $\beta = 1$, tracking (10.26) (0.9250 at $\beta = 1$) to within 0.014. Under (10.24) the same gap is 7 times larger. That is one scalar agreeing, not the whole fitted law certified — but it is the mechanism the proposition predicts, appearing in the fit.

10.6.3 A covariance scalar orders the tested conditions

Proposition 10.2 explains the qualitative difference. A single number makes it quantitative. The scalar used is the per-site KL divergence between the *covariance-matched centred Gaussians* of the contaminated law and of its best first-order chain,

$$\delta = \frac{1}{L} D_{\text{KL}}(\mathcal{N}(0, \Sigma) \parallel \mathcal{N}(0, \Sigma_{\text{CL}})) = \frac{1}{2L} \left[\text{tr}(\Sigma_{\text{CL}}^{-1} \Sigma) - L + \log \frac{\det \Sigma_{\text{CL}}}{\det \Sigma} \right], \quad (10.27)$$

closed-form for both mechanisms and involving no fit and no data. Its status differs by arm, and the distinction is not cosmetic. For the *Gaussian* contaminated laws, (10.27) is the exact distributional KL to the best Gaussian chain — the excess code length that the Gaussian Chow–Liu projection minimises. For the *Laplace* arms the same expression is only a covariance proxy: the true laws are non-Gaussian, their KL involves entropy terms that covariance does not determine, and two non-Gaussian laws with identical covariances

can sit at different true divergences. Where the distinction matters we write δ_{cov} for the proxy reading; it is a second-order misspecification measure, blind to everything beyond covariance, and that is exactly the information the chain estimator’s Gaussian design geometry uses.

Table 10.9: The advantage against the closed-form covariance distance (10.27) from the contaminated law to its best first-order chain — exact KL for the Gaussian rows, covariance proxy δ_{cov} for the Laplace column. Read against their own strength parameters the two mechanisms look unrelated; read against δ they interleave and the sequence is monotone. The entry that carries the argument is $\beta = 1.00$ against $\gamma = 0.05$: a prior in which half of every site’s variance is a shared global constant is *closer* to the chain family, by a factor of 1.7 in nats, than one with the weakest long-range coupling tested — and the measured advantage is ordered the same way. Ratios here are means over the twelve noise levels of the per-level error ratio (a mean of ratios, unlike the pooled headline; the estimand is stated because the two are not interchangeable), against the convolutional and fully connected baselines.

contamination	δ (nats/site)	vs CNN	vs MLP	Laplace, vs CNN
none	0	16.4	29.3	11.1
global, $\beta = 0.10$	4.6×10^{-6}	14.8	22.5	10.9
global, $\beta = 0.25$	1.5×10^{-4}	6.7	11.3	6.2
global, $\beta = 0.50$	1.4×10^{-3}	3.2	4.0	3.0
global, $\beta = 1.00$	6.5×10^{-3}	2.2	1.7	2.1
long-range, $\gamma = 0.05$	1.1×10^{-2}	1.2	1.7	—
long-range, $\gamma = 0.10$	2.2×10^{-2}	0.97	1.2	—
long-range, $\gamma = 0.20$	4.0×10^{-2}	0.88	0.84	—
long-range, $\gamma = 0.40$	6.0×10^{-2}	0.75	0.70	—

Table 10.10: Two controlled violations of the Markov assumption, each evaluated against the appropriate reference: the Gaussian rows against an exact linear-algebra reference for the contaminated model, the Laplace rank-one rows against the validated grid-BP reference. The first two rows of each block are the baseline’s relative score error divided by EM–BP’s, averaged over the noise schedule, so a value above 1 means the structured estimator wins and a bold value means it loses. The third row is the fitted correlation minus the Chow–Liu correlation of the true contaminated law: the distance between the chain that was fitted and the best chain there is. The fourth is EM–BP’s own relative score error at its worst noise level, which says how much of a change in the ratio is the estimator degrading rather than the baseline improving.

	global latent, rank-one, strength β				
	0.00	0.10	0.25	0.50	1.00
Gaussian, vs CNN	16.41	14.84	6.68	3.21	2.16
Gaussian, vs MLP	29.34	22.48	11.28	4.00	1.67
Laplace, vs CNN	11.14	10.92	6.21	2.96	2.09
Laplace, vs MLP	16.51	17.64	10.13	3.67	1.62
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	−0.0025	−0.0008	+0.0008	+0.0051	+0.0144
Gaussian: EM–BP error, worst t	0.019	0.025	0.055	0.134	0.167
Laplace: EM–BP error, worst t	0.022	0.022	0.047	0.132	0.183
	long-range precision coupling, strength γ				
	0.00	0.05	0.10	0.20	0.40
Gaussian, vs CNN	16.07	1.24	0.97	0.88	0.75
Gaussian, vs MLP	27.84	1.66	1.19	0.84	0.70
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	+0.0018	+0.0376	+0.0557	+0.0806	+0.0976
Gaussian: EM–BP error, worst t	0.016	0.298	0.402	0.490	0.579

Table 10.9 and Figure 10.4 report δ beside the measured advantage. Two readings follow, and the second is the one this chapter contributes.

Read against strength, the mechanisms behave differently in kind. At $\beta = 1$ the Gaussian arm is still ahead at every noise level against both baselines, worst ratio 1.06; the Laplace arm is still ahead against the convolutional baseline, worst ratio 1.32, and reaches break-even against the fully connected one at the largest diffusion time, at 0.998.

Under long-range coupling the advantage against the convolutional baseline is gone by $\gamma = 0.1$, having survived $\gamma = 0.05$, and by $\gamma = 0.2$ the estimator loses to both baselines and not to one; at $\gamma = 0.4$ its own relative score error reaches 58%.

Read against δ , the tested conditions collapse onto one empirical curve. The nine contaminated cells interleave — $\beta = 1$ sits between $\gamma = 0$ and $\gamma = 0.05$ — and the advantage is monotone in δ across both mechanisms, over four orders of magnitude in δ and a factor of twenty in the ratio, against both baselines and for both innovation laws. The crossing is reported as the bracket the design actually resolves, between adjacent tested strengths:

$$\delta^* \in [1.1, 2.2] \times 10^{-2} \text{ nats/site (vs CNN)}, \quad [2.2, 4.0] \times 10^{-2} \text{ nats/site (vs MLP)}, \quad (10.28)$$

the CNN ratio falling from 1.2 to 0.97 across its bracket and the MLP one from 1.2 to 0.84 across its own. A straight line through those endpoints crosses at 2.1 and 3.0×10^{-2} respectively, and those interpolations are quoted nowhere as results: each condition here is a single run scored across the schedule, with no seed replication, so no interval can be attached to a crossing and a point estimate would imply a precision the design does not have. Three further scope conditions travel with the bracket. The two mechanisms occupy different parts of the δ axis, so the crossing is bracketed by different conditions in each case; the ratios in this section are schedule means of per-level ratios rather than the pooled estimand of (10.2), which changes their numerical scale though not their ordering; and adjacent strengths differ by roughly a factor of two, which is the resolution of the whole statement. This collapse is an empirical regularity of the two violation paths constructed here, not a sufficiency theorem, and there is a structural reason to expect it to have limits: even between Gaussians, equal KL does not imply equal score geometry — the divergence is a spectral aggregate, while the score error of a chain approximation is $\text{tr}[(Q_{\text{chain}} - Q_{\text{true}}) \Sigma (Q_{\text{chain}} - Q_{\text{true}})^{\text{T}}]$, which depends on *where* the precision error points. Two perturbations of matched δ whose discrepancies live in different spatial modes could in principle separate, and nothing here tests that. Equation (10.28) is interpolation within

the tested conditions, not a universal tolerance for non-Markov structure.

10.7 Why the shape coordinate is the expensive one

Chapter 9 found the innovation shape converging far slower than the autoregressive coefficient, and attributed it to the fourth cumulant being damped by the channel while the second is preserved. That is a claim about the noised likelihood, and it can be checked directly against Fisher information, independently of any optimiser.

10.7.1 What the cumulant law predicts

Cumulants add under independent sums and a Gaussian contributes only to the second, so for $X_t = e^{-t}A + \sqrt{\Delta_t}\varepsilon$,

$$\kappa_2(X_t) = e^{-2t}\kappa_2(A) + \Delta_t = 1, \quad \kappa_4(X_t) = e^{-4t}\kappa_4(A), \quad \kappa_{11}(X_{t,i}, X_{t,i+1}) = e^{-2t}\alpha, \quad (10.29)$$

Now consider what identifies each coordinate. The autoregressive coefficient is second order: it enters through the lag-one cumulant, which the channel attenuates by e^{-2t} . A shape parameter — one that moves the fourth cumulant at exactly fixed variance — is fourth order, and the channel attenuates its signature by e^{-4t} . Reading information as squared sensitivity over sampling variance — a delta-method heuristic, not a likelihood calculation: it treats the cumulant estimator as the inference channel and its variance as approximately constant in t —

$$I_\theta \asymp \frac{\left(\partial_\theta \kappa(X_t)\right)^2}{\text{Var } \hat{\kappa}}, \quad (10.30)$$

and noting that the denominators are t -independent *to leading order* because $\kappa_2(X_t) \equiv 1$ pins the scale of the noised variable at every t (higher moments of the noised law do drift

with t , which is part of what the heuristic omits),

$$\boxed{I_\alpha(t) \asymp e^{-4t}, \quad I_p(t) \asymp e^{-8t}, \quad \frac{I_p(t)}{I_\alpha(t)} \asymp e^{-4t}.} \quad (10.31)$$

The third statement is the robust one: it is a *contrast*, so whatever common factors the leading-order reading omits cancel from it.

10.7.2 The measurement

The prediction is tested on a family in which shape moves at exactly fixed variance. The innovation law is a generalised Gaussian, $f_p(\eta) \propto \exp(-|\eta/s|^p)$, with s fixed by $\text{Var}(\eta) = \sigma_\eta^2 = 1 - \alpha^2$ at every p , so p is a genuine shape coordinate and not a reparameterised scale; $p = 1$ is the Laplace chain of Chapter 7, $p = 2$ Gaussian. For $\theta = (\alpha, \sigma_\eta^2, p)$, Fisher's identity (8.31) supplies the exact per-sequence score of the marginal log-likelihood from one forward-backward pass, and the information matrix is estimated in outer-product form $J = D^{-1} \sum_d g_d g_d^\top$ over $D = 512$ sequences of $L = 32$ sites at $N_g = 401$, $A = 8$. The quantity reported is the *efficient* information $I_{p,(\alpha, \sigma_\eta^2)} = 1/(J^{-1})_{pp}$, what remains about p once the nuisance coordinates must be estimated too — equal to the diagonal entry only when the coordinates are orthogonal and strictly smaller otherwise.

Two things follow from that construction and travel with every number below. An outer-product matrix is the Fisher information only at the true parameter of a correctly specified model, and it is here a D -sample estimate of it, so these are estimates with sampling error and not an exact information calculation. And the reported quantity is an *inverse*, so its meaning degrades with the conditioning of J — which the run records per cell. That conditioning is benign exactly where the information is large and fails exactly where it vanishes: $\kappa(J) = 61$ at $t = 0.05$ on the Laplace chain, a median of 2776 across the sweep, and $1.3e + 07$ in the worst cell ($t = 1.6$, $p = 3$), where the efficient information is itself down at $7e - 07$. The decay these curves measure is therefore resolved at the end where the coordinate is identified, and the far tail is a statement that the information has gone, not a measurement of how much remains.

Table 10.11: How much efficient Fisher information about each coordinate the channel destroys between the smallest and largest diffusion time tested ($t = 0.05$ against $t = 1.6$), and the contrast (10.31) predicts. Entries are ratios of outer-product estimates at $D = 512$ sequences, so only the order is meaningful and two significant digits are printed for that reason. The last column is the quantity the prediction addresses: its geometric mean over the five shapes is 465 against a predicted $e^{4 \times 1.55} = 493$.

true p	excess kurtosis	loss, α	loss, σ_η^2	loss, p	contrast	loss $_p$ /loss $_\alpha$
1.0	+3.00	78×	235×	8.7×10^3		112
1.2	+1.74	87×	392×	2.0×10^4		236
1.5	+0.76	83×	459×	6.4×10^4		776
2.0	0.00	75×	501×	6.5×10^4		867
3.0	−0.58	56×	522×	6.8×10^4		1222
<i>geometric mean of the contrast</i>						465
<i>predicted by (10.31)</i>						493

Figure 10.5 plots the decay against the predicted rates and Table 10.11 reads the endpoints.

10.8 What the chapter establishes

Table 10.12: Each question, the finding, and the conditions the finding is conditional on. Nothing in the right-hand column is a hedge added after the fact; each is a stated feature of the design that a different design would have to remove.

question	finding	conditional on
Value of the chain, against no structure	6.6–10.9× lower pooled error over 8 sizes; more accurate in all but 2 of 1,520 cells	the schedule-pooled paired estimand of (10.2); both arms budget-selected on validation
Value of the <i>model</i> , against locality	2.3–6.1×; EM–BP ahead in every scored cell; the baseline’s error curve shows diminishing returns over four sizes (a fitted floor and a floorless power law are not separable there) while EM–BP’s is still falling	one screened architecture; four sizes; budget grids bind for both arms; no causal split of the gap
Mixture capacity	no tested capacity beats a single Gaussian innovation; $C=16$ resolved <i>worse</i> than $C=8$ at $N = 128$	this grid, M-step, iteration cap and validation rule; not a statement about capacity in the abstract
Tolerance to non-Markov structure	advantage is monotone in one closed-form covariance scalar across the tested conditions; break-even bracketed between adjacent tested strengths, $\delta^* \in [1.1, 2.2] \times 10^{-2}$ (CNN) and $[2.2, 4.0] \times 10^{-2}$ (MLP) nats/site	nine cells, two mechanisms; exact KL only for the Gaussian arms; a measured regularity, not a derived law

Table 10.12 states the four findings at the strength their evidence supports. Three consequences carry into Chapter 11. Supplying the generating family is worth a large factor at every data budget tested, but most of the headline factor does not survive a baseline

that is merely given locality and weight sharing — which bounds how much of it can be attributed specifically to the Markov model, without decomposing causally what the swap of baselines changes. Across the tested violations, the measured advantage ordered monotonically with a covariance scalar computable before any fitting — an empirical regularity of the two paths probed, useful as a screening variable and not established as a decision rule. And the coordinate on which every non-Gaussian claim in this thesis depends is simultaneously the slowest to converge and the poorest in information per observation, which is why the budget disclosures in this chapter are part of its results rather than an appendix to them.

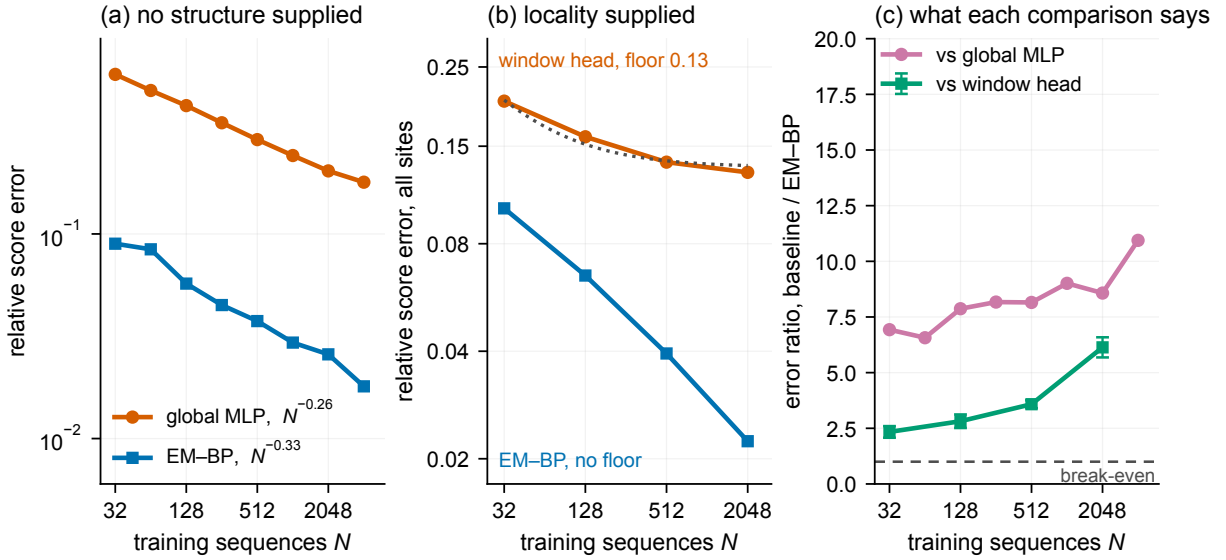


Figure 10.1: The two comparisons, and the difference between them. (a) Headline protocol: schedule-pooled relative score error (10.2) against training-set size for EM-BP and a denoiser given no structure at all, 16 seeds, with the fitted finite-range slopes. (b) All-site protocol: the same quantity for EM-BP and the screened window head, which *is* given locality and weight sharing; the dotted curve is the two-parameter fit (10.22) to the window head, whose fitted floor, 0.13, is indistinguishable from its measured risk at $N = 2048$. Read that as diminishing returns over the four tested sizes and not as a resolved floor: a floorless power law $N^{-0.11}$ fits the same four points with a residual sum only 1.8 times larger at equal parameter count, so two residual degrees of freedom cannot separate a converged curve from a slowly decaying one, and a fitted floor that lands on the last measured point is partly a tautology. EM-BP’s curve, by contrast, is still falling steeply. The two panels are not on a common scale — they use different scoring regions and different test bundles — and no number is carried between them. (c) The two error ratios. The upper curve is the headline 6.6–10.9 $\times$; the lower one is the 2.3–6.1 $\times$ that survives once locality and weight sharing are supplied, and the gap between them is how much of the headline is not attributable specifically to the Markov model. Both grow across the tested sizes, which is a description of eight and four measured points respectively and not a mechanism.

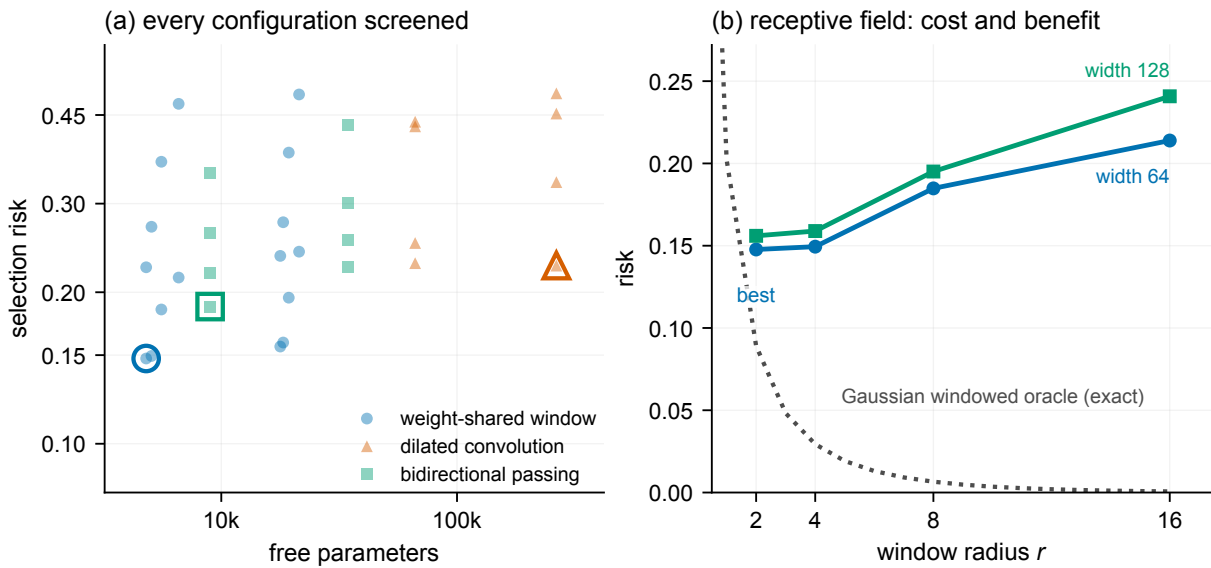


Figure 10.2: (a) Every configuration of the screen, scored on the bulk region with errors pooled over the 4 development seeds and sizes — the selector’s own accumulation; one marker per architecture, the outlined marker being each architecture’s best. The horizontal axis is what makes the comparison readable: the dilated stack sits an order of magnitude to the right and does not reach the window head’s risk anywhere in the screen, so the outcome is not one arm having been starved of capacity. (b) Risk against window radius, at both widths, with the exact Gaussian windowed-oracle error (10.21) on the same axis — the same units, unlike a coefficient-tail fraction, which is deliberately not drawn on a risk axis. The measured curve is minimised at $r = 2$, nearly tied with $r = 4$, and turns upward beyond, as Section 10.4.1 predicts, while the oracle keeps falling geometrically. The vertical gap between the measured heads and the oracle at the same radius is what learning, capacity and the Laplace/Gaussian mismatch cost within a fixed information set.

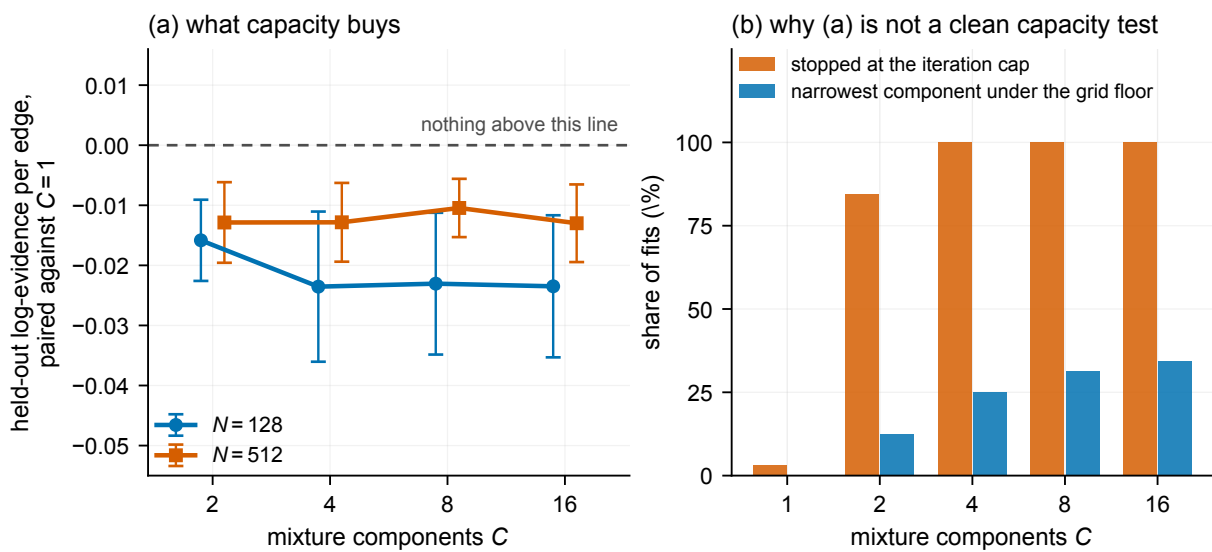


Figure 10.3: (a) Held-out log-evidence per edge, paired within seed against $C = 1$: no tested capacity improves on a single Gaussian innovation at either size, and at $N = 128$ the $C = 16$ deficit is resolved against zero. (b) Why (a) is not a clean capacity test: the share of fits stopped at the iteration cap, and with a narrowest component under the grid's resolution floor, both climb steeply with C .

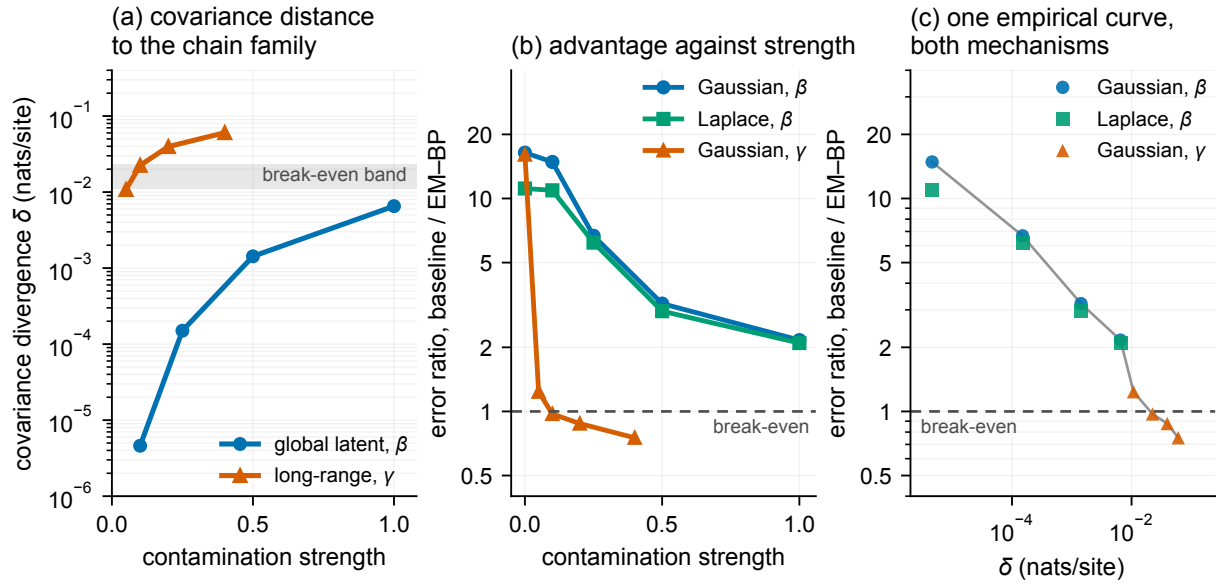


Figure 10.4: (a) The closed-form distance (10.27) to the chain family. The two mechanisms reach the shaded break-even band at strengths that differ by more than an order of magnitude, which is the whole of the difference between them. (b) The measured advantage against strength. Under a global latent it decays smoothly and does not reach the line even where half the marginal variance is a shared constant; under long-range coupling it crosses within the first tested strength. The Laplace and Gaussian curves track each other across the whole rank-one sweep, so the result is not confined to the Gaussian arm; the long-range panel is Gaussian only, since the contaminated Laplace model has no closed-form reference. (c) The same nine points against δ : the two mechanisms interleave and fall on one monotone curve, crossing break-even between the adjacent tested strengths bracketed in (10.28). All curves are schedule means, so they show the direction rather than the worst case, and all fall because the estimator degrades rather than because the baselines improve — Table 10.10 carries EM-BP’s own error alongside.

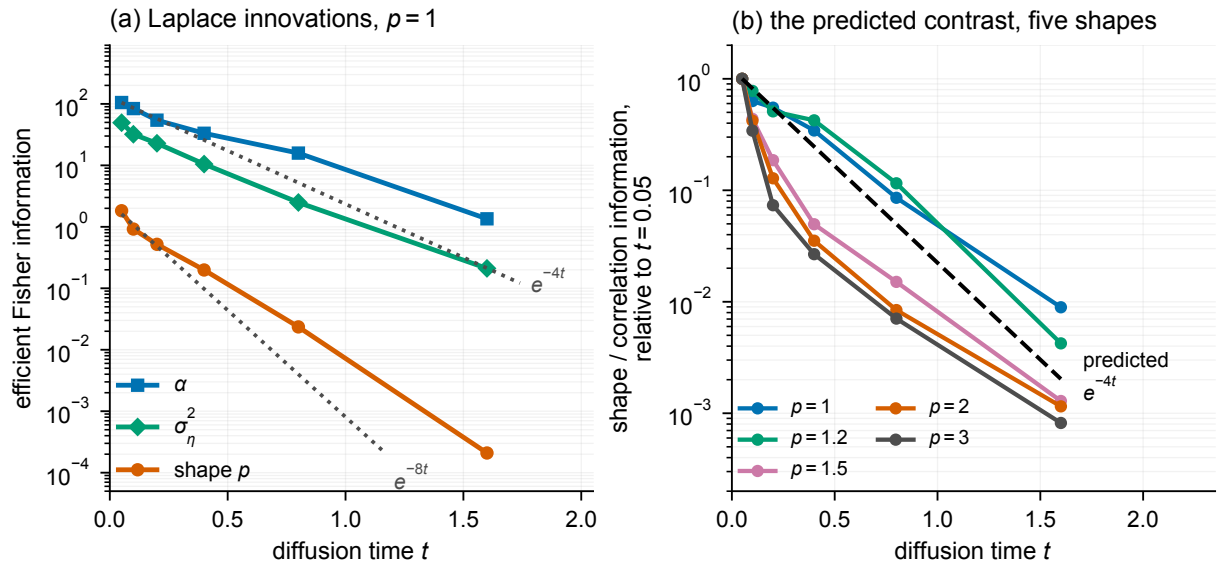


Figure 10.5: (a) Efficient information for each coordinate against diffusion time, at Laplace innovations, with the predicted rates (10.31) as dotted guides. On a linear time axis an exponential is a straight line, so the comparison is read from the slopes. Both measured decays are shallower than predicted by a common offset, which the leading-order reading does not capture. (b) The contrast, which is the prediction that survives the offset: the ratio I_p/I_α for all five shape exponents, normalised at $t = 0.05$, against e^{-4t} . The five curves bracket the prediction, with a mean fitted log-slope of -3.72 against -4 .

Chapter 11

Discussion and Conclusions

11.1 Answers to the research questions

The derivations are in the chapters named; the answers are stated here without repeating them.

RQ1: the joint score and its couplings. Linear and known in closed form, $S(x, t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$. For $\alpha \neq 0$ its couplings have a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one further diagonal per order in t with every off-band entry provably non-zero at every $t > 0$, maximally coupled at intermediate times in the Frobenius measure used to track it, returning to the identity at rate e^{-2t} . A single bulk quantity $q(\alpha, t) \in (0, |\alpha|]$ sets the posterior correlation length. This is a consequence of the model's linearity and not of the chain structure alone (Chapter 5).

RQ2: belief propagation and the Kalman smoother. They are the same computation. The channel corrupts each coordinate independently, so every observation enters through a unary factor, the posterior factor graph stays a tree, and one forward and one backward sweep give all smoothing marginals. On the Gaussian chain the messages close algebraically and the recursion is the Kalman filter followed by RTS smoothing, update by update; the two routes agree to 10^{-15} – 10^{-14} across the tested grid, reproduced by an inde-

pendently written implementation. Tree exactness holds for any innovation law; closure in a finite-dimensional family holds only because the model is Gaussian (Chapter 6).

RQ3: what Gaussian messages cost. Median relative score error 0.27 at $t = 0.05$ on Laplace chains at $\alpha = 0.85$, below 10^{-2} by $t \approx 1$ and below 10^{-3} by $t \approx 2.1$, with direction preserved far better than magnitude. Theorem 7.3 says what that number is: moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model, so the measurement is full posterior inference against the best second-order linear estimator. The *estimator* is therefore fixed by the model class — the same for every innovation law with matching first two moments — while the measured *error* against the true posterior is law-dependent, as the innovation sweep shows: at $\alpha = 0.5$ and $t = 0.05$ it spans 0.07 for a mildly bimodal mixture to 0.94 for a strongly bimodal one at the same second moments (Chapter 7).

RQ4: how fast remote evidence stops mattering. In the bulk of a long chain the influence of evidence at distance d decays exactly as $q(\alpha, t)^d$ — an infinite-chain statement, approximate near the boundaries of a finite one — and the error of the radius- r windowed estimator is measured to decay at a matching geometric rate over the tested radii. These stay two claims: windowing changes the recursion’s boundary conditions in a way the influence argument does not cover, so the second is a numerical law over the tested range. At equal locality budget, truncating the inference beats truncating the score matrix at small t (12.3% against 21.0% at $t = 0.05$) (Chapter 6).

RQ5: recovering an unknown transition. Fisher’s identity turns one forward-backward pass, at each current parameter value, into the exact gradient of the marginal log-likelihood of the functional model, so the E-step is structurally exact for any innovation law and nothing is differentiated through the recursion; the implementation evaluates it on a validated grid, and recovery is demonstrated numerically over the studied regime rather than proved asymptotically. The checks are part of the answer, not a caveat on it: the channel acts unevenly across coordinates, and the fitted innovation shape settles

at a median of 229 updates against 80 for the correlation. A stopping rule read off the correlation trace therefore certifies a kernel that has not converged in the coordinate every non-Gaussian claim depends on, so a fixed iteration budget compares convergence rates and not estimators (Chapters 8 and 9).

RQ6: what supplying the Markov assumption is worth. At an equal number of clean trajectories, $6.6\text{--}10.9\times$ lower schedule-pooled relative score error than networks trained by denoising score matching, across 8 sizes; against a baseline given locality and weight sharing, $2.3\text{--}6.1\times$. The factor is quoted under the aggregation the text defines — squared error and reference energy pooled over the schedule per seed, ratio paired within seed — and Appendix B shows the alternatives, of which the retired mean of per-level ratios is systematically the largest. Three scope conditions travel with those numbers. The arms are asymmetric in supplied information by construction, and in both directions: EM–BP is given the Markov factorisation, while every network arm is given the clean trajectories and redraws the corruption at each gradient step, where EM–BP sees one frozen noised realisation and never the clean signal. So this measures what a correct structural assumption is worth against the standard training recipe for the alternative, per clean trajectory, and it does not rank learning algorithms at a matched information interface. The gap between the two ratios is not a decomposition into architecture and estimation terms, because architecture, capacity, sharing and optimisation change together when the baseline is swapped. And both are ratios of errors at fixed data, not of the data each method needs for fixed error; the curves do not cross in the measured range, so no data-equivalence factor follows (Chapter 10).

11.2 What is new

The exact structure of a chain diffusion score. The joint score of a noised Markov chain is characterised completely for the Gaussian case: its closed form, its eigenbasis at every diffusion time, the law by which its couplings fill and then decay, and the single

parameter that controls posterior influence range. The identification of chain BP with the Kalman–RTS recursion makes the two classical literatures one object rather than two analogous ones.

An exact reading of what Gaussian closure computes. Where previous practice measured the error of moment-projected message passing as the error of an approximation, Theorem 7.3 identifies the estimator it returns. The measured Laplace gap is therefore the price of discarding everything beyond second moments — a property of the model class and not of one implementation. The estimator is shared by every innovation law with the same first two moments; the size of the gap is each law’s own. The validated grid reference that makes the measurement possible is itself a contribution, since the functional recursion has no closed form.

Transition learning, and the measured value of structure. The transition law is estimated from noised sequences alone by exact chain inference inside EM, and the resulting estimator is compared against trained networks at equal data under a protocol that selects both arms on held-out validation. Two boundaries of the advantage are mapped rather than asserted: mixture capacity beyond a single Gaussian innovation buys nothing for this estimator under the grid and cap implemented here, and a chain-structured estimator partly absorbs a rank-one violation of the Markov assumption through its refitted local parameter while failing on a long-range one at a coupling strength of roughly one tenth. The convergence measurement that separates rate from estimator is a methodological result in its own right, and it is what withdrew an earlier reading of the generative comparison.

11.3 Limitations

1. **Model class.** The exact solutions concern a one-dimensional state per frame and linear AR(1) dynamics, extended to a two-dimensional nonlinear state only in Chapter 4. The locality laws are proved for the Gaussian bulk; their non-Gaussian coun-

terparts are only structurally delimited, and no low-rank structure is asserted for the Laplace $L = 2$ posterior Hessian beyond the narrow regime where it holds.

2. **The numerical reference.** The non-Gaussian results are numerical. Grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3h$ on the grid spacing h), and the Laplace closure error is a measured curve rather than a formula.
3. **Optimisation and finite seeds.** The learning results are statistical rather than exact: estimates over a finite number of seeds, reported with paired uncertainty, several resolved at only one of the two training sizes tested. The headline factors are properties of a stated aggregation — the schedule-pooled paired ratio — and move by tens of percent under defensible alternatives (Appendix B), so no exact multiple should be carried out of context. Every capacity cell above $C = 4$ stops at an iteration cap rather than at tolerance, so a statistical disadvantage of larger mixtures is not separated from an optimisation or a discretisation one. The convergence chapter’s fitted slopes are coordinate-wise contraction factors, not Jacobian eigenvalues, and its sample-size table is exploratory. The misspecification study probes exactly two violations of the Markov assumption, with a scalar that is an exact KL only in the Gaussian arms and a covariance proxy in the Laplace ones; it establishes that the two mechanisms are not interchangeable in effect, not a general theory of how much non-Markov structure a chain estimator tolerates.
4. **Generative fidelity, and transfer.** Whether pointwise accuracy and generative fidelity can move in opposite directions within one estimator is unresolved here: the comparison that addresses it evaluates procedures whose fitted shape coordinates are unequally settled, on the single coordinate the generative metric scores, so its reading is withdrawn rather than reversed. More broadly, these are properties of one controlled model. Whether trained score networks on real sequence data inherit them is untested here. The image and video extensions explored in the underlying research package are outside the scope of this thesis and are not reported.

11.4 Future work

Four directions, in the order in which they would most change what this thesis can claim.

The reverse-generation comparison, with converged fits. The withdrawn study integrated the reverse SDE under the estimated and the trained scores, but on fits the convergence analysis shows to be unequally settled. Selecting both arms on a convergence criterion rather than a fixed iteration count would answer the question the experiment was built for, and close the one question this thesis leaves open.

A grid-by-capacity design. The capacity result leaves two mechanisms entangled. A finer grid at fixed capacity would separate fits falling below the grid’s resolution from genuine overfitting, since only the second survives mesh refinement; a budget ladder at fixed capacity would do the same for the iteration cap.

Richer departures from the Markov assumption. Two violations at fixed strengths are probed here. A paired design across strengths, and further violation types, would test whether the finding that the two mechanisms are not interchangeable generalises. Discrete chains are untouched: with a finite alphabet the messages are finite vectors and inference is exact with no closure at all, giving a second solvable benchmark alongside the Gaussian one.

Spatial as well as temporal structure. Whether the structural results survive longer sequences, other innovation laws, and data with spatial structure is the question that decides how far any of this carries beyond the solvable setting. The reading of U-Nets as belief propagation (Section 2.6) suggests where to look: on a hierarchical rather than a chain graph, the same inference argument should apply, with the tree-exactness that makes it work here replaced by an approximation whose cost would have to be measured the way Chapter 7 measures the Gaussian closure.

Bibliography

- Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A. J., Bambrick, J., Bodenstein, S. W., Evans, D. A., Hung, C.-C., O'Neill, M., Reiman, D., Tunyasuvunakool, K., Wu, Z., Žemgulytė, A., and Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500.
- Achilli, B., Benedetti, M., Biroli, G., and Mézard, M. (2026). Theory of speciation transitions in diffusion models with general class structure.
- Achilli, B., Ventura, E., Silvestri, G., Pham, B., Raya, G., Krotov, D., Lucibello, C., and Ambrogioni, L. (2024). Losing dimensions: geometric memorization in generative diffusion.
- Ackley, D. H., Hinton, G. E., and Sejnowski, T. J. (1985). A learning algorithm for Boltzmann machines. *Cognitive Science*, 9(1):147–169.
- Amit, D. J., Gutfreund, H., and Sompolinsky, H. (1985). Storing infinite numbers of patterns in a spin-glass model of neural networks. *Physical Review Letters*, 55(14):1530–1533.
- Anderson, B. D. O. (1982). Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326.
- Bachtis, D., Biroli, G., Decelle, A., and Seoane, B. (2024). Cascade of phase transitions in the training of energy-based models. In *Advances in Neural Information Processing Systems 37*.

- Barahona, F. (1982). On the computational complexity of Ising spin glass models. *Journal of Physics A: Mathematical and General*, 15(10):3241–3253.
- Besag, J. (1974). Spatial interaction and the statistical analysis of lattice systems. *Journal of the Royal Statistical Society: Series B*, 36(2):192–225.
- Biroli, G., Bonnaire, T., de Bortoli, V., and Mézard, M. (2024). Dynamical regimes of diffusion models. *Nature Communications*, 15:9957.
- Biroli, G. and Mézard, M. (2023). Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402.
- Bonnaire, T., Urfin, R., Biroli, G., and Mézard, M. (2025). Why diffusion models don’t memorize: the role of implicit dynamical regularization in training. In *Advances in Neural Information Processing Systems*. Oral presentation; arXiv:2505.17638.
- Brockwell, P. J. and Davis, R. A. (1991). *Time Series: Theory and Methods*. Springer, New York, 2 edition.
- Carbone, D. (2025). Hitchhiker’s guide on the relation of energy-based models with other generative models, sampling and statistical physics: a comprehensive review. *Transactions on Machine Learning Research*.
- Chen, R. T. Q., Rubanova, Y., Bettencourt, J., and Duvenaud, D. (2018). Neural ordinary differential equations. In *Advances in Neural Information Processing Systems 31*.
- Chow, C. K. and Liu, C. N. (1968). Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467.
- Crooks, G. E. (1999). Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences. *Physical Review E*, 60(3):2721–2726.
- Cui, H., Pehlevan, C., and Lu, Y. M. (2025). A solvable model of learning generative diffusion: theory and insights.
- Decelle, A., Fissore, G., and Furtlehner, C. (2017). Spectral dynamics of learning in restricted Boltzmann machines. *EPL (Europhysics Letters)*, 119(6):60001.

- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38.
- Donoho, D. L., Maleki, A., and Montanari, A. (2009). Message-passing algorithms for compressed sensing. *Proceedings of the National Academy of Sciences*, 106(45):18914–18919.
- Doob, J. L. (1942). The Brownian movement and stochastic equations. *Annals of Mathematics*, 43(2):351–369.
- Efron, B. (2011). Tweedie’s formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614.
- Favero, A., Sclocchi, A., Cagnetta, F., Frossard, P., and Wyart, M. (2025a). How compositional generalization and creativity improve as diffusion models are trained. In *International Conference on Machine Learning*.
- Favero, A., Sclocchi, A., and Wyart, M. (2025b). Bigger isn’t always memorizing: early stopping overparameterized diffusion models.
- Gardiner, C. W. (2009). *Stochastic Methods: A Handbook for the Natural and Social Sciences*. Springer, Berlin, 4 edition.
- Gardner, E. (1988). The space of interactions in neural network models. *Journal of Physics A: Mathematical and General*, 21(1):257–270.
- Garnier-Brun, J., Biggio, L., Beltrame, D., Mézard, M., and Saglietti, L. (2026). Biased generalization in diffusion models. Spotlight, International Conference on Machine Learning.
- Garnier-Brun, J., Mézard, M., Moscato, E., and Saglietti, L. (2024). How transformers learn structured data: insights from hierarchical filtering.
- Ghio, D., Dandi, Y., Krzakala, F., and Zdeborová, L. (2024). Sampling with flows, diffusion, and autoregressive neural networks from a spin-glass perspective. *Proceedings of the National Academy of Sciences*, 121(27):e2311810121.

- Goldstein, H., Poole, C. P., and Safko, J. L. (2002). *Classical Mechanics*. Addison-Wesley, San Francisco, 3 edition.
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014). Generative adversarial nets. In *Advances in Neural Information Processing Systems 27*.
- Hinton, G. E. (2002). Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800.
- Ho, J., Jain, A., and Abbeel, P. (2020). Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558.
- Hyvärinen, A. (2005). Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709.
- Ising, E. (1925). Beitrag zur Theorie des Ferromagnetismus. *Zeitschrift für Physik*, 31(1):253–258.
- Jarzynski, C. (1997). Nonequilibrium equality for free energy differences. *Physical Review Letters*, 78(14):2690–2693.
- Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106(4):620–630.
- Jerrum, M. and Sinclair, A. (1993). Polynomial-time approximation algorithms for the Ising model. *SIAM Journal on Computing*, 22(5):1087–1116.
- Kadkhodaie, Z., Guth, F., Simoncelli, E. P., and Mallat, S. (2024). Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations*. Outstanding Paper award.
- Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal*

- of Basic Engineering*, 82(1):35–45.
- Kamb, M. and Ganguli, S. (2024). An analytic theory of creativity in convolutional diffusion models.
- Karras, T., Aittala, M., Aila, T., and Laine, S. (2022). Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 26565–26577.
- Kingma, D. P. and Welling, M. (2014). Auto-encoding variational Bayes. In *International Conference on Learning Representations*.
- Krizhevsky, A., Sutskever, I., and Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems 25*.
- Kschischang, F. R., Frey, B. J., and Loeliger, H.-A. (2001). Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519.
- Lai, C.-H., Song, Y., Kim, D., and Mitsufuji, Y. (2025). The principles of diffusion models: From origins to advances. arXiv:2510.21890.
- Landau, L. D. and Lifshitz, E. M. (1980). *Statistical Physics, Part 1*. Pergamon Press, Oxford, 3 edition.
- LeCun, Y., Bengio, Y., and Hinton, G. (2015). Deep learning. *Nature*, 521:436–444.
- LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., and Huang, F. J. (2006). A tutorial on energy-based learning. In Bakır, G., Hofmann, T., Schölkopf, B., Smola, A. J., Taskar, B., and Vishwanathan, S., editors, *Predicting Structured Data*. MIT Press, Cambridge, MA.
- Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. (2023). Flow matching for generative modeling. In *International Conference on Learning Representations*.
- Malioutov, D. M., Johnson, J. K., and Willsky, A. S. (2006). Walk-sums and belief propagation in Gaussian graphical models. *Journal of Machine Learning Research*, 7:2031–2064.

- Mei, S. (2024). U-nets as belief propagation: Efficient classification, denoising, and diffusion in generative hierarchical models. arXiv:2404.18444.
- Meng, X.-L. and Rubin, D. B. (1993). Maximum likelihood estimation via the ECM algorithm: A general framework. *Biometrika*, 80(2):267–278.
- Mézard, M. (2017). Mean-field message-passing equations in the Hopfield model and its generalizations. *Physical Review E*, 95:022117.
- Mézard, M. and Montanari, A. (2009). *Information, Physics, and Computation*. Oxford University Press, Oxford.
- Mézard, M., Parisi, G., and Virasoro, M. A. (1987). *Spin Glass Theory and Beyond*. World Scientific, Singapore.
- Miyasawa, K. (1961). An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4):181–188.
- Onsager, L. (1944). Crystal statistics. I. A two-dimensional model with an order-disorder transition. *Physical Review*, 65(3-4):117–149.
- Papamakarios, G., Nalisnick, E., Rezende, D. J., Mohamed, S., and Lakshminarayanan, B. (2021). Normalizing flows for probabilistic modeling and inference. *Journal of Machine Learning Research*, 22(57):1–64.
- Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Rauch, H. E., Tung, F., and Striebel, C. T. (1965). Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450.
- Raya, G. and Ambrogioni, L. (2023). Spontaneous symmetry breaking in generative diffusion models. In *Advances in Neural Information Processing Systems*, volume 36.
- Rezende, D. J. and Mohamed, S. (2015). Variational inference with normalizing flows.

- In *Proceedings of the 32nd International Conference on Machine Learning*, pages 1530–1538.
- Risken, H. (1989). *The Fokker–Planck Equation: Methods of Solution and Applications*. Springer, 2nd edition.
- Robbins, H. (1956). An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 157–163.
- Robert, C. P. and Casella, G. (2004). *Monte Carlo Statistical Methods*. Springer, New York, 2 edition.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.
- Ronneberger, O., Fischer, P., and Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241.
- Sclocchi, A., Favero, A., Levi, N. I., and Wyart, M. (2025a). Probing the latent hierarchical structure of data via diffusion models. In *International Conference on Learning Representations*. arXiv:2410.13770.
- Sclocchi, A., Favero, A., and Wyart, M. (2025b). A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 122(1):e2408799121.
- Smolensky, P. (1986). Information processing in dynamical systems: Foundations of harmony theory. In Rumelhart, D. E. and McClelland, J. L., editors, *Parallel Distributed Processing: Explorations in the Microstructure of Cognition, Volume 1: Foundations*, pages 194–281. MIT Press, Cambridge, MA.
- Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., and Ganguli, S. (2015). Deep un-

- supervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 2256–2265.
- Song, Y. and Ermon, S. (2019). Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. (2021). Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*.
- The Royal Swedish Academy of Sciences (2024). The Nobel Prize in Physics 2024. <https://www.nobelprize.org/prizes/physics/2024/summary/>. Last accessed 19 July 2026.
- Thouless, D. J., Anderson, P. W., and Palmer, R. G. (1977). Solution of ‘Solvable model of a spin glass’. *Philosophical Magazine*, 35(3):593–601.
- Tubiana, J. and Monasson, R. (2017). Emergence of compositional representations in restricted Boltzmann machines. *Physical Review Letters*, 118:138301.
- Uhlenbeck, G. E. and Ornstein, L. S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5):823–841.
- van den Oord, A., Kalchbrenner, N., and Kavukcuoglu, K. (2016). Pixel recurrent neural networks. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1747–1756.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.
- Vincent, P. (2011). A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674.
- Weiss, Y. and Freeman, W. T. (2001). Correctness of belief propagation in Gaussian graphical models of arbitrary topology. *Neural Computation*, 13(10):2173–2200.
- Zdeborová, L. and Krzakala, F. (2016). Statistical physics of inference: thresholds and

algorithms. *Advances in Physics*, 65(5):453–552.

Acknowledgements

I thank my supervisor, Marc Mézard, for proposing that belief propagation and diffusion models be made to meet, and for the judgement, at several forks, of which question was the one worth asking. I owe a particular debt to Jérôme Garnier-Brun, who followed this work week by week: many of the checks that this thesis now treats as non-negotiable — validated references, paired comparisons, convergence measured rather than assumed — entered it through his questions.

The experiments were run on the computing cluster of Bocconi University, and I thank the university for access to it and its staff for keeping it running.

Finally, I thank my family, to whom this thesis is dedicated, and my friends, who heard a great deal about Markov chains with patience.

Appendix A

Reproducibility

A.1 Figure-to-code map

Every computed figure in this thesis is built by a script in this folder’s `tools/`: `make_thesis_figures.py` and one `fig_chNN.py` per results chapter, all styled through `tools/figstyle.py` (one palette and layout module, so every figure looks like it came from the same hand; text is embedded as TrueType). The closed-form panels are recomputed from the formulas at build time; the measured panels read the frozen research outputs under `research/outputs/frozen/` directly — the same files the generated tables read, so a figure and the table beside it cannot disagree about a number. Randomness, where used, is seeded. Seven figures are reproduced from the literature with attribution in their captions and are not built here (`fig_song_dog`, `fig_song_sde`, `fig_bachtis_cascade`, `fig_bonnaire_memorize`, `fig_biroli_regimes`, `fig_mezard_bp_messages`, `fig_unet_architecture`); two are *TikZ* schematics drawn in the source.

Table A.1: Every computed figure, the file it is committed as, and the script that builds it. Scripts marked *frozen* read the frozen research outputs; the others recompute closed forms.

Figure	File	Built by
2.3	fig_forward_corruption.pdf	make_thesis_figures, closed forms
3.1	fig_marginal_blindness.pdf	make_thesis_figures, closed forms
4.1	fig_ring_model_dynamics.pdf	make_thesis_figures, simulation
4.2	fig_ring_model_channel.pdf	make_thesis_figures, simulation
4.3	fig_ring_gauge.pdf	make_thesis_figures, simulation
5.1	fig_eigen_relax.pdf	fig_ch05, closed forms
5.2	fig_sparsity_lifecycle.pdf	fig_ch05, closed forms
5.3	fig_markov_survival.pdf	fig_ch05, closed forms
6.1	fig_truncation.pdf	fig_ch06, closed forms
7.1	fig_laplace_nonlinear.pdf	fig_ch07, quadrature
7.2	fig_laplace_metrics.pdf	fig_ch07, exp. 02, frozen
7.3	fig_laplace_closure.pdf	fig_ch07, exp. 02, frozen
8.1	fig_em_grid.pdf	fig_ch08, exp. 01/18, frozen
8.2	fig_em_diagnostics.pdf	fig_ch08, exps. 18 and 06, frozen
8.3	fig_em_innovation.pdf	fig_ch08, exp. 06, frozen
9.1	fig_em_two_rates.pdf	fig_ch09, exp. 27, frozen
10.1	fig_value_of_structure.pdf	fig_ch10, exp. 07/31, frozen
10.2	fig_screening.pdf	fig_ch10, exp. 31 screen, frozen
10.3	fig_capacity.pdf	fig_ch10, exp. 32, frozen
10.4	fig_nonmarkov.pdf	fig_ch10, exp. 21 clean rerun
10.5	fig_information.pdf	fig_ch10, exp. 22, frozen

A.2 The learning experiments of Chapters 8 and 10

The closed-form and quadrature studies above are deterministic and run on a laptop. The estimator comparisons are neither, and they are gated differently.

Provenance. Runs deployed by `hpc/deploy_clean.sh` are certified: the script refuses to build from a dirty tree, ships a `git archive` of the commit rather than the working tree, records the SHA-256 of that tarball, and extracts on the cluster under the commit id so that a run cannot overwrite or edit its own source. The motivating failure is preserved in that script’s header: an earlier two-step stamp-then-rsync path produced `outputs/exp_12_scaled/`, whose recorded commit cannot have run the recorded command, and whose numbers are therefore not recoverable and are not cited in this thesis. `tools/provenance_gate.py` intersects a run’s recorded dirty-file list with the import closure of its own experiment module, so that uncommitted work elsewhere in the tree is distinguished from uncommitted work on the code path that produced the numbers. That distinction does the work below, because the two are not equally serious.

How each run is certified. Not every experiment here was run after `deploy_clean.sh` existed, so four routes appear, in decreasing order of strength. Nothing cited in this thesis hides which route it took. The development record behind each route, which deployments failed, and why each check was added, is in the archived technical compendium.

Certified. `exp_31`, the architecture screen and the confirmatory comparison, and `exp_32`, capacity, were deployed by the script and pass `require_clean` unmodified. Every number in Sections 10.4 and 10.5 comes from them.

Legacy digest, empty intersection. `exp_07`, the headline efficiency comparison, its validation-selection audit and its budget calibration, predates the script. Its records carry a commit, a host, a job id, a resolved configuration and a dirty-file list, but no archive digest, so the plain gate refuses it. What makes it citable is narrower and checkable: the intersection of each run’s recorded dirty list with its own import closure is *empty*, so no uncommitted edit could have reached the code that produced the numbers. `make_tab_efficiency.py` recomputes that intersection on every run and refuses if it is ever non-empty, not resting on a check performed once by hand. The one `exp_07` row where the intersection was *not* empty, $N = 8192$, whose dirty list contained the experiment, the frozen configuration and `src/em.py`, is withdrawn and appears nowhere in this

thesis.

Recorded configuration only. The validation and diagnostic runs — `exp_01` and `exp_18` (grid validation and diagnostics), `exp_02` (the Gaussian-closure sweep), `exp_06` (parameter recovery), `exp_22` (the Fisher-information curves of Section 10.7) and `exp_27` (the settling times of Section 9.3) — predate even the commit recording: their `params` files carry the resolved configuration, host, timestamp and, where run under SLURM, the job id, but an empty revision field. They are reproducible by rerunning the committed experiment code under the recorded configuration (the quadrature studies are deterministic; the rest record their seeds), and the claims that rest on them are stated at matching strength in the chapters — measured curves and orderings, not certified point values. This is the weakest of the four routes and is disclosed as such.

Certified after rerun. `exp_21`, behind Table 10.10 and Figure 10.4, is reported from `outputs/frozen/exp_21_clean`: fifteen cells from two git-archive deployments, every one with an empty dirty list. It replaces an earlier run whose recorded commit did not reconstruct the program, and the Gaussian arm reproduces that run on every number quoted here, to the precision printed. Because the Laplace and Gaussian arms were deployed separately, the rerun spans two commits, so `make_tab_nonmarkov.py` refuses unless no single cell pools commits *and* every file in the experiment’s sixteen-file import closure is byte-identical between them. Both hold, and the check runs on every regeneration.

Generated numbers. No efficiency, structured-baseline or aggregation number in this thesis is typed. Each is defined once as a macro in `thesis/sections/`, written there by one of the thirteen generators below, each reading the recorded outputs directly. Nine live under `research/tools/`; three are thesis-side scripts that emit the macros their own captions quote (`fig_ch10`, `make_laplace_oracle` and `make_build_manifest`). `make_ref_energy` is the fourteenth name and writes no macro at all: it recovers the per-level reference energies of the frozen test bundle that the pooled estimand needs, and the others read what it produces.

```

make_tab_efficiency  make_aggregation_robustness
make_tab_structured  make_aggregation_structured
make_tab_screening   make_convergence_numbers
make_tab_capacity     make_grid_diagnostics
make_tab_nonmarkov    make_ref_energy
fig_ch10              make_laplace_oracle
make_build_manifest

```

The prose then cites the macro rather than the value, so tables, figures and text share a size grid and an aggregation rule by construction and not by inspection.

What Table 10.3 covers. Five qualifications, kept here and not in the caption, which had grown into a methods subsection set in small type. The estimand is the schedule-pooled relative error of Eq. (10.2): per-level squared error and reference score energy are pooled before the square root, per seed, with the per-level reference energies of the frozen test bundle recovered deterministically by `tools/make_ref_energy.py` (the bundle is shared across seeds, so those energies are twelve numbers, validated against `exp_31`’s stored sums). Both arms select their optimisation budget on a disjoint validation bundle, the network its parameterisation *and* training length, EM–BP its iteration count, and that choice agrees with the test-set optimum in 81.1% and 77.6% of cells respectively, so the protocol is symmetric and the selection is close to an oracle without being one. The $N = 4096$ row uses a raised budget, the caps having been set for $N \leq 2048$; rerunning $N = 2048$ at that budget on the same seeds moves its pooled ratio by $+0.53 \pm 0.20$ — small against the ratio’s own size, and in the direction that makes the capped table conservative for the structured arm, so the caps do not carry the comparison. The resolution gate excludes 16 of 1,536 cells, because in those the narrowest fitted mixture component is under two grid cells wide; including them changes no ratio by more than 0.03. The network wins in 2 of the 1,520 reported cells, both at the smallest sizes — Appendix B resolves where the advantage is largest.

What Table 10.7 covers. The confirmatory run scores three regions per cell — all, bulk, centre — of which the table displays two; the third is the predeclared interior slice, and it is the one the architecture screen selects on. That is why the 192 scored cells are $16 \times 4 \times 3$ and not the $16 \times 4 \times 2$ a two-column table suggests. The count is a descriptive diagnostic and not a replication count: regions are a scoring choice applied after fitting, and sizes share nothing but the seed, so the inferential unit behind each displayed row is the 16 paired seeds. Both arms are checkpointed and selected on a validation bundle disjoint from the test bundle, and neither bundle is shared with training.

The architecture screen. Table 10.6 reports it in full. Its source is `screening.csv` under `outputs/frozen/exp_31_screen/` at commit 569a67c, which is the artifact the confirmatory run names in its own recorded configuration (`winners=auto:` that path) rather than a reconstruction. 32 configurations over 4 development seeds, at two training sizes, with learning rates 0.001, 0.003, both output parameterisations, and a checkpoint ladder from 500 to 64000 steps shared across all three architectures. Selection pools squared error and reference energy over the development seeds and sizes before the square root, scored on the bulk region — the same accumulation as the selector inside the confirmatory job, which is what makes the screen table document the screen that ran; the development seeds are numbered from 10000 and share no seed with the confirmatory set, so the reported interval is not a selection artefact. The winner is the window head at radius 2, width 64, learning rate 0.003, eps parameterisation. Parameter counts across the grids are 4.8k–21.4k (window), 9.0k–34.3k (message passing) and 66.4k–264k (dilated convolution): the arm with the most capacity placed last, which is the relevant check on whether the comparison was tuned evenly.

The convergence sweep. The settling times of Section 9.3 come from `outputs/frozen/exp_27_*`, run on 16 August 2026 under SLURM job 628601 and its shards — a recorded-configuration-only run in the sense above, so the gate refuses it by default and the generator passes `allow_legacy`; the route disclosure above goes with it. The quoted

numbers come from the 800-iteration shard, the only run length in the sweep at which no configuration settles at its own cap; the 400-iteration shard is censored by its cap and its settling times are lower bounds, so it is used only for the count of configurations that miss a 40-iteration budget, which censoring cannot inflate. The measurement replaces a figure of “on the order of 2000 iterations” that appeared in earlier drafts and was never measured: it had been carried over from a diagnostic that ran a fit *out to* 2000 iterations to show its per-edge gain had plateaued.

The transition array’s normalisation. Stated in Section 8.3 and repeated here because it determines which probabilistic object the M-step optimises: columns are *not* renormalised under the quadrature weights, so the fitted object is the continuum kernel read through a quadrature rule and the surrogate carries no θ -dependent log-normaliser. The column-mass residual is the measured distance to a normalised finite-state kernel.

Excluded inputs. $N = 8192$ is excluded from both Table 10.3 and Figure 10.1: the intersection of its dirty list with `exp_07`’s import closure contains the experiment, the frozen configuration and `src/em.py` themselves, so the run is not recoverable. The $N = 4096$ rows are retained: their intersection is empty, so no edit in that tree could have moved those numbers.

A.3 Environment

Python ≥ 3.10 with NumPy, SciPy, pandas and Matplotlib. The neural baselines of Chapter 10 are implemented in pure NumPy (`src/nnet.py` for the fully connected denoiser, `src/seq_nets.py` and `src/local_head.py` for the locality-respecting architectures) with an explicit `numpy.random.Generator` driving initialisation and minibatch order, so training is stochastic but reproducible from the recorded seed. Two optional dependencies are used and neither is on the exactness path: CuPy, selected by the `BP_DEVICE` environment variable, runs the same single BP implementation on a GPU, `tests/test_backend_parity.py` asserts the two devices agree, so a device change fails

a test rather than moving a number quietly, and PyTorch, imported lazily in `src/fid.py` alone, for the generative metric. Arithmetic is float64 throughout.

The sweeps of Chapters 8 and 10 run under SLURM from the batch scripts in `hpc/`, one per experiment. The document compiles with a single run of `tectonic main.tex`.

Those are version floors rather than a lock. They do not pin the BLAS and LAPACK implementations behind NumPy, the SciPy special-function versions, or the T_EX bundle, and each of those can move a last digit; the parity test above catches a device change but not a library one. Byte-level reproduction of the frozen outputs is therefore not claimed — only that the pipeline is deterministic given its seeds on a fixed installation, which is what the digests below record.

A.4 Build manifest

The consistency checks this document runs are internal to it, so they cannot tell a reader which sources produced it. This manifest can. It is written at build time by the generator `tools/make_build_manifest.py`, and records the state the artefact was made from:

repository revision	0a8dde0c526f
thesis subtree	last touched 0a8dde0c526f, clean
research subtree	last touched 0a8dde0c526f, clean
generated inputs	f62c391c8e5d7e5b over 19 files
figures	27187a0bfc7d18a1 over 28 files
previous <code>main.pdf</code>	92b69ba834230678
engine	Tectonic 0.16.9

This build was made from the working tree, where the documents sit beside the research package. The documents and the research package live in one repository either way, so the revision is a single fact; the rows below it record the last commit that touched each subtree and whether that subtree carries uncommitted edits, both path-limited, so unrelated work elsewhere in the tree flags neither. Where the repository root *is* the thesis, and in any

repository whose history is a single commit, the three values coincide — not a degenerate manifest but an accurate one: there is only one revision to name.

Three honest caveats. The PDF digest is necessarily that of the build before this one, since writing the manifest changes the input to the current build; it fixes the artefact one step back rather than pretending to hash itself. The recorded revision is likewise one step behind in a repository with history, since a manifest committed in a commit cannot name that commit; in a single-commit publication it names that commit itself. And a manifest fixes the *source* side only: it says which code and which generated inputs produced this document, not that the frozen experiment outputs are themselves certified — that is the route-by-route disclosure of Section [A.2](#), empty revision fields included.

Appendix B

How the Reported Ratios Were Aggregated

A ratio is a nonlinear function of two means, so it depends on the order in which the averaging is done: $\mathbb{E}[X]/\mathbb{E}[Y]$ is not $\mathbb{E}[X/Y]$, and the gap between them grows with the spread. Both ratios reported in this thesis therefore need the same question asked of them, namely whether the number is a property of the estimator or of the estimand chosen to summarise it.

Both were recomputed under four alternative aggregations of the same frozen cells, refitting nothing. The outcome, in one sentence each. For the unstructured comparison the direction is the same under every aggregation; the reported estimand is the schedule-pooled paired per-seed ratio that Eq. (10.2) defines, and the mean of per-level ratios — which earlier drafts reported — is systematically the *largest* of the alternatives (7.0–17.6× against the pooled 6.6–10.9×), which is why naming the estimand matters. For the structure-aware comparison the reported estimand is *not* uniformly the largest, but the floor across all four aggregations is 1.98× with a spread of 0.48, so the reading of a low single-digit multiple at the smallest training set growing to roughly six at the largest survives every one of them.

Both recomputations are reproduced in full below, so that the robustness claim can be checked against the numbers without leaving this document. They are generated

by `tools/make_aggregation_robustness.py` and `tools/make_aggregation_structured.py` from the same frozen outputs that produce the headline table, so the two cannot drift apart; the pooled column of the first reproduces Table 10.3 exactly, which is the consistency check that the alternatives are recomputations of the same cells and not of a different experiment. The same tables, with the longer protocol discussion around them, also appear in the archived technical compendium.

B.1 Does the headline depend on how the ratio was averaged?

A ratio is a nonlinear function of two means, so its value depends on the order of averaging. Table 10.3 reports the schedule-pooled paired per-seed ratio: squared score error and reference score energy are summed over the twelve noise levels before the square root, the ratio is formed within each seed, and the seed is the inferential unit — the twelve levels share a training set and a fitted model. Being the definition stated in the text does not make it the only defensible summary, so here is the same data under four alternatives, with a paired interval on the one the table reports.

Table B.1: The headline under five estimands. (0) is what Table 10.3 reports; (1) is the mean of per-(seed, level) ratios, the summary earlier drafts reported. The interval is a paired bootstrap over seeds, 20,000 resamples; seeds are resampled rather than cells, because cells within a seed share a fitted model.

N	(0) pooled	(1) per-cell	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (0)
32	6.9	7.0	6.8	6.1	6.8	[6.4, 7.5]
64	6.6	7.8	6.6	6.7	6.6	[5.6, 7.6]
128	7.9	8.8	8.0	7.8	8.6	[7.0, 8.8]
256	8.2	9.7	8.8	8.6	9.2	[7.3, 9.0]
512	8.2	9.0	8.6	7.9	8.2	[7.2, 9.2]
1024	9.0	12.5	9.7	10.8	11.1	[7.7, 10.4]
2048	8.6	15.5	10.0	13.2	12.9	[7.4, 9.9]
4096	10.9	17.6	12.4	15.6	16.0	[9.4, 12.5]

Two things follow.

The conclusion does not depend on the choice: no estimand at any size falls below 6.1, every bootstrap interval sits well clear of 1, and no aggregation produces a reversal anywhere.

But the estimands are far from interchangeable, and the mean of cellwise ratios (1) is the largest of the five at every size, reaching 17.6 at the largest size where the pooled estimand reads 10.9. The reason is mechanical: a per-level ratio at a level with a small reference score norm can be very large while contributing little energy, and averaging ratios weights all levels equally. Pooling the energies first weights each level by how much score there is to get wrong, which is exactly what the pooled definition specifies. Earlier drafts reported (1) as the headline; the range 7.0–17.6 they quoted is therefore an artefact of that estimand, and the pooled range 6.6–10.9 replaces it. The ratio of per-seed averaged errors (2) runs 6.6 to 12.4.

No inequality orders these summaries in general: Jensen gives $\mathbb{E}[1/Y] \geq 1/\mathbb{E}[Y]$, but $\mathbb{E}[X/Y] \geq \mathbb{E}[X]/\mathbb{E}[Y]$ does *not* follow for arbitrary positive correlated X and Y . The observed ordering is a property of these cells, which is exactly why the estimand has to be named rather than assumed.

Resolving the average over noise levels shows where any single number hides structure: the per-level ratio is not flat in t but peaks near $t \approx 0.5$ and falls at both ends. Two different “weakest” quantities have to be kept apart. The weakest seed-averaged (N, t) aggregate in the grid is 3.6; the weakest *individual* cell is 0.92, and EM-BP is beaten outright in 2 of the 1,520 cells summarised here. The estimator’s advantage is largest at moderate noise, where the posterior is dominated by neither the likelihood nor the prior — which is where knowing the transition structure should help most.

Table B.2: Table 10.7's all-site ratio under four aggregations of the same cells, with the paired bootstrap interval over seeds. (1) is what the table reports. The conclusion does not depend on the choice: no estimand at any size falls below $1.98\times$, and the largest departure from the reported value is 0.48.

N	(1) per-seed	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (1)
32	2.34	2.00	2.15	1.98	[1.88, 2.86]
128	2.82	2.45	2.61	2.62	[2.29, 3.44]
512	3.58	3.43	3.51	3.59	[3.25, 3.91]
2048	6.14	5.66	5.89	5.93	[5.30, 7.02]